Uniform spanning trees of expanders and complete graphs versus nonnegative probabilistic circuits

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Abstract

Let G be a 2-connected graph with N vertices, M edges, maximum degree at most D and edge expansion at least $h > 0$, and let P be its uniform spanning tree, viewed as a distribution on the M edge indicators. We prove that every smooth decomposable probabilistic circuit with nonnegative weights and leaves of arity at most $2M/3$ whose distribution is within total variation $\varepsilon < 1$ of P has size at least $(1-\varepsilon)^2 e^{cN}$, with an explicit $c = c(D,h) > 0$ that is polynomial in h for fixed D, and at least $8 \cdot 10^{-8}$ for 6-regular Ramanujan graphs. The bound assumes neither determinism nor a condition on girth. It applies, with high probability, to random d-regular graphs for every fixed $d \geq 3$, and to the Ramanujan graphs of Lubotzky, Phillips and Sarnak. By the matrix-tree theorem, P has a generating circuit of size polynomial in N, and all its marginals are ratios of determinants. Since $M = \Theta(N)$, the optimal size of nonnegative smooth decomposable circuits for these distributions is $2^{\Theta(M)}$ at every fixed error below one. The same argument gives $e^{\Omega(N)}$ lower bounds for circuits and monotone arithmetic circuits whose support is exactly the set of spanning trees. For the complete graph K_n, with $M = \binom{n}{2}$ variables and unbounded degree, every circuit of the first kind within total variation $\varepsilon < 1$ of the uniform spanning tree has size at least $(1-\varepsilon)^2 e^{n/146}/(e^3\sqrt{n})$, and the optimal size is $2^{\Theta(n)}$; earlier lower bounds for K_n concerned exact representation or arbitrarily close approximation. The same method gives $2^{\Omega(n)}$ for every graph whose Laplacian spectral gap is at least κn, with a rate depending only on κ, while two cliques joined by a perfect matching show that minimum degree $n/2$ is not enough for it.

The proof bounds, for every balanced split of the edges, the largest Bhattacharyya coefficient between P and a product distribution across the split. We pass to two bits per vertex with edges on both sides, recording whether the tree avoids that vertex's edges on each side, and bound the square-root matrix of their joint law with an adaptive chain rule. The conditional lower bound the chain rule needs comes from Kirchhoff's formula and Rayleigh monotonicity, after conditioning on the stars of a sparse set of vertices; a telescoped flow bound controls it by the spectral gap of the graph left after that set is deleted, and a random deletion with a Peierls-type count over connected sets keeps that graph an expander when G is 2-connected. Some restriction beyond expansion is needed because the uniform spanning tree factorizes over blocks. For bounded-degree expanders we determine exactly when balanced splits give a bound. The degree of our polynomial in h grows with D, and 2-connected hub graphs show that this growth is forced for any bound from the star bits we use; bits for pairs of edges at a vertex give $h^3/(10935D^5)$ on those graphs, and we conjecture that such bits give $c \geq 1/\operatorname{poly}(D, 1/h)$ for all 2-connected bounded-degree expanders. The information-theoretic tools are known: the chain rule is a case of an exercise in Polyanskiy and Wu, and the identification of the optimized affinity with a matrix norm is due to Lapidoth and Pfister. The circuit reduction is from our earlier paper on dimers. For K_n, whose independent sets have one vertex, we instead condition on the tree inside a set of a constant fraction of the vertices,

use a star-deletion bound that is sharp on complete graphs and the exact law of the vertices that this tree leaves isolated, and average over the conditioning. A second, local argument with two bits per short cycle gives much better constants for prisms and squares of expanders (about $3 \cdot 10^{-7}$ for a cubic example) and an $e^{\Omega(L)}$ bound for the $L \times L$ grid. The two arguments are the circuit and cocircuit cases of one argument for uniform bases of matroids. We prove a local form of it for every matroid and a global form for balanced matroids, and obtain $2^{\Omega(n)}$ bounds for a family of binary matroids that are neither graphic nor cographic. Finally, we show that subtraction does not help structured-decomposable circuits. On every 2-connected bounded-degree expander they need size $2^{\Omega(N)}$, without smoothness, at every fixed error $\frac{1}{2}\|P - Q\|_1 \le \varepsilon < \frac{1}{2}$, and at every fixed $\varepsilon < 1$ if the output Q sums to 1, even though Q may take negative values; the threshold $\frac{1}{2}$ is sharp only for unnormalized outputs. One inequality for disjoint permutation blocks of P gives both bounds, and normalized squares of structured circuits obey the nonnegative bound. For unstructured decomposable circuits with subtraction we prove lower bounds only for restricted classes: on bounded-degree expanders, support-exact representation needs size $2^{N^{\Omega(1/d)}}$ at product-depth d and $N^{\Omega(\log N)}$ for formulas, by transferring arguments of Raz and of Raz and Yehudayoff through a large complete minor, and sums of r structured circuits on unrelated vtrees need total size $2^{\Omega_r(N)}$, both for support-exact representation and at error $\varepsilon < \frac{1}{2}$, though with astronomically small constants. The general question stays open. For exact representation it is equivalent to the syntactically multilinear complexity of the spanning-tree polynomial. A small approximating circuit must cancel all but an exponentially small fraction of its mass.

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1 Introduction

Probabilistic circuits (PCs) represent a distribution on $\{0,1\}^M$ by an arithmetic circuit with nonnegative weights. Smoothness and decomposability are the structural conditions under which marginals can be computed in time linear in the circuit size [CVdB20]. Which tractable distributions fall outside this class? We study the uniform spanning tree. By Kirchhoff's matrix-tree theorem [Kir47] its generating polynomial is a determinant, so it has a polynomial-size generating circuit with subtraction, in the sense of Zhang, Juba and Van den Broeck [ZJdB21]. Exact representation of uniform spanning trees of the complete graph by decomposable circuits needs exponential size [MM14, ZHdB20]. Those results concern exact representation or vanishing error on the complete graph, where the number of variables is quadratic in the number of vertices.

We prove lower bounds at constant total-variation error, without assuming determinism, for 2-connected bounded-degree expanders, the complete graph and dense graphs with a linear spectral gap. The two arguments behind these bounds are dual cases of one argument about matroids.

1.1 Main result

For a graph G, the edge expansion is $h(G) = \min\{|\partial S|/|S| : 0 < |S| \le |V(G)|/2\}$, where ∂S is the set of edges with exactly one endpoint in S. A graph is 2-connected if it has at least three vertices and stays connected after deleting any one vertex. Circuit terms are defined in Section 2.1. The size of a circuit is the number of nodes plus the number of wires.

Theorem 1.1 (main theorem)**.** *Fix an integer $D \ge 2$ and a real $h > 0$. Let G be a 2-connected simple graph with N vertices, M edges, maximum degree at most D and edge expansion $h(G) \ge h$,*

and let P be the uniform distribution on the spanning trees of G, viewed as a distribution on $\{0,1\}^{E(G)}$. Let C be a smooth decomposable probabilistic circuit with nonnegative weights over the M edge variables, whose leaves have arity at most $2M/3$, and let Q_C be its normalized distribution. If $\mathrm{TV}(P, Q_C) \le \varepsilon < 1$, then

$$|C| \;\ge\; (1-\varepsilon)^2\, e^{cN}, \qquad c = c(D,h) = \frac{h_1^{5D+4}}{(D+1)^{4(D+1)}\, e^{10(D+1)}}, \qquad h_1 = \min(h,1).$$

If moreover the Laplacian of G has second smallest eigenvalue $\lambda_2(L_G) \ge \lambda > 1$, the same holds with

$$c = c_\lambda(D) = \frac{\lambda\,(D-1)\,(1-1/\lambda)^D}{3D^3(D+\lambda/2)(D^2+1)}.$$

The leaf-arity condition cannot be dropped: a single leaf of arity M can store P. The bound $2M/3$ comes from the circuit reduction (Lemma 3.9). Girth plays no role. Theorem 1.1 applies, with probability tending to one, to random d-regular graphs for every fixed $d \ge 3$, to the Ramanujan graphs of Lubotzky, Phillips and Sarnak [LPS88], and to the prisms and squares of bounded-degree expanders (Corollary 8.1).

Corollary 1.2 (separation)**.** *Fix D, $h > 0$, $\varepsilon < 1$, and a family of graphs G as in Theorem 1.1.*

(a) Every nonnegative smooth decomposable PC with leaves of arity at most $2M/3$ that approximates P within total variation ε has size $2^{\Omega(N)} = 2^{\Omega(M)}$. Some such circuit has size $2^{O(M)}$, so $2^{\Theta(M)}$ is the optimal size.

(b) The generating polynomial $\sum_x P(x) \prod_{e:x_e=1} z_e$ has an arithmetic circuit with subtraction of size polynomial in N, and every marginal probability $P(x_S = a)$ can be computed in polynomial time (Proposition 8.6).

For fixed D the constant $c(D,h)$ is polynomial in h, with $\log(1/c) = O(D\log((D+1)/\min(h,1)))$; the previous version of this paper had $c \le D^{-4D/h}$, exponentially small in D/h. For $D = 3$ and $h = 1/5$ the closed form gives $c \approx 5 \cdot 10^{-41}$ and the optimized parameters of Proposition 4.12 about $5 \cdot 10^{-24}$, compared with $9 \cdot 10^{-44}$ before (Table 1). The spectral constant needs no small parameter at all: for 6-regular Ramanujan graphs, $\lambda_2 \ge 6 - 2\sqrt{5}$ gives $c \ge 8.0 \cdot 10^{-8}$, and $c_\lambda(D) \ge 2.7 \cdot 10^{-9}$ for all d-regular Ramanujan graphs with $6 \le d \le 64$. For Ramanujan graphs $c_\lambda(D)$ decays only polynomially in D, like D^{-4}. The degree of the polynomial in h in $c(D,h)$ grows with D, and for the star bits behind it this is unavoidable: on 2-connected hub graphs they certify at most $O(h^{D/2+1})$ for some balanced split (Corollary 11.12). A second, local argument gives much better constants on two explicit families. The prism $H \square K_2$ consists of two copies of H joined by a perfect matching between corresponding vertices. The square H^2 has the vertices of H, with an edge between any two vertices at distance one or two in H.

Corollary 1.3 (prisms and squares)**.** *Fix an integer $\Delta \ge 2$ and a real $h_0 > 0$. Let H be a connected simple graph with at least three vertices, maximum degree at most Δ and $h(H) \ge h_0$, and let G be $H \square K_2$ or H^2. Then G is 2-connected, and the conclusion of Theorem 1.1 holds with*

$$c = \frac{\min(1,h_0)}{24\,\Delta(\Delta+1)^5(3\Delta+4)} \quad \text{for } G = H\square K_2, \qquad c = \frac{8h_0}{243\,\Delta^9(\Delta-1)(h_0+\Delta^2)} \quad \text{for } G = H^2.$$

These constants come from Theorem 8.2, which applies to every bounded-degree expander whose vertex neighbourhoods are tied together by short cycles (Definition 7.1). For a cubic H

with $h(H) \geq 1/5$, the prism gives $c \geq 3 \cdot 10^{-7}$ (Example 8.4). The same local argument gives $e^{\Omega(L)}$ for the $L \times L$ grid (Corollary 8.5), which is not an expander and is not covered by Theorem 1.1. All the constants are tiny, and in exact computations on small graphs the bounds we prove are far from tight (Appendix A). The results are qualitative. For the bounded-degree expanders the exponent is linear in the number of variables; for the grid it is $\Omega(L) = \Omega(\sqrt{M})$.

Expansion alone is not enough. The uniform spanning tree is the product of the uniform spanning trees of the blocks of G, so a balanced split along blocks has Bhattacharyya coefficient one with a product distribution. Proposition 11.2 shows that such a split exists exactly when no block has more than $2M/3$ edges. Pendant edges give bounded-degree expanders of this kind (Proposition 11.5). For bounded-degree expanders this is the only obstruction: blocks inherit expansion, so if the largest block G_1 has $M_1 > 2M/3$ edges, every balanced split has $\beta^2 \leq e^{-\Omega(M_1 - 2M/3)}$, and for circuits with univariate leaves the bound $e^{\Omega(|V(G_1)|)}$ holds without any threshold (Theorem 11.4). 2-connectivity is the case $G_1 = G$.

We also record exact-support consequences (Corollary 8.7). For every graph as in Theorem 1.1, every nonnegative smooth decomposable PC with the same leaf-arity condition, and every monotone arithmetic circuit, whose support is exactly the set of spanning trees of G, with arbitrary positive values on that set, has size $e^{\Omega(N)}$.

1.2 The complete graph

Both arguments behind Theorem 1.1 need bounded degree. Let μ_n be the uniform spanning tree of K_n, a distribution on $M = \binom{n}{2}$ variables. Section 5 proves that for $n \geq 3$ every balanced split $A|B$ of $E(K_n)$ satisfies

$$-\ln \beta(\mu_n; A|B)^2 \;\geq\; \frac{n}{146} - \frac{1}{2}\ln n - 3$$

(Theorem 5.1), with an explicit sum Φ_n behind it whose exponential rate is $1/144.86$ per vertex. Consequently every smooth decomposable PC with nonnegative weights and leaves of arity at most $2M/3$ with $1 - \mathrm{TV}(\mu_n, Q_C) \geq \delta > 0$ has size at least $\delta^2 e^{n/146}/(e^3\sqrt{n})$ (Corollary 5.2). The arity condition cannot simply be dropped, since a single leaf of arity M represents μ_n. A subset recursion gives an exact deterministic circuit of size $O(n^4 3^n)$ (Proposition 5.11), so at every fixed error the optimal size is $2^{\Theta(n)} = 2^{\Theta(\sqrt{M})}$ (Corollary 5.12). Martens and Medabalimi [MM14] proved a lower bound of the same order for exact representation and for approximation arbitrarily well; ours holds at every fixed total-variation error below one. The proved rate is still far from the truth. Exact computation for $6 \leq n \leq 9$ gives $-\ln \beta^2$ between $0.12n$ and $0.30n$ on the splits we tried, and a Monte Carlo estimate, which is an upper bound on the true rate and is not certified, gives about $0.17n$ for large n (Remark 5.6).

The same method gives $2^{\Omega(n)}$ for every graph whose Laplacian has $\lambda_2 \geq \kappa n$, with a rate of order $\kappa^4 e^{-1.14/\kappa}$ (Theorem 5.8 and Corollary 5.9). For $G(n, \frac{1}{2})$ this applies conditionally on the standard concentration of λ_2, which we do not re-derive. A minimum degree of $n/2$ is not enough for the method: two cliques joined by a perfect matching have a balanced split with $\beta^2 > e^{-2}$ (Proposition 5.10).

1.3 Matroids

A spanning tree contains no cycle and meets every cut. The local argument uses the first fact and the vertex-touch argument the second. Section 9 shows that the two arguments are the circuit and cocircuit cases of one argument for the uniform distribution on the bases of a matroid. Each element of a circuit is certified by a cut that meets the circuit in exactly two elements, and each

element of a cocircuit by a cycle; an exchange argument turns such a frame into a conditional lower bound (Lemma 9.6). This gives a bound for every matroid in terms of compatible framed circuits and cocircuits (Theorems 9.7 and 9.8), and the bound is invariant under matroid duality (Proposition 9.1). For balanced matroids, which include the regular ones, a global version replaces frames by leverages (Theorem 9.11). For a totally unimodular representation the leverages are effective resistances (Lemma 9.12), and applied to vertex stars this recovers Theorem 4.5. As an example outside graphs and their duals we obtain $2^{\Omega(n)}$ bounds for binary non-regular matroids obtained from linked expanders by attaching a Fano plane to every edge (Proposition 9.9). The key lemma of the global version fails for the non-balanced binary matroid S_8 (Section 11); whether the global version holds for all matroids is open.

1.4 Subtraction

The uniform spanning tree has a small circuit with subtraction. Section 10 asks how much of the lower bound survives when subtraction is allowed but the circuit keeps a fixed decomposition structure. For structured-decomposable circuits with real weights, a rank argument adapted from de Colnet and Mengel [dCM21] gives exponential size for exact representation on linked expanders (Theorem 10.3 and Corollary 10.4). One inequality for disjoint permutation blocks of UST (Theorem 10.10) extends this to approximation. If the output Q sums to 1, every fixed error $\frac{1}{2}\|\mathrm{UST} - Q\|_1 \le \varepsilon < 1$ needs size $e^{\Omega(N)}$, although Q may be negative (Theorem 10.12 and Corollary 10.14); the key step is that a matrix of rank r and bounded ℓ_1 mass covers at most a $\sqrt{r/n}$ fraction of an $n \times n$ permutation, a consequence of a rank bound of Ky Fan and Hoffman [FH53]. Without normalization the threshold is $\varepsilon < \frac{1}{2}$, which is sharp because $Q = 0$ has error $\frac{1}{2}$ (Theorem 10.11). The same bounds hold on every 2-connected bounded-degree expander, at any girth and without smoothness (Corollary 10.21), with blocks built from pendant vertices instead of short cycles, and they cover sparse and low-degree approximations (Corollary 10.15). Normalized squares and positive semidefinite forms of structured-decomposable amplitudes obey the same lower bound as nonnegative circuits (Theorem 10.22): for structured circuits the barrier is decomposability, not sign. For unstructured decomposable circuits with subtraction we have lower bounds only for restricted classes. On bounded-degree expanders, support-exact representation needs size $2^{N^{\Omega(1/d)}}$ at product-depth d and $N^{\Omega(\log N)}$ for formulas, and sums of r structured circuits, each on its own vtree, need total size $2^{\Omega(3^{-r}\sqrt{N})}$ (Corollary 10.31). These proofs transfer the restriction arguments of Raz [Raz09] and of Raz and Yehudayoff [RY09] through a large complete minor [KN21, Kri19], after checking that their rank bounds do not depend on the constants of the restriction. For sums of r structured circuits, a packing of far-apart pendant gadgets gives total size $2^{\Omega_r(N)}$ directly in the graph, both for support-exact representation and at error $\varepsilon < \frac{1}{2}$ (Theorem 10.37); its constants are astronomically small, so the minor bound remains better at any realistic size. General circuits stay open. For exact representation the general question is equivalent to the syntactically multilinear complexity of the spanning-tree polynomial (Proposition 10.23), and a small approximating circuit would have to cancel all but an exponentially small fraction of its absolute mass (Theorem 10.38), largely between terms on different splits (Proposition 10.39). We explain why the block method does not yet reach unstructured circuits (Section 10.8).

1.5 What is known and what is new

The ingredients of the proof are known; we credit them where they are used.

- For a joint distribution μ of (X, Y), the largest Bhattacharyya coefficient between μ and a product distribution equals the top singular value of the entrywise square root of μ. This is Lemma 6 of Lapidoth and Pfister [LP19], stated for their order-$\frac{1}{2}$ dependence measure $J_{1/2}$.
- The adaptive chain rule we use (Lemma 3.3) is a case of Exercise I.42(b) in Polyanskiy and Wu [PW25], combined with the previous item. For independent pairs it is the additivity of $J_{1/2}$ [LP19, Lemma 12]. Hellinger distance and its product structure are standard tools for communication lower bounds [BYJKS04, Jay09].
- The reduction from circuits to products across balanced edge splits is the balanced decomposition and the first step of the overlap lemma from our earlier paper on dimers [Bla26, Lemmas 7.1 and 7.2]. The decomposition itself is the standard balancing argument for syntactically multilinear circuits [RY08], also stated for sum-product networks by Martens and Medabalimi [MM14, Thm. 38].
- Kirchhoff's formula for edge marginals, Rayleigh monotonicity, Thomson's principle and the triangle inequality for effective resistance are classical [Kir47, LP16, KR93], as is Mohar's Cheeger-type inequality [Moh89], whose proof we repeat for the lower bound we use (Lemma 4.7). The count of connected sets (Lemma 4.9) is a form of a lemma of Fomin and Villanger [FV12], and the random deletion (Lemma 4.10) is the alteration method with a Peierls-type union bound. Lemma 5.7 and Section 11.5 use the transfer-current theorem of Burton and Pemantle [BP93]. The proof of Theorem 5.1 uses de Caen's bound on sums of squared degrees [dC98], the exact law of the isolated vertices of a uniform labelled forest (a Cayley-type count), and Robbins' form of Stirling's formula [Rob55].
- For matroids we use standard facts from Oxley [Oxl11], the negative correlation of regular matroids due to Feder and Mihail [FM92], and the characterization of balanced binary matroids by Choe and Wagner [CW03].

Constant-error lower bounds against nonnegative decomposable circuits are not new in general. Leland and Choi [LC25] prove one for *deterministic* circuits. A generic $\exp(\Omega(n))$ separation from a random linear code, whose inference is easy by Gaussian elimination, follows from the rank bounds of de Colnet and Mengel [dCM20], as explained in [Bla26, Sec. 10.4]. The dimer paper [Bla26] proves an $\exp(\Theta(\sqrt{n}))$ bound for the uniform domino tiling of the square grid. That paper also lists the robust spanning-tree question for expanders as open [Bla26, Sec. 11, problem 5].

We believe the following are new.

(i) The constant-error $2^{\Omega(N)}$ lower bound against nonnegative smooth decomposable PCs, without determinism, for uniform spanning trees of every 2-connected bounded-degree expander, at any girth. Apart from an earlier version of this paper, which covered prisms and squares of expanders, we know of no constant-error exponential lower bound against nondeterministic decomposable PCs for uniform spanning trees of any family, and none for a distribution of this kind whose inference reduces to determinants.

(ii) The vertex-touch reduction: two bits per vertex that has edges on both sides of the split, conditioning only on the stars of a sparse set of such vertices, and a conditional lower bound from effective resistances, in a telescoped form controlled by λ_2 of the graph left after the sparse set is deleted, with a random deletion that keeps this graph an expander (Lemma 4.3, Theorem 4.5, Lemmas 4.4 and 4.10).

(iii) The local cycle-bit reduction: for spanning trees, two indicator bits per crossing cycle,

together with the spatial Markov property at the edges incident to the cycle, give the conditional non-degeneracy that the chain rule needs (Lemma 6.2 and Theorem 6.6). It gives the better constants of Corollary 1.3 and the grid bound.

(iv) The constant-error $2^{\Omega(n)}$ bound for the uniform spanning tree of K_n with an explicit rate (Theorem 5.1 and Corollary 5.2), its extension to dense graphs (Corollary 5.9), and the averaged conditioning on a rectangle (Lemma 3.7) that lets the proof use a conditioning set that is not independent. We did not find Lemma 3.7 stated in this form, but it is elementary.

(v) The matroid form of both arguments: framed circuits and cocircuits, the local theorem for every matroid, the global theorem for balanced matroids, and the Fano-decorated family (Section 9). We did not find the exchange bound of Lemma 9.3 stated for matroids, but it is also elementary.

(vi) The approximate lower bounds for structured-decomposable circuits with real weights: the permutation-block inequality (Theorem 10.10), which reaches every error below 1 for normalized outputs, and its pendant-vertex form on every 2-connected bounded-degree expander, without smoothness (Corollary 10.21). Lemma 10.8 is classical [FH53].

(vii) The support-exact lower bounds for restricted unstructured signed circuits (Corollary 10.31), and the bounds for sums of structured circuits through packings of pendant gadgets (Theorem 10.37). We did not find them in the literature, but the transfer is short and may be folklore; the rank arguments are those of [Raz09, RY09].

(viii) The limits of the vertex-touch method: on 2-connected hub graphs with expansion h and maximum degree D, star bits certify at most $O(h^{D/2+1})$ for one balanced split, while bits for pairs of edges at a vertex certify $h^3/(10935D^5)$ for that split (Corollary 11.12 and Proposition 11.16). No novelty is claimed for the ingredients.

Our literature search was targeted, not exhaustive. The closest results we found are discussed in Section 12. In particular, Chattopadhyay, Datta and Mukhopadhyay [CDM20] and Chattopadhyay, Datta, Ghosal and Mukhopadhyay [CDGM22] prove exact $2^{\Omega(n)}$ monotone lower bounds for directed spanning-tree polynomials of expanders of large constant degree, and Mészáros and Peled [MP26] prove that the uniform spanning tree of a graph with large minimum degree is close to a product-like process in expected edit distance. Neither statement concerns total variation for circuits; Section 12 explains why neither contradicts nor implies ours.

1.6 Proof outline

Fix a split $E = A \sqcup B$ of the edges. For a product distribution $q = q^A \otimes q^B$ write $\mathrm{BC}(P, q) = \sum_x \sqrt{P(x)q(x)}$, and let $\beta(P; A|B)$ be the largest value over all products. The circuit reduction (Section 3.5) gives

$$|C| \;\geq\; (1-\varepsilon)^2 \Big/ \max_{A|B \text{ balanced}} \beta(P; A|B)^2.$$

So it suffices to show $\beta \leq e^{-cN/2}$ for every split in which both sides have at least $M/3$ edges. The proof of Theorem 1.1 has four steps.

1. *Vertex-touch bits.* Call a vertex w mixed if it has edges in both A and B. Put $a_w = \mathbf{1}[T \text{ uses no } A\text{-edge at } w]$ and $b_w = \mathbf{1}[T \text{ uses no } B\text{-edge at } w]$. The bits a depend only on $T \cap A$ and the bits b only on $T \cap B$. A spanning tree has an edge at every vertex, so $a_w = b_w = 1$ never happens. Data processing gives $\beta(P; A|B) \leq \|\sqrt{\mu}\|_{\mathrm{op}}$, where μ is the joint law of the bits at a set W of mixed vertices (Lemma 3.2).

2. *Chain rule.* If, given every possible value of the earlier pairs, the pair (a_w, b_w) has probability at least p_A of being $(1,0)$ and at least p_B of being $(0,1)$, then the pair contributes a factor $((1+\sqrt{1-4p_Ap_B})/2)^{1/2} \le e^{-p_Ap_B/2}$ to $\|\sqrt{\mu}\|_{\mathrm{op}}$ (Lemmas 3.3 and 3.5). No independence between the pairs is needed.

3. *Conditioning on stars.* Take W independent and condition on the tree on the stars of the other vertices of W. The rest of the tree is a uniform spanning tree of a graph obtained from $G-(W\setminus\{w\})$ by identifying vertices. Deleting the A-edges at w one at a time, Kirchhoff's formula and Rayleigh monotonicity give $\Pr[a_w=1\,|\,\text{past}] \ge \prod_u (1+R_u)^{-1}$, with effective resistances R_u in a graph that contains $G-W$ (Lemma 4.3). Spreading the current from u over all remaining neighbours of w, the product telescopes to $\Pr[a_w=1\,|\,\text{past}] \ge (b/d)(1+1/\lambda_2(L_{G-W}))^{-a}$, where a and b are the numbers of A- and B-edges at w and $d=a+b$ (Lemma 4.4).

4. *Expansion.* Every balanced split of an expander has linearly many mixed vertices (Lemma 4.6). If G is 2-connected, a random sparse set W of them, with the vertices of a few dangerous configurations removed, leaves $G-W$ an expander (Lemma 4.10); the dangerous configurations are counted by a Peierls argument over connected sets (Lemma 4.9). Mohar's inequality turns the expansion of $G-W$ into a bound on λ_2 (Lemma 4.7). If $\lambda_2(L_G)>1$, a set W with pairwise distances at least 3 suffices, since deleting it lowers λ_2 by at most one (Lemma 4.8).

The local argument for Corollary 1.3 replaces step 1 by two bits per short cycle that crosses the split ($u=\mathbf{1}[\text{all } A\text{-edges of the cycle are in } T]$ and v likewise for B; a tree contains no cycle), step 3 by conditioning on everything outside the edges incident to the cycle and counting completions (Lemma 6.2), and step 4 by a packing of short cycles (Lemma 7.3). Cycles and cuts give the two arguments: a tree contains no cycle and meets every cut. Section 9 turns this remark into a theorem about matroids. Only steps 1 and 3 use spanning trees.

For K_n (Section 5) the set W cannot be independent. We condition on the tree edges inside W, keep the set I of vertices of W that these edges do not touch, and average the resulting bounds over the conditioning (Lemma 3.7). Given the conditioning, a star-deletion bound that is sharp on complete graphs gives the kernel of each vertex of I (Lemma 5.3). The law of I for a uniform r-set W is explicit (Lemma 5.5), de Caen's inequality shows that an average vertex has a constant fraction of its edges on each side, and choosing $r\approx 0.38n$ gives the rate. For dense graphs, the transfer-current theorem and Hadamard's inequality replace the exact law of I (Lemma 5.7). The determinant enters only on the upper-bound side.

Organization. Sections 2 and 3 give notation, circuit models and the known tools. Section 4 proves the bound for 2-connected expanders, and Section 5 adapts it to the complete graph. Sections 6 and 7 give the local cycle-bit argument. Section 8 proves the main results, with constants, the families covered, the upper bound and exact-support consequences. Section 9 puts both arguments in the setting of matroids. Section 10 treats circuits with subtraction. Section 11 discusses limitations and variants, including the block structure and the matroid S_8; Sections 12 and 13 cover related work and open questions. Appendix A describes the machine checks.

2 Preliminaries

Graphs. All graphs are finite. $G=(V,E)$ is connected and simple unless we say otherwise, with $N:=|V|\ge 2$, $M:=|E|$ and maximum degree D. For $S\subseteq V$, ∂S is the set of edges with

exactly one endpoint in S. The edge expansion is

$$h(G) := \min\Big\{ \frac{|\partial S|}{|S|} : \ S \subseteq V,\ 0 < |S| \le N/2 \Big\}.$$

Taking $|S| = 1$ shows $h(G) \le D$. A cycle C is a simple cycle with vertex set $V(C)$ and edge set $E(C)$, and $|C| := |E(C)| = |V(C)| \ge 3$. Two cycles are vertex-disjoint if their vertex sets are disjoint. For a cycle C, let

$$I(C) := \{e \in E : \ e \text{ has an endpoint in } V(C)\}$$

be the set of edges incident to C. It contains $E(C)$ and every chord of C.

Spanning trees. Let $\mathcal{T}(G)$ be the set of spanning trees and $\tau(G) := |\mathcal{T}(G)|$. We identify an edge set with its indicator vector in $\{0,1\}^E$. The uniform spanning tree $\mathrm{UST}(G)$ is the uniform distribution on $\mathcal{T}(G)$, viewed as a distribution on $\{0,1\}^E$. We write T for a random tree with this law.

Splits and products. An *edge split* is a partition $E = A \sqcup B$. It is *balanced* if $|A| \ge M/3$ and $|B| \ge M/3$. We write $x = (x_A, x_B)$ for $x \in \{0,1\}^E$. A *product distribution across* $A|B$ is $q = q^A \otimes q^B$, where q^A and q^B are probability distributions on $\{0,1\}^A$ and $\{0,1\}^B$. The product may put mass on non-trees.

Distances. For distributions P, Q on a finite set, $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_x |P(x) - Q(x)|$, so $1 - \mathrm{TV}(P,Q) = \sum_x \min(P(x), Q(x))$. The Bhattacharyya coefficient is $\mathrm{BC}(P,Q) = \sum_x \sqrt{P(x)Q(x)}$. In the normalization of Polyanskiy and Wu, $H^2(P,Q) = 2 - 2\mathrm{BC}(P,Q)$, and the Rényi divergence of order $\frac{1}{2}$ is $D_{1/2}(P\|Q) = -2\log \mathrm{BC}(P,Q)$. The central quantity of this paper is

$$\beta(P; A|B) := \max\{\mathrm{BC}(P, q^A \otimes q^B)\},$$

the maximum over all product distributions across $A|B$.

Matrices. For a nonnegative function μ on a product $\mathcal{X} \times \mathcal{Y}$ of finite sets, $\sqrt{\mu}$ denotes the $\mathcal{X} \times \mathcal{Y}$ matrix with entries $\sqrt{\mu(x,y)}$ (entrywise square root). $\|W\|_{\mathrm{op}}$ is the largest singular value of a matrix W.

2.1 Circuit models

We use the definitions of [Bla26, Sec. 2.1], which we repeat.

Probabilistic circuits. A *probabilistic circuit* (PC) over $x \in \{0,1\}^M$ is a directed acyclic graph with one root and three kinds of node.

- *Leaves* are nonnegative functions of at most a of the variables. The *arity* is a. Indicator leaves $[x_e = 0]$ and $[x_e = 1]$ are the usual choice.
- *Sum nodes* compute a weighted sum of their children with nonnegative weights.
- *Product nodes* multiply their children.

The *scope* X_v of a node v is the set of variables appearing in leaves below v, and v computes a function f_v of x_{X_v}. The circuit computes $f_C := f_{\text{root}} \geq 0$ and represents $Q_C := f_C / \sum_x f_C(x)$. It is *decomposable* if the children of each product node have pairwise disjoint scopes, and *smooth* if the children of each sum node have equal scopes. It is *deterministic* if the children of each sum node have pairwise disjoint supports. We never assume determinism.

Size convention. The size $|C|$ is the number of nodes plus the number of wires. This is the convention under which the decomposition lemma (Lemma 3.9) gives at most $|C|$ terms. For circuits of bounded fan-in, it differs from the node count by a constant factor. Remark 3.10 explains how to obtain the same bound with nodes only.

Arithmetic circuits. A *monotone arithmetic circuit* has binary $+$ and $\times$ gates, inputs z_e ($e \in E$) and nonnegative real constants. It computes a formal polynomial, in which $z_e^2 \neq z_e$. Its size is its number of nodes. The *generating polynomial* of a distribution Q on $\{0,1\}^E$ is $\sum_x Q(x) z^x$, where $z^x := \prod_{e:x_e=1} z_e$. A *probabilistic generating circuit* [ZJdB21] is an arithmetic circuit, with subtraction allowed, that computes it.

3 Known tools

This section collects the tools the proof uses. Apart from the elementary averaging step of Lemma 3.7, which we did not find stated in this form, none of them is new. We give short proofs where this keeps the paper self-contained, and say where each statement comes from.

3.1 Affinity with products is a matrix norm

Let μ be a joint distribution of (X, Y) on a finite product $\mathcal{X} \times \mathcal{Y}$. Write $\beta(\mu; X|Y)$ for the largest value of $\mathrm{BC}(\mu, a \otimes b)$ over probability distributions a on $\mathcal{X}$ and b on $\mathcal{Y}$. For a distribution P on $\{0,1\}^E$ and a split $A|B$ we view P as a joint law of (x_A, x_B), so that $\beta(P; A|B)$ is the quantity defined in Section 2.

Lemma 3.1 (Lapidoth and Pfister [LP19, Lemma 6])**.** *$\beta(\mu; X|Y) = \|\sqrt{\mu}\|_{\text{op}}$, where $\sqrt{\mu}$ is the $\mathcal{X} \times \mathcal{Y}$ matrix with entries $\sqrt{\mu(x,y)}$.*

Proof. For product $a \otimes b$, put $f = \sqrt{a}$ and $g = \sqrt{b}$. These are unit vectors, and $\mathrm{BC}(\mu, a \otimes b) = f^\top \sqrt{\mu}\, g \leq \|\sqrt{\mu}\|_{\text{op}}$. Conversely, $\|\sqrt{\mu}\|_{\text{op}} = \max f^\top \sqrt{\mu}\, g$ over unit vectors, and since $\sqrt{\mu}$ has nonnegative entries the maximum is attained at nonnegative f, g. Then $a = f^2$ and $b = g^2$ are probability distributions with $\mathrm{BC}(\mu, a \otimes b) = \|\sqrt{\mu}\|_{\text{op}}$. □

In the notation of [LP19], $J_{1/2}(X;Y) = -2\log\|\sqrt{\mu}\|_{\text{op}}$ is the Rényi divergence of order $\frac{1}{2}$ between μ and the closest product distribution. The rows and columns of $\sqrt{\mu}$ are indexed by the whole values of X and Y. No independence between coordinates is assumed anywhere.

3.2 Data processing

Lemma 3.2 (data processing)**.** *Let $\phi : \mathcal{X} \to \mathcal{X}'$ and $\psi : \mathcal{Y} \to \mathcal{Y}'$ be maps, and let μ' be the law of $(\phi(X), \psi(Y))$. Then $\beta(\mu; X|Y) \leq \beta(\mu'; \phi(X)|\psi(Y)) = \|\sqrt{\mu'}\|_{\text{op}}$.*

Proof. Let $a \otimes b$ be a product distribution, and let a' and b' be the laws of ϕ under a and of ψ under b. The fibre of (x', y') is $\phi^{-1}(x') \times \psi^{-1}(y')$, and $\sum a(x)b(y)$ over this fibre is $a'(x')b'(y')$. By the Cauchy–Schwarz inequality on each fibre,

$$\mathrm{BC}(\mu, a \otimes b) = \sum_{x',y'} \sum_{\substack{x \in \phi^{-1}(x') \\ y \in \psi^{-1}(y')}} \sqrt{\mu(x,y)}\sqrt{a(x)b(y)} \le \sum_{x',y'} \sqrt{\mu'(x',y')\, a'(x')b'(y')} = \mathrm{BC}(\mu', a' \otimes b').$$

The right side is at most $\beta(\mu')$, and Lemma 3.1 identifies $\beta(\mu')$ with the norm. □

This is the data-processing inequality for $J_{1/2}$, applied on both sides [LP19, Lemma 4]. It is the step that lets us discard almost all of the tree. Coarsening can only increase the affinity with products, so it is the right direction for an upper bound on β. The two maps must act separately on the two sides; this is what keeps the image of a product distribution a product.

3.3 An adaptive chain rule

Let $(X_k, Y_k)_{k=1}^r$ be random variables with values in finite sets $\mathcal{X}_k \times \mathcal{Y}_k$ and joint law μ. Put $X = (X_1, \dots, X_r)$ and $Y = (Y_1, \dots, Y_r)$, and view $\sqrt{\mu}$ as a matrix with rows indexed by values of X and columns by values of Y. For each k and each value π of the joint past $(X_{<k}, Y_{<k})$ that has positive probability, let $K_k(\cdot \mid \pi)$ be the conditional law of (X_k, Y_k), viewed as an $\mathcal{X}_k \times \mathcal{Y}_k$ matrix, and put

$$\gamma_k := \max_{\pi} \|\sqrt{K_k(\cdot \mid \pi)}\|_{\mathrm{op}}.$$

Lemma 3.3 (adaptive chain rule)**.** $\|\sqrt{\mu}\|_{\mathrm{op}} \le \gamma_1 \gamma_2 \cdots \gamma_r$.

Proof. Induction on r. For $r = 1$ the past is empty and $\gamma_1 = \|\sqrt{\mu}\|_{\mathrm{op}}$. Let $r \ge 2$, write $x = (x', x_r)$ and $y = (y', y_r)$, and let μ' be the law of the first $r-1$ pairs. Then $\mu(x,y) = \mu'(x',y')K_r(x_r, y_r \mid x', y')$ when $\mu'(x',y') > 0$, and $\mu(x,y) = 0$ otherwise. Let f, g be unit vectors. For each (x', y') with $\mu'(x',y') > 0$, apply the definition of γ_r to the vectors $f(x', \cdot)$ and $g(y', \cdot)$:

$$\begin{aligned}\sum_{x,y} \sqrt{\mu(x,y)} f(x) g(y) &= \sum_{x',y'} \sqrt{\mu'(x',y')} \sum_{x_r, y_r} \sqrt{K_r(x_r, y_r \mid x', y')}\, f(x', x_r)\, g(y', y_r) \\ &\le \gamma_r \sum_{x',y'} \sqrt{\mu'(x',y')}\, F(x')\, G(y'),\end{aligned}$$

where $F(x') = \|f(x', \cdot)\|_2$ and $G(y') = \|g(y', \cdot)\|_2$. Since $\|F\|_2 = \|f\|_2 = 1$ and $\|G\|_2 = 1$, the right side is at most $\gamma_r \|\sqrt{\mu'}\|_{\mathrm{op}}$. The conditional laws of μ' are those of μ for $k < r$, so induction gives $\|\sqrt{\mu'}\|_{\mathrm{op}} \le \gamma_1 \cdots \gamma_{r-1}$. □

The key point is that F depends only on x' and G only on y', even though the kernel K_r depends on both. By Lemma 3.1, Lemma 3.3 says $J_{1/2}(X;Y) \ge \sum_k \min_\pi J_{1/2}(X_k; Y_k \mid \pi)$, where the minimum is over joint pasts. This is a special case of a known chain rule. Polyanskiy and Wu [PW25, Exercise I.42(b)] ask the reader to show that

$$\mathrm{BC}(P_{Z^n}, Q_{Z^n}) \le \prod_{i=1}^{n} \sup_h \mathrm{BC}(P_{Z_i \mid Z^{i-1}=h},\, Q_{Z_i \mid Z^{i-1}=h})$$

for any two joint laws (they write $1 - \frac{1}{2}H^2$ for BC). Take $Z_i = (X_i, Y_i)$, $P = \mu$, and $Q = a \otimes b$ a product across $X|Y$. Each conditional $Q_{Z_i \mid Z^{i-1}=h}$ is then a product across $X_i|Y_i$, so each factor is

at most γ_i by Lemma 3.1. Maximizing over Q gives Lemma 3.3. For independent pairs the lemma is the additivity of $J_{1/2}$ [LP19, Lemma 12], or the multiplicativity of operator norms under tensor products.

Remark 3.4 (what "adaptive" means). In Lemma 3.3 the kernels may depend on the entire joint past of both coordinates. The coordinates themselves, and their order, are fixed in advance. In our applications the cycles or vertices, and their order, are chosen before the tree is sampled. Choosing the next statistic as a function of both parties' earlier bits would not be covered: such a choice is not a function of one side alone, so Lemma 3.2 would not apply.

The conditional hypothesis cannot be replaced by a hypothesis on marginals. Let (X_1, Y_1) be $(0,1)$ or $(1,0)$ with probability $\frac{1}{2}$ each, and copy it to every later pair. Each pair alone has norm $1/\sqrt{2}$, but $\|\sqrt{\mu}\|_{\mathrm{op}} = 1/\sqrt{2}$ for every r. Lemma 3.3 is consistent with this, since after the first pair the kernels are point masses and $\gamma_k = 1$.

Lemma 3.5 (one forbidden cell)**.** *Let K be a distribution on $\{0,1\}^2$ with $K(1,1) = 0$, $K(1,0) \geq p$ and $K(0,1) \geq p'$. Then $4pp' \leq 1$ and*

$$\|\sqrt{K}\|_{\mathrm{op}}^2 = \frac{1 + \sqrt{1 - 4K(0,1)K(1,0)}}{2} \leq \frac{1 + \sqrt{1 - 4pp'}}{2} \leq 1 - pp'.$$

In particular, for $p' = p$ the bound is $(1 + \sqrt{1 - 4p^2})/2 \leq 1 - p^2$.

Proof. Write $a = K(0,0)$, $b = K(0,1)$, $c = K(1,0)$. The singular values $\sigma_1 \geq \sigma_2$ of $\sqrt{K} = \left(\begin{smallmatrix} \sqrt{a} & \sqrt{b} \\ \sqrt{c} & 0 \end{smallmatrix}\right)$ satisfy $\sigma_1^2 + \sigma_2^2 = a + b + c = 1$ and $\sigma_1\sigma_2 = |\det \sqrt{K}| = \sqrt{bc}$. So σ_1^2 is the larger root of $s^2 - s + bc = 0$, which is the stated formula, and it decreases in $bc \geq pp'$. Also $4bc \leq (b+c)^2 \leq 1$. The last step is $\sqrt{1-x} \leq 1 - x/2$. □

The middle bound is attained by independent copies of the kernel with $K(0,1) = K(1,0) = p$, so for kernels with one forbidden cell, Lemmas 3.3 and 3.5 lose nothing.

The same computation works for every kernel. A rare cell instead of a forbidden one, or any correlation at all, gives a penalty.

Lemma 3.6 (any 2×2 kernel)**.** *Let K be a distribution on $\{0,1\}^2$, write $K_{ij} := K(i,j)$, and put $\Delta := |\sqrt{K_{00}K_{11}} - \sqrt{K_{01}K_{10}}|$. Then $\Delta \leq \frac{1}{2}$ and*

$$\|\sqrt{K}\|_{\mathrm{op}}^2 = \frac{1 + \sqrt{1 - 4\Delta^2}}{2} \leq 1 - \Delta^2.$$

(a) (rare cell) If $K_{11} \leq \delta$, $K_{10} \geq p$ and $K_{01} \geq p'$, then $\Delta \geq (\sqrt{pp'} - \sqrt{\delta})_+$.

(b) (covariance) If $(x,y) \sim K$, then $K_{00}K_{11} - K_{01}K_{10} = \mathrm{Cov}(x,y)$, and Δ is at least $|\mathrm{Cov}(x,y)|/\min(\sigma_x, \sigma_y)$, which is at least $2|\mathrm{Cov}(x,y)|$, where $\sigma_x^2 := \mathrm{Var}(x)$ and $0/0 := 0$.

Proof. As in the proof of Lemma 3.5, the singular values of $\sqrt{K}$ satisfy $\sigma_1^2 + \sigma_2^2 = 1$ and $\sigma_1\sigma_2 = |\det\sqrt{K}| = \Delta$, so $2\Delta \leq 1$ and $\sigma_1^2 = (1+\sqrt{1-4\Delta^2})/2$. (a) $\Delta \geq \sqrt{K_{01}K_{10}} - \sqrt{K_{00}K_{11}} \geq \sqrt{pp'} - \sqrt{\delta}$, since $K_{00} \leq 1$, and $\Delta \geq 0$. (b) $K_{00}K_{11} - K_{01}K_{10} = K_{11}(K_{00} + K_{01} + K_{10} + K_{11}) - (K_{10} + K_{11})(K_{01} + K_{11}) = \Pr[x = y = 1] - \Pr[x = 1]\Pr[y = 1]$. If $s := \sqrt{K_{00}K_{11}} + \sqrt{K_{01}K_{10}} = 0$, both products vanish and $\Delta = \mathrm{Cov} = 0$. Otherwise $\Delta = |\mathrm{Cov}|/s$, and by the Cauchy–Schwarz inequality $s \leq \sqrt{(K_{00} + K_{01})(K_{11} + K_{10})} = \sigma_x$, and likewise $s \leq \sigma_y$. Finally $\sigma_x \leq \frac{1}{2}$. □

Part (a) at $\delta = 0$ is Lemma 3.5. For $\delta \leq pp'/4$ it still gives $\|\sqrt{K}\|_{\mathrm{op}}^2 \leq 1 - pp'/4$. The positive part matters: for the uniform kernel, $p = p' = 0.01$ and $\delta = \frac{1}{4}$, the expression $\sqrt{pp'} - \sqrt{\delta}$ is negative while $\Delta = 0$. Section 11.5 uses part (b) for pairs of edges, whose covariance under the uniform spanning tree is explicit.

3.4 Conditioning on a rectangle

In Section 5 the set of usable coordinates depends on a random configuration that both sides can see. The next lemma conditions on such a configuration and averages the resulting bounds.

Lemma 3.7 (averaged rectangle conditioning)**.** *Let μ be a distribution on $\mathcal{X} \times \mathcal{Y}$, let ϕ and ψ be maps on $\mathcal{X}$ and on $\mathcal{Y}$, and put $Z := (\phi(X), \psi(Y))$. For each value z with $\mu(Z = z) > 0$ let $\mu_z := \mu(\cdot \mid Z = z)$. Then*

$$\beta(\mu; X|Y)^2 \;\le\; \sum_z \mu(Z = z)\, \beta(\mu_z; X|Y)^2.$$

Proof. Let $q = a \otimes b$ be a product distribution. For $z = (z_1, z_2)$ the set $R_z := \phi^{-1}(z_1) \times \psi^{-1}(z_2)$ is a rectangle, and these sets partition $\mathcal{X} \times \mathcal{Y}$. If $q(R_z) > 0$, the restriction of q to R_z, renormalized, is a product distribution q_z. Since μ vanishes outside the cells with $\mu(Z = z) > 0$,

$$\mathrm{BC}(\mu, q) = \sum_z \sqrt{\mu(Z = z)\, q(R_z)}\; \mathrm{BC}(\mu_z, q_z) \le \sum_z \sqrt{\mu(Z = z)\, q(R_z)}\; \beta(\mu_z; X|Y),$$

where the sums run over z with $\mu(Z = z)q(R_z) > 0$. The Cauchy–Schwarz inequality and $\sum_z q(R_z) \le 1$ bound the right side by $(\sum_z \mu(Z = z)\beta(\mu_z; X|Y)^2)^{1/2}$. Maximize over q. □

The bound is an average over z, not a maximum, so a few values of z with $\beta(\mu_z)$ close to 1 cost only their probability. Putting Z in front as the first coordinate of Lemma 3.3 would not work: the chain rule takes the worst past, and if some value of z leaves no usable coordinates, every later kernel can be trivial.

3.5 From circuits to balanced products

The circuit reduction is taken from our dimer paper [Bla26, Sec. 7.1–7.2]. Only its first, distribution-free part is needed here. The matching-profile part of that paper is specific to perfect matchings and plays no role.

Lemma 3.8 (overlap with sums of products; first step of [Bla26, Lemma 7.1])**.** *Let P be any distribution on $\{0,1\}^E$, and let*

$$Q = \sum_{u=1}^{k} g_u(x_{A_u})\, h_u(x_{B_u}),$$

where $g_u, h_u \ge 0$, each $A_u \sqcup B_u = E$ is a balanced split (the splits may differ between terms), and $\sum_x Q(x) = 1$. Then

$$1 - \mathrm{TV}(P, Q) \le \sqrt{k} \max_{A|B \text{ balanced}} \beta(P; A|B).$$

Proof. Drop zero terms and write $g_u h_u = \lambda_u q_u$, where q_u is a product distribution across $A_u|B_u$, $\lambda_u \ge 0$ and $\sum_u \lambda_u = 1$. Pointwise, $\min(P, Q) \le \sqrt{PQ}$ and $\sqrt{\sum_u \lambda_u q_u} \le \sum_u \sqrt{\lambda_u q_u}$. Hence

$$1 - \mathrm{TV}(P, Q) = \sum_x \min(P(x), Q(x)) \le \sum_u \sqrt{\lambda_u}\, \mathrm{BC}(P, q_u) \le \Big(\sum_u \sqrt{\lambda_u}\Big) \max_u \beta(P; A_u|B_u),$$

and $\sum_u \sqrt{\lambda_u} \le \sqrt{k}$ by the Cauchy–Schwarz inequality. □

Mass that Q puts outside the support of P is allowed and never helps. Different terms may use different splits, because the final bound is uniform over balanced splits. This is why determinism is not needed.

Lemma 3.9 (balanced decomposition [Bla26, Lemma 7.2])**.** *Let C be a smooth decomposable PC with nonnegative weights over $x \in \{0,1\}^E$, $|E| = M \geq 2$, whose leaves have arity at most $2M/3$. Then, as functions on $\{0,1\}^E$,*

$$f_C = \sum_{u \in \mathcal{U}} f_u(x_{A_u})\, h_u(x_{E \setminus A_u}),$$

where each $f_u, h_u \geq 0$, each A_u satisfies $M/3 < |A_u| \leq 2M/3$, and $|\mathcal{U}| \leq |C|$, with size counted as nodes plus wires.

Proof. We repeat the proof of [Bla26] with M in place of n. Let X be the scope of the root. If $|X| \leq 2M/3$, pick any $A \supseteq X$ with $M/3 < |A| \leq 2M/3$; then $f_C = f_C(x_A) \cdot 1$ is one term. Otherwise, replace each product node of fan-in d by a chain of $d-1$ binary products. This preserves decomposability and the computed functions, and the number of nodes becomes at most the original number of nodes plus wires, that is, at most $|C|$. Let $\mathcal{U}$ be the set of nodes u with $|X_u| \in (M/3, 2M/3]$, and put $A_u := X_u$. We claim that every node v with $|X_v| > 2M/3$ satisfies

$$f_v = \sum_{u \in \mathcal{U}} f_u\, h_{u,v}, \qquad h_{u,v} \geq 0 \text{ a function of } x_{X_v \setminus X_u},$$

and argue by induction in topological order. No leaf has scope larger than $2M/3$. At a sum node, smoothness gives $X_c = X_v$ for every child c, so $f_v = \sum_c w_c f_c = \sum_u f_u (\sum_c w_c h_{u,c})$. At a product node $v = c_1 \times c_2$ with $|X_{c_1}| \geq |X_{c_2}|$ we have $|X_{c_1}| > M/3$. If $|X_{c_1}| \leq 2M/3$, then $c_1 \in \mathcal{U}$ and $f_v = f_{c_1} f_{c_2}$. Otherwise $f_v = \sum_u f_u\, (h_{u,c_1} f_{c_2})$, and $(X_{c_1} \setminus X_u) \sqcup X_{c_2} = X_v \setminus X_u$ by decomposability. Apply the claim at the root. A function of $x_{X \setminus X_u}$ is also a function of $x_{E \setminus X_u}$. □

This is the standard balancing argument for syntactically multilinear circuits [RY08]. Martens and Medabalimi state the same form with at most s^2 terms for decomposable sum-product networks of size s with univariate leaves [MM14, Thm. 38]; using their version would halve the exponent in our bounds.

Remark 3.10 (counting nodes only). The prefactor $|\mathcal{U}| \leq |C|$ uses the convention that size counts nodes plus wires, because binarizing a product node of fan-in d creates $d-2$ new nodes. If size counts nodes only, the same bound $|\mathcal{U}| \leq$ #nodes holds without binarizing. At a product node v with $|X_v| > 2M/3$, at most one child is high (scope above $2M/3$) and a high child excludes a medium one, since scopes are disjoint and add up to at most M. If a child is high, expand it as above. If a child is medium, select it. If all children are low (scope at most $M/3$), add children to a group one at a time until the union of their scopes first exceeds $M/3$; the union then has size at most $2M/3$. The product of the group is a nonnegative function of a medium scope, and we give it the label v. Every label is then an original node, so there are at most as many terms as nodes.

Corollary 3.11 (circuit reduction)**.** *Let P be a distribution on $\{0,1\}^E$ and let C be as in Lemma 3.9. If $1 - \mathrm{TV}(P, Q_C) \geq \delta > 0$, then*

$$|C| \;\geq\; \delta^2 \Big/ \max_{A|B \text{ balanced}} \beta(P; A|B)^2.$$

Proof. Lemma 3.9, applied to $f_C / \sum_x f_C(x)$, writes Q_C as a sum of $k \leq |C|$ nonnegative products across balanced splits ($|A_u| > M/3$ and $|E \setminus A_u| \geq M/3$). Lemma 3.8 gives $\delta \leq \sqrt{k} \max \beta$. □

4 Vertex-touch bits and 2-connected expanders

This section proves Theorem 1.1. The cycle bits of Section 6 use the fact that a spanning tree contains no cycle. Here we use the dual fact that a spanning tree meets every cut, applied to the smallest cuts, the stars of single vertices. The conditional lower bound that the chain rule needs comes from effective resistances instead of from counting completions, and expansion keeps the resistances bounded at every girth.

4.1 Vertex-touch bits

Fix a split $E = A \sqcup B$. For a vertex w, let $S_A(w)$ and $S_B(w)$ be the sets of edges of A and of B at w, let $S(w) := S_A(w) \cup S_B(w)$ be its star, and let $N_A(w)$ and $N_B(w)$ be the sets of neighbours joined to w by an edge of $S_A(w)$ and of $S_B(w)$. The vertex w is *mixed* if $S_A(w)$ and $S_B(w)$ are both nonempty. For a mixed vertex define

$$a_w(x_A) := \mathbf{1}\big[x_A \cap S_A(w) = \emptyset\big], \qquad b_w(x_B) := \mathbf{1}\big[x_B \cap S_B(w) = \emptyset\big].$$

The bit a_w depends only on the A-part of x and b_w only on the B-part. A spanning tree of a graph with at least two vertices has an edge at every vertex, so $a_w = b_w = 1$ is impossible under $\mathrm{UST}(G)$. This is the forbidden cell of the cycle bits, with "contains all edges" replaced by "contains no edge".

4.2 Electrical facts

Let K be a finite multigraph without loops, with parallel edges kept distinct, and give every edge unit conductance. For a function ϕ on the vertices let $\mathcal{E}_K(\phi) := \sum_e (\phi(e^+) - \phi(e^-))^2$, the sum over the edges e of K with endpoints $e^\pm$. For vertices $x \neq y$ the *effective resistance* is given by the Dirichlet principle,

$$\frac{1}{R_K(x,y)} := \min\{\mathcal{E}_K(\phi) : \ \phi(x) = 1, \ \phi(y) = 0\},$$

with $R_K(x,y) = \infty$ when x and y lie in different components, where the minimum is 0 [LP16, Ch. 2]. Orient each edge e from e^- to e^+. A *flow* is a function ϑ on the edges, with net outflow $(\partial\vartheta)(v) := \sum_{e:e^-=v} \vartheta_e - \sum_{e:e^+=v} \vartheta_e$ at v and energy $\sum_e \vartheta_e^2$; a *unit flow from x to y* has $\partial\vartheta = \mathbf{1}_x - \mathbf{1}_y$. We use six facts.

(E1) (Kirchhoff) If K is connected and e is an edge joining u and w, then $\Pr_{T\sim\mathrm{UST}(K)}[e \in T] = R_K(u,w)$ [Kir47], [LP16, Ch. 4].

(E2) (Rayleigh monotonicity) Deleting edges does not decrease $R_K(x,y)$. Identifying vertices, without identifying x with y, does not increase $R_K(x,y)$.

(E3) (Triangle inequality) $R_K(x,z) \le R_K(x,y) + R_K(y,z)$ [KR93].

(E4) If K is connected, then $R_K(x,y) \le 2/\lambda_2(L_K)$, where $\lambda_2(L_K) > 0$ is the second smallest eigenvalue of the Laplacian L_K.

(E5) (Thomson) If ϑ is a unit flow from x to y, then $R_K(x,y) \le \sum_e \vartheta_e^2$.

(E6) If K is connected and χ is a function on the vertices with $\sum_v \chi(v) = 0$, then some flow ϑ has $\partial\vartheta = \chi$ and energy at most $\|\chi\|^2/\lambda_2(L_K)$.

(E2) and (E4) follow from the definition. Deleting an edge removes a nonnegative term from $\mathcal{E}_K(\phi)$ for every ϕ, so the minimum cannot increase. After identifying vertices, the admissible functions are those of K that are constant on the identified classes, with the same energy since loops contribute 0; a minimum over fewer functions cannot decrease. For (E4), let $\phi(x) = 1$, $\phi(y) = 0$ and let $\psi := \phi - \bar{\phi}\mathbf{1}$ be orthogonal to constants. Then $\mathcal{E}_K(\phi) = \psi^\top L_K \psi \geq \lambda_2(L_K)\|\psi\|^2 \geq \lambda_2(L_K)(\psi(x)^2 + \psi(y)^2) \geq \lambda_2(L_K)/2$, since $\psi(x) - \psi(y) = 1$. The same argument shows that a path of length k from x to y in K gives $R_K(x, y) \leq k$: by the Cauchy–Schwarz inequality the energy on the path is at least $1/k$. For (E5), if $\phi(x) = 1$ and $\phi(y) = 0$, then $1 = \sum_v \phi(v)(\partial\vartheta)(v) = \sum_e \vartheta_e(\phi(e^-) - \phi(e^+)) \leq (\sum_e \vartheta_e^2)^{1/2}\mathcal{E}_K(\phi)^{1/2}$ by the Cauchy–Schwarz inequality; minimize over ϕ. For (E6), L_K is positive semidefinite with kernel the constants, so $L_K\psi = \chi$ for some $\psi \perp \mathbf{1}$. Put $\vartheta_e := \psi(e^-) - \psi(e^+)$. Then $(\partial\vartheta)(v) = \sum_{e \ni v}(\psi(v) - \psi(\text{other end of } e)) = (L_K\psi)(v) = \chi(v)$, and the energy is $\psi^\top L_K \psi = \psi^\top \chi$. Since $\lambda_2(L_K)\|\psi\|^2 \leq \psi^\top L_K \psi = \psi^\top \chi \leq \|\psi\|\|\chi\|$, we get $\|\psi\| \leq \|\chi\|/\lambda_2(L_K)$ and $\psi^\top \chi \leq \|\chi\|^2/\lambda_2(L_K)$.

Lemma 4.1 (deleting one edge)**.** *Let K be connected, let e be an edge joining u and w, and put $R' := R_{K-e}(u, w)$. Then $\Pr_{T \sim \mathrm{UST}(K)}[e \notin T] = 1/(1 + R')$, which is 0 when e is a bridge. If e is not a bridge, then conditionally on $e \notin T$ the tree T is a uniform spanning tree of $K - e$.*

Proof. For $\phi(u) = 1$ and $\phi(w) = 0$ we have $\mathcal{E}_K(\phi) = 1 + \mathcal{E}_{K-e}(\phi)$. Minimizing gives $1/R_K(u, w) = 1 + 1/R'$, with $1/\infty = 0$. By (E1), $\Pr[e \notin T] = 1 - R_K(u, w) = 1 - R'/(1 + R') = 1/(1 + R')$, and the value is 0 when $R' = \infty$, that is, when e is a bridge. The spanning trees of K without e are the spanning trees of $K - e$, so the last claim holds by uniformity. □

4.3 Conditioning on stars

Lemma 4.2 (spatial Markov property)**.** *Let $F \subseteq E$ and $\sigma \subseteq F$ with $\Pr[T \cap F = \sigma] > 0$. Then σ is a forest. Let K_σ be the multigraph obtained from $(V, E \setminus F)$ by identifying the vertices of each connected component of (V, σ) and deleting the loops this creates, with every other edge keeping its identity. Conditionally on $T \cap F = \sigma$, the set $T \setminus F$ is a uniform spanning tree of K_σ.*

Proof. σ lies in a tree, so it is a forest, and (V, σ) has $c = N - |\sigma|$ components, which are the nodes of K_σ. Since T is uniform, $J := T \setminus F$ is uniform over the sets $J \subseteq E \setminus F$ for which $\sigma \cup J$ is a spanning tree. Each component of σ is connected, so $\sigma \cup J$ is connected if and only if J connects the nodes of K_σ, and $|\sigma \cup J| = N - 1$ if and only if $|J| = c - 1$. A connected graph is a tree if and only if it has one edge fewer than vertices. So $\sigma \cup J$ is a spanning tree if and only if J is a connected spanning subgraph of K_σ with $c - 1$ edges, that is, a spanning tree of K_σ. Such a J uses no loop, since $c - 1$ edges that connect c nodes are all needed. □

Lemma 6.1 in Section 6 is the case $F = E \setminus I(C)$ written out explicitly.

For a set W of vertices write $G - W$ for the graph with W and its incident edges deleted, and $G - X$ for the graph with an edge set X deleted.

Lemma 4.3 (touch probabilities)**.** *Let $A|B$ be a split, let W be an independent set of mixed vertices, let $w \in W$, and put $F := \bigcup_{x \in W \setminus \{w\}} S(x)$. For $u \in N_A(w)$ let $R_u := R_{G_w^A}(u, w)$, where $G_w^A := G - (W \setminus \{w\}) - S_A(w)$, and put*

$$p_A(w) := \prod_{u \in N_A(w)} \frac{1}{1 + R_u} \qquad (\text{with } 1/\infty = 0).$$

Define G_w^B and $p_B(w)$ in the same way with B in place of A. Then for every $\sigma \subseteq F$ with $\Pr[T \cap F = \sigma] > 0$,

$$\Pr[T \cap S_A(w) = \emptyset \mid T \cap F = \sigma] \;\geq\; p_A(w), \qquad \Pr[T \cap S_B(w) = \emptyset \mid T \cap F = \sigma] \;\geq\; p_B(w).$$

Moreover, with $\rho_w := \max_{x,y \in N(w)} R_{G-W}(x,y)$, we have $p_A(w)\, p_B(w) \geq (2+\rho_w)^{-\deg(w)}$.

Proof. We prove the bound for A; the case of B is symmetric. Since W is independent, w is not an endpoint of any edge of F, and $E \setminus F$ is the edge set of $G - (W \setminus \{w\})$. By Lemma 4.2, the conditional law of $T \setminus F$ is $\mathrm{UST}(K_\sigma)$. Every $x \in W \setminus \{w\}$ is isolated in $(V, E \setminus F)$ and is identified in K_σ with the other vertices of its component of (V, σ); this component contains a neighbour of x, which is not in W, because T has an edge at x and W is independent. So K_σ is the graph $G - (W \setminus \{w\})$ with the vertices of certain disjoint sets identified, and w is identified with nothing, because no edge of σ has w as an endpoint. The edges of $S_A(w)$ are edges of K_σ, and $T \cap S_A(w) = (T \setminus F) \cap S_A(w)$.

If $p_A(w) = 0$ there is nothing to prove, so assume $p_A(w) > 0$. Then every $u \in N_A(w)$ is joined to w by a path in G_w^A, and the image of this path in $K_\sigma - S_A(w)$ joins (the class of) u to w. Hence $K_\sigma - S_A(w)$ is connected, and so is $K_\sigma - X$ for every $X \subseteq S_A(w)$. List $S_A(w) = \{e_1, \dots, e_k\}$ with $e_j = u_j w$. Applying Lemma 4.1 to $K_\sigma - \{e_1, \dots, e_{j-1}\}$ for $j = 1, \dots, k$, each time conditioning on the previous deletions, which have positive probability, gives

$$\Pr[T \cap S_A(w) = \emptyset \mid T \cap F = \sigma] = \prod_{j=1}^{k} \frac{1}{1 + R_{K_\sigma - \{e_1,\dots,e_j\}}(u_j, w)}.$$

The graph $K_\sigma - \{e_1, \dots, e_j\}$ is $G - (W \setminus \{w\}) - \{e_1, \dots, e_j\}$ with some vertices other than w identified. By (E2), first for the identifications and then for deleting the remaining edges $e_{j+1}, \dots, e_k$,

$$R_{K_\sigma - \{e_1,\dots,e_j\}}(u_j, w) \leq R_{G-(W\setminus\{w\})-\{e_1,\dots,e_j\}}(u_j, w) \leq R_{G_w^A}(u_j, w) = R_{u_j}.$$

This proves the first claim.

For the second, let $u \in N_A(w)$ and pick $v \in N_B(w)$, which exists since w is mixed. Since W is independent, $u, v \notin W$. The graph $G - W$ is a subgraph of G_w^A, and G_w^A contains the edge vw. By (E2), $R_{G_w^A}(u, v) \leq R_{G-W}(u, v) \leq \rho_w$ and $R_{G_w^A}(v, w) \leq 1$, so (E3) gives $R_u \leq 1 + \rho_w$. Hence $p_A(w) \geq (2+\rho_w)^{-|S_A(w)|}$, likewise $p_B(w) \geq (2+\rho_w)^{-|S_B(w)|}$, and $|S_A(w)| + |S_B(w)| = \deg(w)$. □

The lemma conditions only on the stars of the other vertices of W, not on everything outside $S(w)$. Conditioning on everything outside a star can freeze it (Section 11). The price is that the bound depends on resistances in the global graph $G - W$. The next lemma bounds the same product more carefully, by λ_2 of $G - W$, which Section 4.5 controls by expansion.

Lemma 4.4 (telescoped touch bound)**.** *Let $A|B$ be a split of a simple graph G, let W be an independent set of mixed vertices such that $H := G - W$ is connected, and put $\kappa := 1/\lambda_2(L_H)$. Let $w \in W$, let $a := |S_A(w)|$, $b := |S_B(w)|$ and $d := a + b = \deg(w)$, and let F be as in Lemma 4.3. Then for every $\sigma \subseteq F$ with $\Pr[T \cap F = \sigma] > 0$,*

$$\Pr[T \cap S_A(w) = \emptyset \mid T \cap F = \sigma] \;\geq\; \frac{b}{d}(1+\kappa)^{-a}, \qquad \Pr[T \cap S_B(w) = \emptyset \mid T \cap F = \sigma] \;\geq\; \frac{a}{d}(1+\kappa)^{-b}.$$

The product of the two bounds is $(ab/d^2)(1+\kappa)^{-d}$, which is at least $((D-1)/D^2)(1+\kappa)^{-D}$ if $d \leq D$.

The graph H has at least two vertices, since w has neighbours on both sides and none of them is in W, so $\lambda_2(L_H) > 0$.

Proof. We prove the bound for A. Write $S_A(w) = \{e_1, \dots, e_a\}$ with $e_j = u_j w$. As shown in the proof of Lemma 4.3, K_σ is $G - (W \setminus \{w\})$ with the vertices of certain disjoint sets identified, every $x \in W \setminus \{w\}$ is identified with a vertex of H, and w is identified with nothing. Since H is connected and w has an edge of $S_B(w)$ into H, the graph $K_\sigma - S_A(w)$ is connected. The first display in that proof used only this, so

$$\Pr[T \cap S_A(w) = \emptyset \mid T \cap F = \sigma] = \prod_{j=1}^{a} \frac{1}{1 + R_j}, \qquad R_j := R_{K_\sigma - \{e_1,\dots,e_j\}}(u_j, w) \le R_{G_j}(u_j, w),$$

where $G_j := G - (W \setminus \{w\}) - \{e_1, \dots, e_j\}$ and the inequality is (E2) for the identifications. The graph G_j consists of H, the vertex w, and the $m_j := d - j \ge b \ge 1$ remaining edges from w to the set $Y_j \subseteq V(H)$ of its remaining neighbours; $u_j \notin Y_j$ because G is simple. Let $\chi := \mathbf{1}_{u_j} - m_j^{-1}\mathbf{1}_{Y_j}$, a function on $V(H)$ with sum 0 and $\|\chi\|^2 = 1 + 1/m_j$. By (E6) some flow in H has net outflow χ and energy at most $\kappa(1 + 1/m_j)$. Adding a flow of $1/m_j$ along each edge from Y_j to w gives a unit flow from u_j to w in G_j with energy at most $\kappa(1+1/m_j)+1/m_j$. By (E5), $1 + R_j \le (1+\kappa)(m_j+1)/m_j$, and

$$\prod_{j=1}^{a} \frac{1}{1 + R_j} \;\ge\; (1+\kappa)^{-a} \prod_{j=1}^{a} \frac{d-j}{d-j+1} = (1+\kappa)^{-a}\, \frac{d-a}{d} = \frac{b}{d}(1+\kappa)^{-a}.$$

For the last claim, $a, b \ge 1$ gives $ab \ge a + b - 1 = d - 1$, the function $t \mapsto (t-1)/t^2$ decreases for $t \ge 2$, and $2 \le d \le D$. □

Lemma 4.3 routes the current through a single neighbour of w and bounds each factor separately. Spreading it over all remaining neighbours makes the losses telescope to b/d. In the exact checks of Appendix A the smallest ratio of the exact conditional probability to this bound is 1.0, on $K_{3,3}$.

4.4 The bound for one split

Theorem 4.5 (vertex-touch bound for a split)**.** *Let G be a connected graph with $N \ge 2$ vertices, let $E = A \sqcup B$ be any split, and let $W = (w_1, \dots, w_r)$ be an independent set of mixed vertices in a fixed order. Let $p_A(w_k), p_B(w_k) \in [0,1]$ be numbers for which the two conditional bounds of Lemma 4.3 hold for every σ, for instance the quantities defined there or the bounds of Lemma 4.4. Then*

$$\beta(\mathrm{UST}(G); A|B) \;\le\; \prod_{k=1}^{r} \Big(\frac{1 + \sqrt{1 - 4p_A(w_k)p_B(w_k)}}{2} \Big)^{1/2} \;\le\; \exp\Big(-\frac{1}{2} \sum_{k=1}^{r} p_A(w_k)\, p_B(w_k) \Big).$$

Proof. Put $a_k := a_{w_k}(T \cap A)$ and $b_k := b_{w_k}(T \cap B)$. The vector $(a_k)_k$ is a function of $T \cap A$ and $(b_k)_k$ is a function of $T \cap B$, so Lemma 3.2 gives $\beta(\mathrm{UST}(G); A|B) \le \|\sqrt{\mu}\|_{\mathrm{op}}$, where μ is the joint law of (a, b). Apply Lemma 3.3 to the pairs $(X_k, Y_k) = (a_k, b_k)$ in the fixed order. Fix k and a value π of the past $(a_{<k}, b_{<k})$ with positive probability, and let $K = K_k(\cdot \mid \pi)$. Then $K(1,1) = 0$ because T has an edge at w_k. Let $F_k := \bigcup_{j \ne k} S(w_j)$. The past is a function of $T \cap F_k$, so the event $\{(a_{<k}, b_{<k}) = \pi\}$ is a union of events $\{T \cap F_k = \sigma\}$ with positive probability. By the hypothesis on p_A, p_B and the tower property, $K(1,0) = \Pr[a_k = 1 \mid \pi] \ge p_A(w_k)$ and $K(0,1) = \Pr[b_k = 1 \mid \pi] \ge p_B(w_k)$. Lemma 3.5 with $p = p_A(w_k)$ and $p' = p_B(w_k)$

gives $\|\sqrt{K}\|_{\mathrm{op}}^2 \le (1+\sqrt{1-4p_A(w_k)p_B(w_k)})/2$, which bounds γ_k^2 in Lemma 3.3. The last inequality uses $(1+\sqrt{1-4x})/2 \le 1-x \le e^{-x}$. □

The theorem holds for every graph, every split and every independent set of mixed vertices. It needs no expansion and no 2-connectivity; those enter only through the lower bounds on p_A and p_B. With the quantities of Lemma 4.3, it can be evaluated exactly for a given split with one Laplacian solve per vertex of W and side, which is what the checks in Appendix A do.

4.5 Expansion

Lemma 4.6 (many mixed vertices)**.** *Let G have no isolated vertices, maximum degree at most D and $h(G) \ge h > 0$. If $|A| \ge m_0$ and $|B| \ge m_0$, then at least $2hm_0/(D(D+h))$ vertices are mixed.*

Proof. Let Z be the set of mixed vertices, and V_A and V_B the sets of vertices all of whose edges lie in A, respectively in B. Since there are no isolated vertices, $V = V_A \sqcup V_B \sqcup Z$, and no edge joins V_A to V_B. Every edge of A has both endpoints in $V_A \cup Z$, so $D(|V_A| + |Z|) \ge 2|A| \ge 2m_0$, and likewise for V_B. Let X be the smaller of V_A and V_B, so $|X| \le N/2$. Every edge leaving X ends in Z. Hence $D|Z| \ge |\partial X| \ge h|X| \ge h(2m_0/D - |Z|)$, which rearranges to the claim. □

A graph is *2-connected* if it has at least three vertices and $G - \{v\}$ is connected for every vertex v. A 2-connected graph has minimum degree at least 2: if v had a single neighbour y, then $G - \{y\}$ would separate v from the remaining vertices. It also has no bridge: if xy were a bridge, one of x, y would have another neighbour, since $N \ge 3$, and would be a cut vertex. Hence $|\partial S| \ge 2$ for every S with $\emptyset \ne S \ne V$.

The resistances in Lemma 4.4 are controlled by λ_2 of the graph that remains after W is deleted. The first tool turns edge expansion into a spectral gap.

Lemma 4.7 (Cheeger–Mohar lower bound)**.** *Let H be a connected graph with $n \ge 2$ vertices, maximum degree at most Δ and $h(H) \ge i > 0$. Then $i \le \Delta$ and*

$$\lambda_2(L_H) \;\ge\; \mu(\Delta, i) := \Delta - \sqrt{\Delta^2 - i^2} = \frac{i^2}{\Delta + \sqrt{\Delta^2 - i^2}} \;\ge\; \frac{i^2}{2\Delta}.$$

The function $\mu(\Delta, i)$ increases with i and decreases with Δ.

Proof. A single vertex v is admissible in the definition of $h(H)$, so $i \le \deg(v) \le \Delta$. Put $\lambda := \lambda_2(L_H)$. If $\lambda \ge \Delta$, then $\lambda \ge \mu(\Delta, i)$ because $\mu(\Delta, i) \le \Delta$. Assume $\lambda < \Delta$, and let f be an eigenvector for λ. It is nonzero and orthogonal to the constants, so it takes both signs; replacing f by $-f$ if necessary, the set $U := \{f > 0\}$ has $|U| \le n/2$. Let $g := \max(f, 0)$, normalized so that $\sum_v g(v)^2 = 1$. Since $(L_H f)(v) = \lambda f(v)$ for $v \in U$,

$$\lambda = \sum_v g(v)(L_H f)(v) = \sum_{uv \in E(H)} \big(g(u) - g(v)\big)\big(f(u) - f(v)\big) \;\ge\; \sum_{uv \in E(H)} \big(g(u) - g(v)\big)^2 =: Q,$$

because $(g(u) - g(v))(f(u) - f(v)) \ge (g(u) - g(v))^2$ for the positive part g of f. For $t \ge 0$ the level set $\{g^2 > t\}$ lies in U, so it has at most $n/2$ vertices and, if nonempty, at least i times as many boundary edges as vertices. An edge uv lies in $\partial\{g^2 > t\}$ exactly for t in an interval of length $|g(u)^2 - g(v)^2|$, so

$$\sum_{uv \in E(H)} |g(u)^2 - g(v)^2| = \int_0^\infty |\partial\{g^2 > t\}|\, dt \;\ge\; i \int_0^\infty |\{g^2 > t\}|\, dt = i.$$

By the Cauchy–Schwarz inequality and $(g(u)+g(v))^2 = 2g(u)^2 + 2g(v)^2 - (g(u)-g(v))^2$, the left side is at most $\sqrt{Q}\,\sqrt{2\sum_v \deg(v) g(v)^2 - Q} \le \sqrt{Q(2\Delta - Q)}$. The map $t \mapsto t(2\Delta - t)$ increases on $[0, \Delta]$ and $Q \le \lambda < \Delta$, so $i^2 \le \lambda(2\Delta - \lambda)$. With $\lambda < \Delta$ this gives $\lambda \ge \Delta - \sqrt{\Delta^2 - i^2}$. The remaining claims follow from the second expression for μ. □

This is Mohar's argument [Moh89]. We include it because of the case distinction: the upper bound $i \le \sqrt{\lambda_2(2\Delta - \lambda_2)}$ that the argument gives for $\lambda_2 \le \Delta$ can fail when $\lambda_2 > \Delta$. For K_3, $\Delta = 2$, $\lambda_2 = 3$ and $h = 2 > \sqrt{3}$. Only the lower bound on λ_2 is used below.

Lemma 4.8 (deleting a 3-separated set)**.** *Let W be a set of vertices of a graph G with pairwise distances at least 3, and let $H := G - W$ have at least two vertices. Then $h(H) \ge h(G) - 1$ and $\lambda_2(L_H) \ge \lambda_2(L_G) - 1$.*

Proof. W is independent, and every vertex outside W has at most one neighbour in W, since two such neighbours would be at distance at most 2. Let $S \subseteq V(H)$ with $0 < |S| \le |V(H)|/2 \le N/2$. At most $|S|$ edges join S to W, so $|\partial_H S| \ge |\partial_G S| - |S| \ge (h(G) - 1)|S|$. For the second claim let f be a unit vector on $V(H)$ orthogonal to the constants, and extend it by 0 on W to a unit vector $\tilde{f}$ on V, which is orthogonal to the constants on V. Then

$$\lambda_2(L_G) \le \tilde{f}^{\top} L_G \tilde{f} = f^{\top} L_H f + \sum_{v \in V(H)} f(v)^2 \, |N(v) \cap W| \le f^{\top} L_H f + 1.$$

Taking for f an eigenvector of $\lambda_2(L_H)$ gives the claim. □

The second tool counts connected sets. For $S \subseteq V$ let $N(S)$ be the set of vertices outside S with a neighbour in S.

Lemma 4.9 (counting connected sets)**.** *Let u be a vertex of a graph G and let $k \ge 1$ and $j \ge 0$ be integers. At most $\binom{k-1+j}{j}$ connected sets $S \ni u$ have $|S| = k$ and $|N(S)| = j$. Consequently, if $h(G) \ge h > 0$ and $b \ge 1$, at most*

$$F(b) := \binom{\lfloor b/h \rfloor + b + 1}{b+1} - 1 \;\le\; \Lambda^{b+1}, \qquad \Lambda := e\Big(1 + \frac{1}{h}\Big),$$

connected sets $S \ni u$ have $|S| \le N/2$ and $|\partial S| = b$.

Proof. Fix an order of V. Given S, explore it from u: keep a queue, initially (u), a set of discovered vertices, initially $\{u\}$, and a set of rejected vertices, initially empty. Remove the first vertex v of the queue and examine its neighbours in order. A neighbour that is discovered or rejected is skipped. For any other neighbour x, write the bit $\mathbf{1}[x \in S]$; if it is 1, mark x discovered and append it to the queue, and otherwise mark x rejected. Repeat until the queue is empty. Only vertices of S enter the queue, and since S is connected every vertex of S is discovered, so every vertex of $S \setminus \{u\}$ writes exactly one 1. A vertex of $N(S)$ is a neighbour of a vertex of S, so it is examined and writes exactly one 0, and no other vertex is examined. The string therefore has $k - 1$ ones and j zeros. Given G and the order, the string determines the run of the procedure, and hence S, the final set of discovered vertices. So S is determined by a binary string with $k - 1$ ones and j zeros, which proves the first claim.

For the second, $|S| \le N/2$ gives $h|S| \le b$, so $k := |S| \le K := \lfloor b/h \rfloor$, and $j := |N(S)| \le b$ because every vertex of $N(S)$ is the endpoint of an edge of ∂S. By the hockey-stick identity, applied twice,

$$\sum_{k=1}^{K} \sum_{j=0}^{b} \binom{k-1+j}{j} = \sum_{k=1}^{K} \binom{k+b}{b} = \binom{K+b+1}{b+1} - 1.$$

Finally $\binom{K+b+1}{b+1} \le \big(e(K+b+1)/(b+1)\big)^{b+1}$ and $K+b+1 \le (b+1)(1+1/h)$. □

The first claim is a form of a lemma of Fomin and Villanger [FV12], who bound by $\binom{k-1+j}{k-1}$ the number of connected sets of size k containing a given vertex that can be separated from the rest of the graph by deleting j vertices. We include the short proof.

Lemma 4.10 (random sparse deletion)**.** *Let G be a 2-connected graph with N vertices, maximum degree at most D and $h(G) \ge h > 0$, and let $Z \subseteq V$. Let $0 < h' \le h/(D+1)$ and $q \in (0,1]$, put*

$$\theta := \frac{1-h'/h}{D}, \qquad m(b) := \max\{2, \lfloor \theta b \rfloor + 1\}, \qquad \mathcal{S}(q) := D \sum_{b \ge 2} F(b) \binom{b-1}{m(b)-1} q^{m(b)-1},$$

with $F(b)$ as in Lemma 4.9, and suppose $\mathcal{S}(q) \le 1/2$. Then there is an independent set $W \subseteq Z$ with $|W| \ge q|Z|/(2(D+1))$ such that $H := G - W$ has at least two vertices and $h(H) \ge h'$. In particular H is connected.

Proof. Choose an independent set $W_0 \subseteq Z$ greedily; each chosen vertex excludes itself and at most D neighbours, so $|W_0| \ge |Z|/(D+1)$. Let $R \subseteq W_0$ contain each vertex independently with probability q. Call a pair (w, S) *dangerous* if $w \in R$, S is a connected set of G with $|S| \le N/2$, $w \notin S$, $w \in N(S)$, and $|N(S) \cap R| \ge m(b)$ for $b := |\partial S|$. The set S may meet R. Let R^* be R minus every w that lies in a dangerous pair.

Expected size. Fix $w \in W_0$ and condition on $w \in R$; the other vertices are still in R independently, each with probability at most q. If (w, S) is dangerous, then S contains one of the at most D neighbours u of w, and $b \ge 2$ since G has no bridge. For given u and b, Lemma 4.9 gives at most $F(b)$ choices of S. The set $N(S) \setminus \{w\}$ has at most $b-1$ vertices, since $|N(S)| \le b$, and at least $m(b)-1$ of them must lie in R; by a union bound over the subsets of size $m(b)-1$ this has conditional probability at most $\binom{b-1}{m(b)-1} q^{m(b)-1}$. Hence $\Pr[w \notin R^* \mid w \in R] \le \mathcal{S}(q) \le 1/2$, and $\mathbb{E}|R^*| \ge q|W_0|/2$. Fix an outcome with $|R^*| \ge q|W_0|/2$ and put $W := R^*$. It is independent and contained in Z. Every $w \in W$ has at least two neighbours, none in W, and if $W = \emptyset$ then $H = G$; so H has at least two vertices.

Expansion. Suppose some $S \subseteq V(H)$ with $0 < |S| \le |V(H)|/2$ has $|\partial_H S| < h'|S|$. The boundaries in H of the components of $H[S]$ add up to $\partial_H S$, so some component has the same property, and we may assume that S is connected. Then $|S| \le N/2$; let $b := |\partial_G S| \ge h|S|$. Every edge of $\partial_G S$ not in $\partial_H S$ goes to W, so $|\partial_H S| = b - e(S, W)$.

- If $N(S) \cap W = \emptyset$, then $|\partial_H S| = b \ge h|S| \ge h'|S|$, a contradiction.
- If $N(S) \cap W = \{w\}$, then $|\partial_H S| \ge b - D \ge h|S| - D$. Also $|\partial_H S| \ge 1$: the set $V \setminus (S \cup \{w\})$ is nonempty because $|S| \le N/2$ and $N \ge 3$, the graph $G - \{w\}$ is connected, and an edge of $G - \{w\}$ from S to $V \setminus (S \cup \{w\})$ ends outside W, hence in $V(H) \setminus S$. If $|S| \le 1/h'$, then $|\partial_H S| \ge 1 \ge h'|S|$. If $|S| > 1/h'$, then $(h - h')|S| - D > (h/h' - 1) - D \ge 0$ because $h/h' \ge D + 1$, so $|\partial_H S| \ge h|S| - D > h'|S|$. Both are contradictions.
- So $|N(S) \cap W| \ge 2$. Moreover $e(S, W) = b - |\partial_H S| > b - h'|S| \ge b(1 - h'/h)$, and each vertex of W has at most D edges into S, so $|N(S) \cap W| > \theta b$. Hence $|N(S) \cap W| \ge m(b)$.

Pick $w \in N(S) \cap W$. Then $w \in R$, $w \notin S$, and $|N(S) \cap R| \ge |N(S) \cap W| \ge m(b)$, so (w, S) is dangerous and $w \notin R^* = W$, a contradiction. Hence $h(H) \ge h'$, and H is connected because $h' > 0$. □

This is the alteration method in the form of a Peierls-type count. 2-connectivity is used twice: for $b \geq 2$, and in the case of a single neighbour in W. Without it, deleting a vertex of W could cut off a small piece of the graph, R_{G-W} could be infinite, and the touch probabilities could be zero. Proposition 11.2 shows that some restriction of this kind is needed.

4.6 The bound for 2-connected expanders

Lemma 4.11 (easy direction of the Cheeger inequality)**.** *For every graph G, $h(G) \geq \lambda_2(L_G)/2$.*

Proof. Let $0 < |S| \leq N/2$ and $\phi := \mathbf{1}_S - (|S|/N)\mathbf{1}$. Then $\phi \perp \mathbf{1}$, $\phi^\top L_G \phi = |\partial S|$ and $\|\phi\|^2 = |S|(N-|S|)/N$, so the Courant–Fischer theorem gives $|\partial S| \geq \lambda_2(L_G)|S|(N-|S|)/N \geq \lambda_2(L_G)|S|/2$. □

Proposition 4.12 (parametric bounds)**.** *Let G be a 2-connected simple graph with maximum degree at most D and $h(G) \geq h > 0$, and let $A|B$ be a split with $|A| \geq m_0$ and $|B| \geq m_0$, where $m_0 > 0$. Then $\beta(\mathrm{UST}(G); A|B)^2 \leq e^{-\kappa m_0}$ in each of the following cases.*

(i) h' and q are as in Lemma 4.10, with $\mathcal{S}(q) \leq 1/2$, and

$$\kappa = \kappa_P(h', q) := \frac{2h}{D(D+h)} \cdot \frac{q}{2(D+1)} \cdot \frac{D-1}{D^2} \cdot \Big(1 + \frac{1}{\mu(D, h')}\Big)^{-D}.$$

(ii) $\lambda_2(L_G) \geq \lambda > 1$ and

$$\kappa = \kappa_S(\lambda) := \frac{\lambda}{D(D+\lambda/2)} \cdot \frac{1}{D^2+1} \cdot \frac{D-1}{D^2} \cdot \Big(1 - \frac{1}{\lambda}\Big)^{D}.$$

Proof. G has minimum degree at least 2, so $D \geq 2$. Let Z be the set of mixed vertices. Suppose that $W \subseteq Z$ is independent, that $H := G - W$ is connected, and that $\lambda_2(L_H) \geq \ell > 0$. By Lemma 4.4, the numbers $p_A(w) := (b/d)(1+1/\ell)^{-a}$ and $p_B(w) := (a/d)(1+1/\ell)^{-b}$ satisfy the hypothesis of Theorem 4.5, with $p_A(w)p_B(w) \geq ((D-1)/D^2)(1+1/\ell)^{-D}$, so

$$\beta(\mathrm{UST}(G); A|B)^2 \leq \exp\Big(-|W| \, \frac{D-1}{D^2}\Big(1 + \frac{1}{\ell}\Big)^{-D}\Big). \tag{$*$}$$

(i) By Lemma 4.6, $|Z| \geq 2hm_0/(D(D+h))$. Lemma 4.10 gives an independent $W \subseteq Z$ with $|W| \geq q|Z|/(2(D+1))$ such that H is connected, has at least two vertices and $h(H) \geq h'$. H has maximum degree at most D, so Lemma 4.7 gives $\lambda_2(L_H) \geq \mu(D, h')$. Insert into $(*)$.

(ii) By Lemma 4.11, $h(G) \geq \lambda/2$, and the bound of Lemma 4.6 increases with h, so $|Z| \geq \lambda m_0/(D(D+\lambda/2)) > 0$. Choose $W \subseteq Z$ greedily with pairwise distances at least 3. Each chosen vertex excludes the at most $1 + D + D(D-1) = D^2 + 1$ vertices within distance 2 of it, so $|W| \geq |Z|/(D^2+1)$, and W is independent and nonempty. Each $w \in W$ has at least two neighbours, none in W, so H has at least two vertices, and Lemma 4.8 gives $\lambda_2(L_H) \geq \lambda - 1 > 0$. So H is connected, and $(*)$ with $\ell = \lambda - 1$ gives the claim, since $(1+1/(\lambda-1))^{-D} = (1-1/\lambda)^D$. □

If $h > \lambda/2$ is known, $\lambda/2$ can be replaced by h in the first factor of κ_S. The next theorem fixes the parameters in (i).

Theorem 4.13 (2-connected expanders)**.** *Let G be a 2-connected simple graph with N vertices, M edges, maximum degree at most D and $h(G) \geq h > 0$. Put $h_1 := \min(h, 1)$ and*

$$c = c(D, h) := \frac{h_1^{5D+4}}{(D+1)^{4(D+1)}\, e^{10(D+1)}}.$$

For every split $A|B$ with $|A| \geq m_0$ and $|B| \geq m_0$, where $m_0 > 0$,

$$\beta(\mathrm{UST}(G); A|B)^2 \;\leq\; e^{-3cm_0},$$

and for every balanced split $\beta(\mathrm{UST}(G); A|B)^2 \leq e^{-cN}$. If moreover $\lambda_2(L_G) \geq \lambda > 1$, both bounds hold with c replaced by

$$c_\lambda(D) := \frac{\lambda(D-1)(1-1/\lambda)^D}{3D^3(D+\lambda/2)(D^2+1)}.$$

Proof. G has minimum degree at least 2, so $M \geq N$, and a balanced split has $m_0 = M/3 \geq N/3$. The second part is Proposition 4.12(ii), since $\kappa_S(\lambda) = 3c_\lambda(D)$.

For the first part, $h(G) \geq h_1$, so we may apply Proposition 4.12(i) with h_1 in place of h; we write h for h_1, so $h \leq 1$. Take $h' := h/(D+1)$, so $\theta = 1/(D+1)$, and $q := q_0 := (8D\Lambda^3)^{-(D+1)}$ with $\Lambda = e(1+1/h)$.

$\mathcal{S}(q_0) \leq 1/2$. First, $m(b) - 1 = \max\{1, \lfloor \theta b \rfloor\} \geq \theta b/2$: if $\theta b < 2$ the left side is at least 1, and otherwise $\lfloor \theta b \rfloor \geq \theta b - 1 \geq \theta b/2$. So $q_0^{m(b)-1} \leq q_0^{\theta b/2}$. With $\binom{b-1}{m(b)-1} \leq 2^{b-1}$ and $F(b) \leq \Lambda^{b+1}$, the term of index b in $\mathcal{S}(q_0)$ is at most $D\Lambda^{b+1}2^{b-1}q_0^{\theta b/2} = (D\Lambda/2)x^b$, where $x := 2\Lambda q_0^{\theta/2}$. Since $q_0^\theta = 1/(8D\Lambda^3)$, we have $x^2 = 1/(2D\Lambda) \leq 1/16$, so $x \leq 1/4$ and

$$\mathcal{S}(q_0) \leq \frac{D\Lambda}{2} \cdot \frac{x^2}{1-x} \leq \frac{2D\Lambda x^2}{3} = \frac{1}{3}.$$

The constant. Lemma 4.7 gives $1/\mu(D, h') \leq 2D/h'^2 = 2D(D+1)^2/h^2$, and $1 + 2D(D+1)^2/h^2 \leq 3D(D+1)^2/h^2$. Write $n := D+1$. Using $D + h \leq n$ and $\ln \Lambda \leq 1 + \ln 2 + \ln(1/h)$,

$$\ln \frac{3}{\kappa_P(h', q_0)} \leq \ln \frac{3Dn}{2h} + \ln(2n) + n\Big(\ln(8D) + 3 + 3\ln 2 + 3\ln\frac{1}{h}\Big) + \ln\frac{D^2}{D-1} + D\ln\frac{3Dn^2}{h^2}.$$

The coefficient of $\ln(1/h)$ is $1 + 3n + 2D = 5D + 4$. For the rest use $D < n$, $D^2/(D-1) \leq D + 2 = n + 1 \leq e^n$ and $D \leq n - 1$: the terms are at most $\ln\frac{3}{2} + 2\ln n$, $\ln 2 + \ln n$, $n\ln n + n(6\ln 2 + 3)$, n, and $(n-1)(\ln 3 + 3\ln n)$. Their sum is $4n\ln n + n(\ln 3 + 6\ln 2 + 4) \leq 4n\ln n + 10n$. Hence $\kappa_P(h', q_0)/3 \geq h^{5D+4}n^{-4n}e^{-10n} = c$, and Proposition 4.12(i) gives $\beta^2 \leq e^{-3cm_0}$. □

So $\log(1/c) = (5D+4)\log(1/h_1) + O(D\log D) = O(D\log((D+1)/h_1))$ for all $D \geq 2$ and $h > 0$: for fixed D the constant is polynomial in h. The closed form is chosen for simplicity, not sharpness. The previous version of Theorem 4.13 chose W greedily with pairwise distances $s = 2\lceil 2D/h \rceil + 1$ and used a resistance bound $16D^3/h^2$, which gave $c = \frac{2h}{3D^{s+1}(D+h)}(2 + 16D^3/h^2)^{-D} \leq D^{-4D/h}$, exponentially small in D/h. Table 1 compares the constants.

For $D = 3$ and $h = 1/5$ the closed form gives $c \geq 5.2 \cdot 10^{-41}$, the parameters of the proof give $2.4 \cdot 10^{-34}$, and the optimized parameters about $5 \cdot 10^{-24}$; the previous constant was about $9 \cdot 10^{-44}$. At moderate h and large D the closed form is weaker than the previous constant, while the optimized column is stronger in every row. For 6-regular Ramanujan graphs, $\lambda_2 \geq 6 - 2\sqrt{5}$ gives $c \geq 8.0 \cdot 10^{-8}$.

Not every row describes graphs that exist for large N. Bisection-width bounds of Monien and Preis [MP06] give $h \leq 1/3 + o(1)$ for cubic and $h \leq 4/5 + o(1)$ for 4-regular graphs; all rows with $D \leq 4$ lie below these. For random cubic graphs the bisection width is at most $0.139822N$ with high probability [LM23], so $h \leq 0.2797 + o(1)$ for them; a recent preprint of Gaspers [Gas26], which we have not checked, claims bisection bounds that would give $h \leq 0.2796 + o(1)$ for every graph of maximum degree 3 and $h \leq 2/3 + o(1)$ for maximum degree 4. We omit the rows $D = 3$,

D	h	λ	previous	Thm. 4.13	(i) at q_0	(i) optimized	(ii)
3	0.05		306.6	119.1	102.1	73.6	
3	0.1		169.9	105.9	89.5	63.5	
3	0.2		99.2	92.8	77.4	53.6	
4	0.1		274.8	137.5	117.4	83.8	
4	0.2		157.7	120.8	102.1	71.2	
4	0.4		96.1	104.2	87.8	59.0	
6	0.2		290.6	179.2	153.5	107.6	
6	0.5		149.8	148.1	127.1	84.9	
6	0.76		115.7	133.8	116.3	75.1	
6	0.76	1.53	115.6	133.6	116.2	75.0	16.3
8	1.35	2.71	123.6	169.1	150.3	86.2	14.5
10	2	4	135.8	215.5	192.6	98.7	14.4
16	4.13	8.25	184.0	362.7	327.3	144.6	15.2
64	24.06	48.13	628.8	1735.3	1598.3	568.1	19.7

Table 1: Values of $\ln(1/c)$, lower is better, where $\beta^2 \le e^{-cN}$ for balanced splits. "Previous": the constant of the previous version. "Thm.": the closed form $c(D,h)$. "(i) at q_0": Proposition 4.12(i), divided by 3, with the parameters of the proof of Theorem 4.13. "(i) optimized": the same with $\mu(D,h')$, $h' = \frac{ih}{12(D+1)}$ for the best $i \in \{1,\dots,12\}$, and the largest q with $\mathcal{S}(q) \le 1/2$ found by bisection. "(ii)": $c_\lambda(D)$. The last five rows take $\lambda = D - 2\sqrt{D-1}$, the Ramanujan bound, and $h = \lambda/2$. All entries are floating-point evaluations (`verify/constants_touch.py`). For the optimized column the condition $\mathcal{S}(q) \le 1/2$ is also checked only in floating point, with the series truncated at $b = 3000$ and an explicit geometric tail bound; it is not certified by interval arithmetic.

$h = 0.3$ and $D = 4$, $h = 0.67$ for this reason. Ramanujan 6-regular graphs have $h \ge 0.76$ by Lemma 4.11.

The remaining losses are the factor q of the random deletion, the worst-case flow bound, and the use of a sparse set W instead of all mixed vertices. For specific graphs and splits, Theorem 4.5 can be evaluated directly; in the checks of Appendix A, with the resistance bounds of Lemma 4.3, it gives $\beta \le e^{-xN}$ with x between about $5 \cdot 10^{-4}$ and $1.6 \cdot 10^{-3}$ on cubic graphs of girth 8 to 12 and on 2-connected random cubic graphs with N up to 600. These are floating-point evaluations for particular splits, not bounds over all splits. Section 11.4 shows that bounds of the form of Theorem 4.5 cannot give $c \ge \mathrm{poly}(h)$ with a degree independent of D on some graphs, conditionally on an expansion estimate that we do not prove.

5 The complete graph and dense graphs

This section treats the uniform spanning tree of the complete graph K_n and then dense graphs with a linear spectral gap. Neither argument of the previous sections applies directly. The cycle bits give a per-cycle factor $1-O(n^{-3})$, and Theorem 4.5 needs an independent set of vertices, while the independent sets of K_n have one vertex. We keep the vertex-touch bits but drop independence. We condition on the tree edges inside a vertex set W, use only the vertices of W that these edges do not touch, and average over the conditioning with Lemma 3.7. Three ingredients make the constant reasonable for K_n: a star-deletion bound that is sharp on complete graphs (Lemma 5.3), the exact law of the set of vertices of W that the tree leaves isolated inside W (Lemma 5.5), and a choice of W that depends on the split and covers a constant fraction of the vertices.

Throughout, $\mu_n := \mathrm{UST}(K_n)$ is the uniform distribution on the n^{n-2} labelled trees, viewed as a distribution on the $M = n(n-1)/2$ edge indicators, and a balanced split is any partition $E(K_n) = A \sqcup B$ with $|A|, |B| \geq M/3$. The split need not come from a partition of the vertices. For a vertex v write $\deg_A(v)$ and $\deg_B(v)$ for the numbers of edges of A and of B at v. For $0 \leq p \leq \frac{1}{4}$ put

$$\Lambda(p) := \frac{1+\sqrt{1-4p}}{2}, \qquad g(p) := -\ln \Lambda(p),$$

so that Lemma 3.5 reads $\|\sqrt{K}\|_{\mathrm{op}}^2 \leq \Lambda(pp')$. The function Λ is decreasing and concave, and g is increasing and convex, with $g(p) \geq -\ln(1-p) \geq p$. For $m \geq 1$ put $c_m := (1+1/m)^{-m}$, which decreases to $1/e$.

Theorem 5.1 (complete graphs)**.** *Let $n \geq 3$ and let $A|B$ be a balanced split of $E(K_n)$. For every integer $1 \leq r \leq n-2$, with $m := n-r$ and $\Lambda_0 := \Lambda(c_m(1-\frac{1}{m})/9)$,*

$$\beta(\mu_n; A|B)^2 \;\leq\; \Phi_n(r) := \sum_{s=0}^{r} \binom{r}{s} (\Lambda_0 - 1)^s \Big(\frac{m}{n}\Big)^s \Big(1 - \frac{s}{n}\Big)^{r-s-1}.$$

Consequently, for every $n \geq 3$,

$$-\ln \beta(\mu_n; A|B)^2 \;\geq\; 0.0069032\, n - \tfrac{1}{2}\ln n - 2.01 \;\geq\; \frac{n}{146} - \frac{1}{2}\ln n - 3.$$

The coefficient 0.0069032 is $1/144.86$. The sum $\Phi_n(r)$ is $\mathbb{E}[\Lambda_0^{|I|}]$ for the random set I of Lemma 5.5 below, so it depends only on n and r, not on the split.

Corollary 5.2 (circuits for $\mathrm{UST}(K_n)$)**.** *Let $n \geq 3$, and let C be a smooth decomposable probabilistic circuit with nonnegative weights over the $M = n(n-1)/2$ edge variables of K_n,* whose leaves have arity at most $2M/3$, *with size $|C|$ counted as nodes plus wires. If $\mathrm{TV}(\mu_n, Q_C) \leq 1-\delta$ with $\delta > 0$, then*

$$|C| \;\geq\; \frac{\delta^2\, e^{n/146}}{e^3\sqrt{n}}.$$

Proof. Corollary 3.11 gives $|C| \geq \delta^2 / \max \beta^2$, and Theorem 5.1 gives $\max \beta^2 \leq e^3\sqrt{n}\, e^{-n/146}$. □

Determinism and exactness are not assumed, and different product terms of the circuit may use different balanced splits. All constants are absolute. The bound is nontrivial whenever $\delta > e^{3/2} n^{1/4} e^{-n/292}$, so for every fixed error $\mathrm{TV} < 1$ once n is large. The restriction on the arity of leaves cannot simply be dropped: one leaf of arity M represents μ_n exactly. Proposition 5.11 below shows that the exponent $\Theta(n)$ is the truth. Since $M = \Theta(n^2)$, this is $2^{\Theta(\sqrt{M})}$, not exponential in the number of variables.

5.1 Bits, conditioning and sharp star deletion

The first two lemmas hold in any connected simple graph $G = (V, E)$; Section 5.4 uses them again. Fix a split $A|B$ of E and a set W of r vertices, and put $U := V \setminus W$ and $m := |U|$. For $w \in W$ let $S_A^U(w)$ and $S_B^U(w)$ be the sets of edges of A and of B between w and U, and put $d_A(w) := |S_A^U(w)|$, $d_B(w) := |S_B^U(w)|$ and $d_U(w) := d_A(w) + d_B(w)$. Define

$$a_w := \mathbf{1}[T \cap S_A^U(w) = \emptyset], \qquad b_w := \mathbf{1}[T \cap S_B^U(w) = \emptyset].$$

The bit a_w is a function of x_A and b_w is a function of x_B.

Let $E(W)$ be the set of edges with both ends in W, and let $z(T) := T \cap E(W)$. Its two parts $T \cap E(W) \cap A$ and $T \cap E(W) \cap B$ are functions of x_A and of x_B, so z has the form required in Lemma 3.7. For a value z let W'_z be the set of vertices of W that no edge of z touches, and put $I := W'_{z(T)}$, the set of vertices of W that are isolated in the forest $T \cap E(W)$. If $z(T) = z$ and $w \in W'_z$, then T has no edge from w to W, but it has an edge at w, so $a_w = b_w = 1$ is impossible. This is the forbidden cell. Without conditioning on z it is not forbidden, since w may be attached to the tree through W.

Lemma 5.3 (sharp star deletion)**.** *Suppose that $m \geq 2$ and that $G[U]$ is connected, and put $\lambda_U := \lambda_2(L_{G[U]}) > 0$. Fix a value z of $z(T)$ and a vertex $w \in W'_z$ with $d_U(w) \geq 1$. Let $F := \bigcup_{x \in W'_z \setminus \{w\}} (S^U_A(x) \cup S^U_B(x))$ and let $\sigma \subseteq F$ satisfy $\Pr[z(T) = z,\ T \cap F = \sigma] > 0$. Write $\Pr_{z,\sigma}$ for the conditional law given these two events. Then*

$$\Pr_{z,\sigma}[a_w = 1] \ \geq \ p_A(w) := \frac{d_B(w)}{d_U(w)} \Big(1 + \frac{1}{\lambda_U}\Big)^{-d_A(w)},$$
$$\Pr_{z,\sigma}[b_w = 1] \ \geq \ p_B(w) := \frac{d_A(w)}{d_U(w)} \Big(1 + \frac{1}{\lambda_U}\Big)^{-d_B(w)}.$$

In particular $p_w := p_A(w)p_B(w) = \theta_A(w)\theta_B(w)(1 + 1/\lambda_U)^{-d_U(w)}$ with $\theta_A := d_A/d_U$ and $\theta_B := d_B/d_U$. For $G = K_n$ we have $\lambda_U = m = d_U(w)$, and $p_w = c_m \theta_A(w)\theta_B(w)$.

Proof. Write d_A, d_B, d_U for $d_A(w), d_B(w), d_U(w)$, and assume $d_B \geq 1$; otherwise the first bound is 0. Apply Lemma 4.2 with the edge set $E(W) \cup F$ and the configuration $z \cup \sigma$. Conditionally, $T \setminus (E(W) \cup F)$ is a uniform spanning tree of the multigraph K' obtained from $G_0 := G - E(W) - F$ by identifying the vertices of each component of $(V, z \cup \sigma)$ and deleting loops. No edge of $z \cup \sigma$ touches w: z does not because $w \in W'_z$, and every edge of F joins a vertex of $W \setminus \{w\}$ to U. So w is a node of K' by itself, and its edges in K' are exactly its d_U edges to U. The graph K' is connected, since it has a spanning tree.

Let $e_1, \dots, e_{d_A}$ be the edges of $S^U_A(w)$, with endpoints $u_1, \dots, u_{d_A} \in U$, and put $K'_i := K' - e_1 - \cdots - e_i$ and $Y_i := (S^U_A(w) \cup S^U_B(w)) \setminus \{e_1, \dots, e_i\}$, so that $k_i := |Y_i| = d_U - i \geq d_B \geq 1$. Let H_i be the graph on $U \cup \{w\}$ with the edges of $G[U]$ and of Y_i. The graph K'_i arises from $G_0 - e_1 - \cdots - e_i$, which contains H_i as a subgraph, by identifying vertices, never w with another vertex, and deleting loops. By (E2), $R_{K'_i}(w, u_i) \leq R_{H_i}(w, u_i)$, where u_i stands for its image in K'_i.

We bound $R_{H_i}(w, u_i)$ by Thomson's principle (E5). Let $D_i \subseteq U$ be the set of the k_i endpoints of Y_i; it does not contain u_i, because G is simple. Send $1/k_i$ along each edge of Y_i, from w to D_i. By (E6), some flow in $G[U]$ has net outflow $\chi := \mathbf{1}_{D_i}/k_i - \mathbf{1}_{u_i}$ and energy at most $\|\chi\|^2/\lambda_U = (1 + 1/k_i)/\lambda_U$. The two flows use disjoint edges and together form a unit flow from w to u_i, so

$$R_{K'_i}(w, u_i) \ \leq \ R_{H_i}(w, u_i) \ \leq \ \frac{1}{k_i} + \frac{1 + 1/k_i}{\lambda_U}, \qquad 1 + R_{K'_i}(w, u_i) \leq \frac{k_i + 1}{k_i}\Big(1 + \frac{1}{\lambda_U}\Big).$$

In particular u_i and w are connected in K'_i, so e_i is not a bridge of K'_{i-1}, and by induction every K'_i is connected. By Lemma 4.1, applied d_A times, with the tree uniform on K'_{i-1} given $e_1, \dots, e_{i-1} \notin T$,

$$\Pr_{z,\sigma}[a_w = 1] = \prod_{i=1}^{d_A} \frac{1}{1 + R_{K'_i}(w, u_i)} \ \geq \ \Big(1 + \frac{1}{\lambda_U}\Big)^{-d_A} \prod_{i=1}^{d_A} \frac{d_U - i}{d_U - i + 1} = \frac{d_B}{d_U}\Big(1 + \frac{1}{\lambda_U}\Big)^{-d_A}.$$

The bound for b_w is the same with A and B exchanged. For $G = K_n$, $G[U] = K_m$ has $\lambda_2(L_{K_m}) = m$, and every $w \in W$ has $d_U(w) = m$. □

For K_n the bound is sharp: in exact enumeration for $n \le 8$, the smallest ratio of $\Pr_{z,\sigma}[a_w = 1]$ to $p_A(w)$ over all pasts is 1.000 to three decimals (Appendix A). An earlier version of this paper bounded $p_A(w)p_B(w)$ through the event that w is a leaf, by $e^{-1}\theta_A e^{-4\theta_A}$ for $\theta_A \le \frac{1}{2}$, which is smaller by a factor of about 3.7 at $\theta_A = \frac{1}{2}$.

Lemma 5.4 (master bound). *In the setting of Lemma 5.3, put $p_w := 0$ if $d_U(w) = 0$. Then*

$$\beta(\mathrm{UST}(G); A|B)^2 \;\le\; \mathbb{E}\Big[\prod_{w\in I} \Lambda(p_w)\Big], \qquad I = W'_{z(T)}.$$

Proof. Fix z with $\Pr[z(T) = z] > 0$, and order W'_z as $w_1, \dots, w_s$; the order is fixed once z is. Let μ_z be the law of T given $z(T) = z$. By Lemma 3.2, $\beta(\mu_z; A|B)$ is at most the norm of the square root of the joint law of $((a_{w_k})_k, (b_{w_k})_k)$ under μ_z. For the pair (a_{w_k}, b_{w_k}), every value π of the joint past $(a_{w_i}, b_{w_i})_{i<k}$ is a function of $T \cap F$, where F is the set of Lemma 5.3 for $w = w_k$. So the conditional law $K_k(\cdot \,|\, \pi)$ is an average over σ of the laws in Lemma 5.3. It has $K_k(1,1 \,|\, \pi) = 0$, $K_k(1,0 \,|\, \pi) = \Pr[a_{w_k} = 1 \,|\, z, \pi] \ge p_A(w_k)$ and $K_k(0,1 \,|\, \pi) \ge p_B(w_k)$. Lemma 3.5 gives $\gamma_k^2 \le \Lambda(p_{w_k})$, also when $p_{w_k} = 0$, and Lemma 3.3 gives $\beta(\mu_z; A|B)^2 \le \prod_{w\in W'_z} \Lambda(p_w)$. Lemma 3.7 averages over z. □

5.2 The isolated vertices of W in K_n

In K_n the law of I is explicit. Let $\mathsf{T}(x) := \sum_{k\ge1} k^{k-1}x^k/k!$ be the tree function, which converges for $|x| \le 1/e$, has nonnegative coefficients and satisfies $\mathsf{T}(x) = xe^{\mathsf{T}(x)}$, so that $\mathsf{T}(te^{-t}) = t$ for $0 \le t \le 1$.

Lemma 5.5 (isolated vertices). *Let $W \subseteq V(K_n)$ with $|W| = r$ and $m = n - r \ge 1$, let $T \sim \mu_n$ and $I = W'_{z(T)}$.*

(I1) For $S \subseteq W$ with $|S| = s$, $\Pr[S \subseteq I] = (m/n)^s(1 - s/n)^{r-s-1}$.

(I2) For every real Λ,

$$\mathbb{E}[\Lambda^{|I|}] = \sum_{s=0}^{r} \binom{r}{s} (\Lambda - 1)^s \Big(\frac{m}{n}\Big)^s \Big(1 - \frac{s}{n}\Big)^{r-s-1} = \frac{r!}{m\, n^{r-1}} [x^r] \exp\big(m(\mathsf{T}(x) - x) + m\Lambda x\big).$$

(I3) For numbers $\Lambda_w \ge 0$ ($w \in W$) with mean $\bar\Lambda$, $\mathbb{E}\prod_{w\in I} \Lambda_w \le \mathbb{E}[\bar\Lambda^{|I|}]$.

Proof. (I1) The event $S \subseteq I$ says that T avoids the set H_S of edges between S and W. For a simple graph H on V, $L_{K_n - H} = nI - J - L_H$, and L_H commutes with J, so on $\mathbf{1}^\perp$ the eigenvalues of L_{K_n-H} are $n - \mu_i$, where $\mu_1, \dots, \mu_{n-1}$ are the eigenvalues of L_H on $\mathbf{1}^\perp$. By the matrix-tree theorem, $\tau(K_n - H) = n^{-1}\prod_{i=1}^{n-1}(n - \mu_i)$. The graph H_S is the join of a clique on S and an independent set on $W \setminus S$, plus m isolated vertices. For $1 \le s < r$ its eigenvectors are explicit: vectors of sum zero supported on S have eigenvalue r, vectors of sum zero supported on $W \setminus S$ have eigenvalue s, $(r - s)\mathbf{1}_S - s\mathbf{1}_{W\setminus S}$ has eigenvalue r, and the rest of $\mathbf{1}^\perp$ is spanned by vectors that are constant on W, with eigenvalue 0. So $\tau(K_n - H_S) = n^{-1} \cdot n^m(n-r)^s(n-s)^{r-s-1}$; divide by n^{n-2}. The cases $s = 0$ (no edges) and $s = r$ (a clique on W, with eigenvalue r of multiplicity $r - 1$) give the same formula.

(I2) The first equality follows from $\Lambda^{|I|} = \sum_{S\subseteq I}(\Lambda-1)^{|S|}$ and (I1). For the second, fix a forest F with vertex set W and components of sizes $c_1,\dots,c_t$. By Lemma 4.2, the trees with $T\cap E(W)=F$ correspond to the spanning trees of the multigraph Γ_F in which each component of F is one node, joined by c_i parallel edges to each vertex of U, and U carries K_m. Eliminating the t component nodes from the Laplacian of Γ_F, whose diagonal block on them is $\mathrm{diag}(c_i m)$, leaves the Schur complement $L_{K_m} + rI - (r/m)J = (n/m)L_{K_m}$ on U. Deleting the row and column of one vertex of U before eliminating, the matrix-tree theorem gives

$$\tau(\Gamma_F) = \prod_i (c_i m)\cdot\Big(\frac{n}{m}\Big)^{m-1} m^{m-2} = n^{m-1}m^{t-1}\prod_i c_i.$$

So $\Pr[T\cap E(W)=F] = n^{1-r}m^{-1}\prod_i (mc_i)$. There are $\prod_i c_i^{c_i-2}$ forests with given components, and the isolated vertices of F are its components of size 1. Summing over set partitions of W,

$$\mathbb{E}[\Lambda^{|I|}] = \frac{1}{m\,n^{r-1}}\sum_{\pi}\prod_{P\in\pi} w(|P|), \qquad w(1)=m\Lambda,\quad w(c)=mc^{c-1}\ (c\ge 2),$$

and the exponential formula gives $r!\,[x^r]\exp(\sum_c w(c)x^c/c!)$ for the sum, with $\sum_c w(c)x^c/c! = m(\mathsf{T}(x)-x)+m\Lambda x$.

(I3) Permutations of V that fix W as a set preserve μ_n and act transitively on the s-subsets of W, so given $|I|=s$ the set I is uniform among them. Hence $\mathbb{E}\prod_{w\in I}\Lambda_w = \sum_s \Pr[|I|=s]\,e_s(\Lambda)/\binom{r}{s}$, where e_s is the elementary symmetric polynomial, and Maclaurin's inequality [HLP52] gives $e_s(\Lambda)/\binom{r}{s} \le \bar\Lambda^s$. □

5.3 Proof of Theorem 5.1

Choice of W. Put $q_v := \deg_A(v)/(n-1)$ and $a := |A|/M \in [\frac13,\frac23]$. De Caen's inequality [dC98] $\sum_v \deg_A(v)^2 \le |A|(2|A|/(n-1)+n-2)$ and $\sum_v \deg_A(v) = 2|A| = an(n-1)$ give

$$\frac1n\sum_v q_v(1-q_v) = \frac1n\Big(\frac{2|A|}{n-1} - \frac{\sum_v \deg_A(v)^2}{(n-1)^2}\Big) \;\ge\; \frac{a(1-a)\,n}{2(n-1)} \;\ge\; \frac19.$$

Let W be a uniform r-set and w a uniform element of W. Then w is uniform on V, and given w the set U is a uniform m-subset of $V\setminus\{w\}$, so $d_A(w)$ is hypergeometric: m draws from $n-1$ neighbours of which $\deg_A(w)$ are A-neighbours. Its variance formula gives

$$\begin{aligned}\mathbb{E}[\theta_A(w)\theta_B(w)\mid w] &= \frac{\mathbb{E}[d_A(w)(m-d_A(w))\,|\,w]}{m^2}\\ &= q_w(1-q_w)\Big(1-\frac{r-1}{m(n-2)}\Big) \;\ge\; q_w(1-q_w)\Big(1-\frac1m\Big).\end{aligned}$$

Averaging over w, some r-set W has $\frac1r\sum_{w\in W}\theta_A(w)\theta_B(w) \ge (1-\frac1m)/9$. Fix such a W; it depends on the split, and is chosen before the tree is sampled.

Main inequality. By Lemma 5.3, $p_w = c_m\theta_A(w)\theta_B(w) \le \frac14$, and the mean of the p_w over W is at least $p_0 := c_m(1-\frac1m)/9$. Since Λ is concave and decreasing, the mean $\bar\Lambda$ of the numbers $\Lambda(p_w)$ satisfies $\bar\Lambda \le \Lambda(\text{mean of } p_w) \le \Lambda(p_0) = \Lambda_0$. Lemmas 5.4 and 5.5 give

$$\beta(\mu_n;A|B)^2 \;\le\; \mathbb{E}\prod_{w\in I}\Lambda(p_w) \;\le\; \mathbb{E}[\bar\Lambda^{|I|}] \;\le\; \mathbb{E}[\Lambda_0^{|I|}] = \Phi_n(r).$$

Explicit bound. Let $\alpha := 0.38$, $r := \lfloor \alpha n \rfloor$, so that $1 \le r \le n-2$ and $m \ge 0.62n$ for $n \ge 3$, and let $\rho := 0.384\, e^{-0.384} < 1/e$, so that $\mathsf{T}(\rho) = 0.384$. The power series $\exp(m(\mathsf{T}(x)-x)+m\Lambda_0 x)$ has nonnegative coefficients, so its coefficient of x^r is at most its value at ρ divided by ρ^r. Since $c_m \ge 1/e$, $p_0 \ge \frac{1}{9e} - \frac{1}{9em}$, and since $|\Lambda'(p)| = (1-4p)^{-1/2} \le 1.094$ for $p \le \frac{1}{9e}$, we have $\Lambda_0 \le \Lambda_\infty + \frac{1.094}{9em}$ with $\Lambda_\infty := \Lambda(\frac{1}{9e}) = 0.957301\ldots$. With Robbins' bound $\ln r! \le r\ln r - r + \frac{1}{2}\ln(2\pi r) + \frac{1}{12r}$ [Rob55], Lemma 5.5(I2) gives

$$\ln \Phi_n(r) \;\le\; -r\Big(1 + \ln\frac{\rho n}{r}\Big) + mK + \ln\frac{n}{m} + \frac{1}{2}\ln(2\pi r) + \frac{1}{12} + \frac{1.094\,\rho}{9e}, \qquad K := \mathsf{T}(\rho) - \rho + \Lambda_\infty \rho.$$

The function $x \mapsto x(1+\ln(\rho n/x))$ is concave, so on $[\alpha n - 1, \alpha n] \ni r$ it is at least its smaller endpoint value, which is at least $\alpha n(1+\ln(\rho/\alpha)) - (1+\ln(\rho/\alpha))$, since $1+\ln(\rho/\alpha) > 0$. With $m \le (1-\alpha)n + 1$, $m \ge 0.62n$ and $r \le \alpha n$,

$$-\ln \Phi_n(r) \;\ge\; Rn - \tfrac{1}{2}\ln n - C_0, \qquad R := \alpha\Big(1 + \ln\frac{\rho}{\alpha}\Big) - (1-\alpha)K,$$

$$C_0 := \Big(1 + \ln\frac{\rho}{\alpha}\Big) + K - \ln 0.62 + \tfrac{1}{2}\ln(2\pi\alpha) + \tfrac{1}{12} + \frac{1.094\,\rho}{9e}.$$

Numerically $K = 0.37283\ldots$, $R = 0.00690327\ldots > 0.0069032$ and $C_0 = 2.00752\ldots < 2.01$. Finally $0.0069032 > 1/146$. □

Remark 5.6 (what the proof loses, and what is known about the truth). The proof has three relaxations, and their scope matters.

(a) *The scalar average.* After the choice of W the proof replaces the numbers $\Lambda(p_w)$ by the single number Λ_0. For that substitution, Jensen's inequality gives $\Phi_n(r) \ge \Lambda_0^{\mathbb{E}|I|}$, and $\mathbb{E}|I| = r(m/n)(1-1/n)^{r-2} \le n\,\alpha(1-\alpha)e^{-\alpha}e^{2/n}$ with $\alpha = r/n$ by (I1). Since $\max_\alpha \alpha(1-\alpha)e^{-\alpha} = 0.16112\ldots$, at $\alpha = (3-\sqrt{5})/2$, and $g(\frac{1}{9e}) = 0.04364\ldots$, this relaxation cannot certify more than about 0.00703 per vertex, and Theorem 5.1 reaches 98% of that. This ceiling applies to the prescribed uniform average with de Caen's bound only. It is not a ceiling for split-dependent choices of W, for the heterogeneous product of Lemma 5.4, or for single-vertex bits in general. For example, for the quasi-star split (the edges at the sn vertices of largest index form A, with $s = 1-\sqrt{2/3}$), taking W outside the hub vertices with $\alpha = 0.38$ gives a per-vertex kernel $p = \frac{s}{1-\alpha}(1-\frac{s}{1-\alpha})/e$ and, by the same chain of inequalities, an exponent of about 0.0135 per vertex (computed; Appendix A).

(b) *The kernel.* The kernels of Lemma 5.3 satisfy $p_w \le c_m/4$, so any bound of the form $\prod_w \Lambda(p_w)$ over at most n vertices with these kernels certifies at most $n\,g(c_m/4)$, and $g(c_m/4) \to g(\frac{1}{4e}) = 0.1081\ldots$. This is a limit of these kernel bounds, not of every statistic of one vertex.

(c) *The truth.* For each split, $\mathrm{BC}(\mu_n, \mu_A \otimes \mu_B) \le \beta(\mu_n; A|B)$, where $\mu_A \otimes \mu_B$ is the product of the two marginals, so $-2\ln \mathrm{BC}/n$ is an upper bound on the true rate $-\ln \max \beta^2/n$. Exact values for $n \le 7$ and Monte Carlo estimates up to $n = 160$ (not certified) give values decreasing to between 0.17 and 0.18 at $n = 160$ (the spread is Monte Carlo error), for a quasi-star split. So the true rate lies between 0.0069 and, by this estimate, about 0.17; we do not know where. Reaching a rate near 0.17 with the kernels of Lemma 5.3 would, by (b), need statistics that involve several vertices or a different argument. This is relative to that target only: the estimate is an upper bound, and the true rate may be below 0.108.

For $6 \le n \le 9$, exact computation of $\beta(\mu_n; A|B)$ on the splits we tried, including splits found by local search, gives $-\ln \beta/n$ between 0.06 and 0.15 (Appendix A).

The proof works for any balance threshold: if $|A|, |B| \ge aM$ with $0 < a \le \frac{1}{2}$, de Caen's inequality gives $\frac{1}{n}\sum_v q_v(1-q_v) \ge a(1-a)/2$, and the rest goes through with $1/9$ replaced by $a(1-a)/2$.

5.4 Dense graphs

Lemma 5.5 is specific to K_n. For other graphs we average over z with a cruder bound.

Lemma 5.7 (edge products of the tree)**.** *Let G be connected, $F \subseteq E$ and $y_e \ge 1$ for $e \in F$. Then*

$$\mathbb{E} \prod_{e \in T \cap F} y_e \;\le\; \exp\Big(\sum_{e \in F} (y_e - 1)\Pr[e \in T]\Big).$$

Proof. Expand $\prod_{e\in F}(1 + (y_e - 1)\mathbf{1}[e \in T]) = \sum_{S \subseteq F} \prod_{e \in S}(y_e - 1)\,\mathbf{1}[S \subseteq T]$, with nonnegative coefficients. By the transfer-current theorem of Burton and Pemantle [BP93], $\Pr[S \subseteq T] = \det Y_S$ for a positive semidefinite matrix Y_S whose diagonal entries are the probabilities $\Pr[e \in T]$, $e \in S$, and Hadamard's inequality gives $\det Y_S \le \prod_{e \in S} \Pr[e \in T]$. So the expectation is at most $\prod_{e \in F}(1 + (y_e - 1)\Pr[e \in T])$; use $1 + x \le e^x$. □

Theorem 5.8 (dense graphs)**.** *Let G be a connected simple graph on $n \ge 3$ vertices with minimum degree $d_{\min}$, maximum degree $d_{\max}$ and $\lambda_2 := \lambda_2(L_G)$, and let $A|B$ be an edge split with $|A|, |B| \ge a|E|$, where $0 < a \le \frac{1}{2}$. Put $t := a d_{\min}/d_{\max}$ and $\varphi(x) := (e^x - 1)/x$. Let r be an integer with $1 \le r < \lambda_2$ and $m := n - r \ge 2$, and put*

$$\bar p := \frac{m(m-1)}{(n-1)(n-2)} \cdot \frac{d_{\min}\lambda_2\, t(1-t)}{(2 + \lambda_2/d_{\min})\, d_{\max}^2} \cdot e^{-d_{\max}/(\lambda_2 - r)}, \qquad g^* := g\Big(\tfrac{1}{4}\big(1 + \tfrac{1}{m}\big)^{-\max(0,\, d_{\min} - r + 1)}\Big).$$

Then $\bar p \le \frac{1}{4}$ and

$$-\ln \beta(\mathrm{UST}(G); A|B)^2 \;\ge\; r\, g(\bar p)\Big(1 - \frac{2(r-1)\varphi(2g^*)}{\lambda_2}\Big).$$

Proof. Spectral facts. For $U = V \setminus W$ with $|W| = r$, let x be an eigenvector of $L_{G[U]}$ for $\lambda_2(L_{G[U]})$, orthogonal to $\mathbf{1}_U$, extended by 0 to V. It is orthogonal to $\mathbf{1}_V$, and $x^\top L_G x = x^\top L_{G[U]} x + \sum_{u \in U} |N(u) \cap W| x_u^2 \le (\lambda_2(L_{G[U]}) + r)\|x\|^2$, so $\lambda_U := \lambda_2(L_{G[U]}) \ge \lambda_2 - r > 0$ by the Courant–Fischer theorem. In particular $G[U]$ is connected. Also $\lambda_U \le m$, since adding edges to $G[U]$ does not decrease eigenvalues and $\lambda_2(L_{K_m}) = m$. By (E1) and (E4), $\Pr[e \in T] \le 2/\lambda_2$ for every edge.

Choice of W. Let $\theta_v := \deg_A(v)/\deg(v)$ and $\bar\theta := \frac{1}{n}\sum_v \theta_v$. From $\sum_v \deg(v)\theta_v = 2|A| \ge a\sum_v \deg(v) \ge an\, d_{\min}$ and $\sum_v \deg(v)\theta_v \le d_{\max} n\bar\theta$ we get $\bar\theta \ge t$, and in the same way $1 - \bar\theta \ge t$. Let $S := \sum_v \deg(v)\theta_v(1-\theta_v)$. With $x_e := \mathbf{1}[e \in A]$, $(\theta_u - \theta_v)^2 \le 2(\theta_u - x_e)^2 + 2(x_e - \theta_v)^2$ for an edge $e = uv$, and summing over edges gives $\sum_{uv \in E}(\theta_u - \theta_v)^2 \le 2\sum_v \sum_{e \ni v}(x_e - \theta_v)^2 = 2S$. The left side is at least $\lambda_2 \sum_v (\theta_v - \bar\theta)^2$ by the Courant–Fischer theorem. Hence

$$S \;\ge\; d_{\min}\sum_v \theta_v(1-\theta_v) = d_{\min}\Big(n\bar\theta(1-\bar\theta) - \sum_v(\theta_v - \bar\theta)^2\Big) \;\ge\; d_{\min}\Big(nt(1-t) - \frac{2S}{\lambda_2}\Big),$$

so $S \ge n d_{\min}\lambda_2 t(1-t)/(\lambda_2 + 2d_{\min})$ and $\frac{1}{n}\sum_v \deg_A(v)\deg_B(v) \ge \frac{d_{\min}}{n} S \ge d_{\min}\lambda_2 t(1-t)/(2 + \lambda_2/d_{\min})$. If W is a uniform r-set and w a uniform element of W, then U is a uniform m-subset of $V \setminus \{w\}$, and $\mathbb{E}[d_A(w) d_B(w) \mid w] = \deg_A(w)\deg_B(w)\, m(m-1)/((n-1)(n-2))$, since each pair of an A-neighbour and a B-neighbour lies in U with this probability. So some r-set W has $\frac{1}{r}\sum_{w \in W} d_A(w) d_B(w)/d_{\max}^2 \ge \bar p\, e^{d_{\max}/(\lambda_2 - r)}$; fix it.

Kernels. By Lemma 5.3 we have $p_w = \theta_A(w)\theta_B(w)(1+1/\lambda_U)^{-d_U(w)}$, and this is at least $(d_A(w)d_B(w)/d_{\max}^2)\, e^{-d_{\max}/(\lambda_2 - r)}$, using $d_U(w) \le d_{\max}$, $\lambda_U \ge \lambda_2 - r$ and $(1+1/x)^{-d} \ge e^{-d/x}$. So the mean of the p_w over W is at least $\bar{p}$, and $\bar{p} \le \frac{1}{4}$ because every $p_w \le \theta_A\theta_B \le \frac{1}{4}$. Also $d_U(w) \ge d_{\min} - (r-1)$ and $\lambda_U \le m$ give $p_w \le \frac{1}{4}(1+\frac{1}{m})^{-\max(0, d_{\min}-r+1)}$, so $g_w := g(p_w) \le g^*$.

Averaging over z. By Lemma 5.4, $\beta^2 \le \mathbb{E}\prod_{w\in I} e^{-g_w}$. Every $w \in W \setminus I$ is touched by an edge of $z(T)$, and $g_w \ge 0$, so

$$\prod_{w\in I} e^{-g_w} \;\le\; e^{-\sum_{w\in W} g_w} \prod_{uv \in T\cap E(W)} e^{g_u + g_v}.$$

Lemma 5.7 with $y_{uv} := e^{g_u+g_v}$ and $\Pr[e \in T] \le 2/\lambda_2$, then $e^x - 1 \le x\varphi(2g^*)$ for $0 \le x \le 2g^*$, give

$$-\ln \beta^2 \;\ge\; \sum_{w\in W} g_w - \frac{2\varphi(2g^*)}{\lambda_2} \sum_{uv\in E(W)} (g_u + g_v) \;\ge\; \Big(1 - \frac{2(r-1)\varphi(2g^*)}{\lambda_2}\Big) \sum_{w\in W} g_w,$$

since each w lies in at most $r-1$ edges of $E(W)$. If the bracket is negative, the theorem's bound is negative and holds trivially. Otherwise use $\sum_w g_w \ge r\, g(\text{mean of } p_w) \ge r\, g(\bar{p})$, by convexity and monotonicity of g. □

Corollary 5.9 (linear spectral gap)**.** *Let $0 < \kappa \le 1$ and $0 < a \le \frac{1}{2}$. If G has $n \ge \max(20, 18/\kappa)$ vertices and $\lambda_2(L_G) \ge \kappa n$, then every edge split with $|A|, |B| \ge a|E|$ has*

$$-\ln \beta(\mathrm{UST}(G); A|B)^2 \;\ge\; \frac{a\kappa^4 e^{-1.14/\kappa}}{485}\, n.$$

Consequently every smooth decomposable PC with nonnegative weights over the $M = |E|$ edge variables, with leaves of arity at most $2M/3$ and $\mathrm{TV}(\mathrm{UST}(G), Q_C) \le 1 - \delta$, has $|C| \ge \delta^2 \exp(\kappa^4 e^{-1.14/\kappa} n/1455) = \delta^2 2^{\Omega(n)}$.

Proof. Put $\varepsilon := \kappa \ln 2/6$ and $r := \lfloor \varepsilon n \rfloor$, so $1 \le r < \kappa n \le \lambda_2$ and $m \ge (1-\varepsilon)n \ge 0.88n$. The Rayleigh quotient of $\mathbf{1}_v - \mathbf{1}/n$ gives $\lambda_2 \le \frac{n}{n-1}\deg(v)$ for every v, so $d_{\min} \ge \kappa(n-1)$, and $d_{\max} \le n-1$. Hence $d_{\min}/d_{\max} \ge \kappa$, $t \ge a\kappa$ and $t(1-t) \ge a\kappa/2$ (as $t \le \frac{1}{2}$), $d_{\min}\lambda_2/d_{\max}^2 \ge \kappa^2$, $2 + \lambda_2/d_{\min} \le 2 + \frac{n}{n-1} \le 3.5$, $\lambda_2 - r \ge (1 - \frac{\ln 2}{6})\kappa n$, so $d_{\max}/(\lambda_2 - r) \le 1.14/\kappa$, and $m(m-1)/((n-1)(n-2)) \ge \frac{1}{2}$ for $n \ge 20$. So $\bar{p} \ge a\kappa^3 e^{-1.14/\kappa}/14$. Since $g^* \le g(\frac{1}{4}) = \ln 2$ and φ is increasing, $2(r-1)\varphi(2g^*)/\lambda_2 \le 2\varepsilon n \cdot \frac{3}{2\ln 2}/(\kappa n) = \frac{1}{2}$. Theorem 5.8 and $g(p) \ge p$ give $-\ln\beta^2 \ge \frac{r}{2}\bar{p}$, and $r \ge \varepsilon n - 1 \ge \varepsilon n/2$ for $n \ge 18/\kappa$. Finally $\frac{\varepsilon}{4} \cdot \frac{a\kappa^3}{14} = \frac{\ln 2}{336} a\kappa^4 \ge a\kappa^4/485$. The circuit bound is Corollary 3.11 with $a = \frac{1}{3}$. □

The hypothesis is equivalent, up to constants, to linear edge expansion: $h(G) \ge \lambda_2/2$ by Lemma 4.11, and $h(G) \ge cn$ gives $\lambda_2 \ge h^2/(2d_{\max}) \ge c^2 n/2$ by Lemma 4.7. For example, $K_{n/2,n/2}$ has $\lambda_2 = n/2$. For the random graph $G(n, \frac{1}{2})$, the corollary applies with $\kappa = \frac{1}{4}$ *conditionally* on the standard concentration fact that $\lambda_2(L_G) \ge n/4$ with probability tending to one, which we do not re-derive here. The constant is poor. It comes from using $d_{\max}$ in the exponent of Lemma 5.3 and from the crude averaging over z of Lemma 5.7; even for K_n, Theorem 5.8 gives a much smaller rate than Theorem 5.1.

5.5 Minimum degree is not enough

A linear minimum degree does not suffice for the method, because of sparse cuts.

Proposition 5.10 (two cliques and a matching)**.** *Let $h \geq 3$ and let G_h consist of two disjoint copies of K_h joined by a perfect matching, so $n = 2h$, the minimum degree is $h = n/2$, and $\lambda_2(L_{G_h}) = 2$. Let A consist of the edges of the first clique and the matching, and B of the edges of the second clique. This split is balanced, and*

$$\beta(\mathrm{UST}(G_h); A|B)^2 \;\geq\; P_1 := \Pr[|T \cap \mathit{matching}| = 1] = \Big(1 + \frac{2}{h}\Big)^{-(h-1)} \;>\; e^{-2}.$$

Proof. $M = h^2$, $|A| = \binom{h}{2} + h$ and $|B| = \binom{h}{2} \geq h^2/3$ for $h \geq 3$. The Laplacian is $\left(\begin{smallmatrix} L_{K_h}+I & -I \\ -I & L_{K_h}+I \end{smallmatrix}\right)$. On vectors (x, x) it acts as L_{K_h} and on vectors $(x, -x)$ as $L_{K_h} + 2I$, so its spectrum is $\{0, h^{(h-1)}, 2, (h+2)^{(h-1)}\}$ and $\tau(G_h) = h^{h-2}(h+2)^{h-1}$ by the matrix-tree theorem. The trees with exactly one matching edge are a spanning tree of each clique plus one of the h matching edges, so there are $h \cdot h^{2(h-2)}$ of them, and $P_1 = (h/(h+2))^{h-1}$. Also $(h-1)\ln(1+2/h) < 2$. On this event the law of T is uniform over the triples (tree of the first clique, matching edge, tree of the second clique), which is a product distribution q across $A|B$. So $\mathrm{UST}(G_h) \geq P_1 q$ pointwise, and $\mathrm{BC}(\mathrm{UST}(G_h), q) \geq \sqrt{P_1}$. □

So for these graphs the right-hand side of Corollary 3.11, $\delta^2/\max \beta^2$, is less than $\delta^2 e^2$; this is a limit of the reduction, not a lower bound on circuits. Theorem 5.8 applies with $r = 1$ but gives only a weak bound, and Corollary 5.9 gives no uniform exponential rate, because $\lambda_2 = 2$ while n grows. A weaker statement still holds under an extra hypothesis. Marginalizing a smooth decomposable PC to the variables of the first clique (replace each leaf by its sum over the other variables) gives a smooth decomposable PC of no larger size, TV does not increase under marginalization, and the marginal of $\mathrm{UST}(G_h)$ on the first clique is at least $P_1\,\mathrm{UST}(K_h)$ pointwise. So if every leaf touches at most $\frac{2}{3}\binom{h}{2}$ edges of the first clique and $\mathrm{TV}(\mathrm{UST}(G_h), Q_C) \leq P_1 - \delta$, Corollary 5.2 for K_h gives $|C| \geq \delta^2 e^{h/146}/(e^3\sqrt{h})$. This needs TV below about $e^{-2} \approx 0.135$, and the global arity bound $2M/3$ does not imply the clique-specific one. Whether $\mathrm{UST}(G_h)$ needs circuits of size $2^{\Omega(n)}$ at every fixed error $\mathrm{TV} < 1$ is open.

5.6 An exact circuit of size $2^{O(n)}$

Proposition 5.11 (exact upper bound)**.** *μ_n has an exact smooth, decomposable and deterministic probabilistic circuit with nonnegative weights, univariate literal leaves and $O(n^4 3^n)$ nodes plus wires.*

Proof. For a vertex set S and $v \in S$, let $f(S, v)$ be the indicator, a function of the variables $x_{E(S)}$, that the edges of x inside S form a spanning tree of S. The vertex v only guides the recursion. Put $f(\{v\}, v) := 1$. For $|S| \geq 2$ let $s_0 := \min(S \setminus \{v\})$ and

$$f(S, v) = \sum_{\substack{S' \subseteq S\setminus\{v\} \\ s_0 \in S'}} \sum_{u \in S'} f(S', u) \cdot [x_{vu} = 1] \cdot f(S \setminus S', v) \cdot \prod_{e \in E(S', S\setminus S')\setminus\{vu\}} [x_e = 0].$$

If t is a spanning tree of S, let S' be the vertex set of the component of $t - v$ that contains s_0, and let u be the unique vertex of S' adjacent to v in t. Then t restricted to S' and to $S \setminus S'$ are spanning trees, and vu is the only edge of t between S' and $S \setminus S'$, so the summand for (S', u) is 1. Conversely, a summand equal to 1 describes a spanning tree whose decomposition is that summand, so at most one summand is nonzero for every x. By induction on $|S|$, the recursion computes $f(S, v)$, and every sum node is deterministic. The four factors of a summand have the disjoint scopes $E(S')$, $\{vu\}$, $E(S \setminus S')$ and $E(S', S \setminus S') \setminus \{vu\}$, whose union is $E(S)$,

so the circuit is decomposable and smooth. The output is $n^{-(n-2)} f(V, 1)$. There are $\binom{n}{s} s$ pairs (S, v) with $|S| = s$, each with at most $\sum_t \binom{s-2}{t-1} t$ summands of $O(1 + t(s-t))$ wires, so the size is $\sum_s \binom{n}{s} s \cdot O(s^3 2^s) = O(n^4 3^n)$. □

We expect this subset recursion is folklore; we did not find it stated for probabilistic circuits.

Corollary 5.12 (the order of growth)**.** *For every fixed $\delta \in (0, 1]$, the smallest smooth decomposable circuit with nonnegative weights and leaves of arity at most $2M/3$ with $1 - \mathrm{TV}(\mu_n, Q_C) \geq \delta$ has size $2^{\Theta(n)} = 2^{\Theta(\sqrt{M})}$. Equivalently, $\max_{A|B \text{ balanced}} \beta(\mu_n; A|B)^2 = e^{-\Theta(n)}$.*

Proof. The lower bounds are Corollary 5.2 and Theorem 5.1. The circuit of Proposition 5.11 has $\delta = 1$, and Corollary 3.11 applied to it gives $\max \beta^2 \geq 1/|C| = 3^{-n}/O(n^4)$. □

6 Cycle bits and the local lemma

This section gives the local argument behind Theorem 8.2 and Corollary 1.3. It bounds $\beta(\mathrm{UST}(G); A|B)$ for a single split in terms of crossing cycles, the counterpart of Theorem 4.5 with cycles in place of vertices. Section 9 extends it to the circuits of an arbitrary matroid.

Fix a split $E = A \sqcup B$. A cycle C is *mixed* if $E(C)$ meets both A and B. For a mixed cycle define the two bits

$$u_C(x_A) := \mathbf{1}[E(C) \cap A \subseteq x_A], \qquad v_C(x_B) := \mathbf{1}[E(C) \cap B \subseteq x_B],$$

where we identify x_A with the set of edges of A it contains. The bit u_C depends only on the A-part of x and v_C only on the B-part. If x is a spanning tree, then $u_C(x) = v_C(x) = 1$ would mean $E(C) \subseteq x$, which is impossible. So under $\mathrm{UST}(G)$ the cell $(1, 1)$ has probability zero.

6.1 The spatial Markov property at a cycle

Recall that $I(C)$ is the set of edges with an endpoint in $V(C)$. The next lemma describes the conditional law of $T \cap I(C)$ given the rest of the tree. It is a standard consequence of the uniformity of T; we include the proof.

Lemma 6.1 (conditional law near a cycle)**.** *Let C be a cycle of G and let $F \subseteq E \setminus I(C)$ satisfy $\Pr[T \setminus I(C) = F] > 0$. Then F is a forest in which every vertex of C is isolated. Let $K_1, \ldots, K_s$ be the vertex sets of the connected components of the graph $(V \setminus V(C), F)$. Let $\mathcal{M}_F$ be the multigraph with node set $V(C) \cup \{K_1, \ldots, K_s\}$ and one edge for each $e \in I(C)$, joining the nodes that contain the endpoints of e. Then $\mathcal{M}_F$ has no loops, every edge of $\mathcal{M}_F$ has an endpoint in $V(C)$, and conditionally on $T \setminus I(C) = F$, the set $T \cap I(C)$ is uniformly distributed over the edge sets of spanning trees of $\mathcal{M}_F$.*

Proof. F is contained in a tree, so it is a forest. Every edge at a vertex of C lies in $I(C)$, so these vertices are isolated in F, and every edge of $I(C)$ has an endpoint in $V(C)$. Such an endpoint forms a node of its own, so no edge of $\mathcal{M}_F$ is a loop.

Since T is uniform and $T = F \cup J$ with $J := T \cap I(C)$, the conditional law of J is uniform over the sets $J \subseteq I(C)$ for which $F \cup J$ is a spanning tree. We show that these are exactly the spanning trees of $\mathcal{M}_F$. The graph (V, F) has $|V(C)| + s$ components, so $|F| = N - |V(C)| - s$, and $\mathcal{M}_F$ has $|V(C)| + s$ nodes. Each component of F is connected, so $F \cup J$ is connected if and only if J connects $\mathcal{M}_F$. A connected graph on N vertices is a tree if and only if it has $N - 1$

edges, and $|F| + |J| = N - 1$ if and only if $|J| = |V(C)| + s - 1$. So $F \cup J$ is a spanning tree if and only if J is a connected spanning subgraph of $\mathcal{M}_F$ with one edge fewer than the number of nodes, that is, a spanning tree of $\mathcal{M}_F$. □

Let $d_j \geq 0$ be the number of edges of $\mathcal{M}_F$ at the node K_j. Since $\mathcal{M}_F$ has a spanning tree, $d_j \geq 1$ for every j.

Lemma 6.2 (local completion)**.** *Let C be a cycle of G, let $e \in E(C)$, and let F be as in Lemma 6.1. For every $x_0 \in V(C)$,*

$$\Pr\big[\, T \cap E(C) = E(C) \setminus \{e\} \;\big|\; T \setminus I(C) = F \,\big] \;\geq \prod_{y \in V(C) \setminus \{x_0\}} \frac{1}{\deg(y)} \;\geq\; D^{-(|C|-1)}.$$

Proof. By Lemma 6.1 the conditional probability is the fraction of spanning trees J of $\mathcal{M}_F$ with $J \cap E(C) = E(C) \setminus \{e\}$.

Numerator. Let J consist of the path $E(C) \setminus \{e\}$ together with exactly one edge of $\mathcal{M}_F$ at each node K_j. The path spans $V(C)$. Every edge at K_j joins K_j to a vertex of $V(C)$, so J is connected, and it has $|C| - 1 + s$ edges, one fewer than the number of nodes. So J is a spanning tree. It contains no chord of C, so $J \cap E(C) = E(C) \setminus \{e\}$. Distinct choices give distinct sets, so there are $\prod_j d_j$ such trees.

Denominator. Root a spanning tree J of $\mathcal{M}_F$ at x_0. Every node other than x_0 has a unique parent edge in J, which is an edge of $\mathcal{M}_F$ at that node, and J is the set of parent edges. So the map from J to its list of parent edges is injective. The node K_j has d_j possible parent edges. A vertex $y \in V(C) \setminus \{x_0\}$ has at most $\deg(y)$, because all edges of G at y lie in $I(C)$ and these are its edges in $\mathcal{M}_F$. So $\mathcal{M}_F$ has at most $\prod_j d_j \prod_{y \neq x_0} \deg(y)$ spanning trees.

Dividing gives the first inequality, and $|V(C) \setminus \{x_0\}| = |C| - 1$ gives the second. □

Chords of C only enlarge the denominator's choice sets, and parallel edges of $\mathcal{M}_F$ are kept distinct throughout. The case $s = 0$ uses empty products. The bound is uniform over all outside configurations F, and this uniformity is what the chain rule needs.

Remark 6.3. The lemma conditions on T outside *all* edges incident to C, not just outside $E(C)$. This matters. If one conditioned only on $T \setminus E(C)$, the cycle could freeze. For example, let C be a triangle and let a further vertex w be joined to all three vertices of C. Given that all three edges at w are in T, no edge of C can be. Lemma 6.2 leaves those three edges free.

6.2 A framed bound

Lemma 6.2 bounds the probability of a whole configuration on $E(C)$. The chain rule needs less: a lower bound for the event that T contains all edges of C on one side, given the past. The next two lemmas give such a bound in which every vertex of C is charged once, while Lemma 6.2 charges almost every vertex once for each side. Section 9 explains them as the graphic case of a matroid statement: every edge of C is certified by the star of one of its endpoints, a cut that meets C in two edges.

Lemma 6.4 (star bound)**.** *Let G be connected, let $F \subseteq E$ and $\sigma \subseteq F$ satisfy $\Pr[T \cap F = \sigma] > 0$, and let v be a vertex at which no edge of F lies. Then every edge e at v satisfies $\Pr[e \in T \mid T \cap F = \sigma] \geq 1/\deg(v)$.*

Proof. By Lemma 4.2, $T \setminus F$ is a uniform spanning tree of the multigraph K_σ given $T \cap F = \sigma$, and K_σ is connected. No edge of σ lies at v, so v is a node of K_σ by itself, and its edges in K_σ are its $\deg(v)$ edges in G, none of them a loop. By (E1), $\Pr[e \in T \mid T \cap F = \sigma] = R_{K_\sigma}(v, v')$, where v' is the other end of e. The function that is 1 at v and 0 elsewhere has energy $\deg(v)$, so $1/R_{K_\sigma}(v, v') \le \deg(v)$. □

Lemma 6.5 (framed cycles)**.** *Let G be connected and let C be a cycle with vertices $v_1, \dots, v_\ell$ in cyclic order, $\ell \ge 3$, and edges $e_j = v_j v_{j+1}$, indices modulo ℓ. Let $S \subsetneq E(C)$, and let $F \subseteq E$ be a set of edges none of which has an endpoint in $V(C)$. Then for every $\sigma \subseteq F$ with $\Pr[T \cap F = \sigma] > 0$,*

$$\Pr\big[S \subseteq T \mid T \cap F = \sigma\big] \;\ge\; \prod_{j:\ e_j \in S} \frac{1}{\deg(v_{j+1})}.$$

Proof. Since $S \ne E(C)$, the set S is a disjoint union of maximal runs $e_j, e_{j+1}, \dots, e_{j+t}$ of consecutive edges, none of which goes all the way around. List the edges of S as $s_1, \dots, s_k$, run after run, each run in increasing order. If $s_i = e_j$, the only edges of C at v_{j+1} are e_j and e_{j+1}, and e_{j+1} is either not in S or listed after s_i. No edge of F has an endpoint on C. So no edge of $F \cup \{s_1, \dots, s_{i-1}\}$ lies at v_{j+1}. By induction on i, the event $\{T \cap F = \sigma,\ s_1, \dots, s_{i-1} \in T\}$ has positive probability, and Lemma 6.4 with the edge set $F \cup \{s_1, \dots, s_{i-1}\}$ bounds the conditional probability of $s_i \in T$ given it by $1/\deg(v_{j+1})$. Multiply. □

6.3 The bound for one split

Theorem 6.6 (affinity bound for a split)**.** *Let G be a connected simple graph with maximum degree D, let $E = A \sqcup B$ be any split of its edges, and let $C_1, \dots, C_r$ be pairwise vertex-disjoint mixed cycles. For each k put*

$$p_k := \max_{x_0 \in V(C_k)} \prod_{y \in V(C_k) \setminus \{x_0\}} \deg(y)^{-1}, \qquad w_k := \prod_{y \in V(C_k)} \deg(y)^{-1}, \qquad q_k := \max(p_k^2, w_k).$$

Then

$$\beta(\mathrm{UST}(G); A|B) \;\le\; \prod_{k=1}^{r} \Big(\frac{1 + \sqrt{1 - 4q_k}}{2}\Big)^{1/2} \;\le\; \prod_{k=1}^{r} (1 - q_k)^{1/2}.$$

In particular, if every C_k has length at most ℓ, then $\beta(\mathrm{UST}(G); A|B) \le (1 - D^{-\ell})^{r/2}$.

Proof. Let $T \sim \mathrm{UST}(G)$ and put $u_k := u_{C_k}(T \cap A)$ and $v_k := v_{C_k}(T \cap B)$. The vector $u = (u_1, \dots, u_r)$ is a function of $T \cap A$ and v is a function of $T \cap B$. By Lemma 3.2, $\beta(\mathrm{UST}(G); A|B) \le \|\sqrt{\mu}\|_{\mathrm{op}}$, where μ is the joint law of (u, v) with rows indexed by u and columns by v. Order the pairs as $(u_1, v_1), \dots, (u_r, v_r)$; the order is fixed before T is sampled. Orient each C_k cyclically, and let w_k^A and w_k^B be the products of $1/\deg(v)$ over the heads v of the edges of C_k in A and in B; every vertex of C_k is the head of exactly one edge, so $w_k^A w_k^B = w_k$. By Lemmas 3.3 and 3.5, and since $\max(p_k, w_k^A)\max(p_k, w_k^B) \ge q_k$, it suffices to show that for every k and every value π of the past $(u_{<k}, v_{<k})$ with positive probability,

$$\Pr\big[(u_k, v_k) = (1,1) \mid \pi\big] = 0, \qquad \Pr\big[(u_k, v_k) = (0,1) \mid \pi\big] \ge \max(p_k, w_k^B),$$
$$\Pr\big[(u_k, v_k) = (1,0) \mid \pi\big] \ge \max(p_k, w_k^A).$$

The first holds because T contains no cycle. For the others, let $F_k := T \setminus I(C_k)$. An edge of C_j with $j \ne k$ has both endpoints in $V(C_j)$, which is disjoint from $V(C_k)$, so $E(C_j) \cap I(C_k) = \emptyset$.

Hence the past $(u_{<k}, v_{<k})$ is a function of F_k, and the event $\{(u_{<k}, v_{<k}) = \pi\}$ is a union of events $\{F_k = F\}$. Pick $e_A \in E(C_k) \cap A$ and $e_B \in E(C_k) \cap B$. If $T \cap E(C_k) = E(C_k) \setminus \{e_A\}$, then $e_A \notin T$ while every edge of $E(C_k) \cap B$ is in T, so $(u_k, v_k) = (0, 1)$. Likewise $E(C_k) \setminus \{e_B\}$ gives $(1, 0)$. By Lemma 6.2, both events have conditional probability at least p_k given every value F of F_k with positive probability. By the tower property the same bounds hold given π, since conditioning on π averages the conditional laws given $F_k = F$ over the values F compatible with π. For the bounds w_k^A and w_k^B, note that the past is also a function of $T \cap F_k'$ with $F_k' := E(C_1) \cup \cdots \cup E(C_{k-1})$, and no edge of F_k' has an endpoint on C_k. Since $u_k = 1$ forces $v_k = 0$, Lemma 6.5 with $S = E(C_k) \cap A \subsetneq E(C_k)$ gives $\Pr[(u_k, v_k) = (1, 0) \mid T \cap F_k' = \sigma] \geq w_k^A$ for every admissible σ, and the same averaging applies. The bound with B is symmetric. □

Neither of p_k^2 and w_k dominates the other: $w_k \geq p_k^2$ exactly when the product of the $|C_k| - 1$ smallest degrees on C_k is at least the largest one. For cycles of length $\ell \geq 3$ in a graph of maximum degree D, the uniform bound improves from $D^{-2(\ell-1)}$, which an earlier version of this paper used, to $D^{-\ell}$.

The incident sets $I(C_j)$ and $I(C_k)$ may share an edge joining the two cycles. This does not matter, because the bits use cycle edges only. The theorem holds for every split, balanced or not, and every family of vertex-disjoint mixed cycles. The geometric input of Section 7 is a lower bound on r for balanced splits.

Corollary 6.7 (tree rectangles)**.** *Let $R_A \subseteq 2^A$ and $R_B \subseteq 2^B$ be nonempty, and suppose that every set $a \cup b$ with $a \in R_A$, $b \in R_B$ is a spanning tree. Let R be the set of these trees. Then* $\mathrm{UST}(G)(R) \leq \beta(\mathrm{UST}(G); A|B)^2$.

Proof. The map $(a, b) \mapsto a \cup b$ is injective, since $A \cap B = \emptyset$. Let q be the product of the uniform distributions on R_A and R_B, so q is uniform on R. Every element of R has probability $1/\tau(G)$ under $P = \mathrm{UST}(G)$, so $\mathrm{BC}(P, q) = |R|/\sqrt{\tau(G)|R|} = \sqrt{P(R)}$. Now use $\mathrm{BC}(P, q) \leq \beta$. □

7 Geometry: many disjoint crossing cycles

Theorem 6.6 turns vertex-disjoint mixed short cycles into a bound on β. This section shows that, in the graphs of Theorem 8.2, every balanced split has linearly many of them. The condition we need is that short cycles tie together the edges at every vertex.

Definition 7.1 (linked graphs)**.** Let G be a connected graph with $N \geq 3$ vertices, and let $\mathcal{C}$ be a nonempty family of cycles of G. For a vertex x, let Λ_x be the graph whose nodes are the edges at x, with $e \sim f$ when some cycle of $\mathcal{C}$ contains both e and f. The family $\mathcal{C}$ *links* G if Λ_x is connected for every vertex x. We write $\ell := \max_{C \in \mathcal{C}} |C|$ and $\kappa := \max_x |\{C \in \mathcal{C} : x \in V(C)\}|$.

Lemma 7.2. *If $\mathcal{C}$ links G, then every edge of G lies on a cycle of $\mathcal{C}$. In particular G has no bridge, its minimum degree is at least 2, and $1 \leq \kappa < \infty$.*

Proof. Let $e = xy$. Since G is connected with at least three vertices, x or y has degree at least 2, say x. Then Λ_x is connected with at least two nodes, so e is adjacent in Λ_x to some other edge, which means that some cycle of $\mathcal{C}$ contains e. An edge on a cycle is not a bridge, and every vertex has an edge on a cycle, hence degree at least 2. □

The conditions $N \geq 3$ and $\mathcal{C} \neq \emptyset$ exclude K_2, which is vacuously linked by the empty family and has a bridge. Some condition of this kind is needed: Proposition 11.5 shows that pendant edges defeat any bound that is uniform over balanced splits.

7.1 Mixed cycles in linked expanders

Lemma 7.3 (mixed cycles)**.** *Let G be a connected graph with $N \geq 3$ vertices, maximum degree D and edge expansion $h > 0$, linked by a family $\mathcal{C}$ with parameters ℓ, κ. Colour the edges with two colours so that each colour class has at least $m_0 > 0$ edges, and let t be the number of cycles of $\mathcal{C}$ that contain edges of both colours. Then*

$$t \;\geq\; \min\Big\{\frac{hN}{4D\ell},\; \frac{2m_0}{\ell D(1+D/h)}\Big\},$$

and some $\lceil t/(\ell\kappa)\rceil$ of these mixed cycles are pairwise vertex-disjoint.

Proof. Let Z be the set of vertices of mixed cycles of $\mathcal{C}$. Then $|Z| \leq \ell t$.

Good vertices are monochromatic. Let $x \notin Z$. Every cycle of $\mathcal{C}$ through x is monochromatic, so adjacent nodes of Λ_x have the same colour. Since Λ_x is connected, all edges at x have one colour $c(x)$. If $x, y \notin Z$ are adjacent, then $c(x) = c(xy) = c(y)$. Hence for each connected component K of $G - Z$, all edges with an endpoint in K have the same colour.

Case $|Z| \geq hN/(4D)$. Then $t \geq |Z|/\ell \geq hN/(4D\ell)$.

Case $|Z| < hN/(4D)$. Since $h \leq D$, $|Z| < N/4$. Suppose every component of $G - Z$ had at most $N/2$ vertices. If one of them has at least $N/4$ vertices, let U be that component. Otherwise add components one at a time until the union first has at least $N/4$ vertices; this happens because $|V \setminus Z| > 3N/4$, and the union then has fewer than $N/2$ vertices. In both cases U is a union of components with $N/4 \leq |U| \leq N/2$. Every edge of ∂U has its other endpoint in Z, so

$$\frac{hN}{4} \leq h|U| \leq |\partial U| \leq D|Z| < \frac{hN}{4},$$

a contradiction. Hence $G - Z$ has a component K with $|K| > N/2$. Let $S := V \setminus (K \cup Z)$. Then $|S| < N/2$, and every edge of ∂S goes to Z, so $h|S| \leq D|Z|$. Therefore

$$|V \setminus K| = |S| + |Z| \leq (1 + D/h)|Z|.$$

All edges with an endpoint in K have the colour of K, so every edge of the other colour lies inside $V \setminus K$. There are at most $D|V \setminus K|/2 \leq \ell D(1 + D/h)t/2$ such edges, and at least m_0. This gives the second bound.

Packing. Choose mixed cycles greedily. A chosen cycle shares a vertex with at most $\ell\kappa$ cycles of $\mathcal{C}$, itself included, because it has at most ℓ vertices and each lies on at most κ cycles. Discard these and repeat. This produces at least $\lceil t/(\ell\kappa)\rceil$ pairwise vertex-disjoint mixed cycles. □

7.2 Prisms and squares

Lemma 7.4 (prisms)**.** *Let H be a connected simple graph with $n \geq 3$ vertices, maximum degree at most Δ and edge expansion $h(H) \geq h_0 > 0$, and let $G = H \square K_2$. Then $N = 2n$, $M = 2|E(H)| + n \geq N$, $D \leq \Delta + 1$, and $h(G) \geq \min(1, h_0)/3$. The family of 4-cycles*

$$Q_{uv} := (u,0)\,(v,0)\,(v,1)\,(u,1), \qquad uv \in E(H),$$

links G with $\ell = 4$ and $\kappa \leq \Delta$.

Proof. The counts are immediate, and $M \geq 2(n-1) + n \geq 2n$ since H is connected and $n \geq 2$.

Linking. The edges at $(v,0)$ are the rung $(v,0)(v,1)$ and the edges $(v,0)(u,0)$ for $u \sim v$ in H. Each $(v,0)(u,0)$ lies in Q_{uv} together with the rung. So $\Lambda_{(v,0)}$ contains a spanning star centred

at the rung, and is connected. The same holds at $(v,1)$. The vertex (v,i) lies on Q_{uv} for the $\deg_H(v) \le \Delta$ neighbours u of v, and on no other cycle of the family.

Expansion. Let $S = S_0 \times \{0\} \cup S_1 \times \{1\}$ with $0 < |S| \le n$, and put $x = |S_0|$, $y = |S_1|$. By symmetry of the two layers we may assume $x \ge y$. If $x \le n/2$, the edges of the two copies of H give $|\partial S| \ge h_0 x + h_0 y$. If $x > n/2$, then $y \le n - x < n/2$, so the copy of H in layer 1 contributes at least $h_0 y$ boundary edges, and each of the at least $x - y$ vertices $v \in S_0 \setminus S_1$ contributes its rung. So $|\partial S| \ge h_0 y + (x - y)$. If $2y \le x$, this is at least $x/2 \ge (x+y)/3$. If $2y > x$, it is at least $h_0 y \ge h_0(x+y)/3$. In all cases $|\partial S| \ge \min(1, h_0)|S|/3$. □

Lemma 7.5 (squares)**.** *Let H be as in Lemma 7.4 and let $G = H^2$. Then $N = n$, $M \ge N - 1 \ge 2N/3$, $D \le \Delta^2$, and $h(G) \ge h_0$. The family of triangles $\{u, v, w\}$ such that $u\,v\,w$ is a path in H links G with $\ell = 3$ and $\kappa \le 3\Delta(\Delta-1)/2$.*

Proof. A vertex has at most Δ neighbours in H and at most $\Delta(\Delta-1)$ vertices at distance two, so $D \le \Delta^2$. G contains H on the same vertex set, so $|\partial_G S| \ge |\partial_H S|$ for every S, and $h(G) \ge h(H) \ge h_0$. G is connected, so $M \ge N - 1$, and $N - 1 \ge 2N/3$ because $N \ge 3$.

Linking. For a path $u\,v\,w$ in H, the vertices u, v, w are pairwise at distance at most two, so they span a triangle of G. Let x be a vertex. Any two H-edges xu, xw at x lie in the triangle of the path $u\,x\,w$. An edge xw of G that is not in H joins vertices at distance two, so there is a path $x\,v\,w$ in H, and its triangle contains xw and the H-edge xv. So every node of Λ_x is adjacent to an H-edge at x, the H-edges at x are pairwise adjacent, and Λ_x is connected. The family is nonempty because H is connected with $n \ge 3$.

The parameter κ. Assign to each triangle of the family one path that generates it. A triangle through x is assigned a path with centre x or a path with endpoint x. There are at most $\binom{\Delta}{2}$ paths of the first kind and at most $\Delta(\Delta-1)$ of the second, so $\kappa \le 3\Delta(\Delta-1)/2$. □

7.3 The square grid

The grid is not an expander, but it is linked by its unit squares, and an isoperimetric inequality can replace expansion. Let G_L be the $L \times L$ grid with $L \ge 2$, so $N = L^2$, $M = 2L(L-1)$ and $D = 4$. At an interior vertex, the four edges and the four unit squares around it form a 4-cycle in Λ_x; at boundary vertices Λ_x is a path. So the unit squares link G_L with $\ell = 4$ and $\kappa = 4$.

Lemma 7.6 (grid isoperimetry)**.** *For every $S \subseteq V(G_L)$ with $0 < |S| \le L^2/2$, $|\partial S| \ge \min\{2\sqrt{|S|},\, L/2\}$.*

Proof. Call a row full if it lies in S, empty if it misses S, and partial otherwise; the same for columns. A partial row contains two horizontally adjacent vertices, one in S and one not, so it contributes a horizontal edge to ∂S; a partial column contributes a vertical one. Let a and b be the numbers of partial rows and columns. If some row is full and some row is empty, every column is partial, so $|\partial S| \ge L$. If some row is full and no row is empty, then at most $|S|/L \le L/2$ rows are full, so $a \ge L/2$. The same two cases apply to columns. In the remaining case no row and no column is full, so the rows and columns that meet S are partial, $|S| \le ab \le ((a+b)/2)^2$, and $|\partial S| \ge a + b \ge 2\sqrt{|S|}$. □

Lemma 7.7 (mixed squares in the grid)**.** *Let $L \ge 2$. Colour the edges of G_L with two colours so that each colour class has at least $M/3$ edges. Then at least $L/32$ unit squares contain edges of both colours, and some $\lceil L/512 \rceil$ of them are pairwise vertex-disjoint.*

Proof. Let Z be the set of vertices of mixed unit squares and t their number, so $|Z| \le 4t$. As in Lemma 7.3, each component of $G_L - Z$ has all its incident edges of one colour. We show $|Z| \ge L/8$, which gives $t \ge L/32$. Suppose $|Z| < L/8$. Then $|Z| < N/4$. If every component of $G_L - Z$ had at most $N/2$ vertices, the argument of Lemma 7.3 gives a union U of components with $N/4 \le |U| \le N/2$ and $|\partial U| \le 4|Z| < L/2$. Lemma 7.6 gives $|\partial U| \ge \min\{2\sqrt{N/4}, L/2\} = L/2$, a contradiction. So there is a component K with $|K| > N/2$. Let $U_1, U_2, \dots$ be the other components. Each has fewer than $N/2$ vertices, the sets ∂U_i are disjoint, and all their edges end in Z, so $\sum_i |\partial U_i| \le 4|Z| < L/2$. By Lemma 7.6, $|U_i| \le |\partial U_i|^2/4$, hence $\sum_i |U_i| \le (\sum_i |\partial U_i|)^2/4 \le 4|Z|^2$. Every edge of the colour not used at K lies inside $V \setminus K$, which has at most $|Z| + 4|Z|^2$ vertices, so there are at most $2|Z| + 8|Z|^2 < L/4 + L^2/8$ such edges. But there are at least $M/3 = 2L(L-1)/3$, and $L/4 + L^2/8 \le 2L(L-1)/3$ for $L \ge 2$. This is a contradiction. The packing step is as in Lemma 7.3 with $\ell\kappa = 16$. □

8 Main results

8.1 Proof of Theorem 1.1 and the families it covers

Proof of Theorem 1.1. By Theorem 4.13, $\beta(P; A|B)^2 \le e^{-cN}$ for every balanced split. Apply Corollary 3.11 with $\delta = 1 - \varepsilon > 0$. □

Exactly as for Theorem 8.2 below, the bound stays exponential when the overlap $\delta = 1 - \mathrm{TV}$ is itself exponentially small, as long as $\delta^2 \ge e^{-(c-c')N}$ for some $c' > 0$.

Below, *the easy Cheeger direction* is Lemma 4.11: $h(G) \ge \lambda_2(L_G)/2$.

Corollary 8.1 (families)**.** *Each of the following families satisfies the hypotheses of Theorem 1.1 with D and h independent of N.*

(a) For fixed $d \ge 3$, a uniformly random d-regular graph on n vertices, with probability tending to one as $n \to \infty$ (nd even).

(b) The Ramanujan graphs $X^{p,q}$ of Lubotzky, Phillips and Sarnak [LPS88], with $D = p+1$ and $h = (\sqrt{p}-1)^2/2$, for q large enough that they are simple graphs (their girth tends to infinity with q [LPS88]).

(c) The prisms $H \square K_2$ and squares H^2 of connected graphs H with at least three vertices, maximum degree at most Δ and $h(H) \ge h_0 > 0$, with D and h as in Lemmas 7.4 and 7.5.

(d) Every 2-connected graph H with maximum degree at most Δ and $h(H) \ge h_0 > 0$, itself.

Proof. (a) For fixed $d \ge 3$, a random d-regular graph is d-connected with probability tending to one [Wor81a], and there is $h_d > 0$ such that its edge expansion is at least h_d with probability tending to one [Bol88].

(b) The graphs $X^{p,q}$ are connected $(p+1)$-regular Cayley graphs, and every adjacency eigenvalue other than $\pm(p+1)$ has absolute value at most $2\sqrt{p}$ [LPS88]. So $\lambda_2(L) \ge p + 1 - 2\sqrt{p} = (\sqrt{p}-1)^2$, and $h \ge (\sqrt{p}-1)^2/2$ by the easy Cheeger direction. A Cayley graph is vertex-transitive. Every connected graph with at least two vertices has a vertex that is not a cut vertex, for instance a leaf of a spanning tree, and in a vertex-transitive graph every vertex is then a non-cut vertex. So a connected vertex-transitive graph with at least three vertices is 2-connected.

(c) The expansion and degree bounds are Lemmas 7.4 and 7.5. For 2-connectivity, delete a vertex $(x, 0)$ from $H \square K_2$: the copy $H \times \{1\}$ is still connected, and every remaining $(y, 0)$ is joined to $(y, 1)$; the case $(x, 1)$ is symmetric. Delete a vertex v from H^2: every component of $H - \{v\}$

contains an H-neighbour of v, since H is connected, and the H-neighbours of v are pairwise adjacent in H^2, so $H^2 - \{v\}$ is connected.

(d) There is nothing to prove. □

For the graphs of (b) with $p \geq 5$, $\lambda_2(L) \geq (\sqrt{p}-1)^2 > 1$, so Theorem 1.1 holds with the spectral constant $c_\lambda(p+1)$, $\lambda = (\sqrt{p}-1)^2$; for $p = 5$ this is $8.0 \cdot 10^{-8}$. The same applies to random d-regular graphs with $d \geq 6$, with $c_\lambda(d)$ for any fixed $\lambda \in (1, d - 2\sqrt{d-1})$, by Friedman's theorem [Fri08] that all adjacency eigenvalues other than d have absolute value at most $2\sqrt{d-1} + o(1)$ with probability tending to one.

The Ramanujan graphs of [LPS88] have girth of order $\log N$, and random regular graphs have few short cycles: for fixed d and k, the number of cycles of length k converges in distribution to a Poisson variable with mean $(d-1)^k/(2k)$ [Wor81b]. The local argument of Sections 6 and 7 gives nothing there (Section 11), and an earlier version of this paper left this case open.

8.2 Linked expanders

Theorem 8.2 (linked expanders)**.** *Let G be a connected simple graph with $N \geq 3$ vertices, M edges, maximum degree D and edge expansion $h > 0$, linked by a family of cycles with parameters ℓ and κ. Put*

$$r_0 := \frac{1}{\ell\kappa} \min\Big\{ \frac{hN}{4D\ell},\ \frac{2M}{3\ell D(1 + D/h)} \Big\}.$$

Then $\beta(\mathrm{UST}(G); A|B)^2 \leq (1 - D^{-\ell})^{r_0}$ for every balanced split $A|B$. Consequently, if C is a smooth decomposable PC with nonnegative weights over the M edge variables, with leaves of arity at most $2M/3$, and $1 - \mathrm{TV}(\mathrm{UST}(G), Q_C) \geq \delta > 0$, then

$$|C| \ \geq\ \delta^2 (1 - D^{-\ell})^{-r_0} \ \geq\ \delta^2 \exp(r_0\, D^{-\ell}).$$

Proof. Let $A|B$ be balanced and colour A and B with two colours. Each class has at least $M/3$ edges, so Lemma 7.3 with $m_0 = M/3$ gives at least r_0 pairwise vertex-disjoint mixed cycles of length at most ℓ. Theorem 6.6 gives $\beta^2 \leq (1 - D^{-\ell})^{r_0}$. Corollary 3.11 gives the circuit bound, and $-\log(1-x) \geq x$ gives the last inequality. □

With $\delta = 1 - \varepsilon$ and $c := r_0 D^{-\ell}/N$ this is the bound $|C| \geq (1-\varepsilon)^2 e^{cN}$. The bound stays exponential even when the overlap δ is itself exponentially small, as long as $\delta^2 \geq e^{-(c-c')N}$ for some $c' > 0$.

8.3 Prisms and squares: proof of Corollary 1.3

Corollary 8.3 (closed-form constants)**.** *Let H satisfy the hypotheses of Corollary 1.3 and put $h' = \min(1, h_0)$. For $G = H \square K_2$,*

$$r_0\, D^{-\ell} \ \geq\ \frac{h'}{24\,\Delta(\Delta+1)^5(3\Delta+4)}\, N,$$

and for $G = H^2$,

$$r_0\, D^{-\ell} \ \geq\ \frac{8h_0}{243\,\Delta^9(\Delta-1)(h_0+\Delta^2)}\, N.$$

Proof. The quantity $r_0 D^{-\ell}$ increases with h and M and decreases with D and κ, so we may insert the bounds of Lemmas 7.4 and 7.5.

Prism. Here $D \le \Delta + 1$, $\ell = 4$, $\kappa \le \Delta$, $h \ge h'/3$, $M \ge N$ and $D^{-\ell} \ge (\Delta+1)^{-4}$. The first term of the minimum is at least $h'N/(48(\Delta+1))$. The second is at least

$$\frac{2N}{12(\Delta+1)(1+3(\Delta+1)/h')} = \frac{h'N}{6(\Delta+1)(h'+3\Delta+3)} \ge \frac{h'N}{6(\Delta+1)(3\Delta+4)},$$

using $h' \le 1$. Since $6(3\Delta+4) \ge 48$ for $\Delta \ge 2$, the minimum is at least the last expression. Dividing by $\ell\kappa \le 4\Delta$ and multiplying by $D^{-\ell}$ gives the claim.

Square. Here $D \le \Delta^2$, $\ell = 3$, $\kappa \le 3\Delta(\Delta-1)/2$, $h \ge h_0$, $M \ge 2N/3$ and $D^{-\ell} \ge \Delta^{-6}$. The first term is at least $h_0 N/(12\Delta^2)$, and the second is at least

$$\frac{4N}{27\Delta^2(1+\Delta^2/h_0)} = \frac{4h_0 N}{27\Delta^2(h_0+\Delta^2)}.$$

The second is the smaller, since $27(h_0+\Delta^2) \ge 48$. Dividing by $\ell\kappa \le 9\Delta(\Delta-1)/2$ and multiplying by $D^{-\ell}$ gives the claim. □

Proof of Corollary 1.3. G is 2-connected by Corollary 8.1(c). Lemmas 7.4 and 7.5 show that G satisfies the hypotheses of Theorem 8.2. Apply it with $\delta = 1-\varepsilon$ and use Corollary 8.3. □

The closed forms are lossy. A direct evaluation of r_0 is better, as the next example shows.

Example 8.4 (cubic prisms)**.** Let H be 3-regular with $h(H) \ge 1/5$, and $G = H \square K_2$. Then $D = 4$, $M = 2N$, $\ell = 4$, $\kappa = 3$, $h \ge 1/15$ and $D^{-\ell} = 1/256$. The two terms in r_0 are $hN/(4D\ell) \ge N/960$ and $2M/(3\ell D(1+D/h)) \ge N/732$, so $r_0 \ge N/11520$. The per-cycle factor in Theorem 6.6 is at most $(1-4^{-4})^{1/2} = 0.9980449\ldots$; the sharper form $((1+\sqrt{1-4^{-3}})/2)^{1/2} = 0.9980372\ldots$ changes the fifth decimal. So

$$|C| \;\ge\; (1-\varepsilon)^2\,(1-4^{-4})^{-N/11520} \;\ge\; (1-\varepsilon)^2\,e^{3.3974\cdot 10^{-7}N} \;\ge\; (1-\varepsilon)^2\,e^{3\cdot 10^{-7}N}.$$

Here $-\log(1-4^{-4})/11520 = 3.39748\ldots\cdot 10^{-7}$. The closed form of Corollary 8.3 gives only $c \ge 1/4792320 \approx 2.09\cdot 10^{-7}$ for these parameters. Either way the bound exceeds 10 only when N exceeds about $7\cdot 10^6$. The bound of the first version of this paper, with $D^{-2(\ell-1)} = 4^{-6}$ in place of 4^{-4}, was 16 times weaker.

The constants are poor for three reasons: the per-cycle weight $D^{-\ell}$ is crude, the forbidden-cell factor uses two bits per cycle, and the geometric counting and packing lose factors of ℓ, D and κ. In exact computations on five small graphs, the geometric mean of the conditional factors γ_k of Lemma 3.3 was between 0.94 and 0.99 per cycle, while the proved per-cycle bounds on the same instances were between 0.992 and 0.998 (Appendix A). These graphs are far too small to say anything about rates. We made no attempt to optimize.

8.4 The square grid

Corollary 8.5 (grid)**.** *Let $L \ge 2$ and let P be the uniform spanning tree of the $L \times L$ grid, with $M = 2L(L-1)$ edges. Every smooth decomposable PC with nonnegative weights and leaves of arity at most $2M/3$ that satisfies $1 - \mathrm{TV}(P, Q_C) \ge \delta > 0$ has size at least $\delta^2(1-4^{-4})^{-L/512} \ge \delta^2 e^{L/2^{17}}$.*

Proof. Lemma 7.7 gives at least $L/512$ vertex-disjoint mixed unit squares for every balanced split. All degrees are at most 4, so $q_k \ge 4^{-4}$ in Theorem 6.6. Corollary 3.11 finishes the proof. □

The bound is tight up to the constant in the exponent. Process the edges in a fixed order, sweeping the grid with a staircase frontier of $O(L)$ active vertices. A state records the partition of the active vertices induced by the processed edges chosen so far, provided they form a forest in which every component still meets the frontier (a completed spanning component is allowed only at the end). Choosing an edge inside a block is rejected, choosing one across blocks merges them, and a component that would leave the frontier unconnected is rejected. Because the processed region is planar and lies on one side of the frontier, the realizable partitions are noncrossing, so there are at most a Catalan number $2^{O(L)}$ of them; recording arbitrary partitions would give only the weaker bound $2^{O(L \log L)}$. The resulting layered decision diagram has $O(L^2)\, 2^{O(L)}$ nodes. Writing each node at layer i as $[x_i = 0]f_0 + [x_i = 1]f_1$, with all branches kept layered so both children have the same scope, gives a smooth, deterministic, decomposable circuit with univariate leaves of size $2^{O(L)}$ that computes the spanning-tree indicator; dividing by τ gives the uniform spanning tree. For the uniform domino tiling of the same grid, a bound $e^{\Theta(L)}$ was proved in [Bla26] by a much longer argument; Section 11 says why the short argument here does not apply to matchings.

Theorem 1.1 does not cover the grid as stated: $h(G_L)$ is of order $1/L$, so $c(D, h)N$ tends to zero. Theorem 4.5 itself needs no expansion, and we expect a local version to work. If W has pairwise distances at least 3, any two neighbours of $w \in W$ are joined in $G_L - W$ by a path of length at most 4 around w, so $\rho_w \le 4$ and $p_A(w)p_B(w) \ge 6^{-4}$ by Lemma 4.3. What remains is to show that every balanced split has $\Omega(L)$ mixed vertices, using Lemma 7.6 in place of expansion. We have not written this out, so we state the vertex-touch bound for the grid only as a conjecture. It would not improve on Corollary 8.5 beyond constants.

8.5 Upper bounds

Proposition 8.6 (upper bounds)**.** *Let G be any connected graph with N vertices and M edges, and $P = \mathrm{UST}(G)$.*

(a) The generating polynomial $\sum_T P(T) \prod_{e \in T} z_e$ is computed by an arithmetic circuit with subtraction (a probabilistic generating circuit in the sense of [ZJdB21]) of size polynomial in $N + M$.

(b) For disjoint $S_0, S_1 \subseteq E$, the probability $P(S_1 \subseteq T,\ T \cap S_0 = \emptyset)$ is a ratio of two determinants of integer matrices of order less than N, so it can be computed exactly in polynomial time.

(c) Some smooth, deterministic, decomposable PC with univariate leaves represents P exactly and has size $O(M\tau(G)) = O(M2^M)$.

Proof. (a) Let L_z be the weighted Laplacian, with $L_z[x, x] = \sum_{e \ni x} z_e$ and $L_z[x, y] = -\sum_{e=xy} z_e$ for $x \neq y$, and let $L_z^{(v)}$ be L_z with the row and column of a vertex v removed. By the matrix-tree theorem [Kir47], $\det L_z^{(v)} = \sum_T \prod_{e \in T} z_e$. The algorithms of Berkowitz [Ber84] and of Mahajan and Vinay [MV99] compute the determinant of an $n \times n$ matrix by a division-free arithmetic circuit of size polynomial in n. Substituting the linear forms $L_z^{(v)}[x, y]$ and multiplying by the constant $1/\tau(G)$, where $\tau(G) = \det L_{\mathbf{1}}^{(v)}$, gives the circuit.

(b) If S_1 contains a cycle the probability is zero. Otherwise it equals $\tau(G')/\tau(G)$, where G' is the multigraph obtained from G by deleting S_0, contracting S_1 and removing loops. The matrix-tree theorem holds for multigraphs, so both counts are determinants of reduced Laplacians.

(c) Take a sum node with weight $1/\tau(G)$ on each spanning tree T, whose child is the product of the indicator leaves $[x_e = \mathbf{1}[e \in T]]$ over all $e \in E$. □

Since $N-1 \le M \le DN/2$, parts (c) and Theorem 1.1 together say that, for fixed D, h and $\varepsilon < 1$, the optimal size of a nonnegative smooth decomposable PC that ε-approximates P is $2^{\Theta(M)}$. Part (a) is not a nonnegative PC: the determinant circuit uses subtraction, and that is the point of the separation.

8.6 Exact support

Exact representation of a distribution with the same support as P but different probabilities is not the case $\varepsilon = 0$ of Theorem 1.1, since such a distribution can be far from P in total variation. Lower bounds for exact support follow instead from Corollary 6.7 and a rectangle cover.

Corollary 8.7 (exact support)**.** *Let G satisfy the hypotheses of Theorem 1.1, and let c and $\kappa = 3c$ be as in Theorem 4.13.*

(a) Let C be a smooth decomposable PC with nonnegative weights and leaves of arity at most $2M/3$ such that $f_C(x) > 0$ exactly when x is a spanning tree. Then $|C| \ge e^{cN}$.

(b) Let Γ be a monotone arithmetic circuit with fan-in at most two and s nodes computing a polynomial whose monomials are exactly $\prod_{e\in T} z_e$, $T \in \mathcal{T}(G)$, each with a positive coefficient. Then $s \ge e^{2cN/3}$.

The same holds if G satisfies the hypotheses of Theorem 8.2 instead, with $c_0 := r_0 D^{-\ell}/N$ in place of c in (a), and in (b) with c_1 in place of $2c/3$, where c_1 is defined like c_0 with M replaced by $N-1$ in the definition of r_0.

Proof. (a) Lemma 3.9 writes $f_C = \sum_u f_u h_u$ with $k \le |C|$ terms across balanced splits $A_u|B_u$. All terms are nonnegative, so the support of each term lies in $\mathcal{T}(G)$, and the supports cover $\mathcal{T}(G)$. The support of $f_u h_u$ is the rectangle $R_u = \{a \cup b :\ f_u(a) > 0,\ h_u(b) > 0\}$. By Corollary 6.7 and Theorem 4.13,

$$1 \le \sum_u \mathrm{UST}(G)(R_u) \le k \max_u \beta(\mathrm{UST}(G); A_u|B_u)^2 \le |C|\, e^{-cN}.$$

Under the hypotheses of Theorem 8.2, use that theorem instead, with c_0.

(b) Let F be the computed polynomial; it is homogeneous of degree $d = N-1 \ge 2$ with squarefree monomials. Remove gates that compute zero and gates that no longer reach the output. Every remaining gate then computes a nonzero homogeneous multilinear polynomial with nonnegative coefficients, and

$$F = \sum_{j=1}^{r} g_j h_j, \qquad r \le s,$$

where g_j, h_j are nonzero homogeneous polynomials with nonnegative coefficients and $d/3 < \deg g_j \le 2d/3$. These are the cleaning and degree-balanced decomposition lemmas of [Bla26, Lemmas 7.3 and 7.4], whose proofs use only that every monomial of F is squarefree of degree d; the method goes back to [Val80, JS82]. There is no cancellation, so every monomial of $g_j h_j$ is the monomial of a spanning tree. If a variable z_e occurred both in a monomial of g_j and in a monomial of h_j, their product would be a monomial of F containing z_e^2. So the variables of g_j and of h_j are disjoint. Let A_j contain the variables of g_j and none of h_j, and $B_j := E \setminus A_j$. The trees whose monomials occur in $g_j h_j$ form a rectangle R_j across $A_j|B_j$, and both A_j and B_j contain at least $d/3 = (N-1)/3$ edges, since a monomial of g_j has more than $d/3$ edges in A_j

and a monomial of h_j has at least $d/3$ edges in B_j. Theorem 4.13 with $m_0 = (N-1)/3 \geq 2N/9$ bounds each $\beta(\mathrm{UST}(G); A_j|B_j)^2$ by $e^{-2\kappa N/9} = e^{-2cN/3}$. Every monomial of F occurs in some $g_j h_j$, so the R_j cover $\mathcal{T}(G)$, and Corollary 6.7 gives $1 \leq \sum_j \mathrm{UST}(G)(R_j) \leq s\, e^{-2cN/3}$. Under the hypotheses of Theorem 8.2, Lemma 7.3 with $m_0 = (N-1)/3$ and Theorem 6.6 give the bound $e^{-c_1 N}$ instead. □

The coefficients play no role beyond being positive. The fan-in restriction in (b) can be dropped by the grouping of Remark 3.10 with degrees in place of scopes: at a high product gate whose children are all low, multiply children until the degree first exceeds $d/3$, and label the group by that gate, so again $r \leq s$. For prisms and squares of expanders, c_0 and c_1 are positive constants depending only on Δ and h_0, and much larger than $c(D, h)$. Statement (b) is an exact monotone lower bound of the same kind as those of [CDM20, CDGM22], but for the undirected polynomial and for every 2-connected bounded-degree expander; see Section 12.

9 Matroids: circuits and cocircuits

The cycle bits of Section 6 and the vertex-touch bits of Section 4 are instances of the same matroid argument, applied to the cycle matroid $M(G)$ and its dual. A cycle of G is a circuit of $M(G)$, and the frame of Lemma 6.5 consists of vertex stars, which are cuts. A vertex star is a cut, and the touch bound of Lemma 4.3 is certified by effective resistances, that is, by flows, which together with the certified edge are combinations of circuits. This section proves the local (framed) version of the argument for the uniform basis of an arbitrary matroid, and the global (resistance) version for balanced matroids. The only structural input of the local version is orthogonality: a circuit and a cocircuit never meet in exactly one element.

9.1 Background

We use standard matroid terminology [Oxl11]. Let $\mathcal{M}$ be a matroid on a finite set E with $n := |E|$, rank r, set of bases $\mathcal{B}(\mathcal{M})$ and closure operator cl. Write $U(\mathcal{M})$ for the uniform distribution on $\mathcal{B}(\mathcal{M})$, viewed as a distribution on $\{0,1\}^E$, and B for a random basis with this law. The *leverage* of e in $\mathcal{M}$ is $\lambda(e; \mathcal{M}) := \Pr_{U(\mathcal{M})}[e \in B]$. A *cut* is a set $E \setminus Z$ with Z a flat, and a *cycle* is a cut of the dual $\mathcal{M}^*$. Cocircuits are the complements of hyperplanes, so they are cuts, and circuits are cycles. Every flat is an intersection of hyperplanes, so a cut is a union of cocircuits, and a union of cocircuits is a cut. In a graph, the cuts of $M(G)$ are the unions of bonds and the cycles are the unions of circuits; both are wider notions than the binary cycle and cocycle spaces. Every vertex star $\delta(v)$ is a cut of $M(G)$, since the edges not at v span no edge at v.

We use these standard facts [Oxl11, Chapters 1–3]. The bases of $\mathcal{M}^*$ are the complements of the bases of $\mathcal{M}$. If B is a basis and $e \notin B$, then $B + e$ contains a unique circuit $C(e, B)$, and $B - f + e$ is a basis for every $f \in C(e, B)$. A circuit C and a cocircuit D never satisfy $|C \cap D| = 1$ (orthogonality). For disjoint $\sigma, \tau \subseteq E$, the minor $N = \mathcal{M}/\sigma \setminus \tau$ on $E \setminus (\sigma \cup \tau)$ has rank function $r_N(Z) = r(Z \cup \sigma) - r(\sigma)$, so that $x \in \mathrm{cl}_N(Z)$ if and only if $x \in \mathrm{cl}(Z \cup \sigma)$.

A basis contains no nonempty cycle, and it meets every nonempty cut: if $B \subseteq Z$ for a flat Z, then $Z = \mathrm{cl}(Z) \supseteq \mathrm{cl}(B) = E$. For $\mathcal{M} = M(G)$ with G connected, $U(\mathcal{M}) = \mathrm{UST}(G)$.

Proposition 9.1 (duality)**.** *For every split $E = X \sqcup Y$, $\beta(U(\mathcal{M}); X|Y) = \beta(U(\mathcal{M}^*); X|Y)$. A PC for $U(\mathcal{M})$ becomes a PC for $U(\mathcal{M}^*)$ with the same size, scopes, smoothness, decomposability, leaf arity and total variation distance by replacing each leaf input x_e with $1 - x_e$.*

Proof. $B \mapsto E \setminus B$ maps $\mathcal{B}(\mathcal{M})$ onto $\mathcal{B}(\mathcal{M}^*)$ and sends $(B \cap X, B \cap Y)$ to $(X \setminus B, Y \setminus B)$, a bijection on each side separately. So the matrices $\sqrt{U(\mathcal{M})}$ and $\sqrt{U(\mathcal{M}^*)}$ of Lemma 3.1 differ by a permutation of rows and a permutation of columns, and have the same operator norm. The leaf substitution is the bijection $x \mapsto \mathbf{1} - x$ of $\{0,1\}^E$; it changes no scope, and it maps Q_C to the law of $\mathbf{1} - x$ under Q_C, so it preserves total variation distance to the law of $E \setminus B$. □

9.2 Conditioning and exchange

Lemma 9.2 (conditioning is a minor)**.** *Let $F \subseteq E$ and $\sigma \subseteq F$ with $\Pr[B \cap F = \sigma] > 0$, and put $\tau := F \setminus \sigma$. Given $B \cap F = \sigma$, the set $B \setminus F$ is a uniform basis of $\mathcal{M}/\sigma \setminus \tau$.*

Proof. Some basis B_0 satisfies $B_0 \cap F = \sigma$, so σ is independent and $r(E \setminus \tau) = r$. Hence $N := \mathcal{M}/\sigma \setminus \tau$ has rank $r - |\sigma|$, and its independent sets are the sets $I \subseteq E \setminus F$ with $I \cup \sigma$ independent in $\mathcal{M}$. If $B \cap F = \sigma$, then $B \setminus F$ is independent in N and has $r - |\sigma|$ elements, so it is a basis of N. Conversely, if B' is a basis of N, then $B' \cup \sigma$ is independent in $\mathcal{M}$ with r elements, so it is a basis that meets F in σ. The map $B \mapsto B \setminus F$ is therefore a bijection between the bases with $B \cap F = \sigma$ and the bases of N. □

Lemma 9.3 (exchange bound)**.** *If D is a cut of a matroid N and $e \in D$, then $\lambda(e; N) \geq 1/|D|$. Dually, if C is a cycle of N and $e \in C$, then $1 - \lambda(e; N) \geq 1/|C|$.*

Proof. The flat $Z := E(N) \setminus D$ does not contain e. Let $H \supseteq Z$ be a maximal flat with $e \notin H$. If $r_N(H) \leq r(N) - 2$, pick $x \notin \mathrm{cl}_N(H + e)$; then $\mathrm{cl}_N(H + x)$ is a larger flat, and it does not contain e, since otherwise the exchange property would put x in $\mathrm{cl}_N(H + e)$. So H is a hyperplane, and $D' := E(N) \setminus H$ is a cocircuit with $e \in D' \subseteq D$.

Let B be a basis with $e \notin B$. The circuit $C(e, B)$ contains $e \in D'$, so by orthogonality it contains some $f \in D'$ with $f \neq e$, and $f \in B$. Then $B - f + e$ is a basis. Fix one such f for each B. The resulting map from bases without e to bases with e has at most $|D'| - 1$ preimages over each B', namely among the sets $B' - e + f$ with $f \in D' \setminus \{e\}$. So the number of bases without e is at most $(|D'| - 1)$ times the number with e, which gives $\lambda(e; N) \geq 1/|D'| \geq 1/|D|$. The second claim is the first in N^*. □

For $\mathcal{M} = M(G)$ and $D = \delta(v)$ this is $\Pr[e \in T] \geq 1/\deg(v)$, and for a cycle C it is $R(e^+, e^-) \leq (|C| - 1)/|C|$. We did not find Lemma 9.3 stated for matroids, but it is elementary and probably folklore.

Lemma 9.4 (cuts survive minors)**.** *Let D be a cut of $\mathcal{M}$ and $e \in D$, and let $\sigma, \tau \subseteq E \setminus \{e\}$ be disjoint with $D \cap \sigma = \emptyset$. Then $\lambda(e; \mathcal{M}/\sigma \setminus \tau) \geq 1/|D \setminus \tau| \geq 1/|D|$.*

Proof. Put $N := \mathcal{M}/\sigma\setminus\tau$, $H := E\setminus D$, and $Z := H\setminus(\sigma\cup\tau)$. Since $\sigma \subseteq H$, we have $\mathrm{cl}(Z\cup\sigma) \subseteq H$, so $e \notin \mathrm{cl}_N(Z)$. Hence $D_N := E(N)\setminus\mathrm{cl}_N(Z)$ is a cut of N that contains e, and $D_N \subseteq E(N)\setminus Z = D\setminus\tau$. Apply Lemma 9.3 in N. □

9.3 Framed pieces and the local theorem

Definition 9.5 (framed pieces)**.** A *framed piece* of $\mathcal{M}$ is a triple $(K, \pi, (D_e)_{e \in K})$, where π is a cyclic permutation of K with a single cycle, and either

- (circuit type) K is a circuit and every D_e is a cut, or
- (cocircuit type) K is a cocircuit and every D_e is a cycle,

such that $D_e \cap K = \{e, \pi(e)\}$ for every $e \in K$. Its *weight* is $w(K) := \prod_{e\in K} |D_e|^{-1}$ and its *territory* is $\Theta(K) := \bigcup_{e\in K} D_e$, which contains K. Two pieces K, K' are *compatible* if $K' \cap \Theta(K) = \emptyset$ and $K \cap \Theta(K') = \emptyset$. A piece is *mixed* for a split $X|Y$ if it meets both X and Y.

Every circuit K with $|K| \geq 2$ has a frame: for any cyclic order π, put $D_e := E\backslash \mathrm{cl}(K\backslash\{e, \pi(e)\})$. This is a cut; it contains e and $\pi(e)$ because $K - \pi(e)$ and $K - e$ are independent, and it misses the rest of K. Dually, every cocircuit with at least two elements has a frame by cycles. Loops and coloops have no frames, but a one-element set cannot be mixed. So the content of the hypotheses below is that the frames are small and avoid the other pieces.

Graphs. Let $C = v_1e_1v_2e_2\cdots v_ke_kv_1$ be a cycle of a simple graph G with $k \geq 3$, $\pi(e_j) := e_{j+1}$ and $D_{e_j} := \delta(v_{j+1})$, indices mod k. This is a circuit-type piece of $M(G)$ with weight $\prod_{v\in V(C)} \deg(v)^{-1}$. Its territory is the set of edges with an endpoint in $V(C)$, so two cycles are compatible exactly when they are vertex-disjoint.

Lemma 9.6 (frame bound)**.** *Let (K, π, D) be a circuit-type framed piece, let $S \subsetneq K$, and let $F \subseteq E \setminus K$ with $D_e \cap F = \emptyset$ for every $e \in S$. Then for every σ with $\Pr[B \cap F = \sigma] > 0$,*

$$\Pr[S \subseteq B \mid B \cap F = \sigma] \;\geq\; \prod_{e\in S} |D_e|^{-1}.$$

For a cocircuit-type piece the same bound holds for $\Pr[S \cap B = \emptyset \mid B \cap F = \sigma]$.

Proof. Since $S \neq K$ and π is a single cycle, S is a disjoint union of runs $e, \pi(e), \ldots, \pi^i(e)$ whose last element has its successor outside S. List the runs one after another, each in π-order, as $s_1, \ldots, s_k$. Then $\pi(s_j)$ is outside S or later in the list, so $D_{s_j} \cap \{s_1, \ldots, s_{j-1}\} \subseteq D_{s_j} \cap K \setminus \{s_j, \pi(s_j)\} = \emptyset$, and D_{s_j} also avoids F.

Put $\tau := F \setminus \sigma$ and $\sigma_j := \sigma \cup \{s_1, \ldots, s_{j-1}\}$. We show by induction on j that $\Pr[B \cap F = \sigma,\ s_{<j} \subseteq B] > 0$ and that, conditionally on this event, $s_j \in B$ with probability at least $|D_{s_j}|^{-1}$. The event is $\{B \cap (F \cup s_{<j}) = \sigma_j\}$, so by Lemma 9.2 the conditional law of the rest is $U(\mathcal{M}/\sigma_j \setminus \tau)$. Since $D_{s_j} \cap \sigma_j = \emptyset$ and $s_j \notin \sigma_j \cup \tau$, Lemma 9.4 bounds the conditional probability of $s_j \in B$ below by $|D_{s_j}|^{-1} > 0$, which also gives positivity for $j + 1$. Multiplying the factors proves the circuit case.

For the cocircuit case apply the circuit case to $\mathcal{M}^*$ and the basis $E \setminus B$ of $\mathcal{M}^*$: the event $B \cap F = \sigma$ is $(E \setminus B) \cap F = F \setminus \sigma$, and $S \cap B = \emptyset$ is $S \subseteq E \setminus B$. □

For $\mathcal{M} = M(G)$ and star frames, Lemma 9.6 is Lemma 6.5, and Lemmas 9.3 and 9.4 together are Lemma 6.4.

Theorem 9.7 (local bound for one split)**.** *Let $E = X \sqcup Y$ be a split and let $K_1, \ldots, K_r$ be pairwise compatible framed pieces of $\mathcal{M}$, each mixed for $X|Y$. Then*

$$\beta\big(U(\mathcal{M}); X|Y\big)^2 \;\leq\; \prod_{i=1}^{r} \frac{1 + \sqrt{1 - 4w(K_i)}}{2} \;\leq\; \prod_{i=1}^{r} (1 - w(K_i)).$$

Proof. For a circuit-type piece put $u_i := \mathbf{1}[K_i \cap X \subseteq B]$ and $v_i := \mathbf{1}[K_i \cap Y \subseteq B]$; for a cocircuit-type piece put $u_i := \mathbf{1}[K_i \cap X \cap B = \emptyset]$ and $v_i := \mathbf{1}[K_i \cap Y \cap B = \emptyset]$. The value $u_i = v_i = 1$ is impossible, because a basis contains no circuit and meets every cocircuit. Since $u = (u_i)$ is a function of $B \cap X$ and $v = (v_i)$ of $B \cap Y$, Lemma 3.2 gives $\beta(U(\mathcal{M}); X|Y) \leq \|\sqrt{\mathrm{law}(u, v)}\|_{\mathrm{op}}$.

Put $F_i := K_1 \cup \cdots \cup K_{i-1}$. The past $(u_{<i}, v_{<i})$ is a function of $B \cap F_i$, and by compatibility $F_i \cap K_i = \emptyset$ and $D_e \cap F_i = \emptyset$ for every $e \in K_i$. Since K_i is mixed, $K_i \cap X$ and $K_i \cap Y$ are proper subsets of K_i. Lemma 9.6 gives $\Pr[u_i = 1 \mid B \cap F_i = \sigma] \geq p_i := \prod_{e \in K_i \cap X} |D_e|^{-1}$ for every possible σ, and averaging over σ given a past π gives $K_i(1, 0 \,|\, \pi) = \Pr[u_i = 1 \mid \pi] \geq p_i$. In the same way $K_i(0, 1 \,|\, \pi) \geq q_i := \prod_{e \in K_i \cap Y} |D_e|^{-1}$, and $p_i q_i = w(K_i)$. Lemma 3.5 bounds γ_i^2, and Lemma 3.3 finishes the proof. □

Theorem 9.8 (circuit lower bound for matroids)**.** *Let $\mathcal{K}$ be a family of framed pieces of $\mathcal{M}$, each of weight at least w_0, in which each piece is incompatible with at most Γ others. Suppose every balanced split $X|Y$ of E has at least m mixed pieces in $\mathcal{K}$. Let C be a smooth decomposable PC with nonnegative weights over the n element variables, whose leaves have arity at most $2n/3$. If $1 - \mathrm{TV}(U(\mathcal{M}), Q_C) \geq \delta > 0$, then*

$$|C| \;\geq\; \delta^2 (1 - w_0)^{-\lceil m/(\Gamma+1) \rceil} \;\geq\; \delta^2 \exp\!\big(w_0 m/(\Gamma + 1)\big).$$

Proof. Fix a balanced split. Choosing mixed pieces greedily and discarding the at most Γ pieces incompatible with each choice gives $\lceil m/(\Gamma+1) \rceil$ pairwise compatible mixed pieces. Theorem 9.7 bounds β^2 by $(1 - w_0)^{\lceil m/(\Gamma+1) \rceil}$, and Corollary 3.11 gives the first bound. The second uses $-\ln(1 - w_0) \geq w_0$. □

The hypotheses are self-dual: passing to $\mathcal{M}^*$ swaps the two piece types and keeps weights, territories, compatibility and mixed counts, and Proposition 9.1 transfers the conclusion. A convenient sufficient condition: if every piece of $\mathcal{K}$ has at most ℓ elements and frames of size at most d, and every element lies in at most Δ pieces and at most Δ territories, then $w_0 \geq d^{-\ell}$ and $\Gamma \leq \ell(d+1)\Delta$, since $|\Theta(K)| \leq \ell d$. So Theorem 9.8 gives $2^{\Omega(n)}$ whenever ℓ, d, Δ are bounded and every balanced split has $\Omega(n)$ mixed pieces. Section 11 shows that some locality hypothesis of this kind is needed.

9.4 Graphic and cographic matroids

For $\mathcal{M} = M(G)$ with G simple, the star-framed cycles above are framed pieces, and compatibility is vertex-disjointness. With the vertex-disjoint mixed cycles of Lemma 7.3, Theorem 9.7 gives the w_k part of Theorem 6.6 and the bound $\beta^2 \leq (1 - D^{-\ell})^{r_0}$ of Theorem 8.2. Theorem 6.6 also has the alternative weight p_k^2 from Lemma 6.2, which is better for very irregular cycles; it is specific to graphs.

The cographic matroid $M(G)^*$ has $U(M(G)^*)$ equal to the law of the complement of a uniform spanning tree. By Proposition 9.1, every lower bound in this paper holds unchanged for it. When G is nonplanar, $M(G)^*$ is not graphic, but this transfer is a relabeling, not a new result. In $M(G)^*$ the cycles of G are cocircuit-type pieces and the stars are their frames.

9.5 A binary non-regular family

Here is a family to which neither the graph results nor duality applies. Let F_7 be the Fano plane, the rank-3 binary matroid whose seven points are the nonzero vectors of $\mathbb{F}_2^3$ and whose circuits of size three are its seven lines. Its cocircuits are the complements of lines [Oxl11]. Every two lines meet in exactly one point, and every point lies on three lines.

For matroids $\mathcal{M}_1, \mathcal{M}_2$ with $E(\mathcal{M}_1) \cap E(\mathcal{M}_2) = \{p\}$, where p is neither a loop nor a coloop of either, the *2-sum* $\mathcal{M}_1 \oplus_2 \mathcal{M}_2$ on $(E(\mathcal{M}_1) \cup E(\mathcal{M}_2)) \setminus \{p\}$ has as circuits the circuits of $\mathcal{M}_1$ or $\mathcal{M}_2$ that avoid p, together with the sets $(C_1 \cup C_2) \setminus \{p\}$ with $p \in C_1 \cap C_2$ and C_i a circuit of $\mathcal{M}_i$.

Moreover $(\mathcal{M}_1 \oplus_2 \mathcal{M}_2)^* = \mathcal{M}_1^* \oplus_2 \mathcal{M}_2^*$, both parts are minors of the 2-sum, and 2-sums of binary matroids are binary [Oxl11, Section 7.1].

Let G be a simple graph with no bridge. For each edge e take a copy $F^{(e)}$ of F_7 whose point p_e is identified with e, and let $\mathcal{M}_G$ be obtained from $M(G)$ by 2-summing $F^{(e)}$ along e for every edge e. By induction on the number of 2-sums:

(F1) each circuit or cocircuit of $F^{(e)}$ that avoids p_e is a circuit or cocircuit of $\mathcal{M}_G$;

(F2) if C is a cycle of G and $L_e \ni p_e$ is a line of $F^{(e)}$ for each $e \in C$, then $\bigcup_{e\in C}(L_e \setminus \{p_e\})$ is a circuit of $\mathcal{M}_G$;

(F3) if D is a bond of G and $D_f \ni p_f$ is a cocircuit of $F^{(f)}$ for each $f \in D$, then $\bigcup_{f\in D}(D_f \setminus \{p_f\})$ is a cocircuit of $\mathcal{M}_G$.

(F3) is (F2) for the duals, using $(\mathcal{M}_1 \oplus_2 \mathcal{M}_2)^* = \mathcal{M}_1^* \oplus_2 \mathcal{M}_2^*$ and that the circuits of $M(G)^*$ are the bonds of G. The induction applies because each edge lies on a cycle of G and so is never a loop or coloop of the intermediate matroids. The matroid $\mathcal{M}_G$ has $n = 6M$ elements and is binary. It has F_7 as a minor, and F_7 is representable only over fields of characteristic two, while regular matroids are representable over every field and form a minor-closed class [Oxl11]. So $\mathcal{M}_G$ is not regular, and in particular neither graphic nor cographic.

Proposition 9.9 (Fano-decorated linked expanders)**.** *Let G satisfy the hypotheses of Theorem 8.2, and put*

$$\varepsilon := \frac{1}{2}\min\Big\{\frac{hN}{4D\ell^2\kappa M},\ \frac{1}{2\ell^2\kappa D(1+D/h)},\ \frac{1}{12}\Big\}.$$

Then $\beta(U(\mathcal{M}_G); X|Y)^2 \le (1-(12D)^{-\ell})^{\lceil \varepsilon M\rceil}$ for every balanced split $X|Y$ of $E(\mathcal{M}_G)$. Consequently, every smooth decomposable PC with nonnegative weights over the $6M$ element variables, with leaves of arity at most $4M$ and $1-\mathrm{TV}(U(\mathcal{M}_G), Q_C) \ge \delta > 0$, has $|C| \ge \delta^2 \exp(\varepsilon M(12D)^{-\ell})$.

Since $M \le DN/2$, $\varepsilon \ge \frac{1}{2}\min\{h/(2D^2\ell^2\kappa), 1/(2\ell^2\kappa D(1+D/h)), 1/12\}$, so the bound is $2^{\Omega(n)}$ for fixed D, h, ℓ, κ.

Proof. By Lemma 7.2, G has no bridge, so $\mathcal{M}_G$ is defined. We use two kinds of pieces, all of weight at least $(12D)^{-\ell}$; note $\ell \ge 3$.

Local pieces. Let $L = \{x, y, z\}$ be a line of $F^{(e)}$ with $p_e \notin L$, ordered cyclically $x \to y \to z \to x$. By (F1) it is a circuit of $\mathcal{M}_G$. Let $D_x := F^{(e)} \setminus L_z$, where L_z is the line through p_e and z, and define D_y, D_z in the same way. Each is a cocircuit of $F^{(e)}$ avoiding p_e, hence a cut of $\mathcal{M}_G$, and $D_x \cap L = L \setminus L_z = \{x, y\}$ because two lines meet in one point. The weight is $4^{-3} \ge (12D)^{-\ell}$ and the territory lies in $F^{(e)} \setminus \{p_e\}$. If the six points of $F^{(e)} \setminus \{p_e\}$ are not all on one side of the split, some such line is mixed: the four lines avoiding p_e cover these six points, and any two of them meet in a point other than p_e, so if all four were monochromatic they would all have the same colour.

Lifted cycles. Let $C = v_1 e_1 v_2 \cdots v_k e_k v_1$ be a cycle of G, and for each j let $\{p_{e_j}, a_j, b_j\}$ be a line of $F^{(e_j)}$. By (F2), $K := \{a_1, b_1, \ldots, a_k, b_k\}$ is a circuit; order it cyclically as $a_1, b_1, a_2, b_2, \ldots, b_k$. Let $D(a_j) := F^{(e_j)} \setminus L'$ for a line $L' \ni p_{e_j}$ other than $\{p_{e_j}, a_j, b_j\}$; it is a cocircuit by (F1), has four elements, and meets K in $\{a_j, b_j\}$. For $D(b_j)$, take the vertex $v := v_{j+1}$ shared by e_j and e_{j+1}. For every $f \in \delta(v)$ choose a cocircuit $D_f \ni p_f$ of $F^{(f)}$, with $b_j \in D_{e_j} \not\ni a_j$ and $a_{j+1} \in D_{e_{j+1}} \not\ni b_{j+1}$; for example D_{e_j} is the complement of a line through a_j that avoids p_{e_j}. The star $\delta(v)$ is the disjoint union of the bonds $E(v, Q)$ over the components Q of $G - v$, so by (F3) the set $D(b_j) := \bigcup_{f\in\delta(v)}(D_f \setminus \{p_f\})$ is a union of cocircuits, hence a cut. It has $3\deg(v) \le 3D$

elements and meets K in $\{b_j, a_{j+1}\}$, because e_j and e_{j+1} are the only edges of C at v. So K is a framed piece of weight at least $(4 \cdot 3D)^{-k} \geq (12D)^{-\ell}$ when $k \leq \ell$. Its territory lies in the gadgets of edges with an endpoint in $V(C)$, so lifts of vertex-disjoint cycles are compatible.

Counting. Fix a balanced split $X|Y$ of $E(\mathcal{M}_G)$, so $|X|, |Y| \geq 2M$, and call a gadget *cut* if it has points on both sides. Let c be the number of cut gadgets. Local pieces in different gadgets are compatible. If $c \geq \varepsilon M$, we get $c \geq \lceil \varepsilon M \rceil$ compatible mixed local pieces. Otherwise colour each edge e of G by the side containing $F^{(e)} \setminus \{p_e\}$, and colour edges of cut gadgets arbitrarily. Each gadget has six points, and a cut gadget has at most five points in X, so at least $(2M - 5c)/6 \geq M/3 - c > M/4$ gadgets lie entirely in X, using $c < \varepsilon M \leq M/12$; the same holds for Y. Lemma 7.3 with $m_0 = M/4$ gives at least

$$\frac{t}{\ell\kappa} \;\geq\; \min\Big\{\frac{hN}{4D\ell^2\kappa},\; \frac{M}{2\ell^2\kappa D(1 + D/h)}\Big\} \;\geq\; 2\varepsilon M$$

vertex-disjoint mixed cycles of length at most ℓ. Vertex-disjoint cycles are edge-disjoint, so at most $c < \varepsilon M$ of them use an edge with a cut gadget. Each remaining cycle has an edge whose gadget lies in X and one whose gadget lies in Y, so its lift is mixed. This gives more than εM, hence at least $\lceil \varepsilon M \rceil$, compatible mixed lifted pieces. In both cases Theorem 9.7 gives the bound on β^2, and Corollary 3.11 the bound on $|C|$. □

We checked the structure facts (F1)–(F3) that the proof uses, all local pieces, the lifted triangle with its frames, and Theorem 9.7 on random splits, for $G = K_3$ by exact enumeration (Appendix A). The counting step was not checked numerically.

9.6 The global version for balanced matroids

Frames give certificates of size $|D_e|$. For graphs, Section 4 uses effective resistance instead, which can be much better. The matroid analogue of the resistance step needs negative correlation. Following Feder and Mihail [FM92], call $\mathcal{M}$ *balanced* if for every minor N of $\mathcal{M}$ and all distinct e, f, $\Pr_{U(N)}[e, f \in B] \leq \lambda(e; N)\lambda(f; N)$. Regular matroids are balanced [FM92]. A binary matroid is balanced if and only if it has no minor isomorphic to the eight-element matroid S_8 [CW03]. The class is closed under duality, since negative correlation of $\mathbf{1}[e \in B]$ and $\mathbf{1}[f \in B]$ is the same as negative correlation of their complements.

Lemma 9.10 (worst-case minors)**.** *Let $\mathcal{M}$ be balanced, let $F \subseteq E$ and $\sigma \subseteq F$ with $\Pr[B \cap F = \sigma] > 0$, and let $s_1, \dots, s_k \in E \setminus F$ be distinct. Then*

$$\Pr\big[\{s_1, \dots, s_k\} \subseteq B \;\big|\; B \cap F = \sigma\big] \;\geq\; \prod_{j=1}^{k} \lambda(s_j; \mathcal{M}/F/s_{<j}),$$

$$\Pr\big[\{s_1, \dots, s_k\} \cap B = \emptyset \;\big|\; B \cap F = \sigma\big] \;\geq\; \prod_{j=1}^{k} (1 - \lambda(s_j; \mathcal{M} \setminus F \setminus s_{<j})).$$

Proof. If the right side of the first inequality is 0 there is nothing to prove, so assume every factor is positive. Put $\tau := F \setminus \sigma$ and $N_j := \mathcal{M}/\sigma/s_{<j} \setminus \tau$. We claim $\lambda(s_j; N_j) \geq \lambda(s_j; \mathcal{M}/F/s_{<j})$. Turn the elements of τ from deleted into contracted one at a time. Each step compares $N' \setminus f$ with N'/f for a minor N' of $\mathcal{M}$. If f is a loop or coloop of N', the two minors are equal. Otherwise $0 < \lambda(f; N') < 1$ and, by Lemma 9.2, $\lambda(s_j; N'/f) = \Pr_{U(N')}[s_j \in B \mid f \in B]$ and $\lambda(s_j; N' \setminus f) = \Pr_{U(N')}[s_j \in B \mid f \notin B]$. Negative correlation in N' says the first is at most

$\lambda(s_j; N')$, hence at most the second. This proves the claim. Now by induction on j, using Lemma 9.2, the event $\{B \cap F = \sigma,\ s_{<j} \subseteq B\}$ has positive probability and, given it, $s_j \in B$ with probability $\lambda(s_j; N_j) > 0$. Multiplying gives the first inequality. The second is the first for $\mathcal{M}^*$, which is balanced, with $E \setminus B$ in place of B and $\lambda(e; N^*) = 1 - \lambda(e; N)$. □

Only negative correlation of pairs in minors is used, not negative association, in the same way that Section 4 uses only Rayleigh monotonicity.

Theorem 9.11 (global bound for one split)**.** *Let $\mathcal{M}$ be balanced, let $X|Y$ be a split, and let $K_1, \dots, K_r$ be pairwise disjoint mixed circuits or cocircuits in a fixed order. Put $F_i := K_1 \cup \cdots \cup K_{i-1}$ and order each of $K_i \cap X$ and $K_i \cap Y$ arbitrarily. For e in such a half H, with $H_{<e}$ the elements of H before e, put $\phi(e) := \lambda(e; \mathcal{M}/(F_i \cup H_{<e}))$ if K_i is a circuit and $\phi(e) := 1 - \lambda(e; \mathcal{M} \setminus (F_i \cup H_{<e}))$ if K_i is a cocircuit. Then*

$$\beta(U(\mathcal{M}); X|Y)^2 \ \le\ \prod_{i=1}^{r} \frac{1 + \sqrt{1 - 4\phi(K_i)}}{2} \ \le\ \prod_{i=1}^{r} (1 - \phi(K_i)), \qquad \phi(K_i) := \prod_{e \in K_i} \phi(e).$$

Proof. The proof of Theorem 9.7, with Lemma 9.10 in place of Lemma 9.6. Disjointness is all that is needed: the past is a function of $B \cap F_i$ and $F_i \cap K_i = \emptyset$. □

The proof uses only that a basis contains no piece of circuit type and meets every piece of cocircuit type. So Theorem 9.11 also holds with nonempty cycles and cuts in place of circuits and cocircuits.

For regular matroids the factors are resistances. Let A be a totally unimodular real matrix representing $\mathcal{M}$, with columns a_e. For $Z \subseteq E \setminus \{e\}$ put

$$R_Z(e) := \min\Big\{ \|x\|_2^2 :\ x \in \mathbb{R}^{E \setminus (Z \cup \{e\})},\ \sum_f x_f a_f = a_e \Big\} \qquad (\min \emptyset = \infty).$$

Lemma 9.12 (leverage is resistance)**.** *For a regular matroid $\mathcal{M}$ represented by a totally unimodular A, and $Z \subseteq E \setminus \{e\}$, $1 - \lambda(e; \mathcal{M} \setminus Z) = 1/(1 + R_Z(e))$.*

Proof. If e is a coloop of $\mathcal{M} \setminus Z$, then a_e is not in the span of the other columns of $E \setminus Z$, so both sides are 0. Otherwise keep the columns $E \setminus Z$ and a set of rows that is a basis of their row space. This gives a totally unimodular matrix A' of full row rank r' representing $\mathcal{M} \setminus Z$, and the linear constraint in $R_Z(e)$ is unchanged, because the dropped rows are combinations of the kept ones on these columns. By the Cauchy–Binet formula and total unimodularity, $\det(A'A'^{\mathsf{T}}) = \sum_S \det(A'_S)^2$ over r'-sets of columns S is the number of bases of $\mathcal{M} \setminus Z$. Let Q be the same Gram matrix without column e. Then $\det Q$ is the number of bases avoiding e, and Q is invertible because e is not a coloop. By the matrix determinant lemma, $1 - \lambda(e; \mathcal{M} \setminus Z) = \det Q / \det(Q + a'_e a'^{\mathsf{T}}_e) = 1/(1 + a'^{\mathsf{T}}_e Q^{-1} a'_e)$, and $a'^{\mathsf{T}}_e Q^{-1} a'_e$ is the least squared norm of a solution of $\sum_{f \neq e} x_f a'_f = a'_e$, which is $R_Z(e)$. □

For a loop, $R_Z(e) = 0$ and both sides equal 1. For $\mathcal{M} = M(G)$ with a signed incidence matrix, the constraint says that x is a unit flow between the endpoints of e in $G - Z - e$, so $R_Z(e) = R_{G-Z-e}(e^+, e^-)$ by Thomson's principle [LP16]. Any explicit x bounds $R_Z(e)$ from above; for example, m circuits through e that avoid Z, pairwise share only e, and have at most ℓ elements each give $R_Z(e) \le (\ell - 1)/m$, by averaging their signed incidence vectors (kernel vectors of A with entries in $\{0, \pm 1\}$, which exist for every circuit when A is totally unimodular). Dually, $\lambda(e; \mathcal{M}/Z) = 1/(1 + R^*_Z(e))$, with R^* computed in a totally unimodular representation of $\mathcal{M}^*$.

The vertex-touch bits are the cocircuit case. Let G be connected, $X|Y = A|B$ an edge split, and $W = (w_1, \dots, w_r)$ an independent set of mixed vertices. The stars $S(w_k)$ are disjoint mixed cuts. For $s_j = u_j w \in S_A(w)$, Lemma 9.12 turns the factor $\phi(s_j)$ of Theorem 9.11 into $1/(1 + R_{G-F_k-s_{\le j}}(u_j, w))$. The graph $G - F_k - s_{\le j}$ contains $G^A_w = G - (W \setminus \{w\}) - S_A(w)$, so by (E2) this factor is at least $1/(1+R_{u_j})$, and the product over the A-half is at least the quantity $p_A(w)$ of Lemma 4.3; the same holds for B. So Theorem 9.11, applied to the stars, recovers the bound of Theorem 4.5. The graph-specific input of Section 4 is Lemmas 4.4, 4.7 and 4.10, which bound the resistances using expansion; we have no analogue of them for regular matroids.

9.7 The duality behind both arguments

1. *The forbidden cell.* A basis contains no circuit and meets every cocircuit. Its complement, a basis of $\mathcal{M}^*$, contains no cocircuit and meets every circuit. These are the same fact read in $\mathcal{M}$ and in $\mathcal{M}^*$.
2. *The certificate.* Lemma 9.3 uses orthogonality once: the fundamental circuit meets the cocircuit a second time. Frames use it again: $|D_e \cap K| = 2$ is the smallest intersection orthogonality allows. A circuit is certified by cuts, and a cocircuit by cycles.
3. *The two graph arguments.* The cycle bits of Section 6 are circuits of $M(G)$ certified by vertex stars. The vertex-touch bits of Section 4 are vertex stars, which are cuts of $M(G)$, certified by resistances, that is, by flows built from circuits. Each argument's pieces are the other argument's certificates, and the local theorem is invariant under $\mathcal{M} \leftrightarrow \mathcal{M}^*$.

10 Does subtraction help?

The uniform spanning tree has a polynomial-size generating circuit with subtraction (Proposition 8.6(a)), but that circuit is not decomposable. This section asks what subtraction buys when the circuit keeps a decomposable structure. We write $P(x) = \mathbf{1}[x \text{ is a spanning tree}]/\tau$ for $P = \mathrm{UST}(G)$ on $\{0,1\}^E$, where τ is the number of spanning trees, and $\|f\|_1 := \sum_x |f(x)|$. For a split $A|B$ and a real function f on $\{0,1\}^E$, $M_{A,B}(f)$ is the $2^{|A|} \times 2^{|B|}$ matrix $(f(x_A, x_B))$. Cycles are *mixed* if they have edges in A and in B.

For circuits that follow one vtree, subtraction does not help. Exact representation needs size $2^{\Omega(N)}$ (Corollary 10.4). If the output Q is normalized, $\sum_x Q(x) = 1$, but may take negative values, size $e^{\Omega(N)}$ is needed at every fixed error $\frac{1}{2}\|P - Q\|_1 \le \varepsilon < 1$ (Theorem 10.12). Without normalization the threshold is $\varepsilon < \frac{1}{2}$, and it is sharp, since $Q = 0$ has error $\frac{1}{2}$ (Theorem 10.11). Both bounds hold on every 2-connected bounded-degree expander, at every girth and without smoothness (Corollary 10.21), and both come from one inequality for disjoint permutation blocks of P (Theorem 10.10). Normalized squares of structured circuits obey the nonnegative bound (Theorem 10.22). For unstructured circuits we prove lower bounds only for restricted classes (Section 10.6): bounded product-depth and formulas for support-exact representation, and sums of r structured circuits on unrelated vtrees, which need total size $2^{\Omega_r(N)}$ both for support-exact representation and at error $\varepsilon < \frac{1}{2}$. A general unstructured circuit that approximates P must cancel all but an exponentially small fraction of its mass (Theorem 10.38), and Section 10.8 explains why the block method does not reach it. The general question stays open.

10.1 Structured circuits with real weights

A *circuit with real weights* is defined like a PC (Section 2.1), except that leaves are real-valued functions and sum weights are real numbers. Size is the number of nodes plus wires, as before. A *vtree* for E is a rooted tree in which every internal node has exactly two children, and whose leaves are labelled by the blocks of a partition of E into nonempty sets. For a node v, $\mathrm{Var}(v)$ is the union of the blocks below v.

Definition 10.1 (structured circuits)**.** A circuit C with real weights, possibly with several output nodes, is *structured* with respect to a vtree V if every node g has a label $v(g) \in V$ such that: if g is a leaf, then $v(g)$ is a leaf of V and g is a function of $x_{\mathrm{Var}(v(g))}$; if g is a sum node, all its children have label $v(g)$; if g is a product node, then it has exactly two children, labelled by the two children of $v(g)$; and every output node is labelled by the root.

By induction, every node g computes a function of $x_{\mathrm{Var}(v(g))}$; taking $\mathrm{Var}(v(g))$ as the scope of g, a structured circuit is smooth and decomposable. Inputs occur only at vtree leaves, and products follow the vtree exactly. This matters. With the vtree $((x_1, x_2), x_3)$, a circuit whose root multiplies a two-variable input $[x_1 = x_2]$, placed at the internal node $\{x_1, x_2\}$, by an input on x_3 has no node with scope $\{x_1\}$, but $M_{\{x_1\},\{x_2,x_3\}}$ of its function has rank 2, so Lemma 10.2 below would fail for it. The usual definition of structured decomposability asks only that the children of each product node have scopes inside $\mathrm{Var}(v_1)$ and $\mathrm{Var}(v_2)$ for the two children v_1, v_2 of some vtree node. With univariate leaves and singleton blocks, such a circuit can be brought into the form of Definition 10.1 at the cost of a factor $O(M)$ in size. Label each node by the lowest vtree node whose set contains its scope. Then lift every child of a sum or product node to the label its parent requires: along the vtree path, multiply it at each step by the constant function 1 on the sibling subtree, which is computed once for every vtree node by a product tree of the leaves $[x_e = 0] + [x_e = 1]$. On $\{0,1\}^E$ the computed function does not change.

Lemma 10.2 (structured rank; after de Colnet and Mengel)**.** *Let C be structured with respect to V, let $v \in V$, and let G_v be the set of nodes with label v. For every output d of C there are functions $h_{g,d}$ of $x_{E\setminus\mathrm{Var}(v)}$ with*

$$f_d = \sum_{g \in G_v} f_g(x_{\mathrm{Var}(v)})\, h_{g,d}(x_{E\setminus\mathrm{Var}(v)}),$$

so $\operatorname{rank} M_{\mathrm{Var}(v),E\setminus\mathrm{Var}(v)}(f_d) \le |G_v| \le |C|$*. If $M \ge 2$ and every block has at most $2M/3$ elements, some node v has $M/3 < |\mathrm{Var}(v)| \le 2M/3$.*

Proof. Let g be a node whose label u is v or an ancestor of v. We show by induction in topological order that $f_g = \sum_{g' \in G_v} f_{g'} h_{g',g}$ with $h_{g',g}$ a function of $x_{\mathrm{Var}(u)\setminus\mathrm{Var}(v)}$. If $u = v$, take $h_{g,g} = 1$ and $h_{g',g} = 0$ for $g' \neq g$. If $u \neq v$, then u is internal, so g is not a leaf. If g is a sum node, its children have label u, and f_g is a linear combination of their expansions. If g is a product node, its children g_1, g_2 have labels u_1, u_2, the two children of u, and v lies below one of them, say u_1. The expansion of f_{g_1} exists by induction, and f_{g_2} is a function of $x_{\mathrm{Var}(u_2)}$, with $\mathrm{Var}(u_2) \subseteq \mathrm{Var}(u)\setminus\mathrm{Var}(v)$; multiply. Outputs are labelled by the root, which is an ancestor of v or v itself. Each term $f_g h_{g,d}$ has rank one.

For the last claim, start at the root and move to a child with the larger set while the current set has more than $2M/3$ elements. Such a node is internal, since blocks are smaller. The first node v with $|\mathrm{Var}(v)| \le 2M/3$ has a parent u with $|\mathrm{Var}(u)| > 2M/3$, and $|\mathrm{Var}(v)| \ge |\mathrm{Var}(u)|/2 > M/3$. □

This is the rank argument of de Colnet and Mengel [dCM21, Theorem 2 and Lemma 13], who use it, with induced matchings in expanders, to prove exponential lower bounds for structured-decomposable arithmetic circuits that may use negative constants. We state it at a fixed vtree node and for several outputs. Unlike Lemma 3.9, it gives one split for the whole circuit.

Theorem 10.3 (exact rank)**.** *Let $A|B$ be a split of a graph G and let $C_1, \dots, C_m$ be pairwise vertex-disjoint mixed cycles. Then* $\operatorname{rank} M_{A,B}(P) \geq 2^m$.

Proof. Pick $e_i \in E(C_i) \cap A$ and $f_i \in E(C_i) \cap B$. The paths $C_i - e_i$ are vertex-disjoint, so their union is a forest; extend it to a spanning tree T_0. No e_i is in T_0, since it would close C_i, so every f_i is in T_0. Let $F := T_0 \setminus \{f_1, \dots, f_m\}$. For $s, t \in \{0,1\}^m$ let $X_s := (F \cap A) \cup \{e_i : s_i = 1\}$ and $Y_t := (F \cap B) \cup \{f_i : t_i = 1\}$. If $t = \mathbf{1} - s$, then $X_s \cup Y_t$ arises from T_0 by exchanging f_i for e_i for every i with $s_i = 1$. Each exchange gives a tree: the current tree contains $C_i - e_i$, since earlier exchanges touched only other cycles, which are vertex-disjoint from C_i; so the fundamental cycle of e_i is C_i, which contains f_i. If $t \neq \mathbf{1} - s$, then either $s_i = t_i = 1$ for some i, and $X_s \cup Y_t$ contains C_i, or $s_i + t_i \leq 1$ for every i with strict inequality somewhere, and $X_s \cup Y_t$ has fewer than $N - 1$ edges. So the $2^m \times 2^m$ submatrix with rows X_s and columns Y_t is τ^{-1} times a permutation matrix. □

The permutation submatrix plays the part of the induced-matching gadget of de Colnet and Mengel; it is not literally theirs, since their gadget is a function of vertex variables.

Corollary 10.4 (exact representation)**.** *Let G satisfy the hypotheses of Theorem 8.2, with r_0 as there. Every circuit with real weights that is structured with respect to a vtree with blocks of at most $2M/3$ edges and that computes a nonzero multiple of* $\mathrm{UST}(G)$ *has size at least $2^{\lceil r_0 \rceil}$.*

Proof. Take the node v of Lemma 10.2 and the split $A = \mathrm{Var}(v)$, $B = E \setminus A$, which is balanced. As in the proof of Theorem 8.2, Lemma 7.3 with $m_0 = M/3$ gives at least r_0, hence at least $\lceil r_0 \rceil$, pairwise vertex-disjoint mixed cycles. Apply Theorem 10.3 and Lemma 10.2. □

For exact representation by structured circuits this is stronger than the nonnegative bound $(1 - D^{-\ell})^{-r_0}$ of Theorem 8.2.

10.2 Approximation

A small perturbation of a full-rank matrix can have small rank, so exact rank alone says nothing about approximation. What survives perturbation is a large family of disjoint *permutation blocks*: sets of cells on which P is a multiple of a permutation matrix. We build them here from pairs of short cycles, and in Section 10.3 from pendant vertices, and prove one inequality for all such families (Theorem 10.10). For normalized outputs it works at every error below 1; without normalization it stops at $\frac{1}{2}$, which is sharp.

Let $C_1, \dots, C_{2n}$ be pairwise vertex-disjoint mixed cycles, grouped into n pairs (C_i, C_j). On each cycle fix $e \in A$ and $f \in B$ as above; the other edges of the cycle are its *arcs*. For a spanning tree t, the pair (i, j) is *good* if t contains the arcs of both cycles and either $t \cap E(C_i) = E(C_i) \setminus \{f_i\}$ and $t \cap E(C_j) = E(C_j) \setminus \{e_j\}$ (pattern 1), or $t \cap E(C_i) = E(C_i) \setminus \{e_i\}$ and $t \cap E(C_j) = E(C_j) \setminus \{f_j\}$ (pattern 2). Let $\mathcal{G}(t)$ be the set of good pairs, $g(t) := |\mathcal{G}(t)|$, and $F(t)$ the set t minus the four special edges e_i, f_i, e_j, f_j of every good pair. The *block* of t is the set of trees t' with $(\mathcal{G}(t'), F(t')) = (\mathcal{G}(t), F(t))$.

Lemma 10.5 (blocks)**.** *A block with g good pairs contains exactly 2^g trees, one for each choice of pattern at each good pair. Let R be the set of cells (X, Y) in which X is $F \cap A$ plus one of e_i, e_j for each good pair, and Y is $F \cap B$ plus one of f_i, f_j for each good pair. Then $M_{A,B}(P)$ restricted to R is τ^{-1} times a $2^g \times 2^g$ permutation matrix, and the sets R of distinct blocks are disjoint.*

Proof. Let t be in pattern 1 at a good pair. Then $t \supseteq C_i - f_i$, the fundamental cycle of f_i is C_i, which contains e_i, so $t - e_i + f_i$ is a tree. It still contains $C_j - e_j$, because the cycles are vertex-disjoint, so exchanging f_j for e_j gives a tree, in pattern 2 at this pair. No other edge changes, so the set of good pairs and F are unchanged. Hence every choice of patterns occurs in the block, and a tree in the block is determined by F and its patterns.

In R, if the row and column choose (e_i, f_j) or (e_j, f_i) at every good pair, the cell is a tree of the block. If they choose (e_i, f_i) at some pair, the cell contains C_i, because F contains its arcs, and likewise for (e_j, f_j). So each row of R has exactly one tree, and so does each column. For disjointness we recover the block from any cell $(X, Y) \in R$. Let $\mathcal{G}'$ be the set of pairs whose arcs lie in $X \cup Y$ and for which X contains exactly one of e_i, e_j and Y exactly one of f_i, f_j. Every good pair is in $\mathcal{G}'$. For a pair that is not good, the edges of its two cycles in $X \cup Y$ are those of any tree t of the block. If its arcs lie in t, then, since t is acyclic, each of its cycles meets t in the arcs plus e only, f only, or neither. One e and one f in total is possible only as pattern 1 or pattern 2, which would make the pair good. So $\mathcal{G}' = \mathcal{G}$, and F is $X \cup Y$ minus the special edges of $\mathcal{G}$. □

Lemma 10.6 (ℓ_1 distance to low rank)**.** *Let Π be an $n \times n$ permutation matrix and Q a real matrix with* $\operatorname{rank} Q \le r \le n$*. Then $\sum_{a,b} |\Pi_{ab} - Q_{ab}| \ge n - r$.*

Proof. Let $\mathcal{E} := I - \Pi^\top Q$. Then $I - \mathcal{E}$ has rank at most r, so $\mathcal{K} := \ker(I - \mathcal{E})$ has dimension at least $n - r$. With $P_{\mathcal{K}}$ the orthogonal projection onto $\mathcal{K}$, $\mathcal{E} P_{\mathcal{K}} = P_{\mathcal{K}}$, so $n - r \le \operatorname{tr} P_{\mathcal{K}} = \operatorname{tr}(\mathcal{E} P_{\mathcal{K}}) \le \|\mathcal{E}\|_*$, the nuclear norm. Finally $\|\mathcal{E}\|_* = \|\Pi - Q\|_*$, and the nuclear norm of a matrix is at most the sum of the absolute values of its entries, by the triangle inequality over matrices with one nonzero entry. □

Block families. A *permutation block* of P is a set $R \subseteq \{0,1\}^E$ together with a bijection to a product $X \times Y$ of two sets of the same size n_R, under which τP restricted to R becomes an $n_R \times n_R$ permutation matrix; so R contains exactly n_R spanning trees. For a real function Q, $Q|_R$ is the $X \times Y$ matrix obtained in the same way. The sets R of Lemma 10.5 are permutation blocks whose cells are cells of $M_{A,B}$, so $\operatorname{rank} Q|_R \le \operatorname{rank} M_{A,B}(Q)$. For a family $\mathcal{R}$ of pairwise disjoint permutation blocks and an integer $g \ge 0$, let $\mathcal{R}_g$ be the set of blocks with $n_R \ge 2^g$ and

$$\delta_{\mathcal{R}}(g) := 1 - \frac{1}{\tau} \sum_{R \in \mathcal{R}_g} n_R,$$

the probability under P of the trees outside the blocks of $\mathcal{R}_g$. For the blocks of Lemma 10.5, every tree lies in exactly one block, and $\delta(g) = \Pr[g(T) < g]$.

Lemma 10.7 (normalized approximations)**.** *Let Q be a real function on $\{0,1\}^E$ with $\sum_x Q(x) = 1$, write $Q = Q^+ - Q^-$ with $Q^\pm := \max(\pm Q, 0)$, and put $t := \sum_x Q^-(x)$. Then $\|Q\|_1 = 1 + 2t$, $t \le \frac{1}{2}\|P - Q\|_1$, and*

$$\sum_x \min(P(x), Q^+(x)) = 1 - \tfrac{1}{2}\|P - Q\|_1 + t.$$

In particular, $\frac{1}{2}\|P - Q\|_1 \le \varepsilon$ implies $\|Q\|_1 \le 1 + 2\varepsilon$.

Proof. $\sum Q^+ = 1 + t$, so $\|Q\|_1 = 1 + 2t$. Where $Q < 0$ we have $|P - Q| = |P - Q^+| + Q^-$, and elsewhere $|P - Q| = |P - Q^+|$, so $\|P - Q^+\|_1 = \|P - Q\|_1 - t$. Since $\min(a, b) = (a + b - |a - b|)/2$, the sum of the minima is $\frac{1}{2}(1 + (1 + t) - \|P - Q\|_1 + t)$. As $\sum(P - Q) = 0$, the positive part of $P - Q$ has mass $\frac{1}{2}\|P - Q\|_1$, and it is at least Q^- pointwise. □

Lemma 10.8 (row-scaled trace; Ky Fan and Hoffman)**.** *Let B be a real $n \times n$ matrix and $R_i := \sum_j |B_{ij}|$. Then $\sum_{i:R_i>0} |B_{ii}|/R_i \le \operatorname{rank} B$.*

Proof. Let Δ be diagonal with $\Delta_{ii} := \operatorname{sign}(B_{ii})/R_i$ if $R_i > 0$, and $\Delta_{ii} := 0$ otherwise. The matrix $B' := \Delta B$ has rank at most $\operatorname{rank} B$, diagonal entries $|B_{ii}|/R_i$, and absolute row sums at most 1. Every complex eigenvalue λ of B' has $|\lambda| \le 1$: if $B'v = \lambda v$ with $v \neq 0$ and $|v_i|$ maximal, then $|\lambda||v_i| \le \sum_j |B'_{ij}||v_j| \le |v_i|$. The eigenvalue 0 has algebraic multiplicity at least $n - \operatorname{rank} B'$, so at most $\operatorname{rank} B'$ eigenvalues, counted with multiplicity, are nonzero, and $\sum_i |B_{ii}|/R_i = \operatorname{tr} B' \le \operatorname{rank} B' \le \operatorname{rank} B$. □

This is a bound of Ky Fan and Hoffman [FH53]. It says how much of a permutation a matrix of small rank and small mass can cover.

Lemma 10.9 (one block)**.** *Let Q be a real $n \times n$ matrix of rank at most r, π a permutation of $[n]$, $s > 0$, and $\|Q\|_1 := \sum_{i,j} |Q_{ij}|$. Then*

$$\sum_{i=1}^{n} \min(s, Q^+_{i\pi(i)}) \;\le\; \sqrt{s\, r \|Q\|_1}, \qquad \sum_{i=1}^{n} \sqrt{Q^+_{i\pi(i)}} \;\le\; \sqrt{r \|Q\|_1}.$$

The first bound is attained: if $n = rm$ and Q has r diagonal blocks equal to the all-ones $m \times m$ matrix divided by rm^2, then for $\pi = \mathrm{id}$ and $s = 1/(rm^2)$ both sides equal $1/m$.

Proof. Permuting columns changes neither the rank nor $\|Q\|_1$, so let π be the identity. Let I be the set of i with $Q_{ii} > 0$, and put $c_i := \min(s, Q_{ii})$ and $R_i := \sum_j |Q_{ij}|$, so $R_i \ge Q_{ii} \ge c_i$ for $i \in I$. By the Cauchy–Schwarz inequality and Lemma 10.8,

$$\sum_{i \in I} c_i = \sum_{i \in I} \sqrt{\frac{c_i}{R_i}} \sqrt{c_i R_i} \le \Big(\sum_{i \in I} \frac{Q_{ii}}{R_i} \Big)^{1/2} \Big(s \sum_i R_i \Big)^{1/2} \le \sqrt{r\, s \|Q\|_1}.$$

The second bound is the same computation with $\sqrt{Q_{ii}} = \sqrt{Q_{ii}/R_i}\,\sqrt{R_i}$. □

Theorem 10.10 (approximation on permutation blocks)**.** *Let $\mathcal{R}$ be a family of pairwise disjoint permutation blocks of P, let Q be a real function on $\{0,1\}^E$ with $\frac{1}{2}\|P - Q\|_1 \le \varepsilon$, put $r_R := \operatorname{rank} Q|_R$, and let $g \ge 0$ be an integer. Write $t_+ := \max(t, 0)$.*

(a) If $\sum_x Q(x) = 1$, then

$$\big[1 - \varepsilon - \delta_{\mathcal{R}}(g)\big]_+^2 \;\le\; \frac{1}{\tau} \sum_{R \in \mathcal{R}_g} r_R.$$

In particular, if $r_R \le r$ for every $R \in \mathcal{R}_g$, then $r \ge 2^g [1 - \varepsilon - \delta_{\mathcal{R}}(g)]_+^2$.

(b) In general $2\varepsilon \ge \tau^{-1} \sum_{R \in \mathcal{R}} (n_R - r_R)^+$. In particular, if $r_R \le r$ for every $R \in \mathcal{R}_g$, then $2\varepsilon \ge (1 - \delta_{\mathcal{R}}(g))(1 - r/2^g)$.

Proof. (a) Put $\omega := \sum_x \min(P(x), Q^+(x))$, $\delta := \delta_{\mathcal{R}}(g)$, $S := \tau^{-1} \sum_{R \in \mathcal{R}_g} r_R$, and let t be as in Lemma 10.7, so $\omega \ge 1 - \varepsilon + t$ and $\|Q\|_1 = 1 + 2t$. The points outside the blocks of $\mathcal{R}_g$ contribute at most their P-mass, which is δ. In a block $R \in \mathcal{R}_g$, the points where $P > 0$ are the n_R cells

of a permutation, where $P = 1/\tau$, so Lemma 10.9 with $s = 1/\tau$, applied to $Q|_R$, bounds their contribution by $(r_R\|Q|_R\|_1/\tau)^{1/2}$. By the Cauchy–Schwarz inequality and disjointness,

$$1 - \varepsilon + t \le \omega \le \delta + \sum_{R \in \mathcal{R}_g} \Big(\frac{r_R \|Q|_R\|_1}{\tau}\Big)^{1/2} \le \delta + \sqrt{S\|Q\|_1} = \delta + \sqrt{S(1+2t)}.$$

If $a := 1 - \varepsilon - \delta \le 0$ there is nothing to prove. Otherwise $(a+t)^2 \le S(1+2t)$, and $(a+t)^2 - a^2(1+2t) = t^2 + 2at(1-a) \ge 0$ because $0 < a \le 1$, so $S \ge a^2$. For the second claim, the blocks of $\mathcal{R}_g$ are disjoint and each contains at least 2^g trees, so there are at most $\tau 2^{-g}$ of them, and $S \le r2^{-g}$.

(b) Lemma 10.6, applied to $\tau P|_R$ and $\tau Q|_R$ on each block, and disjointness give $2\varepsilon\tau = \|\tau P - \tau Q\|_1 \ge \sum_R (n_R - r_R)^+$. If $r > 2^g$ the second claim is trivial. Otherwise keep the blocks of $\mathcal{R}_g$; for them $n_R - r \ge n_R(1 - r2^{-g})$, and $\tau^{-1}\sum_{R\in\mathcal{R}_g} n_R = 1 - \delta$. □

Normalization moves the threshold from $\frac{1}{2}$ to 1. Without it, (b) is sharp at $\varepsilon = \frac{1}{2}$, since $Q = 0$ has rank 0. With $\sum Q = 1$ the zero function is excluded, the mass of Q is at most $1 + 2\varepsilon$ even though Q may be negative, and Lemma 10.9 says that a matrix of rank r and bounded mass covers at most a $\sqrt{r/n_R}$ fraction of a permutation of size n_R. Negative values do not help Q: each unit of negative mass adds a unit to ω and two units to $\|Q\|_1$, and the last step of the proof shows that this trade never pays. The positive part in (a) cannot be dropped. Take four disjoint triangles joined in a path by three bridges, so $N = 12$, $M = 15$ and $\tau = 81$. On each triangle put an edge e and an arc in A and the third edge f in B, put the bridges in B, and pair the triangles as $(1, 2)$ and $(3, 4)$. Only the 4 trees that are good at both pairs lie in blocks with 2^2 trees, so $\delta(2) = 77/81$. The uniform distribution $Q \equiv 2^{-15}$ has rank 1, $\|Q\|_1 = 1$ and $\varepsilon = 1 - 81/2^{15}$, so $1 - \varepsilon - \delta(2) < 0$, and squaring without the positive part would give the false bound $r \ge 4(1 - \varepsilon - \delta(2))^2 \approx 3.6$. Even the weaker form $r \ge 2^g(1 - \varepsilon - \delta)^2/(1 + 2\varepsilon)$, which uses only $\|Q\|_1 \le 1 + 2\varepsilon$ instead of the identity in Lemma 10.7, would give $r \ge 1.2$ without the positive part.

Theorem 10.11 (approximate rank)**.** *With the pairs as above, let Q be any real function on $\{0,1\}^E$, not necessarily nonnegative or normalized, with* $\operatorname{rank} M_{A,B}(Q) \le r$ *and* $\frac{1}{2}\|P - Q\|_1 \le \varepsilon$. *For every integer $g \ge 0$ with $\alpha_g := \Pr_{T\sim P}[g(T) \ge g] > 0$,*

$$2\varepsilon \;\ge\; \alpha_g\Big(1 - \frac{r}{2^g}\Big), \qquad \textit{equivalently} \qquad r \;\ge\; 2^g\Big(1 - \frac{2\varepsilon}{\alpha_g}\Big).$$

The threshold $\varepsilon < \frac{1}{2}$ is sharp: $Q = 0$ has rank 0 and $\frac{1}{2}\|P - Q\|_1 = \frac{1}{2}$.

Proof. The sets R of Lemma 10.5 form a family of disjoint permutation blocks with $n_R = 2^{g_b}$, where g_b is the number of good pairs of block b, and $\operatorname{rank} Q|_R \le r$. Every tree lies in exactly one block, so $1 - \delta(g) = \alpha_g$. Apply Theorem 10.10(b). □

Theorem 10.12 (approximate rank, normalized)**.** *With the pairs as above, let Q be a real function on $\{0,1\}^E$ with $\sum_x Q(x) = 1$,* $\operatorname{rank} M_{A,B}(Q) \le r$ *and* $\frac{1}{2}\|P - Q\|_1 \le \varepsilon$. *Then for every integer $g \ge 0$,*

$$r \;\ge\; 2^g\,[\alpha_g - \varepsilon]_+^2.$$

Proof. As for Theorem 10.11, with Theorem 10.10(a) and $1 - \varepsilon - \delta(g) = \alpha_g - \varepsilon$. □

Lemma 10.13 (good pairs are frequent)**.** *Let $p_k := D^{-(|C_k|-1)}$, where D is the maximum degree. Conditionally on any event of positive probability that is determined by the sets $T \cap E(C_l)$ for the cycles C_l of the other pairs, the pair (i, j) is good with probability at least $2p_ip_j$. Hence $g(T)$ stochastically dominates the binomial law* $\operatorname{Bin}(n, 2p^2)$, *where $p := \min_k p_k$.*

Proof. Let Π be such an event. No edge of C_l, $l \neq i$, is incident to C_i, so Π and the event $\{T \cap E(C_j) = E(C_j) \setminus \{e_j\}\}$ are functions of $T \setminus I(C_i)$, in the notation of Lemma 6.1. Lemma 6.2, applied to C_i and the edge f_i, and the tower property give

$$\Pr[\text{pattern } 1, \Pi] \;\geq\; p_i \Pr\big[T \cap E(C_j) = E(C_j) \setminus \{e_j\},\ \Pi\big] \;\geq\; p_i p_j \Pr[\Pi],$$

where the second step is the same argument for C_j. Pattern 2 is symmetric, and the two patterns are disjoint. Taking for Π the values of the goodness indicators of the earlier pairs gives conditional probability at least $2p^2$ for each pair given the earlier ones, which yields the domination by the standard sequential coupling. □

Corollary 10.14 (approximation on linked expanders)**.** *Let G satisfy the hypotheses of Theorem 8.2, with r_0 as there, put $p := D^{-(\ell-1)}$ and $n := \lfloor \lceil r_0 \rceil / 2 \rfloor$, and assume $n \geq 1$. Let C be a circuit with real weights, structured with respect to a vtree with blocks of at most $2M/3$ edges, that computes a real function f with $\frac{1}{2}\|\mathrm{UST}(G) - f\|_1 \leq \varepsilon$. Then*

$$|C| \;\geq\; 2^{\lfloor p^2 n \rfloor}\Big(1 - \frac{2\varepsilon}{1 - e^{-p^2 n/4}}\Big), \qquad \text{and if } \sum_x f(x) = 1, \qquad |C| \;\geq\; 2^{\lfloor p^2 n \rfloor}\big[1 - \varepsilon - e^{-p^2 n/4}\big]_+^2.$$

Proof. Lemma 10.2 gives a balanced split $A|B$ with $|C| \geq \operatorname{rank} M_{A,B}(f)$. As in Corollary 10.4 there are at least $2n$ pairwise vertex-disjoint mixed cycles of length at most ℓ; pair them up. Lemma 10.13 gives $p_k \geq p$, and with $g := \lfloor p^2 n \rfloor$ the Chernoff bound gives $1 - \alpha_g \leq \Pr[\mathrm{Bin}(n, 2p^2) < p^2 n] \leq e^{-p^2 n/4}$. Apply Theorems 10.11 and 10.12. □

So with normalization, $|C| \geq (1 - \varepsilon - o(1))^2\, 2^{\lfloor p^2 n \rfloor} = e^{\Omega(N)}$ for every fixed $\varepsilon < 1$, and without it, $|C| \geq (2\eta - o(1))\, 2^{\lfloor p^2 n \rfloor}$ for $\varepsilon \leq \frac{1}{2} - \eta$ with $\eta > 0$ fixed. The normalized bound applies to every circuit whose output f has $\lambda := \sum_x f(x) \neq 0$ and $\frac{1}{2}\|P - f/\lambda\|_1 \leq \varepsilon$, since dividing by λ does not change the rank; the threshold $\frac{1}{2}$ is an artefact of comparing unnormalized outputs, where $f = 0$ is admissible. The exponent is smaller than the nonnegative one, $r_0 D^{-\ell}$, by a factor of about $D^{2-\ell} \ln 2/2$. For the cubic prisms of Example 8.4 ($D = \ell = 4$, $r_0 \geq N/11520$) it is about $N/(1.4 \cdot 10^8)$, against $N/(2.9 \cdot 10^6)$.

Theorem 10.12 needs only the rank of Q at one balanced split, not a circuit, so it also bounds sparse approximations.

Corollary 10.15 (sparse approximations)**.** *Let G, p and n be as in Corollary 10.14. Let Q be a linear combination of S monomials $\prod_{e \in U} x_e$, or of S characters $(-1)^{\sum_{e \in U} x_e}$, with $\sum_x Q(x) = 1$ and $\frac{1}{2}\|P - Q\|_1 \leq \varepsilon$. Then $S \geq 2^{\lfloor p^2 n \rfloor}\big[1 - \varepsilon - e^{-p^2 n/4}\big]_+^2$. In particular, for fixed $\varepsilon < 1$, an approximation of degree at most k needs $\sum_{j \leq k} \binom{M}{j} \geq e^{\Omega(N)}$, so $k = \Omega(N/\log N)$.*

Proof. Each monomial and each character is a product of a function of x_A and a function of x_B, so $\operatorname{rank} M_{A,B}(Q) \leq S$ at every split. Apply Theorem 10.12 as in the proof of Corollary 10.14. For the last claim use $\sum_{j \leq k} \binom{M}{j} \leq (eM/k)^k$ and $M \leq DN/2$. □

The same argument handles a sum of terms that are low-rank across different splits, provided the terms do not cancel each other much.

Proposition 10.16 (several splits)**.** *Let $Q = \sum_{u=1}^{k} Q_u$ be real functions with $\sum_x Q(x) = 1$ and $\frac{1}{2}\|P - Q\|_1 \leq \varepsilon$. For each u let $A_u|B_u$ be a split with $\operatorname{rank} M_{A_u,B_u}(Q_u) \leq r_u$, and $\mathcal{R}_u$ a family of disjoint permutation blocks of P whose cells are cells of M_{A_u,B_u}. Put $\Lambda := \sum_u \|Q_u\|_1$ and $\delta_* := \max_u \delta_{\mathcal{R}_u}(g)$. Then*

$$\big[1 - \varepsilon - k\delta_*\big]_+^2 \;\leq\; \frac{\Lambda}{2^g} \sum_{u=1}^{k} r_u.$$

Proof. Pointwise $Q^+ \le \sum_u Q_u^+$, and $\min(a, b+c) \le \min(a,b) + \min(a,c)$ for $a, b, c \ge 0$, so by Lemma 10.7, $1 - \varepsilon \le \sum_x \min(P, Q^+) \le \sum_u \sum_x \min(P, Q_u^+)$. As in the proof of Theorem 10.10(a), the u-th sum is at most $\delta_* + (r_u \|Q_u\|_1 / 2^g)^{1/2}$. Sum over u and apply the Cauchy–Schwarz inequality, $\sum_u (r_u \|Q_u\|_1)^{1/2} \le (\Lambda \sum_u r_u)^{1/2}$. □

On linked expanders every balanced split carries blocks with $\delta \le e^{-p^2 n/4}$, as in Corollary 10.14. So a sum $f = \sum_u f_u$ of k structured circuits of sizes s_u, on unrelated vtrees, with $\sum f = 1$ and $\frac{1}{2}\|P - f\|_1 \le \varepsilon$, has $\sum_u s_u \ge 2^{\lfloor p^2 n \rfloor} [1 - \varepsilon - k e^{-p^2 n/4}]_+^2 / \Lambda$: a short sum of structured circuits that do not cancel each other much is exponentially large. Theorem 10.37 removes the dependence on Λ at errors below $\frac{1}{2}$.

10.3 All expanders, and circuits that are not smooth

Corollary 10.14 needs short mixed cycles, so it misses expanders of large girth, and it needs smoothness through Lemma 10.2. Both needs go away if the cycles are replaced by single vertices, which is the idea behind Theorem 4.13. Theorem 10.10 only asks for disjoint permutation blocks, and a pendant vertex supplies them without any cycle.

A *gadget* is a connected set $U \subseteq V$ with two distinct marked edges $e_U \in A \cap \partial U$ and $f_U \in B \cap \partial U$, where ∂U is the set of edges with exactly one end in U. A family of gadgets is *admissible* if the sets are pairwise disjoint and no edge joins two of them, so their boundaries are disjoint. A spanning tree T has U *pendant via* b if $T \cap \partial U = \{b\}$; then $T \cap E(U)$ is a spanning tree of $G[U]$, and for every $b' \in \partial U$ the set $T - b + b'$ is again a spanning tree, since U hangs from the rest of T by one edge. Pair the gadgets of an admissible family $U_1, \dots, U_{2n}$ into n pairs (U_i, U_j). A pair is *good* in T if T shows pattern 1, with U_i pendant via e_{U_i} and U_j via f_{U_j}, or pattern 2, with U_i pendant via f_{U_i} and U_j via e_{U_j}. Let $\mathcal{G}(T)$ be the set of good pairs, $g(T) := |\mathcal{G}(T)|$, and $F(T)$ the set T minus the four marked edges of every good pair; a *block* is a set of trees with the same $(\mathcal{G}, F)$.

Lemma 10.17 (pendant blocks)**.** *A block with g good pairs contains exactly 2^g trees, one for each choice of pattern at each good pair. Let R be the set of cells (X, Y) of $M_{A,B}$ in which X consists of $F \cap A$ and one of e_{U_i}, e_{U_j} for each good pair, and Y of $F \cap B$ and one of f_{U_i}, f_{U_j} for each good pair. Then $M_{A,B}(P)$ restricted to R is τ^{-1} times a $2^g \times 2^g$ permutation matrix, and the sets R of distinct blocks are disjoint.*

Proof. Switching the pattern of a good pair is two swaps $b \to b'$ at pendant gadgets. Their boundaries are disjoint from each other and from those of the other gadgets, so the result is a tree with the same $\mathcal{G}$ and F; and the 2^g pattern choices determine the trees of the block.

A cell of R chooses (e_{U_i}, f_{U_j}), (e_{U_j}, f_{U_i}), (e_{U_i}, f_{U_i}) or (e_{U_j}, f_{U_j}) at each good pair. In the first two cases the pair shows pattern 1 or 2. In the last two, one gadget of the pair has no edge of its boundary in the cell, since F contains no other boundary edge of U_i or U_j, so the cell is not a tree. Hence the trees in R are the 2^g trees of the block, one in each row and each column.

Say that a pair *shows a cell pattern* in x if $(x \cap \partial U_i, x \cap \partial U_j)$ is $(\{e_{U_i}\}, \{f_{U_j}\})$, $(\{f_{U_i}\}, \{e_{U_j}\})$, $(\{e_{U_i}, f_{U_i}\}, \emptyset)$ or $(\emptyset, \{e_{U_j}, f_{U_j}\})$. In a cell of R, the good pairs of the block show a cell pattern, and every other pair shows its configuration in the trees of the block, which is not a cell pattern, because the pair is not good and a tree meets every ∂U. So $\mathcal{G}$, and then F, can be read off the cell. □

Lemma 10.18 (pendant probability)**.** *Let $U_1, \dots, U_m$ be an admissible family with union W, such that $H := G - W$ is connected with at least two vertices, and put $\kappa := 1/\lambda_2(L_H)$. Fix a*

gadget U, an edge $b \in \partial U$, and a configuration σ of T on the edges inside and on the boundaries of the other gadgets, with positive probability. Let $d_U := |\partial U|$. Then

$$\Pr[U \text{ pendant via } b \mid \sigma] \;\geq\; D^{-(|U|-1)}\,\frac{1}{d_U}\,(1+\kappa')^{-(d_U-1)},$$

with $\kappa' := \kappa$ if no vertex outside U has two neighbours in U, and $\kappa' := 2\kappa$ in general. In particular, for $U = \{w\}$ the bound is $\frac{1}{d}(1+\kappa)^{-(d-1)}$ with $d = \deg w$.

Proof. By Lemma 4.2, given σ the tree is uniform on K_σ, which is G with the present edges of σ contracted and the absent ones deleted. The edges inside and on the boundary of U are untouched, and each other gadget is merged into vertices of H or removed with its edges. The boundary edges of U end in H, because the family is admissible.

U is spanned. Fix a spanning tree S_0 of $G[U]$ and list its edges so that each joins the earlier ones to a new vertex v. Given that the earlier edges are present, contract them (Lemma 4.2); the next edge has an end v of degree at most D, and identifying all vertices other than v shows, by ((E2)) and ((E1)), that it is present with probability at least $1/D$. If all edges of S_0 are present, then $T \cap E(U) = S_0$. So $\Pr[T \cap E(U) = S_0 \mid \sigma] \geq D^{-(|U|-1)}$.

Pendant, given $T \cap E(U) = S$. For any spanning tree S of $G[U]$, condition on $T \cap E(U) = S$ as well. Contracting S turns U into a vertex u^* of degree d_U, and we must show that $T \cap \partial U = \{b\}$ with probability at least $\frac{1}{d_U}(1+\kappa')^{-(d_U-1)}$. Delete the edges $e_1, \dots, e_{d_U-1}$ of $\partial U \setminus \{b\}$ one at a time, and let $u_j \in V(H)$ be the end of e_j outside U. By Lemma 4.1, the j-th deletion succeeds with probability $1/(1+R_j)$, where R_j is the resistance between u_j and u^* after deleting $e_1, \dots, e_j$. The $m := d_U - j \geq 1$ remaining edges of ∂U end at vertices $y_1, \dots, y_m$ of H. As in the proof of Lemma 4.4, deleting the other gadgets and undoing the identifications only increases resistance ((E2)), and ((E6)) applied in H to $\chi := \mathbf{1}_{u_j} - \frac{1}{m}\sum_k \mathbf{1}_{y_k}$, followed by $\frac{1}{m}$ along each remaining edge, gives by ((E5)) $R_j \leq \kappa\|\chi\|^2 + \frac{1}{m}$. If no vertex outside U has two neighbours in U, the y_k are distinct and differ from u_j, so $\|\chi\|^2 = 1 + \frac{1}{m}$ and $1 + R_j \leq (1+\kappa)(1+\frac{1}{m})$. In general $\|\chi\|^2 \leq \|\chi\|_1\|\chi\|_\infty \leq 2$ and $1 + R_j \leq (1+2\kappa)(1+\frac{1}{m})$. The product of the $d_U - 1$ factors $1/(1+R_j)$ is at least $(1+\kappa')^{-(d_U-1)} \prod_{m=1}^{d_U-1} \frac{m}{m+1} = \frac{1}{d_U}(1+\kappa')^{-(d_U-1)}$. Average over S and multiply by the first part. □

Lemma 10.19 (good pendant pairs are frequent)**.** *Let p_U be the bound of Lemma 10.18 for the marked edges of U. Conditionally on any event of positive probability that is determined by the configurations of T on the gadgets of the other pairs and their boundaries, the pair (U_i, U_j) is good with probability at least $2p_{U_i}p_{U_j}$. Hence $g(T)$ stochastically dominates $\mathrm{Bin}(n, 2p^2)$ with $p := \min_U p_U$.*

Proof. Let Π be such an event. By Lemma 10.18 for U_i, conditioning on the configuration on all other gadgets, and the tower property, $\Pr[\text{pattern } 1, \Pi] \geq p_{U_i} \Pr[U_j \text{ pendant via } f_{U_j}, \Pi] \geq p_{U_i}p_{U_j} \Pr[\Pi]$, where the second step is the same argument for U_j. Pattern 2 is symmetric and disjoint from pattern 1. The domination follows by sequential coupling, as in Lemma 10.13. □

Theorem 10.20 (approximate rank on 2-connected expanders)**.** *Let G be a 2-connected simple graph with maximum degree at most D and $h(G) \geq h > 0$, and let $E = A \sqcup B$ with $|A|, |B| \geq m_0$. Put $h_1 := \min(h, 1)$, $h' := h_1/(D+1)$, $\Lambda := e(1 + 1/h_1)$, $q := (8D\Lambda^3)^{-(D+1)}$,*

$$p := \frac{1}{D}\Big(1 + \frac{1}{\mu(D, h')}\Big)^{-(D-1)}, \qquad a := \frac{qh_1}{D(D+1)(D+h_1)},$$

with μ as in Lemma 4.7. There are $n \ge \lfloor am_0/2 \rfloor$ pairs of single-vertex gadgets with the following property. Let Q be a real function with $\frac{1}{2}\|P-Q\|_1 \le \varepsilon$ and $r := \operatorname{rank} M_{A,B}(Q)$, and put $g := \lfloor p^2 n \rfloor$ and $\delta := e^{-p^2 n/4}$. If $n \ge 1$, then

$$r \;\ge\; 2^g\Big(1 - \frac{2\varepsilon}{1-\delta}\Big), \qquad \text{and if } \sum_x Q(x) = 1, \qquad r \;\ge\; 2^g\,[1-\varepsilon-\delta]_+^2.$$

Moreover $g \ge c_R m_0 - 1$ with $c_R := p^2 a/3$.

Proof. By Lemma 4.6 with h_1 in place of h, the set Z of mixed vertices has $|Z| \ge 2h_1 m_0/(D(D+h_1))$. The proof of Theorem 4.13 shows $\mathcal{S}(q) \le \frac{1}{2}$ for these h' and q, so Lemma 4.10 gives an independent set $W \subseteq Z$ with $|W| \ge q|Z|/(2(D+1)) \ge am_0$ such that $H = G - W$ has at least two vertices and $h(H) \ge h'$; by Lemma 4.7, $\lambda_2(L_H) \ge \mu(D,h')$. For $w \in W$ take the gadget $\{w\}$ with $e_w \in S_A(w)$ and $f_w \in S_B(w)$. The family is admissible because W is independent. Pair up $2n := 2\lfloor |W|/2 \rfloor$ of them. Lemma 10.18 gives $p_{\{w\}} \ge p$, since $\frac{1}{d}(1+\kappa)^{-(d-1)}$ decreases in $d \le D$. By Lemma 10.17 the blocks are disjoint permutation blocks with $n_R = 2^{g_b}$ and rank at most r, and every tree lies in one block; by Lemma 10.19 and the Chernoff bound, $\delta(g) \le \Pr[\mathrm{Bin}(n, 2p^2) < p^2 n] \le \delta$. Apply Theorem 10.10. For the last claim, $n \ge |W|/3$ if $|W| \ge 2$, so $g \ge p^2 am_0/3 - 1$; if $|W| \le 1$, then $am_0 \le 1$ and $c_R m_0 < 1$. □

Corollary 10.21 (structured circuits on 2-connected expanders)**.** *Let G be as in Theorem 10.20 and C a circuit with real weights computing f with $\frac{1}{2}\|P-f\|_1 \le \varepsilon$. Suppose that C is structured with respect to a vtree with blocks of at most $2M/3$ edges, or aligned with a vtree whose leaves are single edges, as defined before Lemma 10.28, in which case C need not be smooth. Then, with p, a, c_R as there, $m_0 := \lceil M/3 \rceil \ge N/3$, $n := \lfloor am_0/2 \rfloor$ and $g := \lfloor p^2 n \rfloor$, if $n \ge 1$,*

$$|C| \;\ge\; 2^g\Big(1 - \frac{2\varepsilon}{1-e^{-p^2 n/4}}\Big), \qquad \text{and if } \sum_x f(x) = 1, \qquad |C| \;\ge\; 2^g\,[1-\varepsilon-e^{-p^2 n/4}]_+^2.$$

So $|C| \ge 2^{\Omega(N)}$ for every fixed $\varepsilon < \frac{1}{2}$, and for every fixed $\varepsilon < 1$ if $\sum f = 1$, with constants depending only on D and h.

Proof. In the first case Lemma 10.2 gives a split with $M/3 < |A| \le 2M/3$ and rank $M_{A,B}(f) \le |C|$. In the second, descend from the root of the vtree to the child with the larger scope until the scope has at most $2M/3$ edges; the scope S of the node reached has $|S| \ge M/3$, and Lemma 10.28 gives $\operatorname{rank} M_{S,E\setminus S}(f) \le |C|$. In both cases both sides have at least m_0 edges, and $M \ge N$ because G has minimum degree at least 2. The bound is monotone in n, so Theorem 10.20 applies with $n = \lfloor am_0/2 \rfloor$ pairs. □

Compared with Corollary 10.14, the graph class grows from linked expanders to every 2-connected bounded-degree expander, the class of Theorem 4.13; this includes the Ramanujan graphs of Lubotzky, Phillips and Sarnak [LPS88], whose girth grows logarithmically. Smoothness is no longer needed. The exponent changes from $p^2 = D^{-2(\ell-1)}$ to $p^2 \approx D^{-2}(1+\kappa)^{-2(D-1)}$, and is again tiny. The same argument applies to every model whose functions have small rank across some balanced split, for instance sums $\sum_j f_{C_j}^2$ of squares of circuits structured with respect to one vtree, as in Theorem 10.22, whose rank at the split of Lemma 10.2 is at most $\sum_j |C_j|^2$.

10.4 Squares and positive semidefinite forms

Squaring a circuit with real weights is another way to use subtraction while keeping a nonnegative model [LSM+24], and sums of squares extend it [LMV25]. Squared circuits with subtraction can be exponentially more succinct than structured nonnegative circuits for some distributions [LSM+24]. For the uniform spanning tree, the structured squared models considered here satisfy the same exponential lower-bound expression as nonnegative circuits; this does not compare the optimal sizes of the two models.

Theorem 10.22 (normalized squares of structured amplitudes)**.** *Let $d = (d_1, \dots, d_K)$ be the outputs of a circuit with real weights of size s, structured with respect to a vtree V with blocks of at most $2M/3$ edges, and let $Q := \|d(x)\|^2/Z$ with $Z := \sum_x \|d(x)\|^2 > 0$. Let $A|B$ be the balanced split of Lemma 10.2, and β_* the largest $\beta(P; A'|B')$ over balanced splits. Then*

$$1 - \mathrm{TV}(P, Q) \;\le\; \mathrm{BC}(P, Q) \;\le\; \beta(P; A|B)\sqrt{s} \;\le\; \beta_*\sqrt{s}.$$

The same holds for $Q = d(x)^\top K d(x)/Z$ with K positive semidefinite. If $Q = \sum_k c_k^2/Z$, where c_k is computed by a structured circuit of size s_k with respect to its own vtree V_k, then $1 - \mathrm{TV}(P, Q) \le \beta_*\sqrt{\sum_k s_k}$.

Proof. Let $D_m := M_{A,B}(d_m)$, and stack them as $\tilde{D} := [D_1|\cdots|D_K]$. By Lemma 10.2 all columns of $\tilde{D}$ lie in the span of the vectors $(f_g(x_A))_{x_A}$, $g \in G_v$, so $\operatorname{rank} \tilde{D} \le s$. Let $S_m(x) := d_m(x)/\|d(x)\|$, with $S_m(x) := 0$ if $d(x) = 0$, let $W := \sqrt{M_{A,B}(P)}^{\circ}$ be the entrywise square root, and let $\tilde{W} := [W \circ S_1|\cdots|W \circ S_K]$ with $\circ$ the entrywise product. Since $W \ge 0$ and $\sum_m S_m^2 \le 1$ entrywise, for vectors y and $z = (z_m)$,

$$|y^\top \tilde{W} z| \le \sum_{a,b} W_{ab}|y_a|\Big(\sum_m z_{m,b}^2\Big)^{1/2} \le \|W\|_{\mathrm{op}}\|y\|\|z\|,$$

so $\|\tilde{W}\|_{\mathrm{op}} \le \|W\|_{\mathrm{op}} = \beta(P; A|B)$ by Lemma 3.1. Hence

$$\sum_x \sqrt{P(x)}\,\|d(x)\| = \langle \tilde{W}, \tilde{D}\rangle \le \|\tilde{W}\|_{\mathrm{op}}\|\tilde{D}\|_* \le \beta(P; A|B)\sqrt{\operatorname{rank}\tilde{D}}\,\|\tilde{D}\|_F \le \beta(P; A|B)\sqrt{s}\sqrt{Z},$$

and dividing by $\sqrt{Z}$ gives the bound on $\mathrm{BC}(P, Q)$. For $K = LL^\top$, replace d by $L^\top d$: the new stacked matrix is $\tilde{D}$ times a matrix, so its rank is still at most $|G_v| \le s$. For several vtrees, $\|(c_k(x))_k\| \le \sum_k |c_k(x)|$, the case $K = 1$ applied to each c_k at its own balanced split gives $\sum_x \sqrt{P(x)}|c_k(x)| \le \beta_*\sqrt{s_k}\|c_k\|_2$, and the Cauchy–Schwarz inequality gives $\sum_k \sqrt{s_k}\|c_k\|_2 \le (\sum_k s_k)^{1/2} Z^{1/2}$. The first inequality of the theorem is $\min(a, b) \le \sqrt{ab}$. □

On the graphs of Theorems 1.1 and 8.2 this is the nonnegative bound: $s \ge (1-\varepsilon)^2 e^{cN}$ and $s \ge (1-\varepsilon)^2(1 - D^{-\ell})^{-r_0}$ respectively, where $\varepsilon = \mathrm{TV}(P, Q) < 1$. It is a statement about the lower-bound expression, not a claim that optimal sizes agree. The theorem covers a single squared structured circuit, as in [LSM+24], explicit sums of squares of structured circuits charged by their total size, and positive semidefinite forms of the outputs of one structured circuit. It does not cover models with nested positive semidefinite units or with exponentially many implicit squares; we have not checked every class of [LMV25] against it.

Inner decomposability is essential. Let $L_r(x)$ be the Laplacian of the subgraph $\{e : x_e = 1\}$ with the row and column of a fixed vertex deleted, and let $H(x) := \mathbf{1}[|x| = N - 1]$. Both have

arithmetic circuits with subtraction of polynomial size, $\det L_r$ by the division-free algorithm of Berkowitz [Ber84] and H by the dynamic program $c_{j,k} = c_{j-1,k}(1 - x_j) + c_{j-1,k-1}x_j$ over the edges. By the matrix-tree theorem, $\det L_r(x)$ counts the spanning trees contained in x, which for $|x| = N - 1$ is $\mathbf{1}[x \text{ is a tree}]$. So $d(x) := \det L_r(x)\, H(x)/\sqrt{\tau}$ satisfies $d(x)^2 = P(x)$ exactly. The amplitude d is small but not decomposable.

10.5 Unstructured circuits

Without a common vtree there is no single split at which to measure rank, and the overlap argument of Lemma 3.8 needs every product term to be nonnegative. We do not know whether small unstructured decomposable circuits with real weights can approximate $\mathrm{UST}(G)$ on expanders. For exact representation the question is a known kind of open problem. A circuit is *syntactically multilinear* if the two children of every product node depend on disjoint sets of variables [RY08]. Let $\mathrm{ST}_G(z) := \sum_T \prod_{e \in T} z_e$ be the spanning-tree polynomial.

Proposition 10.23 (unstructured exact representation)**.** *Let G be a connected graph with M edges and no loops.*

(a) If a smooth decomposable circuit with real weights, indicator leaves and size s computes $\mathrm{UST}(G)$ on $\{0,1\}^E$, then ST_G has a syntactically multilinear arithmetic circuit with binary gates and real constants of size $O(s)$.

(b) If ST_G has a syntactically multilinear arithmetic circuit of size s, with binary gates and real constants, then some smooth decomposable circuit with real weights and indicator leaves of size $O(sM)$ computes $\mathrm{UST}(G)$.

Proof. Write y^0_e and y^1_e for the leaves $[x_e = 0]$ and $[x_e = 1]$, and read the circuit as a formal polynomial in these symbols. Call a polynomial in the symbols of a set $X \subseteq E$ *X-homogeneous* if each of its monomials contains exactly one of y^0_e, y^1_e for every $e \in X$. Such a polynomial is $\sum_{S \subseteq X} c_S \prod_{e \in S} y^1_e \prod_{e \in X \setminus S} y^0_e$, and it is determined both by its values at the indicator points, where it equals c_x, and by its dehomogenization $y^0 = 1$, $y^1 = z$, which is $\sum_S c_S z^S$.

(a) By induction, a node of a smooth decomposable circuit with indicator leaves computes an X-homogeneous polynomial, with X its scope: sums combine polynomials with the same scope, and products multiply polynomials in disjoint sets of symbols. $\mathrm{UST}(G)$ depends on every edge, since G has no loops, so the root has scope E, and its polynomial is $\sum_S \mathrm{UST}(G)(S) \prod_{e \in S} y^1_e \prod_{e \notin S} y^0_e$. Setting $y^0_e = 1$ and $y^1_e = z_e$, and writing each weighted sum with binary gates, gives a syntactically multilinear circuit for ST_G/τ of size $O(s)$; multiply by τ.

(b) Replace each input z_e by y^1_e. For each node g let X_g be the set of variables it depends on. At every sum node g, multiply each child c by $\prod_{e \in X_g \setminus X_c} y^0_e$, and multiply the output by $\prod_{e \in E \setminus X_{\mathrm{root}}} y^0_e$; this costs $O(M)$ nodes per wire. Constant inputs become leaves with empty scope. Products already have disjoint scopes, so the result is smooth and decomposable, every node computes an X_g-homogeneous polynomial, and its dehomogenization is unchanged, because we multiplied only by symbols y^0. At the root, the polynomial is therefore $\sum_S \mathbf{1}[S \text{ is a tree}] \prod_{e \in S} y^1_e \prod_{e \notin S} y^0_e$, which is $\tau\, \mathrm{UST}(G)$ at the indicator points; divide by τ. □

So a superpolynomial lower bound for exact unstructured representation on some expander family would be a superpolynomial lower bound for syntactically multilinear circuits computing an explicit polynomial. We know of none for general circuits; the strongest bound we know of for an explicit polynomial in n variables is $\Omega(n^2/\log^2 n)$, due to Alon, Kumar and Volk [AKV18]. For

restricted unstructured classes, the lower-bound methods for the determinant and the permanent do transfer.

10.6 Restricted unstructured classes

In this subsection a *signed decomposable circuit* is a circuit with real weights whose leaves are constants or functions of one variable, whose product nodes have children with pairwise disjoint scopes, and which need not be smooth. Its *product-depth* is the largest number of product nodes on a path from a leaf to the output, and it is a *formula* if every node has at most one parent. A function f on $\{0,1\}^E$ is *support-exact* for G if $f(x) \neq 0$ exactly when x is a spanning tree. Every nonzero multiple of $\mathrm{UST}(G)$ is support-exact, and so is every edge-weighted spanning-tree measure, but a total-variation approximation usually is not. A signed decomposable circuit computes a multilinear polynomial, which agrees with its function on $\{0,1\}^E$, and for disjoint sets Y, Z of variables the value matrix and the coefficient matrix of a function of $x_{Y\cup Z}$ differ by invertible transforms on each side, so they have the same rank. Fixing some variables to 0 or 1 keeps a circuit in each of the classes below, without increasing size or product-depth.

Lemma 10.24 (cycle restrictions)**.** *Let $C_1, \dots, C_m$ be pairwise vertex-disjoint cycles of a connected graph G, and let $e_i \neq f_i$ be edges of C_i. There is an assignment ρ of values in $\{0,1\}$ to the edges other than $e_1, f_1, \dots, e_m, f_m$ such that an edge set agreeing with ρ is a spanning tree exactly when it contains exactly one of e_i, f_i for every i. So if f is support-exact for G and $f|_\rho$ is the function of (x_Y, x_Z) with $Y = \{e_i\}$ and $Z = \{f_i\}$, then $M_{Y,Z}(f|_\rho)$ has exactly one nonzero entry in each row and each column, and rank 2^m.*

Proof. The paths $C_i - f_i$ are vertex-disjoint, so their union is a forest; extend it to a spanning tree T. Then $f_i \notin T$, since it would close C_i. Let $K := T \setminus \{e_1, \dots, e_m\}$, and let ρ be 1 on K and 0 on the other edges outside $\{e_i, f_i\}$. The graph $H := K \cup \{e_i, f_i\} = T \cup \{f_i\}$ has $N - 1 + m$ edges, and its cycle space is spanned by the fundamental cycles C_i of the f_i. Since the C_i are vertex-disjoint, every cycle of H is one of them, so the spanning trees of H are obtained by deleting exactly one edge from each C_i. Those that contain $K \supseteq C_i \setminus \{e_i, f_i\}$ delete e_i or f_i. □

Lemma 10.25 (minors)**.** *Let H be a connected minor of a connected graph G. There are an injection $\iota : E(H) \to E(G)$ and an assignment ρ of values in $\{0,1\}$ to $E(G) \setminus \iota(E(H))$ such that, for $x \subseteq E(H)$, the set $\iota(x) \cup \rho^{-1}(1)$ is a spanning tree of G exactly when x is a spanning tree of H. So if f is support-exact for G, then $f|_\rho$, read through ι, is support-exact for H.*

Proof. Take a minor model with disjoint connected branch sets B_u, $u \in V(H)$, and enlarge them to a partition of $V(G)$ into connected sets, adding vertices adjacent to a branch set one at a time. Let ρ be 1 on a spanning tree T_u of each $G[B_u]$, choose for each edge uv of H one edge $\iota(uv)$ of G between B_u and B_v, and let ρ be 0 on all other edges. Contracting the trees T_u turns the graph with edge set $\bigcup_u T_u \cup \iota(E(H))$ into H, so, as in the proof of Lemma 4.2, $\bigcup_u T_u \cup \iota(x)$ is a spanning tree exactly when x is a spanning tree of H. □

Lemma 10.25 is about the indicator. For the polynomial ST_G, setting a variable to 1 does not contract the edge: on a triangle, $\mathrm{ST} = ab + ac + bc$ becomes $b + c + bc$ at $a = 1$. We use the complete bipartite graph $K_{t,t}$ as the minor, with rows r and columns c as its two sides and x_{rc} for its edge variables. Two edges r_1c_1 and r_2c_2 with $r_1 \neq r_2$ and $c_1 \neq c_2$ lie on the 4-cycle $r_1c_1r_2c_2$, and two edges rc and rc' lie on the 4-cycle $rcr'c'$ for every other row r'. Raz [Raz09], and Raz and Yehudayoff [RY09], prove their lower bounds for the permanent and the determinant

of a $t \times t$ matrix by random restrictions that send m disjoint pairs of variables, each pair in a common row or column and different pairs in different rows and columns, to variables y_i, z_i, and every other variable to 0 or 1, so that the restricted polynomial is $\prod_i (y_i \pm z_i)$ and has full rank 2^m across $Y|Z$. They then show that a small formula of the relevant kind, after a suitable restriction, has rank less than 2^m. Their constants kill the indicator of spanning trees, since they leave $K_{t,t}$ disconnected. The point of the next proofs is that the part of their argument that bounds the rank depends only on which variables become y_i and z_i, not on the constants, so the constants can be replaced by those of Lemma 10.24.

Theorem 10.26 (bounded product-depth)**.** *Let G be a connected graph with a $K_{t,t}$ minor, and let C be a signed decomposable circuit of product-depth d that computes a function support-exact for G. If $d = o(\log t / \log\log t)$, then $|C| \geq 2^{t^{\Omega(1/d)}}$, with an absolute constant in the Ω.*

Proof. By Lemma 10.25, fixing variables gives a circuit C_0 of size at most $|C|$ and product-depth at most d on the t^2 variables of $K_{t,t}$ whose function f_0 is support-exact for $K_{t,t}$. Each variable occurs in C_0, since flipping an edge of a spanning tree changes f_0 from nonzero to zero. Writing each leaf $a + bx_e$ with one sum node does not change the product-depth, so C_0 is a multilinear arithmetic circuit of product-depth d and size $O(|C|)$, and by [RY09, Lemma 2.1] there is a syntactically multilinear formula Ψ in d-normal form of size at most $(d+1)^2 O(|C|)^{2d+1}$ computing f_0.

Raz and Yehudayoff draw a restriction from a distribution that first chooses the set W of surviving variables with its pairs, each pair in a 2×2 block of its own and different blocks in different rows and columns, then the orientation of each pair (which variable becomes y_i and which z_i), and then the constants. Their Proposition 4.3 [RY09] shows that W is good, a condition on W and the syntactic scopes of Ψ only, with high probability. Their Proposition 4.4 shows that, for good W and with high probability over the orientations, every k-central path of Ψ passes through a product gate whose rank is bounded through the imbalances between y- and z-variables in groups of its children, with $k = t^{\Omega(1/d)}$, provided $|\Psi| < 2^{t^{\varepsilon/d}}$ for a small absolute $\varepsilon > 0$. That rank bound uses only the variable sets of the children, the dimension bound and the multiplicativity of rank, so it holds for every assignment of constants to the variables outside W. (Their weakness is a property of the computed polynomials, and in general it does change with the constants; what is uniform is this particular certificate.) So if $|\Psi| < 2^{t^{\varepsilon/d}}$, there is a choice of W, pairs and orientations such that the restricted formula is k-weak for every assignment of constants outside W. Fix it, and choose the constants by Lemma 10.24, applied to the vertex-disjoint 4-cycles through the pairs: the pairs lie in different rows and columns, and each pair spans a 2×2 block of its own, whose four edges form such a cycle. By [RY09, Theorem 3.1], the restricted polynomial has rank at most $|\Psi| 2^{m-k/2} < 2^m$ across $Y|Z$, while by Lemma 10.24 it has rank 2^m. Hence $|\Psi| \geq 2^{t^{\varepsilon/d}}$, and $|C| \geq (|\Psi|/(d+1)^2)^{1/(2d+1)}/O(1) = 2^{t^{\Omega(1/d)}}$ in the stated range of d. □

Theorem 10.27 (formulas)**.** *Let G be a connected graph with a $K_{t,t}$ minor. A signed decomposable formula that computes a function support-exact for G has size $t^{\Omega(\log t)}$.*

Proof. As for Theorem 10.26, with Raz's argument [Raz09] in place of that of Raz and Yehudayoff. After Lemma 10.25, the formula is a syntactically multilinear formula of size $O(|C|)$ in which every variable of $K_{t,t}$ occurs. Raz's restriction pairs $m = \lceil t^{1/3} \rceil$ variables, each pair in a common row or column and different pairs in different rows and columns, and shows that a formula of size $t^{\varepsilon \log t}$ is, with positive probability over the pairs and their orientations, weak in a sense defined through imbalances between y- and z-variables only, and that weak formulas have rank below 2^m. Neither step depends on the constants. Complete each pair rc, rc' to the 4-cycle $rcr'c'$ with a row r', or a column, used by no pair and by no other cycle; there are enough, since the pairs use at most

$2m \le t - m$ rows and $2m$ columns for large t. Lemma 10.24 then gives constants for which the restricted function has rank 2^m. □

For sums of structured circuits we need a split that is balanced for all the vtrees at once, and a rank bound that does not need smoothness. A circuit with real weights is *aligned* with a vtree V whose leaves are single edges if its leaves are constants or functions of one variable, and every product node has exactly two children, whose scopes lie in $\mathrm{Var}(v_1)$ and $\mathrm{Var}(v_2)$ for the two children v_1, v_2 of some node of V. Sums need not be smooth. Circuits structured as in Definition 10.1, with singleton blocks, are aligned.

Lemma 10.28 (frontier rank)**.** *Let C be aligned with V, let v be a node of V and $S := \mathrm{Var}(v)$. Then* $\operatorname{rank} M_{S,E\setminus S}(f_C) \le |C|$.

Proof. Call a node g inside, outside or straddling as its scope is contained in S, is disjoint from S, or neither; constants are outside. If the output is not straddling, the rank is at most 1. Let I be the set of inside nodes with a straddling parent. We show by induction in topological order that every straddling node g has $f_g = \sum_{u\in I} f_u(x_S)h_{u,g}(x_{E\setminus S}) + h_{0,g}(x_{E\setminus S})$. Leaves do not straddle. For a sum node, add the expansions of the straddling children, the inside children (which lie in I) and the outside children. For a product node, the scopes of its children lie in $\mathrm{Var}(v_1)$ and $\mathrm{Var}(v_2)$ for the children v_1, v_2 of some node u of V. The sets $\mathrm{Var}(\cdot)$ are laminar, $\mathrm{Var}(u)$ meets S and is not contained in it, so v lies strictly below u, say $S \subseteq \mathrm{Var}(v_1)$. Then the second child is outside, and the first is inside, outside or straddling; multiply. So the rank is at most $|I|+1$. Each node of I has its own wire to a straddling parent, and the output is a further node, so $|I| + 1 \le |C|$. □

Lemma 10.29 (splits balanced for several vtrees)**.** *Let $V_1, \dots, V_r$ be vtrees whose leaves are the elements of a set E, and let $E_0 \subseteq E$ with $|E_0| \ge 2$. There are nodes v_k of V_k with sets $S_k := \mathrm{Var}(v_k)$, sets $S'_k \in \{S_k, E \setminus S_k\}$, and disjoint $P, Q \subseteq E_0$ with $|P| = |Q| \ge 3^{-r}|E_0|$, such that $P \subseteq S'_k$ and $Q \cap S'_k = \emptyset$ for every k.*

Proof. For $U \subseteq E_0$ with $|U| \ge 2$ and a vtree V, walk down from the root of V to the child with more elements of U while the current set contains more than $2|U|/3$ of them. The first node v where this fails has $|U|/3 < |\mathrm{Var}(v) \cap U| \le 2|U|/3$, since its parent had more than $2|U|/3$ and v has at least half of them. Apply this to $U = E_0$ in V_1, and let P and Q be equal-sized subsets of $S_1 \cap E_0$ and $E_0 \setminus S_1$, of size $n_1 \ge |E_0|/3$. Given P, Q of size $n \ge 1$, apply it to $U = P \cup Q$ in V_{k+1} with $S := \mathrm{Var}(v)$, and put $a = |P \cap S|$, $b = |P \setminus S|$, $c = |Q \cap S|$, $d = |Q \setminus S|$. Then $a + b = c + d = n$ and $2n/3 < a + c \le 4n/3$. If $\min(a, d)$ and $\min(b, c)$ were both below $n/3$, then one of $a + b$, $c + d$ would be below $2n/3$, or $a + c < 2n/3$, or $a + c > 4n/3$. So one of the pairs $(P \cap S, Q \setminus S)$, with $S' = S$, and $(P \setminus S, Q \cap S)$, with $S' = E \setminus S$, has both sizes at least $n/3$; trim it to equal size $n' \ge n/3$, which is at least 1. The relations for earlier k survive, since P and Q only shrink. □

Fixed balanced splits do not work in place of this adaptive choice: for a partition of E_0 into four equal parts $P_1, \dots, P_4$, the three balanced sets $P_1 \cup P_j$, $j = 2, 3, 4$, admit no large P and Q separated by all three.

Theorem 10.30 (sums of structured circuits)**.** *Let G be a connected graph with a $K_{t,t}$ minor, and let $C_1, \dots, C_r$ be circuits with real weights, C_k aligned with its own vtree V_k whose leaves are the edges of G. If $f_{C_1} + \cdots + f_{C_r}$ is support-exact for G, then*

$$\sum_{k=1}^{r} |C_k| \;\ge\; 2^m, \qquad m \ge t\,3^{-r}/4 - 1.$$

Proof. Restrict by Lemma 10.25 to the edge set E_0 of $K_{t,t}$, viewed inside $E(G)$, and apply Lemma 10.29 to E_0, which gives P and Q of size at least $t^2 3^{-r}$. Choose edges $e_1 \in P$ and $f_1 \in Q$ with no common endpoint, then $e_2 \in P$ and $f_2 \in Q$ avoiding the endpoints of e_1, f_1 and of each other, and so on. After i choices at most $4it$ edges of each set meet a used vertex, so the choice continues while $|Q| > (4i+2)t$, which gives $m \geq (t3^{-r} - 2)/4 \geq t3^{-r}/4 - 1$ pairs. The 4-cycles through e_i and f_i are vertex-disjoint, so Lemma 10.24 gives a restriction ρ under which $f|_\rho$ has rank 2^m across $Y = \{e_i\}$ and $Z = \{f_i\}$. For each k, $Y \subseteq S'_k$ and $Z \cap S'_k = \emptyset$, so $M_{Y,Z}(f_{C_k}|_\rho)$ is a submatrix of $M_{S_k, E\setminus S_k}(f_{C_k})$ or of its transpose, and has rank at most $|C_k|$ by Lemma 10.28. Rank is subadditive. □

To apply these results we need large complete minors in expanders. Krivelevich and Nenadov [KN21] study complete minors in graphs without sparse cuts. The form we use is [Kri19, Theorem 8.4], a joint result with Nenadov whose proof is given in the survey: for every $\alpha > 0$ there are $c, n_0 > 0$ such that every graph on $N \geq n_0$ vertices with maximum degree D and edge expansion at least αD has a complete minor of order $c\sqrt{N}$. With $\alpha := h/D$, deleting edges of the complete minor gives the following.

Corollary 10.31 (unstructured exact representation on expanders)**.** *Fix D and $h > 0$. There are $c', N_0 > 0$ such that every connected graph G with $N \geq N_0$ vertices, maximum degree at most D and $h(G) \geq h$ contains $K_{t,t}$ as a minor with $t \geq c'\sqrt{N}$. Consequently every signed decomposable circuit computing a function support-exact for G, for instance a nonzero multiple of* $\mathrm{UST}(G)$*, has size*

(a) $2^{N^{\Omega(1/d)}}$ *if it has product-depth $d = o(\log N / \log\log N)$;*

(b) $N^{\Omega(\log N)}$ *if it is a formula;*

(c) at least $2^{c'3^{-r}\sqrt{N}/4-1}$ *in total if it is a sum of r circuits, each aligned with its own vtree.*

So depth-3 and depth-4 signed decomposable circuits, of the forms $\Sigma\Pi\Sigma$ and $\Sigma\Pi\Sigma\Pi$, need size $2^{N^{\Omega(1)}}$, and a sum of structured circuits of total size 2^{N^δ} with $\delta < \frac{1}{2}$ needs $r \geq (\frac{1}{2} - \delta - o(1)) \log_3 N$ summands. The same proofs apply to syntactically multilinear circuits computing ST_G, or ST_G times a nonzero constant, with one change: instead of Lemma 10.25, set the variables of the trees T_u in its proof to 1 and the other edges outside $\iota(E(K_{t,t}))$ to 0, and apply the restriction of Lemma 10.24 in G to the cycles obtained from the 4-cycles by joining their edges through the trees T_u. In its proof, choose T to consist of the trees T_u and edges of $\iota(E(K_{t,t}))$, which is possible since contracting the T_u turns the paths $C_i - f_i$ into vertex-disjoint paths of $K_{t,t}$; then the restriction agrees with the first one. Each such cycle C_i contributes the factor $x_{e_i} + x_{f_i} + (|C_i| - 2)x_{e_i}x_{f_i}$, whose 2×2 coefficient matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & |C_i|-2 \end{smallmatrix}\right)$ has rank 2, so the restricted polynomial again has rank 2^m.

Sums of structured circuits without minors. Corollary 10.31(c) loses a square root by passing to a complete minor, and like every restriction argument it says nothing about approximation. Both losses go away for sums of structured circuits if the cycles of Lemma 10.24 are replaced by the pendant gadgets of Section 10.3. The gadgets must now have their marked edges in the sets P and Q of Lemma 10.29, which can be far apart, so single vertices no longer suffice; we build larger gadgets around short paths and keep them far apart so that deleting them preserves expansion. The constants this gives are astronomically small.

Lemma 10.32 (robust short connectors)**.** *Let G have N vertices, maximum degree at most D and $h(G) \geq h > 0$, let $\beta \in (0,1]$, let $X, X' \subseteq V$ with $|X|, |X'| \geq \beta N$, and let $Z \subseteq V$ with $|Z| \leq h\beta N/(4D)$. Put $t_* := \lfloor \ln(1/\beta)/\ln(1+h/(2D)) \rfloor + 1$. Then $G - Z$ contains a path with at most $2t_* + 1$ vertices from $X \setminus Z$ to $X' \setminus Z$.*

Proof. Put $A_0 := X \setminus Z$ and $A_{t+1} := A_t \cup N_{G-Z}(A_t)$. Since $h \leq D$, $|A_0| \geq \beta N(1 - h/(4D)) \geq 3\beta N/4$. If $|A_t| \leq N/2$, then at least $|\partial A_t|/D \geq h|A_t|/D$ vertices outside A_t have a neighbour in A_t, and at most $|Z| \leq h|A_0|/(2D) \leq h|A_t|/(2D)$ of them lie in Z, so $|A_{t+1}| \geq (1 + h/(2D))|A_t|$. As $(1 + h/(2D))^{t_*} > 1/\beta$, this gives $|A_{t_*}| > N/2$; the sets increase, so this also holds if some earlier A_t exceeds $N/2$. The same holds for the sets A'_t grown from $X' \setminus Z$, so A_{t_*} and A'_{t_*} meet, at a vertex within distance t_* of both $X \setminus Z$ and $X' \setminus Z$ in $G - Z$. □

For a nonempty connected $U \subsetneq V$ of a connected graph, let X be the vertex set of a largest component of $G - U$ and $B(U) := V \setminus X$, the *blob* of U.

Lemma 10.33 (blobs)**.** *Let G have maximum degree at most D and $h(G) \geq h > 0$, and let $U \subsetneq V$ be nonempty and connected. Then $G[B(U)]$ is connected, $|B(U) \setminus U| \leq |\partial U|/h \leq D|U|/h$, and every vertex of $B(U)$ is within distance $\lfloor D|U|/h \rfloor$ of U.*

Proof. Every component of $G - U$ has a neighbour in U, since G is connected; this gives the first claim. A component $C \neq X$ of $G - U$ has $|C| \leq |X|$, so $|C| \leq N/2$ and $h|C| \leq |\partial C|$, and every edge of ∂C lies in ∂U. Summing over these components gives the second claim. A vertex of such a C is joined inside C to a neighbour of U by a path with at most $|C| - 1$ edges, so its distance to U is at most $|C| \leq D|U|/h$. □

Lemma 10.34 (bond gadgets)**.** *Let G be simple and 2-connected, $U \subsetneq V$ nonempty and connected, and $e \neq f$ edges each with an end in $B(U)$. Then there is $Y \subseteq B(U)$ with $G[Y]$ and $G - Y$ connected and $e, f \in \partial Y$.*

Proof. Let $K := G/X$ be the multigraph obtained by contracting X to a vertex γ and deleting loops. It is connected and loopless, e and f are edges of K, and $K - \gamma = G[B(U)]$. We find a partition $V(K) = Y \sqcup Z$ with $\gamma \in Z$, $K[Y]$ and $K[Z]$ connected, such that e and f both join Y to Z. This suffices: $Y \subseteq B(U)$, $G[Y] = K[Y]$, and $G - Y$ is connected because $K[Z]$ and $G[X]$ are.

If $|B(U)| = 1$, then $Y := B(U)$ works. Otherwise $|V(K)| \geq 3$ and $K - v$ is connected for every vertex v: $K - \gamma$ is connected by Lemma 10.33, and for $v \in B(U)$, $K - v$ arises from the connected graph $G - v$ by contracting the connected set X. We first find disjoint connected seeds Y_0, Z_0 such that e and f join Y_0 to Z_0. If e and f have the same ends a, a', take $\{a\}$ and $\{a'\}$. If they share exactly one end, $e = xy$ and $f = xz$, take $\{x\}$ and the vertex set of a y–z path in $K - x$. If $e = aa'$ and $f = bb'$ have four distinct ends, no single vertex v meets every path from $\{a, a'\}$ to $\{b, b'\}$, since $K - v$ is connected and keeps a vertex of each pair; by Menger's theorem [Die17, Thm. 3.3.1] there are two disjoint such paths, one starting at a and one at a', and we take their vertex sets. Then, as long as some vertex is unassigned, add an unassigned vertex with a neighbour in one of the two sets to that set. Both sets stay connected. Finally swap the names so that $\gamma \in Z$. □

This is the classical fact that in a 2-connected graph any two edges lie on a common bond [Oxl11, Ch. 4], applied to G/X; the proof above is the one the checks run.

Lemma 10.35 (deleting separated gadgets)**.** *Let G have N vertices, maximum degree at most D and $h(G) \geq h > 0$. Let $Y_1, \ldots, Y_K \subseteq V$ be nonempty with $G - Y_i$ connected, $|Y_i| \leq L$ and $|\partial Y_i| \leq D'$, where $N > 2L$. Suppose $\operatorname{dist}(Y_i, Y_j) \geq \sigma + 1$ for $i \neq j$, where $\sigma \geq 2$ and*

$\lfloor \sigma/2 \rfloor \geq (D'+1)/h$. *Put* $H := G - \bigcup_i Y_i$ *and* $h' := h/(D'+1)$. *Then every* $S \subseteq V(H)$ *with* $0 < |S| \leq |V(H)|/2$ *has* $|\partial_H S| \geq h'|S|$. *If* G *is 2-connected, then* H *has at least two vertices, so* $h(H) \geq h'$, H *is connected, and* $\lambda_2(L_H) \geq \mu(D, h')$ *by Lemma 4.7. The same holds for every subfamily of the* Y_i.

Proof. The H-boundaries of the components of $H[S]$ partition $\partial_H S$, so we may assume that S is connected. Then $|S| \leq N/2$, so $|\partial_G S| \geq h|S|$, and the edges of $\partial_G S \setminus \partial_H S$ go to the sets Y_i. Let j be the number of sets Y_i joined to S by an edge. If $j = 0$, then $\partial_H S = \partial_G S$. If $j = 1$, with set Y, then $|\partial_H S| \geq h|S| - D'$. Also $|\partial_H S| \geq 1$: since $|S| \leq N/2$ and $|Y| < N/2$ there is a vertex outside $S \cup Y$, and since $G - Y$ is connected some edge of $G - Y$ leaves S; its other end is in no other Y_i, so it is in $V(H) \setminus S$. If $|S| \leq 1/h'$ the bound $1 \geq h'|S|$ suffices; otherwise $(h - h')|S| > (h - h')/h' = D'$, so $h|S| - D' > h'|S|$. If $j \geq 2$, pick $u_1, \dots, u_j \in S$ with u_i adjacent to the i-th of these sets; then $\mathrm{dist}_G(u_i, u_{i'}) \geq \sigma - 1$. With $\rho := \lfloor \sigma/2 \rfloor - 1$, the balls of radius ρ around the u_i in $G[S]$ are disjoint, and each has at least $\rho + 1$ vertices, since $G[S]$ is connected and contains a vertex at distance more than ρ from u_i. So $|S| \geq j\lfloor \sigma/2 \rfloor \geq j(D'+1)/h$, the number of edges from S to the Y_i is at most $jD' \leq (h - h')|S|$, and $|\partial_H S| \geq h'|S|$.

For the second claim, if $K \geq 1$ let $N(Y_1)$ be the set of vertices outside Y_1 with a neighbour in Y_1. It lies in $V(H)$, since the other sets are at distance at least 3, and it is nonempty. If it were a single vertex and $V \setminus (Y_1 \cup N(Y_1))$ were nonempty, that vertex would be a cut vertex. Otherwise $|N(Y_1)| \geq 2$ or $N(Y_1) = V \setminus Y_1$, which has $N - L \geq 2$ vertices. Every hypothesis is inherited by subfamilies. □

Theorem 10.36 (packing pendant gadgets)**.** *Let* G *be a simple 2-connected graph with* N *vertices,* M *edges, maximum degree at most* D *and* $h(G) \geq h > 0$, *and let* $\delta \in (0, \frac{1}{2}]$. *Put*

$$\beta := \frac{2\delta}{D}, \quad t_* := \Big\lfloor \frac{\ln(1/\beta)}{\ln(1 + h/(2D))} \Big\rfloor + 1, \quad \ell := 2t_* + 1, \quad L := \Big\lfloor \ell\Big(1 + \frac{D}{h}\Big) \Big\rfloor, \quad D' := DL,$$
$$h' := \frac{h}{D'+1}, \quad \lambda_0 := \mu(D, h'), \quad \sigma := 2\Big\lceil \frac{D'+1}{h} \Big\rceil, \quad s := \sigma + \Big\lfloor \frac{D\ell}{h} \Big\rfloor, \quad c := \frac{h\beta}{8LD^{s+1}}.$$

If $N \geq 2L + 1$, *then for all disjoint* $P, Q \subseteq E$ *with* $|P|, |Q| \geq \delta M$ *there are* $K \geq \lfloor cN \rfloor + 1$ *sets* $Y_1, \dots, Y_K \subseteq V$ *and edges* $e_i \in P$, $f_i \in Q$ *such that*

(1) $G[Y_i]$ *and* $G - Y_i$ *are connected,* $|Y_i| \leq L$, *and* $e_i, f_i \in \partial Y_i$;

(2) $\mathrm{dist}(Y_i, Y_j) \geq \sigma + 1 \geq 3$ *for* $i \neq j$, *so no edge joins two of the sets and no vertex has neighbours in two of them;*

(3) for every $I \subseteq [K]$, *the graph* $H_I := G - \bigcup_{i \in I} Y_i$ *has at least two vertices,* $h(H_I) \geq h'$ *and* $\lambda_2(L_{H_I}) \geq \lambda_0$.

Proof. G has minimum degree at least 2, so $M \geq N$, and a vertex meets at most D edges of P, so the set $V(P)$ of ends of edges of P has $|V(P)| \geq 2|P|/D \geq \beta N$; likewise for Q. Start with $Z_0 := \emptyset$, and while $|Z_k| \leq h\beta N/(4D)$ do the following. Lemma 10.32 with $X = V(P)$, $X' = V(Q)$ and $Z = Z_k$ gives a path $U_{k+1} \subseteq V \setminus Z_k$ with at most ℓ vertices from an end x of an edge $e_{k+1} \in P$ to an end x' of an edge $f_{k+1} \in Q$; $e_{k+1} \neq f_{k+1}$ as $P \cap Q = \emptyset$, and $U_{k+1} \neq V$ as $\ell < N$. Lemma 10.34 gives $Y_{k+1} \subseteq B(U_{k+1})$ with property (1), where $|Y_{k+1}| \leq \ell(1 + D/h)$ by Lemma 10.33, hence $|Y_{k+1}| \leq L$. Put $Z_{k+1} := Z_k \cup \{v : \mathrm{dist}(v, Y_{k+1}) \leq s\}$. Since $D \geq 2$, a set Y has at most $|Y| \sum_{i \leq s} D^i \leq 2LD^s$ vertices within distance s, so $|Z_k| \leq 2kLD^s$, and the step runs for every $k \leq cN$. Hence $K \geq \lfloor cN \rfloor + 1$.

For $j < k$, the path U_k avoids Z_{k-1}, so $\operatorname{dist}(U_k, Y_j) \geq s+1$, and Y_k lies within distance $\lfloor D\ell/h \rfloor$ of U_k by Lemma 10.33. So $\operatorname{dist}(Y_k, Y_j) \geq \sigma + 1$. Finally $|\partial Y_i| \leq DL = D'$, $N > 2L$ and $\lfloor \sigma/2 \rfloor \geq (D'+1)/h$, so Lemma 10.35 gives (3). □

Theorem 10.37 (sums of structured circuits on expanders)**.** *Let G be a simple 2-connected graph with N vertices, maximum degree at most D and $h(G) \geq h > 0$. Let $C_1, \dots, C_r$ be circuits with real weights, C_k aligned with its own vtree V_k whose leaves are the edges of G, and $f := f_{C_1} + \cdots + f_{C_r}$. Take the parameters of Theorem 10.36 with $\delta := 3^{-r}$, assume $N \geq 2L+1$, and put*

$$p := D^{-(L-1)} \frac{1}{DL} \left(1 + \frac{2}{\lambda_0}\right)^{-(DL-1)}, \qquad n := \lfloor cN/2 \rfloor, \qquad g := \lfloor p^2 n \rfloor.$$

(a) If f is support-exact for G, then $\sum_k |C_k| \geq 2^{\lfloor cN \rfloor + 1}$.

(b) If $\frac{1}{2}\|P - f\|_1 \leq \varepsilon$ and $n \geq 1$, then $\sum_k |C_k| \geq 2^g(1 - 2\varepsilon/(1 - e^{-p^2 n/4}))$, and if moreover $\sum_x f(x) = 1$, then $\sum_k |C_k| \geq 2^g[1 - \varepsilon - e^{-p^2 n/4}]_+^2$.

For fixed D and h, $\ln(1/c) = O(r)$ and $\ln(1/p) = O(r \log r)$. So for fixed r the total size is $2^{\Omega(N)}$, both for support-exact representation and at every fixed error $\varepsilon < \frac{1}{2}$, or $\varepsilon < 1$ with normalization.

Proof. Lemma 10.29 with $E_0 := E$ gives nodes v_k, sets S'_k, and disjoint P, Q with $|P| = |Q| \geq 3^{-r}M$ such that $P \subseteq S'_k$ and $Q \cap S'_k = \emptyset$ for every k. Theorem 10.36 gives sets $Y_1, \dots, Y_K$ with $K \geq \lfloor cN \rfloor + 1$. With e_i as e_U and f_i as f_U, they are gadgets for the split $A := S'_1$, $B := E \setminus S'_1$, and they form an admissible family. For every k and every set R of points on which only the marked edges vary, the matrix of f_{C_k} with rows indexed by the values on the e_i and columns by the values on the f_i is a submatrix of $M_{S'_k, E \setminus S'_k}(f_{C_k})$, which is $M_{S_k, E \setminus S_k}(f_{C_k})$ or its transpose; by Lemma 10.28 its rank is at most $|C_k|$, and by subadditivity the corresponding matrix of f has rank at most $\sum_k |C_k|$.

(a) Fix to 1 the edges of a spanning tree of $H_{[K]}$ and of each $G[Y_i]$, and fix to 0 the other edges of $H_{[K]}$ and of the $G[Y_i]$ and the edges of $\partial Y_i \setminus \{e_i, f_i\}$. By (2), every edge of G is now fixed except the e_i and f_i, and these join Y_i to $H_{[K]}$. A completion is a spanning tree if and only if it contains exactly one of e_i, f_i for every i: with neither, Y_i is cut off, and with both, there is a cycle. As f is support-exact, the matrix of the restricted f with rows indexed by $(x_{e_i})_i$ and columns by $(x_{f_i})_i$ has exactly one nonzero entry in each row and each column, so it has rank 2^K.

(b) Pair $2n$ of the gadgets, and let W be their union, so $H := G - W$ has $\lambda_2(L_H) \geq \lambda_0$ by (3). Lemma 10.18 with $\kappa' = 2\kappa \leq 2/\lambda_0$ gives $p_{Y_i} \geq p$, since $|Y_i| \leq L$, $|\partial Y_i| \leq DL$ and $\frac{1}{d}(1+\kappa')^{-(d-1)}$ decreases in d. By Lemmas 10.17 and 10.19, the blocks are disjoint permutation blocks, on each of which only the marked edges vary, and $\delta(g) \leq \Pr[\mathrm{Bin}(n, 2p^2) < p^2 n] \leq e^{-p^2 n/4}$. Apply Theorem 10.10. The orders of $\ln(1/c)$ and $\ln(1/p)$ follow from $\ell, L, \sigma, s = \Theta(r)$ and $1/\lambda_0 = O(L^2)$ by Lemma 4.7. □

Support-exactness fixes only the support of f, not its values, which is why (a) needs only the zero pattern of the restricted function. Theorem 10.37 applies to every 2-connected bounded-degree expander, at every girth, because the gadgets are pendant blobs rather than cycles. Its constants are astronomically small. For $D = 3$, $h = 0.2$ and $r = 1$, the connectors have at most 93 vertices, the gadgets at most 1488, the separation is $\sigma = 44650$, $\log_{10} c \approx -2.2 \cdot 10^4$, and the exponent of (b) is about $p^2 cN/2$ with $\log_{10}(p^2 c/2) \approx -1.1 \cdot 10^5$; each further summand lowers $\log_{10} c$ by about $1.6 \cdot 10^4$. Corollary 10.31(c) has the better dependence on r, 3^{-r} against $e^{-O(r)}$ with a huge constant, and is far better at any realistic N. Comparing exponents to leading order in r, and ignoring the unspecified constant of [Kri19], the bound of (a) wins only when $\ln N$

exceeds about $7 \cdot 10^4\, r$ for $D = 3$ and $h = 0.2$; this is a heuristic comparison, not a theorem. The losses come from worst-case choices: the separation $\sigma \approx 2DL/h$ is needed only for the case $j \geq 2$ of Lemma 10.35, the blocked balls cost D^s, and Lemma 10.18 sees only $\lambda_2(H)$.

These bounds have limits. Theorems 10.26 and 10.27 are for support-exact representation only: a restriction conditions on an event of probability $e^{-\Theta(N)}$, so total-variation guarantees do not survive it, and we claim no approximate version (Section 10.8). Theorem 10.37 survives approximation because it uses disjoint blocks instead of a restriction. None of these bounds covers planar families, such as the grid of Corollary 8.5, which have no $K_{3,3}$ minor and are not expanders, nor bounded-degree graphs that are 2-connected but far from expanders. The minor-based bounds lose a square root in passing from G to $K_{t,t}$; Theorem 10.37 works directly in G, but only for sums of structured circuits and with astronomically small constants. And general signed decomposable circuits, with sharing and unbounded depth, remain open; by Proposition 10.23 they are as hard as syntactically multilinear circuits for ST_G. We have not found these transfers in the literature, but they are short and may be folklore. Closest in spirit to Theorems 10.30 and 10.37 are lower bounds for sums of set-multilinear branching programs with different orders [CKSS24], which use a polynomial designed for the purpose.

10.7 Cancellation

Approximation is at least as hard to rule out as exact representation. We show that a small unstructured circuit must cancel most of its mass.

Theorem 10.38 (cancellation)**.** *Let G be a connected graph with $M \geq 2$ edges, $P = \mathrm{UST}(G)$, and β_* the largest $\beta(P; A|B)$ over balanced splits. Let C be a smooth decomposable circuit with real weights whose leaves are real functions of arity at most $2M/3$, computing f with $\lambda := \sum_x f(x) > 0$. Let C^+ be C with every weight and leaf function replaced by its absolute value, and $Z^+ := \sum_x f_{C^+}(x)$. If $\varepsilon := \frac{1}{2}\|P - f/\lambda\|_1 < 1$, then*

$$|C| \;\geq\; (1-\varepsilon)^2\, \frac{\lambda}{Z^+}\, \frac{1}{\beta_*^2}.$$

Proof. By induction, $f_{C^+} \geq |f_g|$ pointwise at every node g, since $|\sum w_i g_i| \leq \sum |w_i||g_i|$ and $|gh| = |g||h|$. So $Z^+ \geq \sum_x |f(x)| \geq \lambda$. The circuit C^+ is a smooth decomposable PC of size $|C|$ with leaves of arity at most $2M/3$. Let $Q^+ := f_{C^+}/Z^+$. Since $\sqrt{ab} \geq \min(a, b)$ for $a, b \geq 0$,

$$\begin{aligned}
\mathrm{BC}(P, Q^+) &\geq \sum_x \sqrt{P(x)\frac{|f(x)|}{Z^+}} = \sqrt{\frac{\lambda}{Z^+}} \sum_x \sqrt{P(x)\frac{|f(x)|}{\lambda}} \\
&\geq \sqrt{\frac{\lambda}{Z^+}} \sum_x \min\Big(P(x), \frac{f(x)}{\lambda}\Big) = \sqrt{\frac{\lambda}{Z^+}}\,(1-\varepsilon),
\end{aligned}$$

where the last step uses $\min(a, b) = (a + b - |a - b|)/2$ and $\sum P = \sum f/\lambda = 1$. On the other hand, Lemma 3.9 writes f_{C^+} as a sum of at most $|C|$ nonnegative products across balanced splits, and the proof of Lemma 3.8 bounds $\mathrm{BC}(P, Q^+)$ itself by $\beta_* \sqrt{|C|}$. Since $1 - \varepsilon > 0$, squaring gives the claim. □

On the graphs of Theorem 1.1, a circuit of polynomial size with $\varepsilon < 1$ must have $Z^+/\lambda \geq (1-\varepsilon)^2 e^{cN}/|C|$: its absolute-value circuit carries exponentially more mass than the function it computes. Theorem 10.38 does not rule such circuits out, and any lower bound for unstructured circuits with subtraction has to handle them.

The cancellation can be moved into a single subtraction at the output. Let C have binary product nodes (binarizing at most doubles the size), and replace each node v by two nodes v^+, v^-: for a sum $v = \sum_i w_i u_i$, put $v^\pm := \sum_{w_i \geq 0} w_i u_i^\pm + \sum_{w_i<0} |w_i| u_i^\mp$, where an empty sum is a sum node with weight 0 on one child, which keeps the scope; for a product $v = u_1 u_2$, put $v^+ := u_1^+ u_2^+ + u_1^- u_2^-$ and $v^- := u_1^+ u_2^- + u_1^- u_2^+$; for a leaf ℓ, put $\ell^\pm := \max(\pm\ell, 0)$. By induction $f_{v^+} - f_{v^-} = f_v$ and $f_{v^+} + f_{v^-} = f_{v^{\mathrm{abs}}}$, where v^{abs} is the copy of v in C^+, and the scopes, so smoothness and decomposability, are unchanged. So $f_C = g - h$ for nonnegative smooth decomposable circuits g, h that share their nodes, of total size $O(|C|)$, with $\sum g + \sum h = Z^+$; conversely $g - h$ is a circuit of size $|g| + |h| + O(1)$ with these masses. A signed circuit with cancellation ratio $K := Z^+/\lambda$ is therefore the same, up to a constant factor in size, as a difference of two nonnegative circuits with total mass $K\lambda$, and Theorem 10.38 reads $\log|C| + \log K \geq \log(1/\beta_*^2) - 2\log\frac{1}{1-\varepsilon}$, which is $cN - O(1)$ on the graphs of Theorem 1.1. This trade-off is for smooth circuits with $\lambda > 0$ that approximate $\mathrm{UST}(G)$ in total variation. It says nothing about circuits for ST_G or about support-exact circuits, whose masses are unrelated to those of $\mathrm{UST}(G)$: on a triangle, $\mathrm{ST}_G(1,1,1) = 3$ while the indicator of spanning trees vanishes at the all-ones point. Without a bound on K, by Proposition 10.23, the problem is as hard as lower bounds for general syntactically multilinear circuits.

Proposition 10.16 gives a version of Theorem 10.38 that counts only cancellation between terms on different splits. The proof of Lemma 3.9 is syntactic: with real weights it gives the same expansion $f_C = \sum_{u \in \mathcal{U}} f_u h_u$, with real f_u, h_u. Binarize C and fix all choices in that proof. Identify a split $A|B$ with $B|A$, group the terms by their split π, $f_\pi := \sum_{u:\, A_u|E\setminus A_u = \pi} f_u h_u$, and put $\Lambda_\times := \sum_\pi \|f_\pi\|_1/\lambda$, where $\lambda := \sum_x f_C(x)$.

Proposition 10.39 (cancellation across splits)**.** *Let G, p and n be as in Corollary 10.14, and $g := \lfloor p^2 n \rfloor$. Let C be a smooth decomposable circuit with real weights whose leaves have arity at most $2M/3$, with $\lambda > 0$ and $\frac{1}{2}\|P - f_C/\lambda\|_1 \leq \varepsilon$. Then, with $\Lambda_\times$ defined from the fixed decomposition above,*

$$\left[1 - \varepsilon - |C| e^{-p^2 n/4}\right]_+^2 \;\leq\; |C|\, \Lambda_\times\, 2^{-g}.$$

Proof. $\operatorname{rank} M_\pi(f_\pi)$ is at most the number r_π of terms on π, and $\sum_\pi r_\pi = |\mathcal{U}| \leq |C|$; in particular there are at most $|C|$ splits. Every split $M/3 < |A_u| \leq 2M/3$ carries blocks with $\delta \leq e^{-p^2 n/4}$, as in the proof of Corollary 10.14. Apply Proposition 10.16 to the functions f_π/λ. □

Running the same decomposition on C^+ gives $|f_u| \leq f_u^+$ and $|h_u| \leq h_u^+$ pointwise with $\sum_u f_u^+ h_u^+ = f_{C^+}$, so $\Lambda_\times \leq Z^+/\lambda$. For $\varepsilon < 1$ fixed and $|C| \leq e^{p^2 n/8}$, Proposition 10.39 gives $\log|C| + \log\Lambda_\times \geq g\ln 2 - 2\ln\frac{1}{1-\varepsilon} - o(1)$: a small approximating circuit must cancel between different splits, while terms on the same split may cancel each other at no cost. For a structured circuit whose choices follow the vtree, all terms lie on one split, so $\Lambda_\times = \|f_C\|_1/\lambda \leq 1 + 2\varepsilon$ by Lemma 10.7, and the proposition recovers Theorem 10.12 up to the constants. It refines the bookkeeping of Theorem 10.38 rather than dominating it, since its exponent $p^2 n$ is much smaller than cN. Unlike Z^+, $\Lambda_\times$ depends on the decomposition and not only on the computed function: for a normalized product distribution H, the circuit $(K+1)H - KH$ has $\Lambda_\times = 1$ if both copies of H are binarized alike and $2K+1$ if they are binarized along different balanced splits. So $\Lambda_\times$ is unbounded already at product-depth 1.

Remark 10.40 (slices do not lower cancellation). One might hope to split into a low-cancellation case, handled by Theorem 10.38, and a high-cancellation case handled by restricting to slices, as in Section 10.6. This fails. Let ρ range over the assignments to a fixed set of edges, and let Z_ρ^+ and λ_ρ be the sums of f_{C^+} and of f_C over the slice of ρ. Then $\sum_\rho Z_\rho^+ = Z^+$ and $\sum_\rho \lambda_\rho = \lambda$, so slicing

redistributes cancellation without removing it, and padding puts it on every slice: if C computes P with nonnegative weights, then $C+cU-cU$, with U the uniform distribution on $\{0,1\}^E$ (a product of M sums) and $c := 2^{M+N}$, computes the same function with size $|C| + O(M)$, and every slice that fixes $k \le M$ edges and has P-mass $P(\rho) > 0$ has $Z^+_\rho/\lambda_\rho = 1 + 2c2^{-k}/P(\rho) \ge 1 + 2^{N+1}$. Nor can slices of a signed approximation be normalized: replacing P on one slice ρ by $P - 2P(\rho)u$, with u a distribution on the slice, gives error $P(\rho)$, which may be tiny, and slice mass $\lambda_\rho = -P(\rho) < 0$. For rank arguments neither issue arises, because Theorem 10.10 sums over disjoint blocks without normalizing them.

10.8 What blocks the unstructured case

For unstructured circuits, approximate versions of Theorems 10.26 and 10.27 remain open. Theorem 10.10 does not need a common split, so it reduces them to a statement about ranks inside blocks.

Proposition 10.41 (transfer)**.** *Let $\mathcal{R}$ be a family of disjoint permutation blocks of P with total P-mass at least $1-\delta$, and let Q be a real function with $\sum_x Q(x) = 1$, $\frac{1}{2}\|P-Q\|_1 \le \varepsilon$ and $\operatorname{rank} Q|_R \le \eta n_R$ for every $R \in \mathcal{R}$. Then $[1-\varepsilon-\delta]^2_+ \le \eta$.*

Proof. Theorem 10.10(a) with $g = 0$, and $\tau^{-1}\sum_R r_R \le \eta\tau^{-1}\sum_R n_R \le \eta$. □

So exact rank deficiency in most blocks, rather than in one restriction, would give lower bounds at every error $\varepsilon < 1$. The restriction arguments of Section 10.6 do not supply it, and two obstructions show why simple variants cannot.

Norm-type measures bound mass, not size. Let ν be a nonnegative, subadditive and absolutely homogeneous functional on real functions on $\{0,1\}^E$, such as a trace norm or γ_2 norm at a fixed split. Balanced product terms are closed under scaling, so $\sup \nu$ over them is 0 or ∞, and a termwise certificate $\nu(T) \le \kappa\, m(T)$ for a homogeneous mass m gives only $\sum_u m(T_u) \ge \nu(Q)/\kappa$ for a decomposition $Q = \sum_u T_u$: a lower bound on the mass of the decomposition, not on the number of terms. Turning it into a size bound needs a bound on the mass, which is the role of Z^+/λ in Theorem 10.38, and circuits have no such bound in terms of size: $\prod_{i\le n}((1+Kx_i)+(-Kx_i))$ computes 1 with $O(n)$ wires and product-depth 1 for every $K \ge 0$, while its absolute-value circuit has value $(1+2K)^n$ at the all-ones point. This does not exclude scale-invariant ratios or arguments that use the approximation more cleverly.

Proposition 10.42 (one circuit fills every block of a fixed family)**.** *Fix a split $A|B$, pairwise vertex-disjoint mixed cycles $C_1,\dots,C_m$ with $e_k \in A$ and $f_k \in B$ on C_k, and any pairing, and let the blocks be those of Lemma 10.5. Put $\Psi(x) := \prod_{k\le m}(1 - \prod_{a\in E(C_k)} x_a)$. Then Ψ is computed by a smooth decomposable circuit with real weights, product-depth 2 and $O(M)$ wires, and $\Psi = \tau P$ on every set R of Lemma 10.5; in particular $\operatorname{rank}\Psi|_R = n_R$ is full.*

Proof. Write each factor as $\prod_{a\in E(C_k)} 1_a - \prod_{a\in E(C_k)} x_a$, with 1_a the constant 1 on scope $\{a\}$, and multiply the factors with constant leaves 1_a for the edges on no cycle. On a cell of R, a cycle outside the good pairs has its edges fixed by F, as in a spanning tree, so it is not complete and its factor is 1. At a good pair (i,j) both arcs are present, so C_i is complete exactly when the cell chooses (e_i,f_i) and C_j exactly when it chooses (e_j,f_j). So the product is 1 on the tree cells of R and 0 on the others. □

So no assertion of rank deficiency that ignores the approximation can hold on a fixed family of cycle blocks, whatever the pairing and orientation. This does not contradict Theorem 10.12:

Ψ is structured only for vtrees that keep each cycle together, and the mixed cycles at a balanced split of such a vtree are different cycles. Nor does it give a small approximation: Ψ/τ is not normalized, and on the 8-prism with four square gadgets its total mass is about 86. Restriction proofs in the style of Raz and of Raz and Yehudayoff instead choose the pairing at random. Pair blocks pay for disjointness with double visibility: each row bit of a pair is visible through two edges, e_i and e_j, and at a balanced product of fan-in two with a random placement of the special edges the visible-bit count is about $\frac{3}{2}g$, which exceeds the g bits of a block, so counting visible bits gives nothing there; this blocks that counting argument, not every proof. Blocks of single cycles see each bit once but overlap, and then the off-support mass of Q can sit on cells shared by many blocks. We know no block design with both properties. A sufficient condition in the spirit of Proposition 10.41 would be a circuit-independent distribution over block systems from a packing of short cycles such that, for every circuit of product-depth d and size s, some system has $\operatorname{rank} Q|_R \le s\, 2^{-g_R^{a/d}} n_R$ on all its blocks outside P-mass $\exp(-N^{b_d})$, with $a > 0$ independent of d; on graphs with $N^{1-o(1)}$ disjoint pairs of cycles of length at most $c \log_D N + 1$, where $c < \frac{1}{2}$, this would give $s \ge 2^{N^{\Omega(1/d)}}$ at every $\varepsilon < 1$ for normalized outputs. Such graphs are a restricted class: Lemma 10.13 keeps polynomially many good pairs only for $c < \frac{1}{2}$, which excludes expanders of girth $(\frac{4}{3} - o(1)) \log_{D-1} N$. We state this only as a direction; we have no evidence that it holds.

11 Limits of the method and two variants

11.1 Blocks: why expansion alone is not enough

All our bounds go through $\max_{A|B} \beta(P; A|B)$ over balanced splits. The block structure of G decides when this quantity equals one.

A *block* of a connected graph is a maximal connected subgraph that has no cut vertex of its own. Every edge lies in exactly one block, a bridge forms a block with one edge, and a block with at least two edges is 2-connected. We use three standard facts [Die17, Ch. 3]. Every cycle lies in a single block. The block graph is a tree; by induction on the number of blocks, removing the vertices of a leaf block other than its cut vertex, this gives $\sum_i (|V(G_i)| - 1) = N - 1$ for the blocks $G_1, \dots, G_m$. Any two edges of a 2-connected graph lie on a common cycle; this follows from Menger's theorem applied to the graph in which the two edges are subdivided, which is again 2-connected.

Lemma 11.1 (block factorization)**.** *An edge set T is a spanning tree of G if and only if $T \cap E(G_i)$ is a spanning tree of G_i for every block G_i. Hence* $\mathrm{UST}(G)$ *is the product of the laws* $\mathrm{UST}(G_i)$ *on the edge sets $E(G_i)$.*

Proof. Write $T_i := T \cap E(G_i)$. Every cycle lies in one block, so T is acyclic if and only if every T_i is. If every T_i is a spanning tree of G_i, then T is acyclic with $\sum_i (|V(G_i)| - 1) = N - 1$ edges, so it is a spanning tree. Conversely, if T is a spanning tree, then each T_i is acyclic, so $|T_i| \le |V(G_i)| - 1$, and $\sum_i |T_i| = N - 1$ forces equality; an acyclic graph with one edge fewer than vertices is a spanning tree. The uniform distribution on a product set is the product of the uniform distributions. □

Proposition 11.2 (the block criterion)**.** *Let G be a connected simple graph with M edges and blocks $G_1, \dots, G_m$, and let $A|B$ be a split. Then $\beta(\mathrm{UST}(G); A|B) = 1$ if and only if every block has all its edges in A or all in B. Consequently, some balanced split has $\beta = 1$ if and only if some union of blocks has between $M/3$ and $2M/3$ edges, and this happens if and only if no block has more than $2M/3$ edges.*

Proof. If every block lies on one side, Lemma 11.1 makes $T \cap A$ and $T \cap B$ independent, so $P = \mathrm{UST}(G)$ is itself a product across $A|B$ and $\beta = \mathrm{BC}(P, P) = 1$. Conversely, suppose $\beta = 1$. The maximum defining β is attained (Lemma 3.1), and $\mathrm{BC}(P, q) = 1$ for probability distributions only if $P = q$, by the equality case of the Cauchy–Schwarz inequality. So P is a product across $A|B$, and its support $\mathcal{T}(G)$ is a rectangle: if $a \cup b$ and $a' \cup b'$ are spanning trees with $a, a' \subseteq A$ and $b, b' \subseteq B$, then $a' \cup b$ is a spanning tree. Suppose some block contains edges $e \in A$ and $f \in B$, and let C be a cycle through both. Grow the path $E(C) \setminus \{e\}$ greedily to a spanning tree $T_1 = (E(C) \setminus \{e\}) \cup X$. No edge of X has both endpoints in $V(C)$, since such an edge closes a cycle with the path. So $T_2 := (E(C) \setminus \{f\}) \cup X$ is also connected with $N - 1$ edges, hence a spanning tree. Now $(T_2 \cap A) \cup (T_1 \cap B)$ contains every edge of C in A, because those differ from f, and every edge of C in B, because those differ from e. So it contains the cycle C and is not a tree, a contradiction.

For the second statement, a balanced split with $\beta = 1$ is a union of blocks with between $M/3$ and $2M/3$ edges on side A. If some block has more than $2M/3$ edges, every union of blocks has more than $2M/3$ or fewer than $M/3$ edges. If every block has at most $2M/3$ edges, either some block has at least $M/3$, or all have fewer and we add blocks one at a time until the total first reaches $M/3$, at which point it is below $2M/3$. □

A positive fraction of edges in small blocks is not enough to give $\beta = 1$. Let G be K_4 with one pendant edge. The blocks have 6 and 1 edges, $M = 7$, and a balanced side has 3 or 4 edges, so no union of blocks is balanced and $\beta < 1$ for every balanced split. Exhaustive enumeration gives $\max \beta = 0.75$ over balanced splits, attained for instance when A is a triangle of K_4 (Appendix A). For bounded-degree expanders the block structure decides everything, because blocks inherit expansion.

Lemma 11.3 (blocks inherit expansion)**.** *Let G be connected and let G_1 be a block of G with at least two edges. Then $h(G_1) \geq h(G)$.*

Proof. First, every edge of G with both endpoints in $V(G_1)$ lies in G_1: such an edge $xy \notin E(G_1)$, together with a path of G_1 from y to x, which has length at least two since G is simple, would form a cycle with edges in two blocks. Second, every component K of $G - V(G_1)$ has exactly one neighbour v_K in $V(G_1)$: it has one because G is connected, and if it had two, $x \neq y$, a path from x through K to y and a path of G_1 from y to x would form a cycle with edges in two blocks.

Let $S \subseteq V(G_1)$ with $0 < |S| \leq |V(G_1)|/2$, and let S' be the union of S and all components K with $v_K \in S$. By the two facts, the edges of G leaving S' are exactly the edges of G_1 leaving S, so $|\partial_G S'| = |\partial_{G_1} S|$. If $|S'| \leq N/2$, then $|\partial_{G_1} S| \geq h(G)|S'| \geq h(G)|S|$. Otherwise $V \setminus S'$ is nonempty, has at most $N/2$ vertices, contains $V(G_1) \setminus S$ and has the same boundary, so $|\partial_{G_1} S| \geq h(G)|V(G_1) \setminus S| \geq h(G)|S|$. □

Theorem 11.4 (block dichotomy)**.** *Let G be a connected simple graph with N vertices, $M \geq 2$ edges, maximum degree at most D and $h(G) \geq h > 0$, let G_1 be a block with the largest number M_1 of edges, and let $c = c(D, h)$ be as in Theorem 4.13.*

(a) If $M_1 \leq 2M/3$, some balanced split has $\beta = 1$, and a product node with two leaves of arity at most $2M/3$ represents $\mathrm{UST}(G)$ exactly.

(b) If $M_1 > 2M/3$, then every balanced split satisfies $\beta(\mathrm{UST}(G); A|B)^2 \leq e^{-3c(M_1 - 2M/3)}$, and some balanced split has $\beta(\mathrm{UST}(G); A|B)^2 \geq 2^{-(M_1 - 2M/3) - 1}$.

(c) *Let C be a smooth decomposable PC with nonnegative weights, each of whose leaves depends on at most $2M_1/3$ variables of $E(G_1)$, for instance a circuit with univariate leaves. If $M_1 \geq 2$ and $\mathrm{TV}(\mathrm{UST}(G), Q_C) \leq \varepsilon < 1$, then $|C| \geq (1-\varepsilon)^2 e^{c|V(G_1)|}$.*

Proof. (a) By Proposition 11.2 there is a union U of blocks with between $M/3$ and $2M/3$ edges, and $\mathrm{UST}(G)$ is the product of its marginals on U and on $E \setminus U$ by Lemma 11.1. Store each marginal in one leaf.

(b) Since $M_1 > 2M/3 \geq 4/3$, G_1 has at least two edges, so it is 2-connected, with maximum degree at most D and $h(G_1) \geq h$ by Lemma 11.3. Let $A|B$ be balanced. The restrictions $x_A \mapsto x_{A \cap E(G_1)}$ and $x_B \mapsto x_{B \cap E(G_1)}$ act on the two sides separately, so Lemma 3.2 bounds $\beta(\mathrm{UST}(G); A|B)$ by β of the law of $T \cap E(G_1)$ across $A \cap E(G_1) \,|\, B \cap E(G_1)$, and this law is $\mathrm{UST}(G_1)$ by Lemma 11.1. Since $|B| \leq 2M/3$, the set $A \cap E(G_1)$ has at least $M_1 - 2M/3$ edges, and likewise for B. Apply Theorem 4.13 to G_1 with $m_0 = M_1 - 2M/3$.

For the lower bound, let $k := \lceil M/3 \rceil - (M - M_1)$, so $1 \leq k \leq M_1 - 2M/3 + 1$. Let A consist of all edges outside G_1 and k edges of G_1, and let $A_1 := A \cap E(G_1)$ and $B := E \setminus A \subseteq E(G_1)$. Then $|A| = \lceil M/3 \rceil \leq 2M/3$ since $M \geq 2$, so the split is balanced. Write $P = P_0 \otimes P_1$ with P_0 the law on $E \setminus E(G_1)$ and $P_1 = \mathrm{UST}(G_1)$ (Lemma 11.1). Let a^* be a most likely value of x_{A_1} under P_1, so $P_1(x_{A_1} = a^*) \geq 2^{-k}$, and take the product $q := (P_0 \otimes \delta_{a^*}) \otimes P_1(x_B \in \cdot \,|\, x_{A_1} = a^*)$. Then

$$\mathrm{BC}(P, q) = \sum_{x_B} \sqrt{P_1(a^*, x_B) \, \frac{P_1(a^*, x_B)}{P_1(x_{A_1} = a^*)}} = \sqrt{P_1(x_{A_1} = a^*)} \geq 2^{-k/2}.$$

(c) Replace every leaf of C by its sum over the values of its variables outside $E(G_1)$. Each scope X_v becomes $X_v \cap E(G_1)$, so smoothness and decomposability are preserved and the size does not grow. By induction over the circuit, using smoothness at sum nodes and decomposability at product nodes, the new circuit C_1 computes the sum of f_C over the variables outside $E(G_1)$, so Q_{C_1} is the marginal of Q_C on $E(G_1)$; this is the standard computation of marginals in such circuits [CVdB20]. Its leaves have arity at most $2M_1/3$. Total variation does not increase under taking marginals, and the marginal of $\mathrm{UST}(G)$ on $E(G_1)$ is $\mathrm{UST}(G_1)$. Apply Theorem 1.1 to G_1, which is 2-connected with maximum degree at most D and $h(G_1) \geq h$ by Lemma 11.3. □

So for bounded-degree expanders, and circuits whose leaves have arity up to $2M/3$, the bound from balanced splits is $e^{-\Theta(M_1 - 2M/3)}$ when $M_1 > 2M/3$ and trivial otherwise, and in the trivial case a constant-size circuit exists. With univariate leaves the threshold disappears: by (c), one 2-connected block with linearly many vertices already forces exponential size. 2-connectivity of G is the case $M_1 = M$ of (b).

For example, attach a pendant edge to every vertex of a 2-connected d-regular expander on n vertices. Then $M = (d/2+1)n$ and $M_1 = dn/2$, so $M_1 - 2M/3 = (d-4)n/6$. For $d \geq 5$, part (b) gives exponential decay of β, and for $d \leq 4$ the pendant edges form a balanced union of blocks. The next proposition is the case $d \leq 4$, with the expansion of G checked.

Proposition 11.5 (pendant edges)**.** *Let H be a connected graph with $v \geq 2$ vertices, at most $2v$ edges, maximum degree Δ and $h(H) \geq h_0 > 0$. Let G be obtained by attaching a new vertex of degree one to every vertex of H. Then G has maximum degree $\Delta+1$ and $h(G) \geq \min(1, h_0)/3$. Let A be the set of pendant edges and $B = E(H)$. Then $A|B$ is balanced and $\beta(\mathrm{UST}(G); A|B) = 1$.*

Proof. Every spanning tree contains every pendant edge, so $\mathrm{UST}(G)$ is the product of the point mass on $x_A = \mathbf{1}$ and the law of $\mathrm{UST}(H)$ on x_B, and $\beta = 1$. The split is balanced because $|A| = v$, $|B| = |E(H)|$, $v - 1 \leq |E(H)| \leq 2v$ and $v \geq 2$. For the expansion, let S consist of a vertices of

H and b pendant vertices, with $0 < a + b \le v$. A pendant edge is in ∂S when exactly one of its endpoints is in S; there are at least $|a - b|$ such edges. If $a \le v/2$, then $|\partial S| \ge h_0 a + |a - b|$, which is at least $h_0(a + b)/3$ if $b \le 2a$ and at least $b - a \ge (a + b)/3$ if $b > 2a$. If $a > v/2$, then $v - a \ge b$, so $|\partial S| \ge h_0(v - a) + (a - b) \ge h_0 b + (a - b)$. This is at least $(a + b)/3$ if $a \ge 2b$ and at least $h_0(a + b)/3$ if $a < 2b$. □

So expansion and bounded degree do not suffice for any bound that is uniform over balanced splits. Here the obstruction is real for circuits with leaves of arity $2M/3$: since $|E(H)| \le 2M/3$, one leaf that stores $\mathrm{UST}(H)$, multiplied by the indicator leaves $[x_e = 1]$ of the pendant edges, represents $\mathrm{UST}(G)$ exactly with size $O(v)$. For circuits with univariate leaves it is not: if H is 2-connected, Theorem 11.4(c) gives $|C| \ge (1-\varepsilon)^2 e^{c(\Delta, h_0)v}$. The linking condition of Definition 7.1 excludes bridges (Lemma 7.2), and more generally forces every edge onto a short cycle.

11.2 Frozen regions and large girth

Lemma 6.2 conditions on everything outside a region, uniformly over the outside configuration. The next observation shows that the region must contain a cycle.

Proposition 11.6 (acyclic regions freeze)**.** *Let $I \subseteq E$ be acyclic, and let T_0 be a spanning tree with $I \subseteq T_0$. Then, conditionally on $T \setminus I = T_0 \setminus I$, we have $T = T_0$ almost surely.*

Proof. A completion $J \subseteq I$ must satisfy $|J| = N - 1 - |T_0 \setminus I| = |I|$, so $J = I$. □

So a bound like Lemma 6.2, which must hold for every outside configuration, needs a region that contains a cycle. In a graph of girth g, any such region spans at least g vertices, and at most N/g vertex-disjoint cycles exist. With the regions $I(C)$ used here, Theorem 6.6 gives an exponent of at most about $(N/g)D^{-2(g-1)}$ in a D-regular graph. This tends to zero once $g \ge (\frac{1}{2} + \eta) \log_D N$ for a fixed $\eta > 0$, which is the regime of the best explicit expanders [LPS88]. Random regular graphs have very few short cycles, so the local route gives nothing there either.

A star is acyclic too, so conditioning on everything outside a star can also freeze it. Given $T \setminus S(w)$, the rest of T must join w to each component of $(V \setminus \{w\}, T \setminus S(w))$ by exactly one edge, so if every such component meets the star in exactly one edge, $T \cap S(w)$ is determined. Section 4 avoids this in two ways. Lemma 4.3 conditions only on the stars of the other vertices of a sparse set W, not on everything outside $S(w)$. And it bounds the conditional probability by effective resistances in $G - W$, which depend on the whole graph and which the spectral gap of $G - W$ controls at every girth (Lemmas 4.4 and 4.10). This is how Theorem 1.1 covers random regular graphs and Ramanujan graphs, which an earlier version of this paper listed as open.

11.3 Unbounded degree

On the complete graph K_n the cycle bits give almost nothing: for triangles Theorem 6.6 gives $q_k = (n-1)^{-3}$, at most $n/3$ vertex-disjoint triangles exist, and the resulting exponent is $O(n^{-2})$. Longer cycles are worse. Theorem 4.5 needs an independent set W, and the independent sets of K_n have one vertex. Section 5 gets around both problems. It conditions on the tree edges inside a set W and averages (Lemma 3.7), so independence is not needed, and its star-deletion bound (Lemma 5.3) has the form of Lemma 4.4 with the spectral gap of $G - W$. In K_n that gap is $n - r$, of the order of the degree, so the factor $(1 + 1/\lambda_U)^{-d_A}$ stays above e^{-1} instead of decaying exponentially in the degree. This gives $2^{\Omega(n)}$ for every balanced split of K_n (Theorem 5.1) and for every graph with $\lambda_2(L_G) \ge \kappa n$ (Corollary 5.9), and Proposition 5.10 shows that minimum

degree $n/2$ alone does not suffice. Between bounded degree and linear spectral gap we have no good bound. Theorem 5.8 conditions on fewer than λ_2 vertices, so it gives little unless λ_2 grows linearly. The constant $c(D,h)$ of Theorem 4.13 decays like $(D+1)^{-4(D+1)}e^{-10(D+1)}$ for fixed $h \geq 1$, and even the spectral constant $c_\lambda(D)$ decays like D^{-4}. The lower bounds of [MM14, ZHdB20] for $\mathrm{UST}(K_n)$ are discussed in Section 12.

11.4 How far star bits can go

The constant of Theorem 4.13 is polynomial in h for fixed D, but the degree $5D+4$ grows with D. We show that some growth is forced for every bound obtained from star bits. Call a *star-bit certificate* any bound $\beta(\mathrm{UST}(G); A|B) \leq \prod_k \gamma_k$ obtained from Lemmas 3.2 and 3.3 applied to the bits (a_{w_k}, b_{w_k}) of an ordered set $(w_1, \dots, w_r)$ of mixed vertices, independent or not. Theorem 4.5 bounds the γ_k of such a certificate further.

Lemma 11.7 (limits of star bits)**.** *In a star-bit certificate, $\gamma_k^2 \geq 1 - \pi_k$ with $\pi_k := \min(\Pr[a_{w_k} = 1], \Pr[b_{w_k} = 1])$. Any numbers $p_A(w), p_B(w)$ admissible in Theorem 4.5 satisfy $p_A(w) \leq \Pr[a_w = 1]$ and $p_B(w) \leq \Pr[b_w = 1]$.*

Proof. If $N \geq 2$, no spanning tree avoids the whole star of w, so every conditional law K of (a_{w_k}, b_{w_k}) has $K(1,1) = 0$. The row $a = 0$ of $\sqrt{K}$ has squared norm $K(0,0)+K(0,1) = 1-K(1,\cdot)$, and the operator norm is at least the norm of any row. The conditional probability of $a_{w_k} = 1$ given the past averages to $\Pr[a_{w_k} = 1]$, so some past with positive probability has it at most $\Pr[a_{w_k} = 1]$; there $\gamma_k^2 \geq 1 - \Pr[a_{w_k} = 1]$. Columns give the same with b. For the second claim, average the conditional bound of Lemma 4.3 over σ. □

Lemma 11.8 (bare paths)**.** *Let w be a vertex, and suppose that each edge $e_j = wu_{j,1}$ of $S_A(w)$, $1 \leq j \leq a$, starts a path $w, u_{j,1}, \dots, u_{j,L}$ whose interior vertices $u_{j,1}, \dots, u_{j,L-1}$ have degree 2 in G. Then $\Pr[a_w = 1] \leq L^{-a}$.*

Proof. Let $G_j := G - \{e_1, \dots, e_j\}$. If some e_j is a bridge of G_{j-1}, then $\Pr[a_w = 1] = 0$. Otherwise, conditionally on $e_1, \dots, e_{j-1} \notin T$, the tree is uniform on G_{j-1}, and by Kirchhoff's formula and the parallel law $e_j \notin T$ has probability $1 - R_{G_{j-1}}(u_{j,1}, w) = 1/(1 + R_{G_j}(u_{j,1}, w))$. In G_j the vertex $u_{j,1}$ is joined to the rest of the graph only through the path $u_{j,1}, \dots, u_{j,L}$, so a unit flow from $u_{j,1}$ to w sends a unit through each of its $L-1$ edges, and $R_{G_j}(u_{j,1}, w) \geq L - 1$ by Thomson's principle (fact (E5)). Multiply the a factors, each at most $1/L$. □

The *hub graph* $G_{L,n} = G_{L,n}(a,b)$ has integer parameters $a, b \geq 2$, $L \geq 2$ and $n \geq 2$, with an and bn even. Fix $h_0 > 0$ and two 2-connected cubic graphs X_A and X_B, the cores, with exactly an and bn vertices and edge expansion at least h_0. Such cores exist for all large even sizes with an absolute h_0: a random cubic graph is 3-connected and has edge expansion at least an absolute constant with probability tending to one [Wor81a, Bol88]. Add n new vertices, the hubs. Join each hub by a paths of length L to vertices of X_A and by b paths of length L to vertices of X_B, with new interior vertices, so that every core vertex is the endpoint of exactly one path. Let A be $E(X_A)$ together with the edges of the paths to X_A, and B the rest. Then $G_{L,n}$ is 2-connected: after deleting one vertex, every other vertex is still joined to a core, each core stays connected, and the two cores stay joined through a hub other than the deleted vertex. Its maximum degree is $D = \max(a+b, 4)$, it has $N = n(1 + (a+b)L)$ vertices, the hubs are exactly its mixed vertices, and for $a = b$ the split is balanced, with $|A| = |B|$. The $L-1$ interior vertices of one path have two boundary edges, so $h(G_{L,n}) \leq 2/(L-1)$.

Proposition 11.9 (hub graphs)**.** *On $G_{L,n}$ with the split $A|B$, every star-bit certificate satisfies $\prod_k \gamma_k^2 \geq \exp(-\frac{4}{3} n L^{-\min(a,b)})$, and every bound of Theorem 4.5 on β^2 is at least $\exp(-4\ln 2 \cdot nL^{-(a+b)})$. So a bound $\beta^2 \leq e^{-cN}$ proved by star bits has $c \leq \frac{4}{3} L^{-\min(a,b)}/((a+b)(L-1))$, and one proved by Theorem 4.5 has $c \leq 4\ln 2 \cdot L^{-(a+b)}/((a+b)(L-1))$.*

Proof. The mixed vertices are the n hubs, so a certificate has at most n factors. Every edge at a hub starts a bare path of length L, so by Lemma 11.8 $\Pr[a_w = 1] \leq L^{-a}$ and $\Pr[b_w = 1] \leq L^{-b}$. By Lemma 11.7, $\gamma_k^2 \geq 1 - \pi_k$ with $\pi_k \leq L^{-\min(a,b)} \leq \frac{1}{4}$, and $-\ln(1-\pi) \leq \frac{4}{3}\pi$ for $\pi \leq \frac{1}{4}$. For Theorem 4.5, $p_A p_B \leq L^{-(a+b)} \leq \frac{1}{16}$, and $f(x) := -\ln \frac{1+\sqrt{1-4x}}{2}$ is convex with $f(0) = 0$ and $f(\frac{1}{4}) = \ln 2$, so $f(x) \leq 4\ln 2 \cdot x$ on $[0, \frac{1}{4}]$; the bound of the theorem on β^2 is $\exp(-\sum_k f(p_A p_B))$. Divide the exponents by $N \geq n(a+b)(L-1)$. □

With $a = b = D/2$ the caps are $O(L^{-D/2}/(DL))$ and $O(L^{-D}/(DL))$. To turn them into statements about h we need the matching lower bound $h(G_{L,n}) = \Omega(1/L)$, uniformly in n. This is where the pinned core sizes matter: if the cores were much larger than an and bn, the cut around X_A would make $h(G_{L,n})L$ tend to 0. We first bound the expansion of a subdivision.

Lemma 11.10 (subdivided edges)**.** *Let H be a connected graph, $F \subseteq E(H)$ and $L \geq 1$, and let G arise from H by replacing each $f \in F$ by a path of length $\ell_f \in [1, L]$ with new interior vertices. Let $\nu := \max_{\emptyset \neq S_0 \subseteq V(H)} e_F(S_0)/|S_0|$, where $e_F(S_0)$ counts the edges of F with both ends in S_0. Then*

$$h(G) \;\geq\; \frac{h(H)}{1 + (L-1)(\nu + h(H))}.$$

Proof. For $L = 1$, $G = H$. Let $L \geq 2$, $\emptyset \neq S \subseteq V(G)$ and $S_0 := S \cap V(H)$, and suppose first that $|S_0| \leq |V(H)|/2$. Call a path of F inside, crossing or outside if two, one or none of its ends lie in S_0. Each of the $c := e_H(S_0, V(H) \setminus S_0)$ edges of H leaving S_0 gives a boundary edge of S in G, on its own path if it is subdivided. Each of the o outside paths that have an interior vertex in S gives two more, on that path. So $|\partial_G S| \geq c + 2o$. The vertices of S are those of S_0, at most $(L-1)e_F(S_0) \leq (L-1)\nu|S_0|$ interior vertices of inside paths, at most $(L-1)c$ of crossing paths and at most $(L-1)o$ of outside paths. If $S_0 = \emptyset$, then $c = 0$ and the ratio is at least $2/(L-1)$. Otherwise $|S_0| \leq c/h(H)$, and

$$\frac{|\partial_G S|}{|S|} \;\geq\; \frac{c + 2o}{c\,((1 + (L-1)\nu)/h(H) + L - 1) + (L-1)o} \;\geq\; \min\Big(\frac{h(H)}{1 + (L-1)(\nu + h(H))},\; \frac{2}{L-1}\Big),$$

and the first term is the smaller, since it is below $1/(L-1)$. If $|S_0| > |V(H)|/2$ and $|S| \leq |V(G)|/2$, the complement S^c has $|S^c \cap V(H)| < |V(H)|/2$ and $|S^c| \geq |S|$, so $|\partial S|/|S| \geq |\partial S^c|/|S^c|$, which the first case bounds. □

Proposition 11.11 (expansion of hub graphs)**.** *Let $h_1 := \min(h_0, 1)$ and $m := \min(a, b)$. For all $L \geq 2$ and $n \geq 2$,*

$$\frac{2h_1 m}{5(a+b+1)L} \;\leq\; h(G_{L,n}) \;\leq\; \frac{2}{L-1}.$$

Proof. The upper bound was shown above. Let H be $G_{1,n}$, in which every path is a single edge; each core vertex has exactly one hub neighbour $\eta(x)$. We first show $h(H) \geq \eta_0 := h_1 m/(2(a+b+1))$. Let $0 < |S| \leq |V(H)|/2 = n(a+b+1)/2$, and put $\alpha := |S \cap X_A|/(an)$, $\alpha' := |S \cap X_B|/(bn)$ and $\kappa := |S \cap \text{hubs}|/n$, so $|S|/n = a\alpha + b\alpha' + \kappa$. An edge $x\eta(x)$ with $x \in X_A$ leaves S when exactly one of $x, \eta(x)$ is in S. There are $a\alpha n$ vertices $x \in X_A \cap S$ and $a\kappa n$ vertices $x \in X_A$ with

$\eta(x) \in S$, so at least $an|\alpha - \kappa|$ such edges leave S, and at least $h_0 an \min(\alpha, 1-\alpha)$ edges of X_A do. With the same count for X_B,

$$\frac{|\partial S|}{n} \;\geq\; a\big[h_1 \min(\alpha, 1-\alpha) + |\alpha - \kappa|\big] + b\big[h_1 \min(\alpha', 1-\alpha') + |\alpha' - \kappa|\big].$$

If $\alpha, \alpha' \leq \frac{1}{2}$, each bracket is at least $h_1 \max(\alpha, \kappa)$, because $h_1 \leq 1$, so $|\partial S|/n \geq \frac{h_1}{2}(a\alpha + b\alpha' + (a+b)\kappa) \geq \frac{h_1}{2}|S|/n$. If $\alpha > \frac{1}{2} \geq \alpha'$, then either $\kappa \leq \alpha - \frac{1}{4}$ and the first bracket is at least $\frac{1}{4}$, or $\kappa > \frac{1}{4}$ and the second is at least $h_1\kappa > h_1/4$; so $|\partial S|/n \geq h_1 m/4$, while $|S|/n \leq (a+b+1)/2$. The case $\alpha' > \frac{1}{2} \geq \alpha$ is symmetric. If $\alpha, \alpha' > \frac{1}{2}$, then $\kappa < \frac{1}{2}$ because $|S| \leq |V(H)|/2$, and the first bracket is at least $\frac{1}{4}$ if $\kappa \leq \frac{1}{4}$ and at least $h_1(1-\alpha) + h_1(\alpha - \kappa) \geq h_1/2$ otherwise; again $|\partial S|/n \geq h_1 m/4$. In all cases $|\partial S|/|S| \geq \eta_0$.

$G_{L,n}$ arises from H by subdividing the hub–core edges F into paths of length L. Each edge of F has a core end that lies on no other edge of F, so $e_F(S_0) \leq |S_0|$ and $\nu \leq 1$. Lemma 11.10 gives $h(G_{L,n}) \geq h(H)/(1 + (L-1)(1+h(H))) \geq h(H)/(L(1+h(H)))$. The function $x/(1+x)$ increases and $\eta_0 \leq \frac{1}{4}$, so $h(G_{L,n}) \geq \eta_0/((1+\eta_0)L) \geq 4\eta_0/(5L)$. □

Corollary 11.12 (star bits are not polynomial in h)**.** *Let $\kappa_0 := 2h_1 m/(5(a+b+1))$ and $h := h(G_{L,n})$. On $G_{L,n}$ with the split $A|B$, a bound $\beta^2 \leq e^{-cN}$ proved by star bits has $c \leq \frac{8}{3(a+b)}(h/\kappa_0)^{m+1}$, and one proved by Theorem 4.5 has $c \leq \frac{8\ln 2}{a+b}(h/\kappa_0)^{a+b+1}$. For $a = b = D/2$ these are $O_D(h^{D/2+1})$ and $O_D(h^{D+1})$, while $h \to 0$ as $L \to \infty$, uniformly in n.*

Proof. By Proposition 11.11, $L \geq \kappa_0/h$, so $L^{-k} \leq (h/\kappa_0)^k$, and $1/(L-1) \leq 2/L \leq 2h/\kappa_0$. Insert into Proposition 11.9. □

So star bits cannot give $c(D,h) \geq \mathrm{poly}(h)$ with a degree independent of D, even on 2-connected, balanced, bounded-degree graphs, and the exponent $5D+4$ of Theorem 4.13 is within a factor of about 5 of what Theorem 4.5 can give. This is a limit of the bits, not of β. The cycle bits do worse on these graphs. A mixed cycle must change colour at two hubs, and at each it traverses a whole path of each colour, so it has at least $4(L-1)$ vertices of degree 2, and the quantity q_k of Theorem 6.6 is at most $2^{-4(L-1)}$; that theorem gives at most $\exp(-\Omega(1/h))$ here. Edge bits do better, as we show next.

11.5 Edge pairs on the hub graphs

In $G_{L,n}$ the two sides interact only at the hubs, and the first edges of an A-path and a B-path at a hub are negatively correlated under the uniform spanning tree. Their conditional law given a past is again the law of two edges under a uniform spanning tree, so Lemma 3.6(b) applies at every step. No forbidden cell is needed.

Lemma 11.13 (two edges at a vertex)**.** *Let K be a connected multigraph, $T \sim \mathrm{UST}(K)$, and let e, f be distinct edges with a common endpoint w and other endpoints u and v (possibly $u = v$). Put $q_e := \Pr[e \notin T]$, $q_f := \Pr[f \notin T]$ and*

$$Y := \tfrac{1}{2}\big(R_K(u,w) + R_K(v,w) - R_K(u,v)\big), \qquad R_K(u,u) := 0.$$

Then $Y \geq 0$, $\mathrm{Cov}(\mathbf{1}[e \in T], \mathbf{1}[f \in T]) = -Y^2$, and the law K_{ef} of the two indicators has $\Delta \geq 2Y^2$, and $\Delta \geq Y^2/(2\sqrt{q_e q_f})$ if $q_e q_f > 0$, with Δ as in Lemma 3.6.

Proof. Orient e from u to w and f from v to w. By the transfer-current theorem [BP93], $\Pr[e, f \in T] = \Pr[e \in T]\Pr[f \in T] - Y(e,f)Y(f,e)$, where $Y(e,f)$ is the current through f when a unit current enters at u and leaves at w, and $Y(f,e) = Y(e,f)$ for unit conductances [LP16, Ch. 4]. Let L_w be the Laplacian of K with the row and column of w deleted. The potential of that current with $\phi(w) = 0$ is $\phi = L_w^{-1}\mathbf{1}_u$, so $Y(e,f) = \phi(v) = (L_w^{-1})_{vu}$. This is nonnegative, because L_w^{-1} is the Green function of the random walk killed at w, and expanding $R_K(u,v) = (\mathbf{1}_u - \mathbf{1}_v)^\top L_w^{-1}(\mathbf{1}_u - \mathbf{1}_v)$ with $R_K(u,w) = (L_w^{-1})_{uu}$ and $R_K(v,w) = (L_w^{-1})_{vv}$ gives $(L_w^{-1})_{uv} = Y$. So the covariance is $-Y^2$, and Lemma 3.6(b) gives $\Delta \geq 2Y^2$. For the second bound, $K_{01} \leq q_e$, $K_{10} \leq q_f$, $K_{11} \leq 1$ and $K_{00} = q_e q_f - Y^2 \leq q_e q_f$, where K_{00} is the probability that neither edge is in T, so $\sqrt{K_{00}K_{11}} + \sqrt{K_{01}K_{10}} \leq 2\sqrt{q_e q_f}$. □

Lemma 11.14 (a resistance criterion)**.** *In the setting of Lemma 11.13, let ρ be the smaller of $R_{K-e}(u,w)$ and $R_{K-f}(v,w)$, let $R' := R_{K-w}(u,v)$, with the value ∞ for disconnected vertices. Then*

$$Y \;\geq\; \frac{\rho}{1+\rho} - \frac{R'}{2+R'} = \frac{2\rho - R'}{(1+\rho)(2+R')},$$

where $\infty/(c+\infty) := 1$.

Proof. By the parallel law, $R_K(u,w) = R_{K-e}(u,w)/(1 + R_{K-e}(u,w)) \geq \rho/(1+\rho)$, and likewise for v. Deleting the edges at w other than e and f does not decrease $R_K(u,v)$ by (E2), and leaves the network $K - w$ in parallel with the path u, w, v of resistance 2, so $R_K(u,v) \leq 2R'/(2+R')$. Insert into the definition of Y. □

Theorem 11.15 (edge-pair chain)**.** *Let G be a connected graph with a split $A|B$, and let $w_1, \dots, w_r$ be vertices, each with edges $e_k = u_k w_k \in A$ and $f_k = v_k w_k \in B$, where all $2r$ edges are distinct. For a pattern σ of in and out on $\{e_j, f_j : j < k\}$ with positive probability, let K_σ be the multigraph of Lemma 4.2, and let Y_σ and $q_{e,\sigma}, q_{f,\sigma}$ be the quantities of Lemma 11.13 for e_k, f_k in K_σ when neither is a loop there. Put $B_\sigma := 0$ if one of the two bits $\mathbf{1}[e_k \in T]$, $\mathbf{1}[f_k \in T]$ is constant under $\mathrm{UST}(K_\sigma)$, and otherwise $B_\sigma := Y_\sigma^2 \max\bigl(2,\, 1/(2\sqrt{q_{e,\sigma}q_{f,\sigma}})\bigr)$, and let $\Delta_k := \min_\sigma B_\sigma$. Then*

$$\beta\bigl(\mathrm{UST}(G); A|B\bigr)^2 \;\leq\; \prod_{k=1}^{r}(1-\Delta_k^2) \;\leq\; \exp\Bigl(-\sum_k \Delta_k^2\Bigr).$$

Proof. The bits $x_k := \mathbf{1}[e_k \in T]$ are functions of x_A, and $y_k := \mathbf{1}[f_k \in T]$ of x_B. By Lemma 3.2, β is at most the norm of the square root of the joint law of the bits, which Lemma 3.3 bounds by $\prod_k \gamma_k$. A joint past of positive probability is a pattern σ as in the statement, and given it, $T \setminus \{e_j, f_j : j < k\}$ is a uniform spanning tree of K_σ by Lemma 4.2. If one of the bits is constant given σ (for instance because e_k or f_k became a loop), the conditional law has norm at most 1 and $B_\sigma = 0$. Otherwise both marginals lie strictly between 0 and 1, so $q_{e,\sigma}q_{f,\sigma} > 0$, and Lemmas 11.13 and 3.6 bound the squared norm by $1 - B_\sigma^2$. So $\gamma_k^2 \leq 1 - \Delta_k^2$. □

No novelty is claimed for the ingredients: the theorem combines our chain rule with the transfer-current theorem and the spatial Markov property. On the hub graphs it gives a bound of degree 3 in $1/h$ for the gadget's own split.

Proposition 11.16 (edge pairs on hub graphs)**.** *Let $a = b \geq 2$, $k := 2a - 1$, $n \geq 13$ and $L \geq 3$, and let λ_A, λ_B be the Laplacian spectral gaps of the cores, with $L \geq 4k(1/\lambda_A + 1/\lambda_B)$. At each hub*

w let e_w and f_w be the first edges of one A-path and one B-path. Then for every hub, in every order, and for every past σ on the pairs of the other hubs,

$$Y_\sigma \;\ge\; \frac{1}{3(a+b)L}, \qquad q_{e,\sigma}, q_{f,\sigma} < \frac{1}{L},$$

and

$$\beta\big(\mathrm{UST}(G_{L,n}); A|B\big)^2 \;\le\; e^{-c_{\mathrm{split}} N}, \qquad c_{\mathrm{split}} \;\ge\; \frac{1}{405(a+b)^5 L^3} \;\ge\; \frac{h^3}{10935\, D^5},$$

where $h = h(G_{L,n})$ and $D = a+b$. The hypothesis on L holds for $L \ge 48k/h_0^2$, since $\lambda_A, \lambda_B \ge h_0^2/6$ by Lemma 4.7.

Proof. Fix a hub w with $e = uw$ and $f = vw$, where u and v are the first interior vertices of the paths P_e to $x_u \in X_A$ and P_f to $y_v \in X_B$. The past only involves first edges at other hubs, so in K_σ neither e nor f is a loop, and w and the interiors of its paths are untouched. Apply Lemma 11.14 in K_σ.

The bound on ρ. In $K_\sigma - e$ the vertex u hangs from x_u by a path of $L-1$ edges, so $R_{K_\sigma - e}(u,w) = L - 1 + R_{K_\sigma - e}(x_u, w)$. Identify, for $1 \le j \le L-1$, the k vertices at distance j from w on the other k paths at w, and identify all vertices off these paths, other than w, with x_u. This does not increase the resistance by (E2), and leaves L bundles of k parallel edges in series. So $R_{K_\sigma - e}(x_u, w) \ge L/k$, and the same holds for v. Hence $\rho \ge L - 1 + L/k$, and $q_e = 1/(1 + R_{K_\sigma - e}(u,w)) < 1/L$ by Lemma 4.1, and likewise for q_f.

The bound on R'. Build a unit flow from u to v in $K_\sigma - w$: along P_e to x_u, with energy $L-1$; inside X_A from x_u to the uniform distribution on the core ends of the $(n-1)(a-1)$ A-paths of the other hubs that carry no edge of a pair, with energy at most $2/\lambda_A$ by (E6); along these paths to their hubs, with energy $L/((n-1)(a-1))$; from each of these hubs out along its $b-1$ free B-paths, with energy $L/((n-1)(b-1))$; inside X_B to y_v, with energy at most $2/\lambda_B$; and along P_f to v. The past deletes or contracts only first edges of pair paths at other hubs. The flow uses none of them, and contracting an edge merges two vertices at which the flow is conserved, so it is still a unit flow from u to v in $K_\sigma - w$, and by (E5), $R' \le 2(L-1) + R_{\mathrm{mid}}$ with

$$R_{\mathrm{mid}} := \frac{2}{\lambda_A} + \frac{2}{\lambda_B} + \frac{L}{n-1}\Big(\frac{1}{a-1} + \frac{1}{b-1}\Big) \;\le\; \frac{L}{2k} + \frac{2L}{(n-1)(a-1)} \;\le\; \frac{L}{k},$$

using the hypothesis on L and $4k \le 12(a-1) \le (n-1)(a-1)$.

Combination. The bound of Lemma 11.14 increases in ρ and decreases in R', so

$$Y_\sigma \;\ge\; \frac{2L/k - R_{\mathrm{mid}}}{(L + L/k)(2L + R_{\mathrm{mid}})} \;\ge\; \frac{L/k}{(L+L/k)(2L+L/k)} = \frac{k}{(k+1)(2k+1)L} \;\ge\; \frac{1}{3(a+b)L}.$$

Lemma 11.13 with $q_e q_f < L^{-2}$ gives $B_\sigma \ge Y_\sigma^2 L/2 \ge 1/(18(a+b)^2 L)$, and Theorem 11.15 over the n hubs gives $\beta^2 \le \exp(-n/(324(a+b)^4 L^2))$. Finally $N = n(1 + (a+b)L) \le \frac{5}{4} n(a+b)L$, and $L \le 3/h$ since $h \le 2/(L-1) \le 3/L$ for $L \ge 3$. □

On the hub graphs with $a = b = D/2$ the three certificates give, for this split, exponents of order $h^{D/2+1}$ (star bits, Corollary 11.12), $\exp(-\Omega(1/h))$ (cycle bits) and h^3/D^5 (edge pairs). Proposition 11.16 covers only the gadget's own split, and only L above a threshold that depends on the cores; it says nothing about other balanced splits of these graphs. Whether edge pairs give $c \ge 1/\operatorname{poly}(D, 1/h)$ for every balanced split of every 2-connected expander is Conjecture 13.1 in Section 13. The numbers are not small. For $a = b = 2$ and cubic cores with $h_0 \approx 0.1$, the hypothesis needs L in the tens of thousands.

11.6 What is and is not used

The local argument uses three properties of the uniform spanning tree: a forbidden local pattern (no cycle), the spatial Markov property in the form of Lemma 6.1, and the fact that, given everything outside the edges incident to a short cycle, each path obtained by deleting one edge of the cycle keeps probability bounded below. The vertex-touch argument uses the dual forbidden pattern (no isolated vertex), the spatial Markov property (Lemma 4.2), and Kirchhoff's formula $\Pr[e \in T] = R(e^+, e^-)$ for a single edge, with Rayleigh monotonicity. Kirchhoff's formula is the one-edge case of the transfer-current theorem of Burton and Pemantle [BP93], which makes the edge set of the uniform spanning tree a determinantal process. Neither argument uses the determinantal structure beyond this one-edge case, and the determinant itself appears only in the upper bound. The transfer-current theorem is used beyond the one-edge case in two places: through Hadamard's inequality in Lemma 5.7, and for two edges in Section 11.5.

We do not know how to run this argument for uniform perfect matchings. There, conditioning on the outside of a small region can force the matching inside it. For example, in the grid, if the outside leaves unmatched exactly two vertices next to a unit square, and these are adjacent to two neighbouring corners of the square, the completion is unique. The dimer paper [Bla26] used a global argument with kernel estimates instead. Its bound $e^{\Theta(\sqrt{n})}$ for n edges is $e^{\Theta(L)}$ on the $L \times L$ grid, the same order as Corollary 8.5.

Remark 11.17 (weighted spanning trees). Let $P_w(T) \propto \prod_{e \in T} w_e$ with $w_e \in [w_{\min}, w_{\max}]$, $w_{\min} > 0$. In the proof of Lemma 6.2, the numerator is at least $w_{\min}^{|C|-1} \prod_j W_j$, where W_j is the total weight of the edges at K_j, and the parent-edge encoding bounds the denominator by $\prod_j W_j \prod_{y \neq x_0} \sum_{e \ni y} w_e$. So the local bound becomes $(w_{\min}/(D w_{\max}))^{|C|-1}$. In Lemma 6.4, Kirchhoff's formula reads $\Pr[e \in T] = w_e R(e^+, e^-)$ and the indicator of v has energy $\sum_{f \ni v} w_f$, so the bound becomes $w_e / \sum_{f \ni v} w_f \geq w_{\min}/(D w_{\max})$. So Theorem 6.6 holds with $p_k := (w_{\min}/(D w_{\max}))^{|C_k|-1}$ and $w_k := (w_{\min}/(D w_{\max}))^{|C_k|}$, and Theorem 8.2 with $(w_{\min}/(D w_{\max}))^{\ell}$ in place of $D^{-\ell}$. The vertex-touch argument also extends. With conductances w_e, Kirchhoff's formula reads $\Pr[e \in T] = w_e R(e^+, e^-)$, Lemma 4.1 becomes $\Pr[e \notin T] = 1/(1 + w_e R')$, and the weighted resistances are at most $1/w_{\min}$ times the unweighted ones. The proof of Lemma 4.3 then gives $p_A(w) p_B(w) \geq (1 + (w_{\max}/w_{\min})(1 + \rho_w))^{-\deg(w)}$, and the weighted analogue of Theorem 4.5 holds with these bounds. We have not redone the constants of Theorem 4.13 in the weighted case.

11.7 Uniform spanning forests

Let $\mathrm{USF}(G)$ be the uniform distribution over *all* spanning forests of G, that is, all acyclic subsets of E, with any number of components. (We do not mean rooted forests, forests with a prescribed number of components, or the uniform spanning forest measures of infinite graphs.) It also forbids cycles, but Lemma 6.2 does not carry over unchanged. On a triangle there are seven forests, so for each edge e the probability that the forest is $E(C) \setminus \{e\}$ is $1/7$, below the value $1/4 = D^{-(|C|-1)}$ of Lemma 6.2. A different local bound holds.

Proposition 11.18 (forest local bound). *Let $\Phi \sim \mathrm{USF}(G)$, let C be a cycle, $e \in E(C)$ and $x_0 \in V(C)$, and let $F \subseteq E \setminus I(C)$ be a forest. Then*

$$\Pr\big[\, \Phi \cap E(C) = E(C) \setminus \{e\} \;\big|\; \Phi \setminus I(C) = F \,\big] \;\geq \prod_{y \in V(C) \setminus \{x_0\}} \frac{1}{\deg(y) + 1} \;\geq\; (D+1)^{-(|C|-1)}.$$

Consequently Theorem 6.6 holds for $\mathrm{USF}(G)$ with q_k replaced by $\big(\max_{x_0} \prod_{y \neq x_0} (\deg(y) + 1)^{-1}\big)^2$, and Theorem 8.2 holds for $\mathrm{USF}(G)$ with $(D+1)^{-2(\ell-1)}$ in place of $D^{-\ell}$.

Proof. Define $K_1, \dots, K_s$ and $\mathcal{M}_F$ as in Lemma 6.1, and let $d_j \geq 0$ be the degree of K_j in $\mathcal{M}_F$. For $J \subseteq I(C)$, the set $F \cup J$ is acyclic if and only if J is acyclic in $\mathcal{M}_F$. Indeed, a cycle of $F \cup J$ must use edges of J, since F is acyclic, and contracting the components of F maps it to a closed trail in $\mathcal{M}_F$ that uses each of its J-edges once, which contains a cycle. Conversely, a cycle of $\mathcal{M}_F$ lifts to a cycle of $F \cup J$ by joining consecutive edges through the connected components of F. Since Φ is uniform, the conditional law of $\Phi \cap I(C)$ is uniform over the forests of $\mathcal{M}_F$.

Numerator. Take the path $E(C) \setminus \{e\}$, and for each K_j either no edge or one of its d_j edges. Each such set is a forest, so there are at least $\prod_j (d_j + 1)$ forests with $J \cap E(C) = E(C) \setminus \{e\}$.

Denominator. Root each component of a forest J of $\mathcal{M}_F$ at x_0 if it contains x_0, and at any fixed node otherwise. Record for each node other than x_0 its parent edge, or the symbol "root". The forest J is the set of recorded edges, so the encoding is injective. The node K_j has at most $d_j + 1$ possible records and $y \in V(C) \setminus \{x_0\}$ has at most $\deg(y) + 1$. So there are at most $\prod_j (d_j + 1) \prod_{y \neq x_0} (\deg(y) + 1)$ forests.

The part of the proof of Theorem 6.6 that uses Lemma 6.2 uses only that Φ contains no cycle, that the past is a function of $\Phi \setminus I(C_k)$, and the local bound. The star bound of Lemma 6.4, and with it w_k, does not carry over: a forest need not meet a star. □

This bound has been checked exactly on small graphs (Appendix A). It gives a $2^{\Omega(N)}$ lower bound for approximating $\mathrm{USF}(G)$ on prisms and squares of expanders, but this is not a separation. Counting spanning forests is #P-hard for general graphs [JVW90], and we know of no polynomial-size generating circuit for USF on these families. The vertex-touch argument of Section 4 does not apply to USF: a spanning forest can leave a vertex isolated, so $a_w = b_w = 1$ is allowed.

11.8 Matroids: balance and locality

Two hypotheses of Section 9 cannot simply be dropped.

Balance. Lemma 9.10 needs negative correlation in minors, and it can fail without it. Let S_8 be the binary matroid represented over $\mathbb{F}_2$ by $[I_4 \,|\, Q]$, where Q has rows 0111, 1011, 1101, 1111. It has 48 bases. For e the fourth unit vector and f the all-ones column, 28 bases contain e, 20 contain f and 12 contain both, and $12 \cdot 48 = 576 > 560 = 28 \cdot 20$. So e and f are positively correlated and S_8 is not balanced, in line with the theorem of Choe and Wagner [CW03] that a binary matroid is balanced exactly when it has no S_8 minor. Huh, Schröter and Wang [HSW22] show that positive correlation between two elements of a matroid is bounded by a constant factor, but it does occur. An exhaustive check over all $|F|, |S| \leq 2$ in S_8 finds 4 of 1624 cases in which one of the two inequalities of Lemma 9.10 fails, the worst with ratio 14/15 between the two sides (Appendix A). We do not know whether Theorem 9.11 itself fails for some matroid that is not balanced. The local Theorems 9.7 and 9.8 hold for every matroid.

Locality. Theorem 9.8 needs pieces with small frames that interfere with few other pieces, and matroid connectivity does not provide them. In the uniform matroid $U_{k,2k}$, every k of the $2k$ elements form a basis. It is connected, since any two elements lie on a common circuit of size $k+1$. Each frame $E \setminus \mathrm{cl}(K \setminus \{e, \pi(e)\})$ of a circuit K has $k+1$ elements, so every territory is E. For a split $X|Y$, the law of $(B \cap X, B \cap Y)$ is uniform on pairs with $|B \cap X| + |B \cap Y| = k$, so $\sqrt{\mu}$ is block diagonal with constant blocks indexed by $j = |B \cap X|$, and

$$\beta(U(U_{k,2k}); X|Y)^2 \;=\; \max_j \Pr[|B \cap X| = j] \;\geq\; \frac{1}{k+1}.$$

For $k = 3, \dots, 7$ and $|X| = k$ the value is between 0.35 and 0.52. Indeed $U(U_{k,2k})$ has a deterministic smooth decomposable PC with $O(k^2)$ nodes, which reads the elements in order and tracks

how many have been chosen. So the hypotheses of Theorem 9.8 are not implied by connectivity.

12 Related work

Our literature search was targeted, not exhaustive. The comparisons below refer to the statements we checked in the cited sources.

12.1 Spanning trees and determinantal distributions in probabilistic circuits

Martens and Medabalimi [MM14] prove that the uniform distribution on spanning trees of the complete graph K_m cannot be represented by decomposable and complete (smooth) sum-product networks of size below $2^{m/30240}$ for $m \geq 20$, and the same holds for approximation "arbitrarily well" by networks of bounded size (their Theorem 37). Their proof writes the network as a sum of s^2 "weak" functions and shows that each is nonzero on only an exponentially small fraction of the trees; it is a support argument. It does not give a bound at a fixed total-variation error, and the exponent is the number of vertices, which is the square root of the number of variables. Zhang, Holtzen and Van den Broeck [ZHdB20] restate this result (their Theorem 5), observe that the uniform spanning tree of K_n is a determinantal point process (Theorem 4), and prove that some determinantal point processes have no polynomial-size deterministic PC. Zhang, Juba and Van den Broeck [ZJdB21] introduce probabilistic generating circuits, which allow subtraction, show that they subsume determinantal point processes, and cite the two results above for the corresponding separation from decomposable PCs. Agarwal and Bläser [AB24] and Broadrick, Zhang and Van den Broeck [BZdB24] study the power of such circuits further.

Theorem 1.1 instead gives a fixed-error bound without assuming determinism or exact representation. It holds for every 2-connected bounded-degree expander, with an exponent linear in the number of variables. For K_n itself, Theorem 5.1 gives a bound at every fixed total-variation error below one, for circuits whose leaves have arity at most $2M/3$, with an exponent of the same order $\Theta(n) = \Theta(\sqrt{M})$ as theirs. The constants are not directly comparable, since the settings differ, but for the record their bound is $e^{m \ln 2/30240} \approx e^{m/43600}$ and ours is $e^{n/146}$ up to the factor $\delta^2/(e^3\sqrt{n})$. Their theorem does not give a bound at a fixed error, and ours does not cover leaves of large arity. Proposition 5.11 shows that this order is optimal. For dense graphs (Corollary 5.9) we found no earlier circuit lower bound for the uniform spanning tree.

12.2 Constant-error lower bounds for probabilistic circuits

Leland and Choi [LC25] prove that a *deterministic* decomposable PC that approximates a certain distribution within total variation $\varepsilon \leq \frac{1}{16} - \Omega(1/\mathrm{poly}(n^2))$ has size $2^{\Omega(n)}$ (their Theorem 4.1). Their input is an $n \times n$ array, so this is $2^{\Omega(\sqrt{N})}$ in the number $N = n^2$ of variables. The target is built from Sauerhoff's function [Sau03] and is itself represented by a polynomial-size nondeterministic decomposable PC. Their proof prunes inputs of small probability, which uses determinism, and reduces to weak approximation by d-DNNFs. Our lower bound does not assume determinism, and our target has a polynomial-size generating circuit with subtraction rather than a small nonnegative one.

The dimer paper [Bla26] proves an $\exp(\Theta(\sqrt{n}))$ constant-error bound for uniform domino tilings of the square grid with n edges. Its Section 10.4 also notes a stronger generic separation: the uniform distribution on a suitable random linear code, which has easy inference by Gaussian

elimination, needs $\exp(\Omega(n))$ size at constant error, by the rank bounds of de Colnet and Mengel [dCM20]. That example is generic rather than natural, and its tractability comes from linear algebra over $\mathbb{F}_2$. De Colnet and Mengel study weak and strong approximation in knowledge compilation, where a Boolean function is approximated by another Boolean function. Their results do not give ours as a black-box corollary. A distribution within small total variation of P can have full support (take $(1-\delta)P + \delta U$ with U uniform), so approximating P says nothing about approximating its support as a set. Their robust transfer also uses covers by disjoint rectangles, which is why it targets deterministic circuits, while Lemma 3.8 allows overlapping terms with different splits.

Squared and subtractive circuits [LSM+24, WdB25] and sums of squares [LMV25] also go beyond nonnegative PCs. Loconte et al. [LSM+24] show that squared circuits with subtraction can be exponentially smaller than structured nonnegative ones for some distributions; Theorem 10.22 shows that for the uniform spanning tree, squared circuits with structured amplitudes satisfy the same exponential lower-bound expression as nonnegative circuits. Section 10 uses the rank method of de Colnet and Mengel [dCM21], who use it, with induced matchings in expanders, for exponential lower bounds on structured-decomposable arithmetic circuits that may use negative constants; our contribution there is the permutation-block inequality that makes the rank bound robust to approximation (Theorem 10.10), at every error below 1 for normalized outputs, and its pendant-vertex form for all 2-connected bounded-degree expanders (Corollary 10.21). The rank bound behind Lemma 10.9 is due to Ky Fan and Hoffman [FH53]. We found no earlier lower bound of this kind for approximation by circuits with subtraction; the nearest results we know are the monotone bounds for spanning-tree polynomials discussed below and the work on circuits with negative weights cited above [AB24, BZdB24]. We did not search the literature on border complexity, which studies approximation by limits of small circuits, and a related result may exist there.

12.3 Multilinear circuits and formulas

Raz [Raz09] proved that multilinear formulas for the permanent and the determinant of a $t \times t$ matrix have size $t^{\Omega(\log t)}$, and Raz and Yehudayoff [RY09] proved $2^{t^{\Omega(1/d)}}$ for multilinear circuits of product-depth d. Both use random restrictions that pair variables in common rows or columns and a partial-derivative rank. Section 10.6 transfers these arguments to the indicator of spanning trees of any graph with a $K_{t,t}$ minor. Bounded-degree expanders have complete minors of order $\Omega(\sqrt{N})$ by Krivelevich and Nenadov [KN21] (see [Kri19, Section 8]). For general syntactically multilinear circuits computing an explicit polynomial, the best lower bound we know is the $\Omega(n^2/\log^2 n)$ of Alon, Kumar and Volk [AKV18]. Sums of structured objects with different orders have been studied for set-multilinear branching programs by Chatterjee, Kush, Saraf and Shpilka [CKSS24]; Theorem 10.30 is a simpler statement of this kind for the spanning-tree indicator, through a frontier rank lemma and a split chosen adaptively for all vtrees. Theorem 10.37 replaces the complete minor by a packing of pendant gadgets, which gives a linear exponent and an approximate version, at the cost of astronomically small constants.

12.4 Knowledge compilation and communication complexity

Bova, Capelli, Mengel and Slivovsky [BCMS16, Thm. 6] show that a DNNF of size s yields a cover of its models by at most s balanced rectangles, and use multi-partition communication complexity [ĎHJ+04] to prove DNNF lower bounds. Sauerhoff [Sau03] studies approximation of

Boolean functions by such rectangles. Corollary 8.7 is a rectangle-cover argument of this kind, with the rectangle bound coming from Corollary 6.7.

The information-complexity method of Bar-Yossef, Jayram, Kumar and Sivakumar [BYJKS04] uses Hellinger distance, the Bhattacharyya coefficient for products of distributions, and the cut-and-paste property of protocols. See also later work such as Jayram's [Jay09]. Braun and Pokutta [BP16] use Hellinger-type arguments for robust nonnegative-rank bounds for unique disjointness. Our chain rule (Lemma 3.3) is a case of Exercise I.42(b) of Polyanskiy and Wu [PW25], and the identification of the optimized affinity with a matrix norm is Lemma 6 of Lapidoth and Pfister [LP19]. The forbidden outcomes $u_i = v_i = 1$ of our cycle bits and $a_w = b_w = 1$ of our vertex-touch bits have the form of set disjointness, which connects the argument to the Hellinger methods for disjointness and AND [BYJKS04, Jay09]. We did not find the combination with a spatial Markov property that we use here. The argument is short, however, and similar arguments may appear elsewhere.

12.5 Monotone arithmetic circuits

Jerrum and Snir [JS82] proved $2^{\Omega(n)}$ monotone lower bounds for the directed spanning-tree polynomial of the complete graph, which has $\Theta(n^2)$ variables. Chattopadhyay, Datta, Ghosal and Mukhopadhyay [CDGM22, Thm. 1.1] prove that for a sufficiently large constant d and every d-regular expander with second eigenvalue at most $d^{1-\epsilon}$, every monotone circuit for the directed spanning-tree polynomial (one variable per arc, trees directed towards a root) has size $2^{\Omega(n)}$. The same theorem, with the same proof idea, appears in the earlier report of Chattopadhyay, Datta and Mukhopadhyay [CDM20, Thm. 1.1]. This is the closest exact analogue of our result that we found. It differs in three ways: it is exact rather than approximate; it concerns the directed polynomial, from which the undirected one is obtained by identifying the two arc variables of each edge (so lower bounds do not transfer from directed to undirected); and it needs large degree and a strong spectral condition, while our graphs have any fixed degree but must be 2-connected. Related monotone lower bounds via communication complexity are in [CDM21]. Since the undirected polynomial is a substitution instance of the directed one, a monotone lower bound for the undirected polynomial implies one for the directed polynomial of the same graph, but not conversely. So Corollary 8.7(b) also gives directed bounds for our families. For the undirected polynomial, Jukna and Seiwert [JS19, Thm. 2] prove a $2^{\Omega(\sqrt{n})}$ monotone arithmetic lower bound on the complete graph K_n, and the exact uniform-spanning-tree lower bound of Martens and Medabalimi [MM14, Thm. 37] also transfers to monotone circuits for the complete-graph polynomial, since replacing z_e by the leaf $[x_e = 1]$ and padding with $[x_e = 0]$ turns such a circuit into a smooth decomposable PC with polynomial overhead. So exponential exact lower bounds for undirected spanning-tree polynomials are not new in general. What Corollary 8.7(b) adds is an exponent linear in the number of variables for the undirected polynomial of every 2-connected bounded-degree expander, including cubic ones, with a very small constant. A targeted search found no earlier result of this form, but we cannot rule one out.

12.6 Matroids

Our matroid terminology and the standard facts we use are from Oxley [Oxl11]. Feder and Mihail [FM92] introduced balanced matroids and proved that regular matroids are balanced; Section 9 uses only the negative correlation of pairs in minors. Choe and Wagner [CW03] characterize the balanced binary matroids as those with no S_8 minor, and Huh, Schröter and Wang [HSW22] bound the correlations in arbitrary matroids. We did not find the exchange bound of Lemma 9.3

or the resistance formula of Lemma 9.12 stated for matroids. Both are short consequences of standard facts (basis exchange and orthogonality, and the Cauchy–Binet formula), and we expect them to be known. On the side of circuit lower bounds, Cavalar, Fabris, Mukhopadhyay, Srinivasan and Yehudayoff [CFM+25] construct an explicit matroid whose associated monotone Boolean function needs large monotone Boolean circuits, which answers an open problem of Jukna and Seiwert (see [CFM+25]). That result concerns monotone Boolean circuits for a function defined by a matroid, not distributions on bases; we found no lower bound for probabilistic circuits representing uniform bases of matroids beyond the graphic case.

12.7 Product approximations of uniform spanning trees

Mészáros and Peled [MP26] prove that for a graph of minimum degree $k \geq 2$ there is a coupling of the uniform spanning tree with the random 1-out subgraph (each vertex picks one incident edge independently) whose expected number of differing edges is $O(n \log(k)/\sqrt{k})$. They prove analogous couplings for a wide class of determinantal processes. This does not conflict with our result, though the 1-out subgraph is close to a product distribution. Their bound is on the expected number of edits, a Wasserstein-type distance, not on total variation; a small expected edit distance per vertex is compatible with total variation close to one. The 1-out law is a product over vertices, not over a split of the edges, since two vertices can choose the same edge. And at fixed degree the bound is a fixed fraction of n, which is small only when k is large. Our result is for fixed degree and total variation.

12.8 Expanders

For background on expanders and explicit constructions see Hoory, Linial and Wigderson [HLW06] and Lubotzky, Phillips and Sarnak [LPS88]. Our geometric lemma (Lemma 7.3) is a giant-component argument of a standard kind. Lemma 4.7 is Mohar's Cheeger-type inequality [Moh89], and Lemma 4.4 combines Thomson's principle with the standard spectral bound on effective resistance. The count of connected sets in Lemma 4.9 is the binomial bound of Fomin and Villanger [FV12], and the random deletion of Lemma 4.10 is the alteration method combined with a Peierls-type sum over connected sets. Bisection widths of regular graphs, which bound h from above, are studied by Monien and Preis [MP06] and, for random cubic graphs, by Lichev and Mitsche [LM23]. For random regular graphs we use the connectivity result of Wormald [Wor81a] and the expansion bound of Bollobás [Bol88].

13 Open questions

1. *Large girth (answered).* An earlier version of this paper asked whether the uniform spanning tree of a bounded-degree expander of girth $\Omega(\log N)$, such as the Ramanujan graphs of Lubotzky, Phillips and Sarnak [LPS88], needs $2^{\Omega(N)}$-size nonnegative decomposable PCs at constant total-variation error. Theorem 1.1 answers this for every such graph that is 2-connected, which includes the Ramanujan graphs of [LPS88] and random d-regular graphs with probability tending to one (Corollary 8.1). For bounded-degree expanders that are not 2-connected, Theorem 11.4 describes what balanced splits can give in terms of the largest block. It remains open whether 2-connected bounded-degree graphs with much weaker expansion, such as $h(G) \to 0$ slowly, still need superpolynomial size; the grid (Corollary 8.5) is one such case.

2. *Unbounded degree (partly answered).* An earlier version asked for a constant-error exponential lower bound for $\mathrm{UST}(K_n)$. Corollary 5.2 gives one, of the optimal order $2^{\Theta(n)}$ (Corollary 5.12), and Corollary 5.9 gives $2^{\Omega(n)}$ for every graph with $\lambda_2(L_G) \geq \kappa n$, with a rate of order $\kappa^4 e^{-1.14/\kappa}$. A minimum degree of $n/2$ alone does not make balanced splits weakly correlated (Proposition 5.10). Between bounded degree and a linear spectral gap we have no good bound. In particular, for expanders whose degree grows with N, and for products or powers of expanders, $\ln(1/c(D,h))$ grows like $D \log D$ even for fixed h, the spectral constant $c_\lambda(D)$ decays like D^{-4}, the cycle weights are $D^{-\ell}$, and Theorem 5.8 conditions on fewer than λ_2 vertices. How does the exponent depend on the degree? And does UST of the two cliques of Proposition 5.10 need circuits of size $2^{\Omega(n)}$ at every fixed error?

3. *Constants.* The constant in Example 8.4 is about $3 \cdot 10^{-7}$, $c(D,h)$ in Theorem 1.1 is $5.2 \cdot 10^{-41}$ in closed form and about $5 \cdot 10^{-24}$ with optimized parameters for $D = 3$ and $h = 1/5$ (Table 1), and $8.0 \cdot 10^{-8}$ for 6-regular Ramanujan graphs. For K_n, Theorem 5.1 gives $-\ln \beta^2 \geq n/146 - O(\log n)$, while exact computations for $n \leq 9$ give $-\ln \beta^2$ between $0.12n$ and $0.30n$ on the splits we tried, and a Monte Carlo upper estimate of the true rate is about $0.17n$ (Remark 5.6). Evaluated directly for particular splits, Theorem 4.5 gives exponents of order $10^{-3}N$ on random cubic graphs (Appendix A), so most of the loss is in the worst-case steps. What is the correct exponent for, say, cubic Ramanujan graphs, their prisms, the $L \times L$ grid, or K_n? Natural improvements are a sharper local lemma, using all mixed cycles or all mixed vertices rather than a sparse subfamily, typical rather than worst-case resistances, and a direct bound on $\|\sqrt{\mu}\|_{\mathrm{op}}$ for the joint law of all the bits. Is $c(D,h) \geq \mathrm{poly}(h)$ with a degree independent of D, or at least $c \geq 1/\mathrm{poly}(D, 1/h)$? Star bits cannot show either: on the hub graphs of Section 11.4, which are 2-connected with maximum degree D, they certify at most $O(h^{D/2+1})$ for one balanced split (Corollary 11.12). Edge pairs give $h^3/(10935D^5)$ for that split (Proposition 11.16), and Conjecture 13.1 below would give a polynomial constant for every split.

4. *Other models.* Does the bound extend to nonnegative circuits that are decomposable but not smooth, with a size measure independent of smoothing? For structured circuits Corollary 10.21 already does this, with a smaller exponent. To structured or deterministic variants with sharper constants? To positive semidefinite circuit models not covered by Theorem 10.22 [LMV25, WdB25]?

5. *Unstructured subtraction (partly answered).* Must a smooth decomposable circuit with real weights and univariate leaves, with no common vtree, that computes Q with $\frac{1}{2}\|\mathrm{UST}(G) - Q\|_1 \leq \varepsilon < \frac{1}{2}$ on a bounded-degree 2-connected expander have size $e^{\Omega(N)}$? Even superpolynomial? With a common vtree, Theorem 10.12 and Corollary 10.21 answer this for every error below 1 when Q sums to 1. Without one, Corollary 10.31 answers the support-exact question for bounded product-depth and for formulas, on expanders with large complete minors, and Theorem 10.37 answers the support-exact and the approximate question for sums of a bounded number of structured circuits, on every 2-connected bounded-degree expander. This leaves general circuits, where exact representation is as hard as lower bounds for syntactically multilinear circuits computing the spanning-tree polynomial (Proposition 10.23), approximate versions of Theorems 10.26 and 10.27, and planar families such as the grid. Section 10.8 shows why norm-type measures and fixed cycle blocks cannot give the approximate versions. Is there a support-exact lower bound for the grid? Can the constants of Theorem 10.37 be made reasonable? By Theorem 10.38, a small approximating circuit would have to cancel all but an exponentially small fraction of its absolute mass.

6. *Other determinantal distributions.* The local proof uses only a forbidden local pattern and a spatial Markov property, and the vertex-touch proof adds Kirchhoff's formula for one edge. Section 9 carries both over to uniform bases of matroids. Which other distributions with determinant-based inference, such as rooted spanning forests, whose weighted count is $\det(L + qI)$, admit the same argument on expander-like families?

7. *Matroids.* Does Theorem 9.11, or some global version of it, hold for every matroid? Lemma 9.10 fails for S_8 (Section 11), but we do not know whether the theorem fails. For regular matroids, is there an analogue of Lemmas 4.4 and 4.10: a condition playing the role of expansion that keeps the resistances $R_Z(e)$ of Lemma 9.12 bounded after a sparse set is deleted? Which natural matroid families, beyond graphs, their duals and the Fano-decorated family, satisfy the hypotheses of Theorem 9.8?

The edge-pair chain of Theorem 11.15 suggests one route to polynomial constants. For a vertex w with an edge $e = uw \in A$ and an edge $f = vw \in B$, and a pattern σ on earlier pairs, let $Y_\sigma(e, f)$ be the quantity of Lemma 11.13 in the graph K_σ of Theorem 11.15.

Conjecture 13.1. *There is a polynomial P such that for every 2-connected graph G with maximum degree D and $h(G) \geq h > 0$, and every balanced split $A|B$, there are vertices $w_1, \dots, w_r$ with $r \geq N/P(D, 1/h)$ and edges $e_k \in A$ and $f_k \in B$ at w_k, all $2r$ edges distinct, such that, for every k and every pattern σ on the earlier pairs with positive probability, neither e_k nor f_k is a loop of K_σ and $Y_\sigma(e_k, f_k) \geq 1/P(D, 1/h)$.*

By Theorem 11.15 and Lemma 11.13, Conjecture 13.1 would give $c \geq 1/\operatorname{poly}(D, 1/h)$ for all 2-connected bounded-degree expanders. The 2-connectivity is needed. Attach a triangle at each vertex of a cubic expander H on n vertices, with two new vertices per triangle. The result has $N = 3n$, $M = 9n/2$, maximum degree 5 and $h(G) \geq \min(h(H)/3, 1)$, and the split $A = E(H)$, $B =$ the triangle edges is balanced. The uniform spanning tree is an independent product of a spanning tree of H and one spanning tree of each triangle, so $\beta(\mathrm{UST}(G); A|B) = 1$ and every Y between an edge of A and an edge of B vanishes. For random regular graphs with $D \in \{3, 4\}$ and $N \leq 1000$, sampled values of Y_σ on spread-out mixed vertices stayed within a factor 0.81 of the tree-like value $(D-2)/(D(D-1))$; this is numerical evidence for fixed graphs only, and we have no proof.

A Machine checks

None of the proofs depends on these computations. They test the lemmas on small graphs, recompute the constants in exact arithmetic, and show how loose the bounds are. Small graphs say nothing about rates in N, and the computations are not evidence for the asymptotic statements.

The scripts are in the `verify/` folder of `https://github.com/chrisb4096-alt/expander-spanning-trees`, with their output in `verify/out/`. The command `verify/run_all.sh` compiles the C++ helper with `g++`, runs every script with `uv run`, saves the output, and stops at the first failure. Every script exits with a nonzero status if a checked inequality fails, so a run that prints `ALL CHECKS PASSED` has checked everything listed below. The Python scripts use NumPy, SciPy and NetworkX where noted, and otherwise only the standard library.

`constants.py` The constants of Corollary 8.3 and Example 8.4, and the vertex-touch constant of the previous version of this paper (the column "previous" of Table 1), in exact rational arithmetic. For the previous constant at $D = 3$ and $h = 1/5$ it checks $s = 61$ and $q =$

10802^{-3} and prints $8.66 \cdot 10^{-44}$; it also prints it for four other pairs. For the cubic prism with $h(H) \geq 1/5$ it checks $t \geq N/960$ (the other term is $N/732$), $r_0 \geq N/11520$ and $r_0 D^{-\ell} \geq N/2949120$ with $D^{-\ell} = 1/256$, and it prints the per-cycle factors 0.9980449639 and 0.9980372592, the exponent $3.397482 \cdot 10^{-7}$, and the closed form $1/4792320$ of Corollary 8.3 at these parameters. It checks that the closed forms of Corollary 8.3 lie below the exact value of $r_0 D^{-\ell}/N$ at 77 pairs (Δ, h_0) with $2 \leq \Delta \leq 12$.

`check_tensorization.py` Lemmas 3.3 and 3.5 (NumPy). It builds 6000 random laws on $(\{0,1\}^2)^r$ by the chain rule, with conditionals that depend on the whole past, have no mass on $(1,1)$ and mass at least p on $(0,1)$ and $(1,0)$, including the extreme points of this polytope. It checks $\|\sqrt{\mu}\|_{\mathrm{op}} \leq \prod_k \gamma_k \leq ((1+\sqrt{1-4p^2})/2)^{r/2} \leq (1-p^2)^{r/2}$ up to floating-point tolerance 10^{-12}.

`check_chain.py` and `enum_trees.cpp` The whole chain of Theorem 6.6 on five graphs: the prisms over K_4, C_6 and $K_{3,3}$, the square of C_{10}, and the 4×4 grid, with 3456 to 405000 spanning trees, enumerated by the C++ helper (NumPy, SciPy). For each graph it first checks Lemma 6.2 for every cycle of length at most 4, every edge on it and every outside configuration; the smallest ratio of the true conditional probability to the bound is between 1.92 and 2.67. It then takes five balanced splits per graph, three random and two found by local search to make β large, and checks

$$\beta(\mathrm{UST}(G); A|B) \leq \|\sqrt{\mu}\|_{\mathrm{op}} \leq \prod_k \gamma_k \leq \prod_k \Big(\frac{1+\sqrt{1-4q_k}}{2}\Big)^{1/2} \leq \prod_k (1-q_k)^{1/2},$$
$$q_k = \max(p_k^2, w_k),$$

for a greedily chosen family of disjoint mixed cycles. For every cycle and every past it also checks the two frame bounds behind Lemma 6.5: the conditional probabilities of $(1,0)$ and $(0,1)$ are at least $\max(p_k, w_k^A)$ and $\max(p_k, w_k^B)$, where w_k^A is the product of $\deg(v)^{-1}$ over the heads v of the A-edges of C_k in a fixed cyclic orientation, and likewise for B, so that $w_k^A w_k^B = w_k$. On these 25 instances the geometric mean of the γ_k is between 0.94 and 0.99, while the proved bound per cycle is between 0.992 and 0.998.

`check_forest.py` Lemma 6.2 and Proposition 11.18 in exact integer arithmetic on eight small graphs with 3 to 10 vertices, for every cycle of length at most 4, every edge on it and every outside configuration. It also checks the triangle: 7 forests, conditional probability $1/7$, below the tree bound $1/4$ and above the forest bound $1/9$.

`independent_check.py` A separate implementation written for review (NumPy, NetworkX), sharing no code with the other scripts. It checks the chain inequality on 2805 cases (405 exhaustive depth-two extreme-point assignments and 2400 random chains); Lemma 6.2 for cycles of length at most 4 on all connected graphs with 3 to 5 vertices and four further graphs (3858 conditional inequalities, smallest ratio to the bound 4/3); the chain of Theorem 6.6 for single cycles and ordered pairs of disjoint cycles on 2894 balanced splits; and the geometric facts behind Lemmas 7.3, 7.4 and 7.5, namely the expansion, linking and κ bounds and the count of mixed cycles, for the prisms and squares of all connected graphs with 3 or 4 vertices and of C_5 (18 graphs, 103878 balanced splits). Its line about the forest counterexample records a known value and is not a computation; `check_forest.py` computes it.

`check_blocks.py` The block results and the electrical lemmas, by exhaustive enumeration (NumPy, NetworkX). For six small graphs with 6 to 12 edges and nontrivial block structure

(K_4 with a pendant edge or a pendant path, two triangles or two copies of K_4 sharing a vertex, a 4-cycle and a triangle sharing a vertex, and the prism over K_3), it checks Proposition 11.2 on every split: $\beta = 1$ exactly when every block lies on one side, and a balanced split with $\beta = 1$ exists exactly when no block has more than $2M/3$ edges. For K_4 with a pendant edge the largest β over the 70 balanced splits is 0.75, attained by 40 of them, including the splits where A is a triangle of K_4. It checks the lower bound of Theorem 11.4(b) on the three graphs with $M_1 > 2M/3$, and Lemma 11.3 on eight graphs, with h computed exactly over all vertex subsets. For twelve 2-connected graphs with 5 to 18 vertices it checks, for every vertex w, the one-vertex deletion bounds of the previous version of this paper, $h(G-w) \geq h(G)/(2D)$ (smallest ratio 4) and $R_{G-w} \leq 16D^3/h(G)^2$, which the current proof no longer uses, and the inequality $h^2 \leq 2D\lambda_2$ of Lemma 4.7 for G and every $G - w$. Finally it checks Lemma 4.1 for every edge of four graphs.

`check_vertex_exact.py` The whole chain of Theorem 4.5 on nine 2-connected graphs: $K_{3,3}$, the cube, the Petersen, Heawood, Möbius–Kantor and Pappus graphs, two cubic graphs with 14 and 16 vertices and a 4-regular graph with 12 vertices, with 81 to 1259712 spanning trees (NumPy, SciPy, NetworkX). The last three are `networkx` random regular graphs, resampled until 2-connected. For four balanced splits per graph (two random, one from a Fiedler bisection, one found by local search to make β large) and a greedy independent set W of mixed vertices, it checks

$$\beta(\mathrm{UST}(G); A|B) \leq \|\sqrt{\mu}\|_{\mathrm{op}} \leq \prod_k \gamma_k \leq \prod_k \Big(\frac{1+\sqrt{1-4p_A(w_k)p_B(w_k)}}{2}\Big)^{1/2},$$

Lemma 4.3 for every configuration on the stars of $W \setminus \{w\}$, and the inequality $p_A(w)p_B(w) \geq (2+\rho_w)^{-\deg(w)}$. On the 36 instances, β is between 0.25 and 0.73, $\prod_k \gamma_k$ between 0.90 and 0.98, and the proved bound between 0.96 and 1. On every graph the smallest ratio of a true conditional probability to the bound of Lemma 4.3 is 1 to four decimal places, so the lemma is essentially tight there. In 6 instances $G - W$ is disconnected, some $p_A(w)$ is 0 and the proved bound is the trivial value 1. In the other 30 it also checks Lemma 4.4: every true conditional probability is at least $(b/d)(1+\kappa)^{-a}$, with $\kappa = 1/\lambda_2(L_{G-W})$, and the resulting bound of Theorem 4.5 is above $\prod_k \gamma_k$. The bound on β is between 0.971 and 0.9999; the smallest ratio of a true probability to the bound is 1.0, attained on $K_{3,3}$, and the smallest ratio per instance is at most 2.83.

`check_vertex_medium.py` Lemma 4.3 and the exponent of Theorem 4.5 on larger graphs where trees cannot be enumerated: the Tutte–Coxeter graph ($N = 30$, girth 8), the Foster graph ($N = 90$, girth 10), the Tutte 12-cage ($N = 126$, girth 12), and two `networkx` random cubic graphs with 200 and 600 vertices, resampled until 2-connected (NumPy, NetworkX). For three balanced splits per graph (random, Fiedler bisection, a ball grown from a vertex), with W chosen greedily at pairwise distance at least 3, it samples 20 trees by Wilson's algorithm, conditions on the stars of $W \setminus \{w\}$ for a random $w \in W$, and computes the conditional probabilities of Lemma 4.3 exactly as ratios of determinants. All 600 comparisons hold, with smallest ratio 1.0006. The resulting bound $\beta \leq e^{-xN}$ for the split has x between $5.4 \cdot 10^{-4}$ and $1.6 \cdot 10^{-3}$. These values are floating-point evaluations for fifteen particular splits, not bounds over all splits, and the random graphs are not evidence about random regular graphs in general.

`check_deletion.py` The lemmas of Section 4.5 (NumPy, NetworkX). It checks Lemma 4.7 on 152 connected graphs (complete graphs, a path, a cycle, K_4 minus an edge, the named graphs

above and all their one-vertex deletions), with h computed exactly; the smallest ratio of λ_2 to $\mu(\Delta, h)$ is 1.5. It also prints the triangle, for which the inequality $h \le \sqrt{\lambda_2(2\Delta - \lambda_2)}$ fails. It checks Lemma 4.9 by listing every connected set that contains a fixed vertex in seven graphs (29 to 10649 sets); the largest ratio of the number of sets with given k and j to $\binom{k-1+j}{j}$ is exactly 1. It checks Lemma 4.8 for greedy 3-separated sets in nine graphs with exact h, and its spectral part on a random 6-regular and a random 8-regular graph with 60 vertices (largest drop of λ_2: 0.25 and 0.02). The spectral part is tight on $K_{3,3}$ and the cube. Finally it checks the deterministic step of Lemma 4.10, $h(G - R^*) \ge h/(D+1)$, for 30 random sets R on each of eight graphs with h computed exactly. On graphs this small every threshold $m(b)$ is 2, so only the case of one neighbour in R, the case that uses 2-connectivity, occurs.

`constants_touch.py` The constants of Theorem 4.13 in floating point (standard library only). It checks that the closed form $c(D, h)$ is at most one third of the bound of Proposition 4.12(i) at the parameters of the proof, for $2 \le D \le 300$ and 400 values of h, and that $\mathcal{S}(q_0) \le 1/3$ for $D \in \{2, 3, 4, 6, 10, 32\}$ and five values of h between 0.01 and 1; the largest value found is $2.7 \cdot 10^{-7}$. It prints Table 1 and the value $c \ge 8.0 \cdot 10^{-8}$ for 6-regular Ramanujan graphs. At the Ramanujan bound $\lambda = D - 2\sqrt{D-1}$ it checks $c_\lambda(D) \ge 2.7 \cdot 10^{-9}$ for $6 \le D \le 64$ (the minimum, $2.79 \cdot 10^{-9}$, is at $D = 64$), and it prints $D^4 c_\lambda(D)$, which lies between 0.047 and 0.062 for $64 \le D \le 200$.

`check_complete.py` Section 5 (NumPy, SciPy, NetworkX, mpmath). In interval arithmetic it checks the constants of Theorem 5.1: $K = 0.37283\ldots$, $R \in [0.0069032712, 0.0069032713]$, so $1/R = 144.86$ and $R > 0.0069032 > 1/146$, $C_0 \le 2.007515 < 2.01$, and $|\Lambda'| \le 1.094$ where the proof needs it; and the numbers of Remark 5.6: the ceiling 0.007031 (so the theorem reaches 98% of it), $g(1/(4e)) = 0.1081$ and the quasi-star exponent 0.01357. It evaluates $\Phi_n(\lfloor 0.38n \rfloor)$ exactly for $3 \le n \le 300$ and the bound used in the proof for $3 \le n \le 10^6$; the smallest slack against $0.0069032\, n - \frac{1}{2}\ln n - 2.01$ is 2.56 and 1.08, at $n = 3$ and $n = 8$. It checks Lemma 5.5 (I1)–(I3) by enumerating all trees for $n = 7$ and $n = 8$ (largest error $2.2 \cdot 10^{-16}$), and (I2) by sampling at $n = 300$. It checks that Lemma 5.3 is sharp on K_n: for $n = 6, 7, 8$, three kinds of split and sets W of two and three vertices, the smallest ratio of $\Pr_{z,\sigma}[a_w = 1]$ to $p_A(w)$ over all z and σ is 1.000000. For $6 \le n \le 9$ it enumerates all n^{n-2} trees and, for 18 balanced splits (random, a clique, a star, a complete bipartite graph, and for $n \le 7$ a split found by local search to make β large) and sets W of 1 to 3 vertices, checks the whole chain of Lemma 5.4,

$$\beta \le \Big(\sum_z \Pr(z)\beta(\mu_z)^2\Big)^{1/2} \le \Big(\sum_z \Pr(z)\|\sqrt{\mu_z^{\mathrm{bits}}}\|_{\mathrm{op}}^2\Big)^{1/2} \le \Big(\mathbb{E}\prod_{w\in I}\Lambda(p_w)\Big)^{1/2} \le \Big(\mathbb{E}\,\bar{\Lambda}^{|I|}\Big)^{1/2} \le 1,$$

where $\bar{\Lambda}$ is the mean of the $\Lambda(p_w)$, together with the same chain for each z separately and the forbidden cell. The values of $-\ln\beta/n$ lie between 0.061 and 0.147. It checks Proposition 5.10 for $h = 3, 4, 5$ (the tree count, P_1, $\lambda_2 = 2$ and balance), Lemma 5.7 on 150 random instances on five small graphs, and the arithmetic of Corollary 5.9. For the quasi-star split it computes the Bhattacharyya coefficient with the product of the marginals exactly for $n = 6, 7$, where it is below β, and by 300 Prüfer samples at $n = 40$ and $n = 160$ (Remark 5.6(c); the sampled values are not certified). Finally it checks that the recursion of Proposition 5.11 generates

every labelled tree exactly once for $n \le 7$ and counts n^{n-2} trees for $n \le 12$, and prints its size, 95300245 nodes and wires for $n = 12$, below $n^4 3^n$.

`check_edge_pairs.py` Lemma 3.6 and the edge-pair results of Sections 11.4 and 11.5 (NumPy, SciPy, NetworkX). It checks the norm formula, the bound $\Delta \le \frac{1}{2}$, item (a) and the covariance identity and bounds of item (b) on 20000 random kernels, a fifth of them with a zero cell. For all 172 pairs of edges at a common vertex in seven small graphs, one with parallel edges, it checks the law of Lemma 11.13 against enumeration, the two lower bounds on Δ, and the resistance bound of Lemma 11.14. On five graphs with 10 to 12 vertices and three random splits each it checks the chain of Theorem 11.15 for up to four pairs: for each of 1069 pasts σ the conditional law of the two bits equals the law computed in K_σ, $\Delta \ge B_\sigma$, and β is below the norm of the square-root matrix of the pair bits, which is below the product of the factors $(1 - \Delta_k^2)^{1/2}$. For hub graphs it checks Lemma 11.10 on 40 random subdivisions with h computed exactly (smallest ratio to the bound 1), Proposition 11.11 on $G_{2,2}(2,2)$ and $G_{2,2}(2,3)$ (18 and 22 vertices, exact h), Lemma 11.8 at the hubs of $G_{L,2}(2,2)$ for $L \in \{2,3,5,8\}$, and Proposition 11.16 on $G_{65,13}(2,2)$ with random cubic cores ($N = 3393$, L the smallest value allowed), for 36 pasts at four hubs: the smallest Y_σ is $3.3 \cdot 10^{-3}$, above the bound $1.3 \cdot 10^{-3}$. It checks the example after Conjecture 13.1 on K_4 and $K_{3,3}$ with a triangle at each vertex (1296 and 59049 trees): $\beta = 1$, every covariance between an edge of A and an adjacent edge of B vanishes, and the stated expansion, degree and balance hold. Finally, as evidence for the conjecture only, it computes Y_σ for random regular graphs with $D \in \{3,4\}$ and $N \in \{200, 1000\}$, at up to 30 mixed vertices at pairwise distance at least 4, with sampled pasts; the smallest ratio to $(D-2)/(D(D-1))$ is 0.91, and the script fails if it drops below 0.81.

`check_matroid.py` Section 9 and the matroid part of Section 11, by enumerating bases (NumPy). The library is $M(K_4)$, $M(K_{3,3})$, $M(K_5)$, $M^*(K_{3,3})$, R_{10}, F_7, F_7^*, $AG(3,2)$, S_8 and $U_{2,4}$, with 6 to 162 bases. It checks Proposition 9.1 on 60 splits (largest difference $4 \cdot 10^{-16}$); Lemma 9.3 on 4879 pairs of a cut and an element, in each matroid and its dual (smallest value of $\lambda(e)|D|$ is 1.5); Lemma 9.6 for random framed circuits and cocircuits, random S and random F avoiding the frames (1273 conditional probabilities, smallest ratio to the bound 1.07); Theorem 9.7 with greedily chosen compatible pieces on 80 splits; Lemma 9.10 exhaustively for $|F|, |S| \le 2$, with no violation in the eight balanced matroids and 4 violations in 1624 cases for S_8, worst ratio 14/15; the positively correlated pair of S_8 ($576 > 560$); Lemma 9.12 for totally unimodular representations of R_{10} and $M(K_4)$ (largest error $2.4 \cdot 10^{-15}$, including coloop cases); Theorem 9.11 on 30 splits of regular matroids; and $\beta^2 = \max_j \Pr[|B \cap X| = j]$ for $U_{k,2k}$, $3 \le k \le 7$. For the Fano-decorated K_3 (18 elements, 9216 bases) it checks the facts (F1)–(F3) for all choices of lines and cocircuits, the six-point colouring fact used for local pieces, all local pieces with their frames, the lifted triangle with its frames (weight 24^{-3}, which equals the bound $(12D)^{-\ell}$ there), and Theorem 9.7 on six splits. The counting step of Proposition 9.9 is not checked.

`check_signed.py` Section 10, by enumerating spanning trees (NumPy, NetworkX) on $K_{3,3}$, the cube, the Petersen graph, the prisms over C_5, C_6 and C_8 and the Möbius ladder with 12 vertices (81 to 150528 trees), with three random balanced splits per graph (one for the prism over C_8) and a greedy family of 1 to 4 vertex-disjoint mixed cycles of length at most 6. It checks that the submatrix in the proof of Theorem 10.3 is a permutation matrix and that $\operatorname{rank} M_{A,B}(P) \ge 2^m$ (observed ranks 10 to 519). With the cycles paired, it checks Lemma 10.5 on every block, Theorem 10.11 for the best approximations of ranks 1, 2, 4

and 8, and, for their normalizations, Lemma 10.7 and Theorem 10.12, and Lemma 10.13 in exact arithmetic, conditionally on the intersection of T with every cycle of the other pairs (smallest ratio to the bound 7.6; the conditioning is nontrivial only for the prism over C_8, which has two pairs). It checks the central inequality in the proof of Theorem 10.22, $\sum_x \sqrt{P(x)}\|d(x)\| \le \beta\sqrt{s}\sqrt{Z}$, for random amplitudes of rank at most $s \in \{1, 2, 4, 8\}$ and for truncated singular value decompositions of the square-root matrix, where it is attained at $s = 1$. It checks that the reduced Laplacian determinant is 1 on sampled spanning trees and 0 on random non-tree sets of $N - 1$ edges, and Lemma 10.6 on 3000 random instances, including its equality case.

`check_signed_restricted.py` Section 10.6 and the sign split (NumPy, NetworkX). On $K_{t,t}$ with $5 \le t \le 15$ it places the pairs of the proofs of Theorems 10.26 and 10.27: with the constants of Raz and Yehudayoff the restricted indicator is identically 0, while the constants of Lemma 10.24 give $\prod_i [y_i \ne z_i]$ with rank 2^m, also for pairs in a common row or column completed by spare rows and columns. On random cubic graphs with 20 to 60 vertices and 4 or 5 disjoint cycles it checks Lemma 10.24 for the indicator and for random signed support-exact values (rank 2^m in both cases), and that the restriction of the spanning-tree polynomial is $\prod_i (x_{e_i} + x_{f_i} + (|C_i| - 2)x_{e_i}x_{f_i})$. It checks Lemma 10.25 for a $K_{3,3}$ minor of the Petersen graph and eight random minors of random cubic graphs, and Lemma 10.28 on 840 pairs of a random aligned circuit with real constants and a node of its vtree. For Lemma 10.29 and the packing step of Theorem 10.30 it takes $r \le 4$ random vtrees on the edges of $K_{t,t}$, $t \in \{10, 16, 24\}$, and checks $|P| = |Q| \ge t^2 3^{-r}$, the separation of P and Q in every vtree, at least $t3^{-r}/4 - 1$ disjoint four-cycles, and full rank of the result when $m \le 6$. For the sign split after Theorem 10.38 it checks, on 40 random signed smooth decomposable circuits with real leaf functions, that $f = g - h$, $g + h = f_{C^+}$ and $g, h \ge 0$, with scopes unchanged and size at most $6|C|$; the first inequality in the proof of that theorem, $\mathrm{BC}(P, Q^+) \ge (1 - \varepsilon)\sqrt{\lambda/Z^+}$, on 20000 random instances with f_{C^+} replaced by any function at least $\max(f, 0)$; and the triangle, where $\mathrm{ST}_G(1, 1, 1) = 3$ while the indicator of spanning trees vanishes.

`check_signed_approx.py` The approximation results of Section 10 (NumPy, NetworkX). On 2879 random scaled matrices of ranks 1 to 8, some with zero rows, and on all sign patterns of 3×3 matrices of rank one, it checks Lemma 10.8 and both bounds of Lemma 10.9, including the equality case of r blocks $J_m/(rm^2)$. On 3000 random signed Q with $\sum Q = 1$ it checks the identity and both bounds of Lemma 10.7. On $K_{3,3}$, the cube, the Petersen and Heawood graphs, the Möbius–Kantor graph and random cubic graphs with 14 and 16 vertices, with three balanced splits each, it enumerates spanning trees and checks Lemmas 10.17, 10.18 and 10.19 for one-vertex gadgets: every block has 2^g trees, its rectangle is a permutation, the rectangles are disjoint, the conditional pendant bound holds for every configuration of the other stars, and the pair bound holds. On the graphs with at most 14 vertices, for truncated singular value decompositions of $M_{A,B}(P)$, with and without a random rank-one perturbation, it checks Theorem 10.10(b) and, after normalization, part (a) and Lemma 10.9 on every block (smallest slack 0.125). It checks the four-triangle example after Theorem 10.10, Proposition 10.42 on every cell of every block of prisms with square and hexagon gadgets, including the zero cells, and the slice ratios of Remark 10.40 on K_4 (smallest $513 \ge 33$).

`check_signed_packing.py` Theorems 10.36 and 10.37 (NumPy, SciPy, NetworkX). It checks the construction of Lemma 10.34 for all 29943 edge pairs of 170 random 2-connected graphs with at most 11 vertices, against brute force up to 9 vertices, and Lemmas 10.32, 10.33

and 10.35 (one gadget) against the exact edge expansion of small graphs. It runs the greedy packing of Theorem 10.36 on random cubic and 4-regular graphs, cubic graphs with pockets, truncated cubic graphs and prisms with 200 to 2400 vertices, for far-apart edge sets and for the sets of Lemma 10.29, and reports the gadget density and the connectivity and λ_2 of the remaining graph for separations 2, 4 and 6; these runs are numerical only, since the proof's separation is far larger. On small graphs with absorbed gadgets it enumerates spanning trees and checks the restriction of part (a) (rank 2^K), the pair blocks, and Lemma 10.18 with $\kappa' = 2\kappa$. It also evaluates the constants of Theorems 10.36 and 10.37 quoted after the latter.

The local lemma is loose by a factor of about 2 on these graphs, and the chain bound by much more: the proved factor per cycle, $1 - \Omega(D^{-\ell})$, is far closer to 1 than the observed one. This is the main reason for the small constants in Section 8. For the vertex-touch argument the per-vertex lemmas are close to tight, and the losses in $c(D, h)$ come from the worst-case steps of Theorem 4.13: the random deletion, the count of connected sets, and the spectral bound on resistances.

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